

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

$$S = \int_a^b f(x) dx$$

$$z = r \cdot \text{cis}(\theta)$$

INTERNATIONAL BACCALAUREATE

$$P(A|B) = P(A \cap B) / P(B)$$

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$${}^nC_r = n! / (r!(n-r)!)$$

$$V = \pi \cdot \int_a^b [f(x)]^2 dx$$

$$S_{\infty} = u_1 / (1 - r)$$

$$\Delta = b^2 - 4ac$$

$$a^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$$

COURSE
Mathematics AA HL

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Ultimate Handbook

ULTIMATE HANDBOOK

FORMULA SHEET · DERIVATIONS · WORKED EXAMPLES · EXAM MASTERY

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Chapter 1

Number and Algebra

1.1 Scientific Notation

Definition 1.1.1 (Scientific Notation / Standard Form). A positive real number is written in *scientific notation* (also called *standard form*) as

$$a \times 10^k,$$

where $1 \leq a < 10$ and $k \in \mathbb{Z}$. For negative numbers we write $-a \times 10^k$ with the same conditions on a and k .

Scientific Notation Conversion Rules

- **Large numbers** (≥ 1): count how many places the decimal point moves *left* to get a ; this is k .

$$345,000 = 3.45 \times 10^5$$

- **Small numbers** ($0 < x < 1$): count how many places the decimal point moves *right*; this is $-k$.

$$0.0072 = 7.2 \times 10^{-3}$$

- **Significant figures**: the value of a carries the required precision.

$$0.00506237 \approx 5.06 \times 10^{-3} \quad (3 \text{ s.f.})$$

Calculator vs. Non-Calculator Approach

- **Paper 1 (non-calculator)**: You must be comfortable converting by hand. Move the decimal point, count the shifts. Multiplication and division of numbers in scientific notation use the laws of indices: multiply the a -values and add the powers.
- **Paper 2 (calculator)**: Use the GDC. However, you must still write your answer in correct scientific notation on the paper. On the TI-Nspire, press **Ctrl + Enter** to evaluate, and the calculator will display the result.
- **IB warning**: never write “ 34.5×10^4 ” as your final answer. The coefficient *must* satisfy $1 \leq a < 10$.

Worked Example: Scientific Notation Multiplication

Question. The speed of light is approximately 3.00×10^8 m/s. How far does light travel in one year (a light-year), in metres? Give your answer in scientific notation to 3 significant figures.

Solution.

$$\begin{aligned}
 \text{Distance} &= c \times t \\
 &= (3.00 \times 10^8) \times (365.25 \times 24 \times 3600) \\
 &= 3.00 \times 10^8 \times 31,557,600 \\
 &= 3.00 \times 10^8 \times 3.15576 \times 10^7 \\
 &= 9.46728 \times 10^{15} \\
 &\approx 9.47 \times 10^{15} \text{ m} \quad (3 \text{ s.f.})
 \end{aligned}$$

Worked Example: Converting Very Small Numbers

Question. The charge of an electron is approximately 0.000,000,000,000,000,16 coulombs. Express this in scientific notation.

Solution.

$$0.000,000,000,000,000,16 = 1.6 \times 10^{-19} \text{ C}$$

The decimal point moved 19 places to the right, so $k = -19$.

1.2 Arithmetic Sequences and Series

Definition 1.2.1 (Arithmetic Sequence). An **arithmetic sequence** (or *arithmetic progression*) is a sequence in which each term differs from the previous term by a constant called the **common difference** d . That is,

$$u_{n+1} - u_n = d \quad \text{for all } n \geq 1.$$

The sequence takes the form $u_1, u_1 + d, u_1 + 2d, u_1 + 3d, \dots$

Arithmetic Sequence: n th Term

$$u_n = u_1 + (n - 1)d$$

where u_1 is the first term, d is the common difference, and $n \in \mathbb{N}$.

Derivation of the n th Term Formula

Starting from the recursive definition $u_{n+1} = u_n + d$:

$$\begin{aligned}
u_1 &= u_1 \\
u_2 &= u_1 + d \\
u_3 &= u_2 + d = u_1 + 2d \\
u_4 &= u_3 + d = u_1 + 3d \\
&\vdots \\
u_n &= u_1 + \underbrace{d + d + \cdots + d}_{n-1 \text{ times}} \\
&= u_1 + (n-1)d
\end{aligned}$$

We see that the coefficient of d is always one less than the index n .

Arithmetic Series: Sum of First n Terms

$$S_n = \frac{n}{2}(2u_1 + (n-1)d) \quad (1.1)$$

$$= \frac{n}{2}(u_1 + u_n) \quad (1.2)$$

Derivation of the Sum Formula (Pairing Method)

Write out S_n forwards and backwards, then add:

$$\begin{aligned}
S_n &= u_1 + (u_1 + d) + (u_1 + 2d) + \cdots + (u_1 + (n-1)d) \\
S_n &= u_n + (u_n - d) + (u_n - 2d) + \cdots + (u_n - (n-1)d)
\end{aligned}$$

Adding the two rows term by term:

$$\begin{aligned}
2S_n &= (u_1 + u_n) + (u_1 + u_n) + \cdots + (u_1 + u_n) \\
&= n(u_1 + u_n)
\end{aligned}$$

Therefore

$$S_n = \frac{n}{2}(u_1 + u_n).$$

Substituting $u_n = u_1 + (n-1)d$ into the second factor gives the alternative form:

$$S_n = \frac{n}{2}(2u_1 + (n-1)d).$$

Useful Sigma Notation Manipulations

Let $\sum$ denote summation. The following properties are frequently tested on IB exams:

$$\sum_{r=1}^n k = kn \quad (\text{constant out of the sum})$$

$$\sum_{r=1}^n (ar^2 + br + c) = a \sum_{r=1}^n r^2 + b \sum_{r=1}^n r + c \sum_{r=1}^n 1 \quad (\text{linearity})$$

$$\sum_{r=1}^n f(r) = \sum_{r=1}^m f(r) + \sum_{r=m+1}^n f(r) \quad (\text{splitting the sum})$$

$$\sum_{r=k+1}^n f(r) = \sum_{r=1}^n f(r) - \sum_{r=1}^k f(r) \quad (\text{shifting/lowering the index})$$

Sigma Notation Exam Strategy

Common IB trick: You may be asked to find $\sum_{r=21}^{50} u_r$ for an arithmetic sequence. Use the splitting property:

$$\sum_{r=21}^{50} u_r = \sum_{r=1}^{50} u_r - \sum_{r=1}^{20} u_r = S_{50} - S_{20}.$$

Worked Example: Arithmetic Series with Sigma Notation

Question. An arithmetic sequence has first term 5 and common difference 3. Find $\sum_{r=16}^{40} u_r$.

Solution.

$$\begin{aligned} \sum_{r=16}^{40} u_r &= \sum_{r=1}^{40} u_r - \sum_{r=1}^{15} u_r \\ S_{40} &= \frac{40}{2}(2(5) + (40-1)(3)) = 20(10 + 117) = 20 \times 127 = 2540 \\ S_{15} &= \frac{15}{2}(2(5) + (15-1)(3)) = \frac{15}{2}(10 + 42) = \frac{15 \times 52}{2} = 390 \\ \sum_{r=16}^{40} u_r &= 2540 - 390 = 2150 \end{aligned}$$

Worked Example: Using Both Forms of S_n

Question. The sum of the first 20 terms of an arithmetic sequence is 440, and the sum of the first 40 terms is 1700. Find u_1 and d .

Solution. Using $S_n = \frac{n}{2}(2u_1 + (n-1)d)$:

$$S_{20} = \frac{20}{2}(2u_1 + 19d) = 440 \implies 2u_1 + 19d = 44 \quad (1.3)$$

$$S_{40} = \frac{40}{2}(2u_1 + 39d) = 1700 \implies 2u_1 + 39d = 85 \quad (1.4)$$

Subtracting (1.3) from (1.4):

$$20d = 41 \implies d = 2.05.$$

Substituting back: $2u_1 + 19(2.05) = 44$, so $2u_1 + 38.95 = 44$, giving $2u_1 = 5.05$, hence $u_1 = 2.525$.

1.3 Geometric Sequences and Series

Definition 1.3.1 (Geometric Sequence). A **geometric sequence** (or *geometric progression*) is a sequence in which the ratio of consecutive terms is constant. This constant is called the **common ratio** r :

$$\frac{u_{n+1}}{u_n} = r \quad \text{for all } n \geq 1.$$

Geometric Sequence: n th Term

$$u_n = u_1 r^{n-1}$$

Geometric Series: Sum of First n Terms ($r \neq 1$)

$$S_n = \frac{u_1(r^n - 1)}{r - 1} = \frac{u_1(1 - r^n)}{1 - r}$$

Both forms are equivalent; choose the one with the positive denominator to minimise sign errors.

Derivation of the Geometric Sum Formula

Write out S_n and multiply by r :

$$\begin{aligned} S_n &= u_1 + u_1 r + u_1 r^2 + \cdots + u_1 r^{n-1} \\ rS_n &= \quad u_1 r + u_1 r^2 + \cdots + u_1 r^{n-1} + u_1 r^n \end{aligned}$$

Subtracting the second line from the first (most terms cancel):

$$\begin{aligned} S_n - rS_n &= u_1 - u_1 r^n \\ S_n(1 - r) &= u_1(1 - r^n) \\ S_n &= \frac{u_1(1 - r^n)}{1 - r} \end{aligned}$$

Infinite Geometric Series

If $|r| < 1$, then $r^n \rightarrow 0$ as $n \rightarrow \infty$, and the sum to infinity is

$$S_\infty = \frac{u_1}{1 - r}$$

The series **converges** if and only if $|r| < 1$. If $|r| \geq 1$, the series **diverges** (the partial sums grow without bound or oscillate).

Proof of Convergence: $r^n \rightarrow 0$ when $|r| < 1$ **Case 1:** $0 < r < 1$.Write $r = \frac{1}{1+h}$ for some $h > 0$. Then by the binomial theorem (or Bernoulli's inequality):

$$r^n = \frac{1}{(1+h)^n} \leq \frac{1}{1+nh}.$$

As $n \rightarrow \infty$, the denominator $1+nh \rightarrow \infty$, so $r^n \rightarrow 0$.**Case 2:** $-1 < r < 0$.Then $|r| < 1$, so $|r^n| = |r|^n \rightarrow 0$ by Case 1, which implies $r^n \rightarrow 0$.**Divergence when $|r| \geq 1$:**

- If $r = 1$: $S_n = nu_1 \rightarrow \pm\infty$ (unless $u_1 = 0$).
- If $r = -1$: S_n oscillates between u_1 and 0, never settling.
- If $r > 1$: $r^n \rightarrow \infty$, so $|S_n| \rightarrow \infty$.
- If $r < -1$: $|r^n| \rightarrow \infty$ with alternating sign, so S_n diverges by oscillation with increasing magnitude.

IB Exam Tip: Checking Convergence

Always state the condition $|r| < 1$ before using S_∞ . If the question asks “find the sum to infinity” and $|r| \geq 1$, you must write “the series diverges; no finite sum exists.” Marks are awarded for recognising convergence.

Worked Example: Geometric Series with Sum to Infinity**Question.** A geometric sequence has first term 8 and second term -4 . Find the sum to infinity.**Solution.**

$$r = \frac{u_2}{u_1} = \frac{-4}{8} = -0.5.$$

Since $|r| = 0.5 < 1$, the series converges.

$$S_\infty = \frac{u_1}{1-r} = \frac{8}{1-(-0.5)} = \frac{8}{1.5} = \frac{16}{3}.$$

Worked Example: Finding n in a Geometric Sum**Question.** The sum of the first n terms of a geometric sequence with $u_1 = 3$ and $r = 2$ is 3069. Find n .**Solution.**

$$S_n = \frac{3(2^n - 1)}{2 - 1} = 3069$$

$$2^n - 1 = 1029$$

$$2^n = 1028.$$

Since $2^{10} = 1024 \neq 1028$, we check: $n = \frac{\ln 1028}{\ln 2} \approx 10.003$. For integer n this does not work, so we recheck. Actually:

$$\frac{3(2^n - 1)}{1} = 3069 \implies 2^n = 1029 + 1 = 1030? \quad \text{No.}$$

Let us redo: $3(2^n - 1) = 3069$, so $2^n - 1 = 1023$, thus $2^n = 1024 = 2^{10}$, giving $n = 10$.

Worked Example: Mixed Arithmetic–Geometric

Question. Three numbers a, b, c form both an arithmetic sequence and a geometric sequence. Prove that $a = b = c$.

Solution. Since a, b, c are arithmetic: $2b = a + c$.

Since a, b, c are geometric: $b^2 = ac$.

From the arithmetic condition, $a + c = 2b$. Squaring:

$$(a + c)^2 = 4b^2 \implies a^2 + 2ac + c^2 = 4ac \implies a^2 - 2ac + c^2 = 0 \implies (a - c)^2 = 0.$$

Hence $a = c$, and substituting back, $2b = 2a$, so $b = a$. Therefore $a = b = c$.

1.4 Financial Mathematics

Compound Interest

When a principal P is invested at an annual interest rate of r (as a decimal), compounded n times per year for t years, the accumulated amount is

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

For annual compounding ($n = 1$):

$$A = P(1 + r)^t$$

Compound Interest as a Geometric Sequence

Starting with principal P and annual rate r (compounded yearly):

$$\text{After 1 year: } A_1 = P + Pr = P(1 + r)$$

$$\text{After 2 years: } A_2 = A_1(1 + r) = P(1 + r)^2$$

$$\text{After 3 years: } A_3 = A_2(1 + r) = P(1 + r)^3$$

$$\vdots$$

$$\text{After } t \text{ years: } A_t = P(1 + r)^t$$

This is a geometric sequence with first term $P(1 + r)$ and common ratio $(1 + r)$, but evaluated at the t -th term starting from P . Each year, the amount is multiplied by the factor $(1 + r)$.

Annuity: Present Value

An **annuity** is a series of equal periodic payments FV (future value of each payment). The **present value** PV of an annuity paying $\$PMT$ per period for n periods at periodic rate i is

$$PV = \frac{PMT}{i} (1 - (1 + i)^{-n})$$

Annuity: Future Value

The **future value** (accumulated amount) is

$$FV = \frac{PMT}{i} ((1 + i)^n - 1)$$

Derivation of the Annuity Formulas

Consider an annuity with payment PMT at the *end* of each period, interest rate i per period, for n periods.

Future Value derivation. The first payment earns interest for $n - 1$ periods, the second for $n - 2$ periods, etc.:

$$FV = PMT(1 + i)^{n-1} + PMT(1 + i)^{n-2} + \cdots + PMT(1 + i) + PMT.$$

This is a geometric series with first term PMT and common ratio $(1 + i)$:

$$FV = PMT \cdot \frac{(1 + i)^n - 1}{(1 + i) - 1} = \frac{PMT}{i} ((1 + i)^n - 1).$$

Present Value derivation. Discount each payment back to time 0:

$$PV = \frac{PMT}{1 + i} + \frac{PMT}{(1 + i)^2} + \cdots + \frac{PMT}{(1 + i)^n}.$$

This is a geometric series with first term $\frac{PMT}{1 + i}$ and common ratio $\frac{1}{1 + i}$:

$$PV = \frac{PMT}{1 + i} \cdot \frac{1 - \left(\frac{1}{1 + i}\right)^n}{1 - \frac{1}{1 + i}} = \frac{PMT}{1 + i} \cdot \frac{1 - (1 + i)^{-n}}{\frac{i}{1 + i}} = \frac{PMT}{i} (1 - (1 + i)^{-n}).$$

Loan Amortization: Monthly Payment

For a loan of principal L at annual rate r compounded monthly over N months, the monthly payment PMT is

$$PMT = \frac{L \cdot i}{1 - (1 + i)^{-N}}$$

where $i = \frac{r}{12}$ is the monthly interest rate.

Derivation of the Monthly Payment Formula

Setting $PV = L$ (the loan amount equals the present value of all payments):

$$L = \frac{PMT}{i} (1 - (1 + i)^{-N}).$$

Solving for PMT :

$$PMT = \frac{L \cdot i}{1 - (1 + i)^{-N}}.$$

Depreciation

Straight-line depreciation:

$$V_t = V_0 - dt$$

where V_0 is the initial value and d is the depreciation per period.

Compound (reducing-balance) depreciation:

$$V_t = V_0(1 - r)^t$$

where r is the depreciation rate per period. This is identical in form to compound interest but with a decreasing factor $(1 - r) < 1$.

IB Exam Rounding Rules for Financial Mathematics

- **NEVER round intermediate steps.** Store exact values on your GDC and round only the final answer.
- If the question asks for the answer “to the nearest dollar” or “to 2 decimal places,” apply rounding *only* to the final answer.
- Common trap: calculating the monthly payment, rounding it, then using the rounded value to find total interest paid. This gives a wrong answer.
- The IB markscheme will accept answers within a specified tolerance (typically the last digit may differ by 1 due to rounding).

Worked Example: Loan Repayment

Question. A mortgage of \$250,000 is taken out at an annual interest rate of 4.8%, compounded monthly, to be repaid over 25 years. Find the monthly payment and the total interest paid.

Solution.

$$L = 250000, \quad r = 0.048, \quad i = \frac{0.048}{12} = 0.004, \quad N = 25 \times 12 = 300.$$

$$PMT = \frac{250000 \times 0.004}{1 - (1.004)^{-300}} = \frac{1000}{1 - 0.30097\dots} = \frac{1000}{0.69903\dots} \approx \$1430.21$$

Total amount repaid: $300 \times 1430.21 = \$429,063$.

Total interest: $429,063 - 250,000 = \$179,063$.

Worked Example: Compound Depreciation

Question. A car is purchased for \$35,000 and depreciates at 15% per year. Find its value after 6 years and determine when its value falls below \$10,000.

Solution.

$$V_6 = 35000(1 - 0.15)^6 = 35000(0.85)^6 \approx 35000 \times 0.3771 \approx \$13,200.$$

For $V_t < 10000$:

$$\begin{aligned} 35000(0.85)^t &< 10000 \\ (0.85)^t &< \frac{2}{7} \\ t \ln(0.85) &< \ln(2/7) \\ t &> \frac{\ln(2/7)}{\ln(0.85)} \approx \frac{-1.2528}{-0.1625} \approx 7.71. \end{aligned}$$

So the value falls below \$10,000 after **8 complete years**.

1.5 Exponents and Logarithms

1.5.1 Exponent Laws

Fundamental Laws of Exponents

For real base $a > 0$ and real exponents m, n :

$$a^m \times a^n = a^{m+n} \quad (1.5)$$

$$\frac{a^m}{a^n} = a^{m-n} \quad (1.6)$$

$$(a^m)^n = a^{mn} \quad (1.7)$$

$$a^{-n} = \frac{1}{a^n} \quad (1.8)$$

$$a^{1/n} = \sqrt[n]{a} \quad (1.9)$$

$$a^{m/n} = (\sqrt[n]{a})^m = \sqrt[n]{a^m} \quad (1.10)$$

$$a^0 = 1 \quad (a \neq 0) \quad (1.11)$$

Rational Exponents Explained

We define $a^{1/n} = \sqrt[n]{a}$ for $a > 0$ and $n \in \mathbb{N}$. Then:

Fractional exponents: $a^{m/n} = (a^{1/n})^m$.

Verification. Using law (1.7):

$$(a^{1/n})^m = a^{m/n} = a^{(m)(1/n)}.$$

We can also verify by raising both sides of $a^{1/n} = \sqrt[n]{a}$ to the power n :

$$(a^{1/n})^n = a^{n/n} = a^1 = a = (\sqrt[n]{a})^n. \quad \checkmark$$

1.5.2 Logarithm Laws

Definition 1.5.1 (Logarithm). For $a > 0$, $a \neq 1$, the logarithm base a is defined by

$$\log_a x = y \iff a^y = x.$$

The function $\log_a x$ is the inverse of a^x .

Laws of Logarithms

For $a > 0$, $a \neq 1$, and $x, y > 0$:

$$\log_a(xy) = \log_a x + \log_a y \quad (1.12)$$

$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y \quad (1.13)$$

$$\log_a(x^m) = m \log_a x \quad (1.14)$$

$$\log_a 1 = 0 \quad (1.15)$$

$$\log_a a = 1 \quad (1.16)$$

Derivation of Logarithm Laws from Exponent Laws

Product law: Let $\log_a x = m$ and $\log_a y = n$. Then $a^m = x$ and $a^n = y$.

$$xy = a^m \cdot a^n = a^{m+n} \quad (\text{by (1.5)})$$

So $\log_a(xy) = m + n = \log_a x + \log_a y$.

Quotient law: Similarly, $\frac{x}{y} = \frac{a^m}{a^n} = a^{m-n}$ (by (1.6)).

So $\log_a(x/y) = m - n = \log_a x - \log_a y$.

Power law: Let $\log_a x = m$, so $a^m = x$. Then

$$x^k = (a^m)^k = a^{mk} \quad (\text{by (1.7)}).$$

So $\log_a(x^k) = mk = k \log_a x$.

Change of Base Formula

$$\log_b x = \frac{\log_a x}{\log_a b}$$

In practice, use natural logarithms or base 10:

$$\log_b x = \frac{\ln x}{\ln b} = \frac{\log_{10} x}{\log_{10} b}.$$

Proof of the Change of Base Formula

Let $y = \log_b x$. By definition, $b^y = x$.

Take $\log_a$ of both sides:

$$\log_a(b^y) = \log_a x.$$

Using the power law (1.14):

$$y \log_a b = \log_a x.$$

Therefore

$$y = \frac{\log_a x}{\log_a b},$$

which is $\log_b x = \frac{\log_a x}{\log_a b}$.

1.5.3 Solving Exponential and Logarithmic Equations

Strategy for Exponential Equations

1. Isolate the exponential term.
2. Take $\ln$ (or $\log$) of both sides.
3. Use the power law to bring the exponent down.
4. Solve the resulting linear (or polynomial) equation.
5. **Always check** that solutions satisfy the domain of the original equation.

Strategy for Logarithmic Equations

1. Combine all log terms using log laws (if on the same side).
2. Or exponentiate both sides to remove the logarithm.
3. Solve the resulting equation.
4. **Check all solutions in the original equation** — extraneous solutions can arise from squaring or from the domain restriction $\log_a x$ requires $x > 0$.

Worked Example: Exponential Equation

Question. Solve $3^{2x-1} = 5^{x+2}$. Give your answer to 3 significant figures.

Solution.

$$\begin{aligned}
 3^{2x-1} &= 5^{x+2} \\
 \ln(3^{2x-1}) &= \ln(5^{x+2}) \\
 (2x-1)\ln 3 &= (x+2)\ln 5 \\
 2x\ln 3 - \ln 3 &= x\ln 5 + 2\ln 5 \\
 2x\ln 3 - x\ln 5 &= \ln 3 + 2\ln 5 \\
 x(2\ln 3 - \ln 5) &= \ln 3 + 2\ln 5 \\
 x &= \frac{\ln 3 + 2\ln 5}{2\ln 3 - \ln 5} = \frac{\ln 3 + \ln 25}{\ln 9 - \ln 5} = \frac{\ln 75}{\ln(9/5)} \approx \frac{4.317}{0.5878} \approx 7.34 \quad (3 \text{ s.f.})
 \end{aligned}$$

Worked Example: Logarithmic Equation

Question. Solve $\log_2(x+3) + \log_2(x-1) = 4$.

Solution. Domain: $x+3 > 0$ and $x-1 > 0$, so $x > 1$.

$$\begin{aligned}
 \log_2((x+3)(x-1)) &= 4 \\
 (x+3)(x-1) &= 2^4 = 16 \\
 x^2 + 2x - 3 &= 16 \\
 x^2 + 2x - 19 &= 0 \\
 x &= \frac{-2 \pm \sqrt{4+76}}{2} = \frac{-2 \pm \sqrt{80}}{2} = -1 \pm 2\sqrt{5}.
 \end{aligned}$$

Since $x > 1$, we take $x = -1 + 2\sqrt{5} \approx 3.47$. The other root $-1 - 2\sqrt{5} \approx -5.47$ is rejected (fails the domain condition).

Worked Example: Hidden Quadratic in Exponent

Question. Solve $4^x + 2^{x+1} - 3 = 0$.

Solution. Note that $4^x = (2^2)^x = 2^{2x} = (2^x)^2$. Let $u = 2^x$:

$$\begin{aligned}u^2 + 2u - 3 &= 0 \\(u + 3)(u - 1) &= 0.\end{aligned}$$

Since $u = 2^x > 0$, we reject $u = -3$. So $u = 1$, giving $2^x = 1$, hence $x = 0$.

1.6 Proof

1.6.1 Direct Proof

Structure of a Direct Proof

1. **State** what you are going to prove clearly.
2. Start from known facts, definitions, or given conditions.
3. Use logical deductions (algebra, known theorems) to reach the conclusion.
4. End with a clear concluding statement such as “hence proved” or “as required.”

Worked Example: Direct Proof

Question. Prove that the product of two odd integers is odd.

Solution. Let $a = 2m + 1$ and $b = 2n + 1$ be two odd integers, where $m, n \in \mathbb{Z}$.
Then

$$ab = (2m + 1)(2n + 1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1.$$

Since $2mn + m + n \in \mathbb{Z}$, the product ab is of the form $2k + 1$ with $k \in \mathbb{Z}$, which is odd. ■

1.6.2 Proof by Contradiction

Structure of Proof by Contradiction

1. **Assume** the negation of the statement to be proved.
2. **Derive** a logical contradiction (something absurd or false).
3. **Conclude** that the assumption must be false, so the original statement is true.

Worked Example: Proof by Contradiction

Question. Prove that $\sqrt{2}$ is irrational.

Solution. Assume, for contradiction, that $\sqrt{2}$ is rational. Then

$$\sqrt{2} = \frac{p}{q},$$

where $p, q \in \mathbb{Z}^+$, $\gcd(p, q) = 1$ (the fraction is in lowest terms).

Squaring: $2 = \frac{p^2}{q^2}$, so $p^2 = 2q^2$.

This means p^2 is even, so p is even (since the square of an odd number is odd). Write $p = 2k$. Then $(2k)^2 = 2q^2$, so $4k^2 = 2q^2$, giving $q^2 = 2k^2$.

This means q^2 is even, so q is even.

But if both p and q are even, then $\gcd(p, q) \geq 2$, contradicting $\gcd(p, q) = 1$.

Therefore the assumption is false: $\sqrt{2}$ is irrational. ■

1.6.3 Counterexamples

How to Find and Present Counterexamples

- A **counterexample** is a single specific case that disproves a general statement.
- To disprove “for all x , $P(x)$,” find one x_0 such that $P(x_0)$ is false.
- **Present clearly:** state the proposed claim, give the counterexample, show the computation, and conclude.
- **IB warning:** a counterexample must be explicit. “There exists some value” is not enough — you must state the value.

Worked Example: Counterexample

Question. A student claims: “If n^2 is divisible by 4, then n is divisible by 4.” Disprove this statement.

Solution. Consider $n = 6$. Then $n^2 = 36$, and 36 is divisible by 4 (since $36/4 = 9$). However, 6 is *not* divisible by 4 (since $6/4 = 1.5$). This is a counterexample, so the claim is false.

1.6.4 Mathematical Induction

Structure of Proof by Mathematical Induction

To prove a statement $P(n)$ for all integers $n \geq n_0$:

Basis step. Verify $P(n_0)$ is true (usually $n_0 = 1$).

Inductive hypothesis.

Assume $P(k)$ is true for some arbitrary $k \geq n_0$.

Inductive step.

Using the hypothesis, prove $P(k + 1)$ is true.

Conclusion. By the principle of mathematical induction, $P(n)$ is true for all $n \geq n_0$.

IB Mark Allocation for Proofs

- **M1** (Method): Correctly setting up the proof structure (stating basis step, hypothesis, etc.).
- **A1** (Answer): Correct algebraic manipulation within each step.
- **R1** (Reasoning): The concluding statement “by mathematical induction” or “hence proved for all n .”

- **Common loss of marks:** Forgetting to state the inductive hypothesis explicitly; forgetting the concluding statement; algebraic errors in the inductive step.

Worked Example: Induction — Sum Formula

Question. Prove by induction that $\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \in \mathbb{N}$.

Solution.

Basis step ($n = 1$): $\sum_{r=1}^1 r^2 = 1$, and $\frac{1 \times 2 \times 3}{6} = 1$. ✓

Inductive hypothesis: Assume for some $k \geq 1$:

$$\sum_{r=1}^k r^2 = \frac{k(k+1)(2k+1)}{6}. \quad (\text{IH})$$

Inductive step: We must show

$$\sum_{r=1}^{k+1} r^2 = \frac{(k+1)(k+2)(2k+3)}{6}.$$

$$\begin{aligned} \sum_{r=1}^{k+1} r^2 &= \sum_{r=1}^k r^2 + (k+1)^2 \\ &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \quad (\text{by IH}) \\ &= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \\ &= \frac{(k+1)[k(2k+1) + 6(k+1)]}{6} \\ &= \frac{(k+1)(2k^2 + k + 6k + 6)}{6} \\ &= \frac{(k+1)(2k^2 + 7k + 6)}{6}. \end{aligned}$$

Factorising the quadratic: $2k^2 + 7k + 6 = (k+2)(2k+3)$. Therefore:

$$\sum_{r=1}^{k+1} r^2 = \frac{(k+1)(k+2)(2k+3)}{6}. \quad \checkmark$$

Conclusion. By the principle of mathematical induction, the formula holds for all $n \in \mathbb{N}$. ■

Worked Example: Induction — Divisibility

Question. Prove by induction that $3^{2n} - 1$ is divisible by 8 for all $n \in \mathbb{N}$.

Solution.

Basis step ($n = 1$): $3^2 - 1 = 9 - 1 = 8$, which is divisible by 8. ✓

Inductive hypothesis: Assume $3^{2k} - 1 = 8m$ for some integer m .

Inductive step: Consider $n = k + 1$:

$$\begin{aligned}
 3^{2(k+1)} - 1 &= 3^{2k+2} - 1 = 9 \cdot 3^{2k} - 1 \\
 &= 9 \cdot 3^{2k} - 9 + 8 \\
 &= 9(3^{2k} - 1) + 8 \\
 &= 9 \cdot 8m + 8 \quad (\text{by IH}) \\
 &= 8(9m + 1).
 \end{aligned}$$

Since $9m + 1 \in \mathbb{Z}$, $3^{2(k+1)} - 1$ is divisible by 8. ✓

Conclusion. By mathematical induction, $3^{2n} - 1$ is divisible by 8 for all $n \in \mathbb{N}$. ■

1.7 Binomial Theorem

Definition 1.7.1 (Binomial Coefficient). For non-negative integers n and r with $0 \leq r \leq n$:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

This counts the number of ways to choose r objects from n distinct objects.

The Binomial Theorem (Positive Integer n)

$$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$$

Expanding:

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \cdots + \binom{n}{n-1} a b^{n-1} + b^n.$$

General term:

$$T_{r+1} = \binom{n}{r} a^{n-r} b^r$$

is the term containing b^r (or equivalently, the $(r + 1)$ -th term).

Proof of the Binomial Theorem by Induction

Claim: $(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$ for all $n \in \mathbb{N}_0$.

Basis step ($n = 0$): $(a + b)^0 = 1$ and $\binom{0}{0} a^0 b^0 = 1$. ✓

Basis step ($n = 1$): $(a + b)^1 = a + b$ and $\binom{1}{0} a^1 b^0 + \binom{1}{1} a^0 b^1 = a + b$. ✓

Inductive hypothesis: Assume for some $k \geq 1$:

$$(a + b)^k = \sum_{r=0}^k \binom{k}{r} a^{k-r} b^r. \quad (\text{IH})$$

Inductive step:

$$\begin{aligned}
 (a+b)^{k+1} &= (a+b)(a+b)^k \\
 &= (a+b) \sum_{r=0}^k \binom{k}{r} a^{k-r} b^r \\
 &= \sum_{r=0}^k \binom{k}{r} a^{k+1-r} b^r + \sum_{r=0}^k \binom{k}{r} a^{k-r} b^{r+1}.
 \end{aligned}$$

Shift the index in the second sum (let $s = r + 1$, so $r = s - 1$):

$$\begin{aligned}
 &= \sum_{r=0}^k \binom{k}{r} a^{k+1-r} b^r + \sum_{s=1}^{k+1} \binom{k}{s-1} a^{k+1-s} b^s \\
 &= a^{k+1} + \sum_{r=1}^k \binom{k}{r} a^{k+1-r} b^r + \sum_{s=1}^k \binom{k}{s-1} a^{k+1-s} b^s + b^{k+1} \\
 &= a^{k+1} + \sum_{r=1}^k \left[\binom{k}{r} + \binom{k}{r-1} \right] a^{k+1-r} b^r + b^{k+1}.
 \end{aligned}$$

Using Pascal's identity $\binom{k}{r} + \binom{k}{r-1} = \binom{k+1}{r}$:

$$(a+b)^{k+1} = \binom{k+1}{0} a^{k+1} + \sum_{r=1}^k \binom{k+1}{r} a^{k+1-r} b^r + \binom{k+1}{k+1} b^{k+1} = \sum_{r=0}^{k+1} \binom{k+1}{r} a^{k+1-r} b^r. \quad \checkmark$$

Conclusion. By induction, the binomial theorem holds for all $n \in \mathbb{N}_0$. ■

Pascal's Triangle Connection

- Row n of Pascal's triangle gives the coefficients of $(a+b)^n$.
- The identity $\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$ is why adjacent entries in Pascal's triangle sum to the entry below.
- For small n (up to $n = 5$ or 6), writing out Pascal's triangle is faster than computing $\binom{n}{r}$ individually.

Extension: Binomial Expansion for Rational n

For $n \in \mathbb{Q}$ (not necessarily a positive integer) and $|x| < 1$:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \frac{n(n-1)(n-2)}{3!} x^3 + \cdots = \sum_{r=0}^{\infty} \binom{n}{r} x^r$$

where the generalised binomial coefficient is

$$\binom{n}{r} = \frac{n(n-1)(n-2) \cdots (n-r+1)}{r!}.$$

Validity: The expansion converges when $|x| < 1$. When n is a positive integer, it reduces to the finite expansion (the series terminates after $r = n$).

Binomial Coefficient Properties

$$\binom{n}{r} = \binom{n}{n-r} \quad (\text{Symmetry})$$

$$\sum_{r=0}^n \binom{n}{r} = 2^n \quad (\text{Sum of row})$$

$$\sum_{r=0}^n (-1)^r \binom{n}{r} = 0 \quad (\text{Alternating sum})$$

Proof of Sum of Row = 2^n

Set $a = 1$, $b = 1$ in the binomial theorem:

$$(1 + 1)^n = 2^n = \sum_{r=0}^n \binom{n}{r} 1^{n-r} 1^r = \sum_{r=0}^n \binom{n}{r}.$$

Similarly, for the alternating sum, set $a = 1$, $b = -1$:

$$(1 - 1)^n = 0 = \sum_{r=0}^n \binom{n}{r} 1^{n-r} (-1)^r = \sum_{r=0}^n (-1)^r \binom{n}{r}.$$

Worked Example: Finding a Specific Term

Question. Find the coefficient of x^3 in the expansion of $(2 - 3x)^7$.

Solution. The general term is

$$T_{r+1} = \binom{7}{r} (2)^{7-r} (-3x)^r = \binom{7}{r} 2^{7-r} (-3)^r x^r.$$

We need $r = 3$:

$$\text{Coefficient} = \binom{7}{3} 2^4 (-3)^3 = 35 \times 16 \times (-27) = -15120.$$

Worked Example: Infinite Binomial Expansion

Question. Find the coefficient of x^3 in the expansion of $\frac{1}{(1 - 2x)^3}$ up to and including the term in x^3 .

Solution. Write $\frac{1}{(1 - 2x)^3} = (1 - 2x)^{-3}$. Using the generalised binomial theorem with $n = -3$ and x replaced by $-2x$ (valid when $|-2x| < 1$, i.e. $|x| < 0.5$):

$$\begin{aligned} (1 + y)^{-3} &= 1 + (-3)y + \frac{(-3)(-4)}{2!} y^2 + \frac{(-3)(-4)(-5)}{3!} y^3 + \dots \\ &= 1 - 3y + 6y^2 - 10y^3 + \dots \end{aligned}$$

Substituting $y = -2x$:

$$\begin{aligned} (1 - 2x)^{-3} &= 1 - 3(-2x) + 6(-2x)^2 - 10(-2x)^3 + \dots \\ &= 1 + 6x + 24x^2 + 80x^3 + \dots \end{aligned}$$

The coefficient of x^3 is **80**.

Worked Example: Binomial Approximation

Question. Use the binomial expansion to approximate $\sqrt{1.02}$ to 5 decimal places.

Solution. $\sqrt{1.02} = (1.02)^{1/2} = (1 + 0.02)^{1/2}$.

$$\begin{aligned}(1+x)^{1/2} &= 1 + \frac{1}{2}x + \frac{\frac{1}{2} \cdot (-\frac{1}{2})}{2!}x^2 + \frac{\frac{1}{2} \cdot (-\frac{1}{2}) \cdot (-\frac{3}{2})}{3!}x^3 \\ &= 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \dots\end{aligned}$$

With $x = 0.02$:

$$\begin{aligned}(1.02)^{1/2} &\approx 1 + 0.01 - \frac{0.0004}{8} + \frac{0.000008}{16} \\ &= 1 + 0.01 - 0.00005 + 0.0000005 \\ &= 1.0099505 \\ &\approx 1.00995.\end{aligned}$$

1.8 Counting

Definition 1.8.1 (Factorial).

$$n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1, \quad n \in \mathbb{N}.$$

By convention, $0! = 1$.

Permutations

The number of ways to arrange r objects chosen from n distinct objects, **where order matters**:

$${}^n P_r = \frac{n!}{(n-r)!}$$

Equivalently written as $P(n, r)$.

Combinations

The number of ways to choose r objects from n distinct objects, **where order does not matter**:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Equivalently written as ${}^n C_r$ or $C(n, r)$.

Derivation of Permutation and Combination Formulas

Permutations. We have n distinct objects and r positions to fill.

- Position 1: n choices.
- Position 2: $n - 1$ remaining choices.
- Position 3: $n - 2$ remaining choices.
- ...
- Position r : $n - (r - 1) = n - r + 1$ remaining choices.

By the multiplication principle:

$${}^nP_r = n(n-1)(n-2)\cdots(n-r+1) = \frac{n!}{(n-r)!}.$$

Combinations. Each combination of r objects can be arranged in $r!$ different orders (permutations). Therefore:

$${}^nP_r = \binom{n}{r} \times r!$$

Solving:

$$\binom{n}{r} = \frac{{}^nP_r}{r!} = \frac{n!}{r!(n-r)!}.$$

Circular Arrangements

The number of distinct ways to arrange n distinct objects in a circle is

$$(n-1)!$$

Reason: Fix one person's position to break the rotational symmetry. The remaining $(n-1)$ people can be arranged in $(n-1)!$ ways.

With reflection equivalence (e.g. a necklace that can be flipped):

$$\frac{(n-1)!}{2} \quad \text{for } n \geq 3.$$

Arrangements with Restrictions

- **Items must be together:** Treat the group as a single “block.” If k items must sit together, treat them as one unit, giving $(n-k+1)!$ arrangements of the blocks, then multiply by $k!$ for internal arrangements of the block.

$$\text{Total} = (n-k+1)! \times k!$$

- **Items must NOT be together:** Use the complement rule.

$$(\text{Items not together}) = \text{Total arrangements} - (\text{Items together})$$

For two specific people not sitting together in a line of n : $n! - (n-1)! \times 2!$.

- **Items in specific positions:** Place the restricted items first, then arrange the rest.

IB Exam Strategies for Counting Problems

- **Identify** whether order matters (permutation) or not (combination).
- **Draw** a simple diagram or list small cases to verify your reasoning.
- **Look for** “at least” or “at most” phrasing — these often require summing cases or using complements.
- **Check** your answer makes sense: ${}^nC_r \leq {}^nP_r \leq n^r$.
- **Common pitfall:** confusing permutation and combination. Ask yourself: “if I swap two elements, is it a different arrangement?”

- For complex multi-step problems, use the **multiplication principle**: if step 1 has a options and step 2 has b options, the total is $a \times b$.

Worked Example: Committee Selection

Question. A committee of 5 people is to be selected from 7 men and 6 women. How many committees can be formed if the committee must contain at least 2 women?

Solution. Count by cases (number of women on the committee):

$$2 \text{ women, 3 men: } \binom{6}{2} \times \binom{7}{3} = 15 \times 35 = 525$$

$$3 \text{ women, 2 men: } \binom{6}{3} \times \binom{7}{2} = 20 \times 21 = 420$$

$$4 \text{ women, 1 man: } \binom{6}{4} \times \binom{7}{1} = 15 \times 7 = 105$$

$$5 \text{ women, 0 men: } \binom{6}{5} \times \binom{7}{0} = 6 \times 1 = 6$$

$$\text{Total} = 525 + 420 + 105 + 6 = 1056.$$

Alternative (complement method):

$$\text{Total: } \binom{13}{5} = 1287$$

$$0 \text{ women: } \binom{7}{5} = 21$$

$$1 \text{ woman: } \binom{6}{1} \binom{7}{4} = 6 \times 35 = 210$$

$$\text{Answer} = 1287 - 21 - 210 = 1056. \quad \checkmark$$

Worked Example: Circular Arrangement with Restrictions

Question. In how many ways can 5 boys and 3 girls sit around a circular table if two particular girls, Anna and Bella, must not sit next to each other?

Solution. Total circular arrangements of 8 people: $(8 - 1)! = 7! = 5040$.

Arrangements where Anna and Bella ARE together: Treat them as a block. We have 7 units (the block + 6 others). Circular arrangements: $(7 - 1)! = 6! = 720$. The block has $2! = 2$ internal orderings. So: $720 \times 2 = 1440$.

Answer:

$$5040 - 1440 = 3600.$$

Worked Example: Seating with Restrictions

Question. How many ways can the letters of the word ARRANGE be arranged?

Solution. The word ARRANGE has 7 letters with repetitions: A appears twice, R appears twice.

$$\text{Arrangements} = \frac{7!}{2! \times 2!} = \frac{5040}{4} = 1260.$$

1.9 Partial Fractions

Partial fractions decompose a rational function $\frac{P(x)}{Q(x)}$ (where $\deg P < \deg Q$) into a sum of simpler fractions. This technique is essential for integration.

Partial Fractions: Distinct Linear Factors

For

$$\frac{P(x)}{(x-a)(x-b)(x-c)},$$

write

$$\frac{P(x)}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}.$$

Partial Fractions: Repeated Linear Factors

For

$$\frac{P(x)}{(x-a)^2(x-b)},$$

write

$$\frac{P(x)}{(x-a)^2(x-b)} = \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}.$$

Important: Include *both* $\frac{A}{x-a}$ and $\frac{B}{(x-a)^2}$ — a common mistake is to omit the first term.

Partial Fractions: Irreducible Quadratic Factor

If $x^2 + bx + c$ has no real roots (discriminant < 0), it is irreducible. For

$$\frac{P(x)}{(x-a)(x^2+bx+c)},$$

write

$$\frac{P(x)}{(x-a)(x^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}.$$

Note: The numerator over the quadratic is a *linear* expression $Bx + C$, not just a constant.

Method for Finding Coefficients

1. **Multiply through** by the denominator to clear fractions.
2. **Substitute convenient values** of x (the roots of the denominator) to find some coefficients immediately.
3. **Equate coefficients** of like powers of x to set up simultaneous equations for the remaining coefficients.
4. **Check** your answer by recombining the partial fractions.

Connection to Integration

Partial fractions are primarily used in IB Math AA HL to integrate rational functions. For example:

$$\int \frac{1}{(x-1)(x+2)} dx = \int \frac{1/3}{x-1} dx + \int \frac{-1/3}{x+2} dx = \frac{1}{3} \ln|x-1| - \frac{1}{3} \ln|x+2| + C.$$

The **logarithm integration rule** $\int \frac{1}{ax+b} dx = \frac{1}{a} \ln |ax+b| + C$ applies directly to each partial fraction with a linear denominator.

Worked Example: Distinct Linear Factors

Question. Express $\frac{5x+7}{(x+1)(x-2)}$ in partial fractions.

Solution. Write

$$\frac{5x+7}{(x+1)(x-2)} = \frac{A}{x+1} + \frac{B}{x-2}.$$

Multiplying through by $(x+1)(x-2)$:

$$5x+7 = A(x-2) + B(x+1).$$

Substitute $x = -1$: $5(-1) + 7 = A(-3) + B(0)$, so $2 = -3A$, giving $A = -\frac{2}{3}$.

Substitute $x = 2$: $5(2) + 7 = A(0) + B(3)$, so $17 = 3B$, giving $B = \frac{17}{3}$.

$$\frac{5x+7}{(x+1)(x-2)} = \frac{-2/3}{x+1} + \frac{17/3}{x-2}.$$

Worked Example: Repeated Factors

Question. Express $\frac{2x^2+3x+1}{(x-1)^2(x+1)}$ in partial fractions.

Solution. Write

$$\frac{2x^2+3x+1}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1}.$$

Multiplying through by $(x-1)^2(x+1)$:

$$2x^2+3x+1 = A(x-1)(x+1) + B(x+1) + C(x-1)^2.$$

Substitute $x = 1$: $2 + 3 + 1 = A(0) + B(2) + C(0)$, so $B = 3$.

Substitute $x = -1$: $2 - 3 + 1 = A(0) + B(0) + C(4)$, so $0 = 4C$, giving $C = 0$.

Equate coefficients of x^2 : $2 = A + C = A + 0$, so $A = 2$.

$$\frac{2x^2+3x+1}{(x-1)^2(x+1)} = \frac{2}{x-1} + \frac{3}{(x-1)^2}.$$

Worked Example: Improper Fraction — Long Division First

Question. Express $\frac{x^3+1}{x^2-1}$ in partial fractions.

Solution. Since the degree of the numerator (3) exceeds the degree of the denominator (2), first perform polynomial long division:

$$\frac{x^3+1}{x^2-1} = x + \frac{x+1}{x^2-1} = x + \frac{x+1}{(x-1)(x+1)} = x + \frac{1}{x-1}.$$

Note that $\frac{x+1}{(x-1)(x+1)} = \frac{1}{x-1}$ (after cancellation, valid for $x \neq -1$).

1.10 Complex Numbers

1.10.1 Definitions and Basic Operations

Definition 1.10.1 (Imaginary Unit). The **imaginary unit** i is defined by $i^2 = -1$. A **complex number** is a number of the form

$$z = a + bi,$$

where $a, b \in \mathbb{R}$. Here $a = \operatorname{Re}(z)$ is the real part and $b = \operatorname{Im}(z)$ is the imaginary part.

Definition 1.10.2 (Complex Conjugate). The **complex conjugate** of $z = a + bi$ is

$$\bar{z} = a - bi.$$

Definition 1.10.3 (Modulus). The **modulus** (absolute value) of $z = a + bi$ is

$$|z| = \sqrt{a^2 + b^2}.$$

Definition 1.10.4 (Argument). The **argument** of $z = a + bi$ is the angle $\theta = \arg(z)$ measured from the positive real axis to the line joining the origin to z , where

$$\theta = \arctan\left(\frac{b}{a}\right).$$

Quadrant awareness is essential. Use the signs of a and b to determine the correct quadrant:

- $a > 0, b > 0$: first quadrant, $\theta \in (0, \pi/2)$.
- $a < 0, b > 0$: second quadrant, $\theta \in (\pi/2, \pi)$.
- $a < 0, b < 0$: third quadrant, $\theta \in (-\pi, -\pi/2)$.
- $a > 0, b < 0$: fourth quadrant, $\theta \in (-\pi/2, 0)$.

The **principal argument** $\operatorname{Arg}(z)$ satisfies $-\pi < \operatorname{Arg}(z) \leq \pi$.

Complex Conjugate Properties

$$\begin{aligned} z\bar{z} &= |z|^2 = a^2 + b^2 \\ \overline{z+w} &= \bar{z} + \bar{w} \\ \overline{zw} &= \bar{z}\bar{w} \\ z + \bar{z} &= 2\operatorname{Re}(z) \\ z - \bar{z} &= 2i\operatorname{Im}(z) \end{aligned}$$

1.10.2 The Complex Plane (Argand Diagram)

Working with the Argand Diagram

- The horizontal axis represents $\operatorname{Re}(z)$ and the vertical axis represents $\operatorname{Im}(z)$.
- The modulus $|z|$ is the distance from the origin to the point representing z .
- The argument $\arg(z)$ is the angle from the positive real axis, measured anticlockwise (positive) or clockwise (negative).

- Geometric interpretation: $|z - w|$ is the distance between the points representing z and w on the Argand diagram.
- $|z - a| = r$ represents a circle with centre a and radius r .
- $\arg(z - a) = \theta$ represents a half-line from a at angle θ .

1.10.3 Polar and Exponential Forms

Polar Form

Any non-zero complex number $z = a + bi$ can be written as

$$z = r(\cos \theta + i \sin \theta) = r \operatorname{cis} \theta$$

where $r = |z| = \sqrt{a^2 + b^2}$ and $\theta = \arg(z)$.

Euler's Formula and Exponential Form

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Therefore the **Euler form** (exponential form) of z is

$$z = r e^{i\theta}$$

Connection: Euler's Formula via Taylor Series

Euler's formula can be understood by comparing the Taylor series expansions:

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \\ \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \end{aligned}$$

Substituting $x = i\theta$ into the series for e^x :

$$\begin{aligned} e^{i\theta} &= 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \cdots \\ &= 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \frac{i\theta^5}{5!} - \cdots \\ &= \underbrace{\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right)}_{=\cos \theta} + i \underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right)}_{=\sin \theta} \\ &= \cos \theta + i \sin \theta. \end{aligned}$$

This gives the celebrated identity $e^{i\theta} = \cos \theta + i \sin \theta$. Setting $\theta = \pi$ yields Euler's identity: $e^{i\pi} + 1 = 0$.

Key Identity: Euler's Identity

$$e^{i\pi} + 1 = 0$$

This connects five fundamental constants: e , i , π , 1 , and 0 .

1.10.4 De Moivre's Theorem

Theorem 1.10.1 (De Moivre's Theorem). *For any integer n and any complex number $z = r \operatorname{cis} \theta$:*

$$z^n = r^n \operatorname{cis}(n\theta) = r^n (\cos(n\theta) + i \sin(n\theta))$$

Equivalently, in Euler form: $(r e^{i\theta})^n = r^n e^{in\theta}$.

Proof of De Moivre's Theorem by Induction. Claim: For $z = r \operatorname{cis} \theta$ and all $n \in \mathbb{N}$, $z^n = r^n \operatorname{cis}(n\theta)$.

Basis step ($n = 1$): $z^1 = r \operatorname{cis} \theta = r^1 \operatorname{cis}(1 \cdot \theta)$. ✓

Inductive hypothesis: Assume $z^k = r^k \operatorname{cis}(k\theta)$ for some $k \geq 1$.

Inductive step:

$$\begin{aligned} z^{k+1} &= z^k \cdot z \\ &= r^k \operatorname{cis}(k\theta) \cdot r \operatorname{cis} \theta \quad (\text{by IH}) \\ &= r^{k+1} \operatorname{cis}(k\theta) \operatorname{cis}(\theta). \end{aligned}$$

Using the compound angle formulas:

$$\begin{aligned} \operatorname{cis}(k\theta) \operatorname{cis}(\theta) &= (\cos(k\theta) + i \sin(k\theta)) (\cos \theta + i \sin \theta) \\ &= \cos(k\theta) \cos \theta - \sin(k\theta) \sin \theta \\ &\quad + i(\sin(k\theta) \cos \theta + \cos(k\theta) \sin \theta) \\ &= \cos((k+1)\theta) + i \sin((k+1)\theta) \\ &= \operatorname{cis}((k+1)\theta). \end{aligned}$$

Therefore $z^{k+1} = r^{k+1} \operatorname{cis}((k+1)\theta)$. ✓

Conclusion. By mathematical induction, De Moivre's theorem holds for all $n \in \mathbb{N}$. The extension to negative integers follows since $z^{-n} = \frac{1}{z^n}$. □

1.10.5 Roots of Complex Numbers

Theorem 1.10.2 (Roots of a Complex Number). *The equation $z^n = w$, where $w = R \operatorname{cis} \alpha \neq 0$ and $n \in \mathbb{N}$, has exactly n distinct solutions:*

$$z_k = R^{1/n} \operatorname{cis}\left(\frac{\alpha + 2k\pi}{n}\right), \quad k = 0, 1, 2, \dots, n-1$$

*These n roots are **equally spaced** around a circle of radius $R^{1/n}$ centred at the origin, with consecutive roots separated by an angle of $\frac{2\pi}{n}$.*

Derivation of the Roots Formula

Let $z = r \operatorname{cis} \theta$ and $w = R \operatorname{cis} \alpha$. We want $z^n = w$.

By De Moivre's theorem: $r^n \operatorname{cis}(n\theta) = R \operatorname{cis} \alpha$.

Equating moduli: $r^n = R$, so $r = R^{1/n}$.

Equating arguments: $n\theta = \alpha + 2k\pi$ for $k \in \mathbb{Z}$ (since the argument is periodic with period 2π).

Therefore

$$\theta = \frac{\alpha + 2k\pi}{n}.$$

The distinct values are obtained for $k = 0, 1, 2, \dots, n-1$. For $k = n$, we get

$$\theta = \frac{\alpha + 2n\pi}{n} = \frac{\alpha}{n} + 2\pi,$$

which gives the same complex number as $k = 0$. So there are exactly n distinct roots.

Sum of Roots of Unity

The n th roots of unity (solutions to $z^n = 1$) are $\operatorname{cis}\left(\frac{2k\pi}{n}\right)$ for $k = 0, 1, \dots, n-1$. Their sum is:

$$\sum_{k=0}^{n-1} \operatorname{cis}\left(\frac{2k\pi}{n}\right) = 0.$$

This is because they form a geometric series with ratio $\operatorname{cis}(2\pi/n) \neq 1$, and their sum equals $\frac{1 - r^n}{1 - r} = \frac{1 - 1}{1 - r} = 0$.

Worked Example: Converting to Polar Form

Question. Express $z = -1 + \sqrt{3}i$ in polar and exponential form.

Solution.

$$r = |z| = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{1 + 3} = 2.$$

$$\theta = \arctan\left(\frac{\sqrt{3}}{-1}\right).$$

Since z is in the second quadrant ($a < 0$, $b > 0$): $\theta = \pi - \arctan(\sqrt{3}) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$.

$$z = 2 \operatorname{cis}\left(\frac{2\pi}{3}\right) = 2e^{2\pi i/3}.$$

Worked Example: De Moivre for Large Powers

Question. Find $(1 + i)^8$.

Solution. First convert $1 + i$ to polar form:

$$r = \sqrt{1^2 + 1^2} = \sqrt{2}, \quad \theta = \arctan(1) = \frac{\pi}{4}.$$

$$(1 + i)^8 = (\sqrt{2} \operatorname{cis}(\pi/4))^8 = (\sqrt{2})^8 \operatorname{cis}(8 \cdot \pi/4) = 16 \operatorname{cis}(2\pi) = 16(\cos 2\pi + i \sin 2\pi) = 16.$$

Worked Example: Finding Cube Roots

Question. Find all cube roots of $z = -8i$.

Solution. Express $w = -8i$ in polar form:

$$R = |-8i| = 8, \quad \alpha = \arg(-8i) = -\frac{\pi}{2}.$$

The cube roots are

$$z_k = 8^{1/3} \operatorname{cis}\left(\frac{-\pi/2 + 2k\pi}{3}\right) = 2 \operatorname{cis}\left(\frac{-\pi + 4k\pi}{6}\right),$$

for $k = 0, 1, 2$:

$$k = 0: \quad z_0 = 2 \operatorname{cis}\left(-\frac{\pi}{6}\right) = 2 \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = \sqrt{3} - i.$$

$$k = 1: \quad z_1 = 2 \operatorname{cis}\left(\frac{3\pi}{6}\right) = 2 \operatorname{cis}\left(\frac{\pi}{2}\right) = 2i.$$

$$k = 2: \quad z_2 = 2 \operatorname{cis}\left(\frac{7\pi}{6}\right) = 2 \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = -\sqrt{3} - i.$$

These three roots are equally spaced at 120° intervals on a circle of radius 2.

Worked Example: Geometric Locus on the Argand Diagram

Question. Sketch the locus of z satisfying $|z - 2 - i| = 3$ and find the maximum value of $\arg(z)$ for points on this locus.

Solution. The equation $|z - (2 + i)| = 3$ represents a circle with centre $2 + i$ (the point $(2, 1)$ on the Argand diagram) and radius 3.

The maximum argument occurs at the point on the circle where the tangent from the origin touches the circle in the upper half-plane. The argument of the centre is $\alpha = \arctan(1/2)$. The angle between the line from the origin to the centre and the tangent is $\beta = \arcsin(3/\sqrt{5})$, where $\sqrt{5}$ is the distance from the origin to the centre.

$$\max \arg(z) = \alpha + \beta = \arctan\left(\frac{1}{2}\right) + \arcsin\left(\frac{3}{\sqrt{5}}\right).$$

Since $\arcsin(3/\sqrt{5})$ requires $3/\sqrt{5} \leq 1$, i.e. $\sqrt{5} \geq 3$, but $\sqrt{5} \approx 2.236 < 3$. This means the origin lies *inside* the circle. In this case, the maximum argument is found from the tangent from the origin to the circle.

Actually, since the origin is at distance $\sqrt{2^2 + 1^2} = \sqrt{5} \approx 2.236$ from the centre, and the radius is $3 > \sqrt{5}$, the origin lies *inside* the circle. The maximum argument is achieved at the point where a line from the origin is tangent to the circle. Using the relationship:

$$\max \arg(z) = \arctan\left(\frac{1}{2}\right) + \arccos\left(\frac{\sqrt{5}}{3}\right).$$

IB Exam Tips for Complex Numbers

- **Polar form is essential** for multiplication, division, and powers. Always convert before applying De Moivre.
- **Cartesian form** is best for addition and subtraction.

- When asked for roots, always give *all* n roots and verify they are equally spaced.
- **Show working** for the argument — do not just write the calculator value. Show which quadrant z is in.
- For locus questions, clearly label the centre and radius, and indicate the relevant region.
- When simplifying $\frac{z}{w}$, multiply numerator and denominator by $\bar{w}$ to rationalise: $\frac{z}{w} = \frac{z\bar{w}}{w\bar{w}} = \frac{z\bar{w}}{|w|^2}$.
- **Common error:** confusing the argument of a product: $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$ (not the product of the arguments).

Chapter 2

Functions

Functions form the backbone of the IB Mathematics AA HL syllabus. Nearly every topic—from calculus to probability—relies on a solid understanding of functions. This chapter covers every function type examined at HL, from linear fundamentals through to solving inequalities with absolute values and rational expressions. Every formula is boxed, every key derivation is shown, and every subtopic includes worked examples in the IB exam style.

2.1 Function Fundamentals

Definition 2.1.1 (Function). A function f from a set A to a set B , written $f: A \rightarrow B$, is a rule that assigns to each element $x \in A$ **exactly one** element $f(x) \in B$.

- **Domain:** the set of all valid inputs, A .
- **Codomain:** the set B into which outputs are mapped.
- **Range (image):** the subset of B consisting of all actual outputs:

$$\text{Range}(f) = \{f(x) \mid x \in A\} \subseteq B$$

Definition 2.1.2 (Function Notation). The value of f at x is written $f(x)$. In applied contexts, common notations include:

- $v(t)$ — velocity as a function of time
- $s(t)$ — displacement as a function of time
- $P(n)$ — population as a function of n
- **Composite notation:** $(f \circ g)(x) = f(g(x))$

Vertical Line Test

A relation represented graphically is a function if and only if **every vertical line** intersects the graph **at most once**.

Definition 2.1.3 (One-to-One (Injective) Function). A function f is one-to-one (injective) if

$$f(x_1) = f(x_2) \implies x_1 = x_2$$

*Equivalently: distinct inputs produce distinct outputs. Graphically, **every horizontal line** intersects the graph at most once (horizontal line test).*

Definition 2.1.4 (Onto (Surjective) Function). A function $f: A \rightarrow B$ is onto (surjective) if every element of B is mapped to by at least one element of A :

$$\forall y \in B, \exists x \in A \text{ such that } f(x) = y$$

This means $\text{Range}(f) = B$.

Definition 2.1.5 (Bijective Function). A function that is **both injective and surjective** is called bijective (a one-to-one correspondence).

Example 1: Determining if a Relation is a Function

Let $f = \{(1, 3), (2, 5), (3, 5), (4, 7)\}$ and $g = \{(1, 2), (2, 3), (1, 4)\}$. Determine whether f and g are functions.

Solution:

f : each x -value $\{1, 2, 3, 4\}$ appears exactly once. Therefore f is a function. Since $f(2) = f(3) = 5$ while $2 \neq 3$, f is *not* one-to-one.

g : the input $x = 1$ is mapped to both 2 and 4. Therefore g is *not* a function.

Example 2: Finding Domain and Range

Find the natural domain and range of $f(x) = \sqrt{4 - x^2}$.

Solution:

For the square root to be defined (over $\mathbb{R}$), we require

$$4 - x^2 \geq 0 \implies x^2 \leq 4 \implies -2 \leq x \leq 2$$

Therefore, $\text{Domain}(f) = [-2, 2]$.

Since $x^2 \geq 0$ for all x , we have $4 - x^2 \leq 4$, so $f(x) = \sqrt{4 - x^2} \leq 2$. The minimum value is $f(2) = f(-2) = 0$. Therefore, $\text{Range}(f) = [0, 2]$.

2.2 Inverse and Composite Functions

2.2.1 Inverse Functions

Definition 2.2.1 (Inverse Function). The inverse of a function f , denoted f^{-1} , is the function that “undoes” f :

$$f^{-1}(f(x)) = x \quad \text{for all } x \in \text{Domain}(f)$$

$$f(f^{-1}(y)) = y \quad \text{for all } y \in \text{Range}(f)$$

Theorem 2.2.1 (Existence of Inverse). A function f has an inverse function f^{-1} if and only if f is **one-to-one** (injective). If f is not injective, we must **restrict its domain** to a subset on which it is injective.

Algebraic Method for Finding f^{-1}

Given $y = f(x)$:

1. Replace $f(x)$ with y .
2. **Swap** x and y .
3. Solve for y in terms of x .
4. Replace y with $f^{-1}(x)$.

5. State the domain of f^{-1} (equals the range of f).

Graphical Interpretation

The graph of $y = f^{-1}(x)$ is the **reflection** of the graph of $y = f(x)$ in the line $y = x$.

Key relationships:

- $\text{Domain}(f^{-1}) = \text{Range}(f)$
- $\text{Range}(f^{-1}) = \text{Domain}(f)$
- If (a, b) lies on f , then (b, a) lies on f^{-1}

Example 1: Finding an Inverse Function

Let $f(x) = \frac{2x+3}{x-1}$ for $x > 1$. Find f^{-1} and state its domain.

Solution:

Set $y = \frac{2x+3}{x-1}$.

Swap x and y :

$$x = \frac{2y+3}{y-1}$$

Solve for y :

$$x(y-1) = 2y+3$$

$$xy - x = 2y + 3$$

$$xy - 2y = x + 3$$

$$y(x-2) = x+3$$

$$y = \frac{x+3}{x-2}$$

Therefore $f^{-1}(x) = \frac{x+3}{x-2}$ with domain $x > 2$ (since the range of f is $y > 2$, which can be verified by noting $f(x) = 2 + \frac{5}{x-1}$).

Example 2: Restricting Domain for Invertibility

The function $f(x) = x^2 - 4x + 3$ does not have an inverse on its natural domain. Restrict the domain so that f^{-1} exists.

Solution:

Complete the square:

$$f(x) = (x-2)^2 - 1$$

The vertex is at $x = 2$. For f to be one-to-one, restrict to one side of the vertex:

- Option A: $\text{Domain}(f) = [2, \infty)$, then $\text{Range}(f) = [-1, \infty)$
- Option B: $\text{Domain}(f) = (-\infty, 2]$, then $\text{Range}(f) = [-1, \infty)$

Using Option A, set $y = (x - 2)^2 - 1$, swap and solve:

$$\begin{aligned}x &= (y - 2)^2 - 1 \\x + 1 &= (y - 2)^2 \\\sqrt{x + 1} &= y - 2 \quad (\text{since } y \geq 2) \\y &= 2 + \sqrt{x + 1}\end{aligned}$$

So $f^{-1}(x) = 2 + \sqrt{x + 1}$ with domain $[-1, \infty)$.

2.2.2 Composite Functions

Definition 2.2.2 (Composite Function). *Given functions f and g , the composite $f \circ g$ is defined by*

$$(f \circ g)(x) = f(g(x))$$

This means: first apply g , then apply f to the result.

Domain of a Composite Function

The domain of $(f \circ g)(x)$ is the set of all x in the domain of g such that $g(x)$ is in the domain of f :

$$\text{Domain}(f \circ g) = \{x \in \text{Domain}(g) \mid g(x) \in \text{Domain}(f)\}$$

IB Tip: “Hence” Questions

When a question says “hence find f^{-1} ” after asking you to find $(f \circ g)(x) = x$, it is telling you that $g = f^{-1}$. IB examiners frequently use this structure:

1. Part (a): find $(f \circ g)(x)$ and simplify to x .
2. Part (b): “Hence write down $f^{-1}(x)$.”

The answer to part (b) is simply $g(x)$.

Common IB Error

Composite functions are **not** commutative in general: $(f \circ g)(x) \neq (g \circ f)(x)$. Always check which function is applied first. Reading right to left: $f(g(x))$ means g first, then f .

Example 3: Composite Functions and Domain

Let $f(x) = \sqrt{x - 1}$ and $g(x) = x^2 + 3$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$, stating the domain of each.

Solution:

Part 1: $(f \circ g)(x) = f(g(x)) = f(x^2 + 3) = \sqrt{x^2 + 3 - 1} = \sqrt{x^2 + 2}$.

Domain: we need $g(x) \geq 1$ (the domain condition of f), i.e. $x^2 + 3 \geq 1$, i.e. $x^2 \geq -2$, which is always true. Since $\text{Domain}(g) = \mathbb{R}$, we have $\text{Domain}(f \circ g) = \mathbb{R}$.

Part 2: $(g \circ f)(x) = g(f(x)) = g(\sqrt{x - 1}) = (\sqrt{x - 1})^2 + 3 = x - 1 + 3 = x + 2$.

Domain: we need $x - 1 \geq 0$ (domain of f), so $\text{Domain}(g \circ f) = [1, \infty)$.

Note: The simplified expressions $\sqrt{x^2 + 2}$ and $x + 2$ are different functions, confirming $f \circ g \neq g \circ f$.

Example 4: “Hence” Inverse Question

Let $f(x) = 2x + 1$ and $g(x) = \frac{x-1}{2}$.

(a) Find $(f \circ g)(x)$.

(b) Hence write down $f^{-1}(x)$.

Solution:

(a)

$$(f \circ g)(x) = f\left(\frac{x-1}{2}\right) = 2\left(\frac{x-1}{2}\right) + 1 = (x-1) + 1 = x$$

(b) Since $(f \circ g)(x) = x$, we have $g = f^{-1}$, so

$$f^{-1}(x) = \frac{x-1}{2}$$

2.3 Linear Functions

Definition 2.3.1 (Linear Function). A linear function has the form

$$y = mx + c$$

where m is the **gradient** (slope) and c is the **y-intercept**.

Key Linear Formulas

Gradient (two-point form):

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Midpoint:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Distance:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Parallel lines: $m_1 = m_2$

Perpendicular lines: $m_1 \times m_2 = -1$

Derivation of the Perpendicular Gradient Condition

If line L_1 has gradient m and line L_2 is perpendicular to L_1 , then the angle between them is 90° . Using the tangent of the angle between two lines:

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

Setting $\theta = 90^\circ$, we have $\tan 90^\circ$ undefined, so the denominator must be zero:

$$1 + m_1 m_2 = 0 \implies m_1 m_2 = -1$$

Therefore, if $m_2 = -\frac{1}{m_1}$, the lines are perpendicular.

IB Tip: Coordinate Geometry

In Paper 2, coordinate geometry questions frequently combine the distance formula with gradients. Common question structures:

- “Find the equation of the perpendicular bisector of AB .” — find midpoint of AB , find gradient of AB , use negative reciprocal, substitute midpoint.
- “Show that ABC is a right-angled triangle.” — show two gradients multiply to -1 .

Example 1: Equation of a Perpendicular Bisector

The points $A(1, 3)$ and $B(7, -1)$ are given. Find the equation of the perpendicular bisector of AB .

Solution:

Step 1: Midpoint of AB :

$$M = \left(\frac{1+7}{2}, \frac{3+(-1)}{2} \right) = (4, 1)$$

Step 2: Gradient of AB :

$$m_{AB} = \frac{-1-3}{7-1} = \frac{-4}{6} = -\frac{2}{3}$$

Step 3: Gradient of perpendicular bisector:

$$m_{\perp} = -\frac{1}{m_{AB}} = -\frac{1}{-2/3} = \frac{3}{2}$$

Step 4: Equation through $M(4, 1)$:

$$y - 1 = \frac{3}{2}(x - 4) \implies y = \frac{3}{2}x - 5$$

Example 2: Right-Angled Triangle

Show that the triangle with vertices $A(0, 0)$, $B(4, 2)$, $C(1, 7)$ is right-angled.

Solution:

Compute the gradients:

$$m_{AB} = \frac{2-0}{4-0} = \frac{1}{2}$$

$$m_{AC} = \frac{7-0}{1-0} = 7$$

$$m_{BC} = \frac{7-2}{1-4} = \frac{5}{-3} = -\frac{5}{3}$$

Check products of pairs:

$$m_{AB} \times m_{AC} = \frac{1}{2} \times 7 = \frac{7}{2} \neq -1$$

$$m_{AB} \times m_{BC} = \frac{1}{2} \times \left(-\frac{5}{3}\right) = -\frac{5}{6} \neq -1$$

$$m_{AC} \times m_{BC} = 7 \times \left(-\frac{5}{3}\right) = -\frac{35}{3} \neq -1$$

None of the products equal -1 , so no two sides are perpendicular. Alternatively, use the Pythagorean theorem:

$$AB^2 = 4^2 + 2^2 = 20$$

$$AC^2 = 1^2 + 7^2 = 50$$

$$BC^2 = 3^2 + 5^2 = 34$$

Check: $20 + 34 = 54 \neq 50$. Since $AB^2 + BC^2 \neq AC^2$, the triangle is *not* right-angled at B . Checking $AB^2 + AC^2 = 70 \neq 34$ and $AC^2 + BC^2 = 84 \neq 20$, we confirm $\triangle ABC$ is **not** right-angled.

2.4 Quadratic Functions

Definition 2.4.1 (Quadratic Function). A quadratic function is a polynomial of degree 2. It can be expressed in **three forms**:

Three Forms of a Quadratic

1. Standard (general) form:

$$f(x) = ax^2 + bx + c, \quad a \neq 0$$

2. Factored (intercept) form:

$$f(x) = a(x - p)(x - q)$$

where p and q are the x -intercepts (roots).

3. Vertex form:

$$f(x) = a(x - h)^2 + k$$

where (h, k) is the vertex of the parabola.

2.4.1 Completing the Square

Completing the Square: Full Derivation

Start with $f(x) = ax^2 + bx + c$ where $a \neq 0$.

Step 1: Factor out a from the first two terms:

$$f(x) = a\left(x^2 + \frac{b}{a}x\right) + c$$

Step 2: Take half the coefficient of x and square it:

$$\left(\frac{b/a}{2}\right)^2 = \frac{b^2}{4a^2}$$

Step 3: Add and subtract this inside the brackets:

$$\begin{aligned}
 f(x) &= a \left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} - \frac{b^2}{4a^2} \right) + c \\
 &= a \left(x + \frac{b}{2a} \right)^2 - a \cdot \frac{b^2}{4a^2} + c \\
 &= a \left(x + \frac{b}{2a} \right)^2 - \frac{b^2}{4a} + c \\
 &= a \left(x + \frac{b}{2a} \right)^2 + \frac{4ac - b^2}{4a}
 \end{aligned}$$

Comparing with $f(x) = a(x - h)^2 + k$, we identify:

$$h = -\frac{b}{2a}, \quad k = \frac{4ac - b^2}{4a} = c - \frac{b^2}{4a}$$

2.4.2 The Discriminant

The Discriminant

For $f(x) = ax^2 + bx + c$, the *discriminant* is

$$\Delta = b^2 - 4ac$$

Theorem 2.4.1 (Discriminant Analysis). 1. $\Delta > 0$: Two distinct real roots. The parabola crosses the x -axis at two points.

2. $\Delta = 0$: One repeated real root (a double root). The parabola is tangent to the x -axis at the vertex.

3. $\Delta < 0$: No real roots. The parabola does not intersect the x -axis.

2.4.3 Quadratic Formula

Derivation of the Quadratic Formula

Start with the vertex form derived by completing the square:

$$a \left(x + \frac{b}{2a} \right)^2 = \frac{b^2 - 4ac}{4a}$$

Divide both sides by a :

$$\left(x + \frac{b}{2a} \right)^2 = \frac{b^2 - 4ac}{4a^2}$$

Take square roots:

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

Therefore:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Quadratic Formula

$$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2.4.4 Vieta's Formulas for Quadratics

Theorem 2.4.2 (Vieta's Formulas — Quadratic). If $ax^2 + bx + c = 0$ has roots α and β , then:

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

Proof of Vieta's Formulas

Starting from the factored form:

$$a(x - \alpha)(x - \beta) = 0$$

Expanding:

$$a(x^2 - (\alpha + \beta)x + \alpha\beta) = ax^2 - a(\alpha + \beta)x + a\alpha\beta = 0$$

Comparing with $ax^2 + bx + c = 0$:

$$-a(\alpha + \beta) = b \implies \alpha + \beta = -\frac{b}{a}$$

$$a\alpha\beta = c \implies \alpha\beta = \frac{c}{a}$$

IB Tip: Converting Between Forms

Standard to vertex: Complete the square (see derivation above).

Standard to factored: Factor or use the quadratic formula to find roots, then write $f(x) = a(x - \alpha)(x - \beta)$.

Factored to standard: Expand: $a(x - p)(x - q) = ax^2 - a(p + q)x + apq$.

Factored to vertex: Expand to standard, then complete the square. Or note that $h = \frac{p + q}{2}$ (midpoint of roots).

Vertex to standard: Expand: $a(x - h)^2 + k = ax^2 - 2ahx + (ah^2 + k)$.

Example 1: Discriminant and Nature of Roots

For $f(x) = 2x^2 - 6x + k$, find the values of k for which the equation $f(x) = 0$ has: **(a)** two distinct real roots, **(b)** a repeated root, **(c)** no real roots.

Solution:

$a = 2$, $b = -6$, $c = k$. The discriminant is:

$$\Delta = (-6)^2 - 4(2)(k) = 36 - 8k$$

(a) Two distinct roots: $\Delta > 0 \implies 36 - 8k > 0 \implies k < \frac{36}{8} = \frac{9}{2}$.

(b) Repeated root: $\Delta = 0 \implies 36 - 8k = 0 \implies k = \frac{9}{2}$.

(c) No real roots: $\Delta < 0 \implies 36 - 8k < 0 \implies k > \frac{9}{2}$.

Example 2: Completing the Square and Vertex

Express $f(x) = -3x^2 + 12x - 7$ in vertex form. Hence state the maximum value and the value of x at which it occurs.

Solution:

$$\begin{aligned} f(x) &= -3x^2 + 12x - 7 \\ &= -3(x^2 - 4x) - 7 \\ &= -3(x^2 - 4x + 4 - 4) - 7 \\ &= -3(x - 2)^2 + 12 - 7 \\ &= -3(x - 2)^2 + 5 \end{aligned}$$

Vertex form: $f(x) = -3(x - 2)^2 + 5$. Since $a = -3 < 0$, the parabola opens downward. The maximum value is $k = 5$, occurring at $x = h = 2$.

Example 3: Vieta's Formulas Application

The quadratic equation $2x^2 + 5x - 3 = 0$ has roots α and β . Without solving the equation, find: (a) $\alpha + \beta$, (b) $\alpha\beta$, (c) $\alpha^2 + \beta^2$, (d) $\frac{1}{\alpha} + \frac{1}{\beta}$.

Solution:

By Vieta's formulas: $\alpha + \beta = -\frac{5}{2}$ and $\alpha\beta = -\frac{3}{2}$.

(c)

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(-\frac{5}{2}\right)^2 - 2\left(-\frac{3}{2}\right) = \frac{25}{4} + 3 = \frac{37}{4}$$

(d)

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-5/2}{-3/2} = \frac{5}{3}$$

2.5 Polynomial Functions

Definition 2.5.1 (Polynomial Function). A polynomial function *has the form*

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where $n \in \mathbb{N}_0$ is the **degree**, $a_n \neq 0$ is the **leading coefficient**, and a_0 is the **constant term**.

2.5.1 Factor and Remainder Theorems

Theorem 2.5.1 (Factor Theorem). Let $P(x)$ be a polynomial. Then:

$$P(a) = 0 \iff (x - a) \text{ is a factor of } P(x)$$

Theorem 2.5.2 (Remainder Theorem). When a polynomial $P(x)$ is divided by $(x - a)$, the remainder is $P(a)$.

Proof of the Remainder Theorem. By the polynomial division algorithm, dividing $P(x)$ by $(x - a)$ gives:

$$P(x) = (x - a) \cdot Q(x) + R$$

where $Q(x)$ is the quotient and R is a constant (since $\deg(x - a) = 1$).

Substituting $x = a$:

$$P(a) = (a - a) \cdot Q(a) + R = 0 \cdot Q(a) + R = R$$

Therefore the remainder is $R = P(a)$. □

IB Tip: Using the Remainder Theorem

The remainder theorem is frequently tested in two ways:

1. “Find the remainder when $P(x)$ is divided by $(x - a)$ ” — simply evaluate $P(a)$.
2. “Find the value of k such that $(x - a)$ is a factor” — set $P(a) = 0$ and solve for the unknown.

2.5.2 Vieta's Formulas for Cubics and Quartics

Theorem 2.5.3 (Vieta's Formulas — Cubic). *If $ax^3 + bx^2 + cx + d = 0$ has roots α, β, γ , then:*

$$\begin{aligned}\alpha + \beta + \gamma &= -\frac{b}{a} \\ \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} \\ \alpha\beta\gamma &= -\frac{d}{a}\end{aligned}$$

Theorem 2.5.4 (Vieta's Formulas — Quartic). *If $ax^4 + bx^3 + cx^2 + dx + e = 0$ has roots $\alpha, \beta, \gamma, \delta$, then:*

$$\begin{aligned}\sum \alpha &= -\frac{b}{a} \\ \sum \alpha\beta &= \frac{c}{a} \\ \sum \alpha\beta\gamma &= -\frac{d}{a} \\ \alpha\beta\gamma\delta &= \frac{e}{a}\end{aligned}$$

where the sums are taken over all distinct combinations.

Proof of Vieta's Formulas for a Cubic

Start from the factored form:

$$a(x - \alpha)(x - \beta)(x - \gamma) = 0$$

Expanding step by step:

$$\begin{aligned}(x - \alpha)(x - \beta)(x - \gamma) &= (x^2 - (\alpha + \beta)x + \alpha\beta)(x - \gamma) \\ &= x^3 - \gamma x^2 - (\alpha + \beta)x^2 + (\alpha + \beta)\gamma x + \alpha\beta x - \alpha\beta\gamma \\ &= x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \alpha\gamma + \beta\gamma)x - \alpha\beta\gamma\end{aligned}$$

Multiplying by a and comparing with $ax^3 + bx^2 + cx + d = 0$:

$$\begin{aligned} -a(\alpha + \beta + \gamma) &= b \implies \alpha + \beta + \gamma = -\frac{b}{a} \\ a(\alpha\beta + \alpha\gamma + \beta\gamma) &= c \implies \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} \\ -a\alpha\beta\gamma &= d \implies \alpha\beta\gamma = -\frac{d}{a} \end{aligned}$$

The proof for quartics follows the same pattern by expanding $(x - \alpha)(x - \beta)(x - \gamma)(x - \delta)$.

2.5.3 Graph Behaviour of Polynomials

End Behaviour of Polynomials

For $P(x) = a_n x^n + \dots$:

- If n is **even** and $a_n > 0$: both ends go $\uparrow$
- If n is **even** and $a_n < 0$: both ends go $\downarrow$
- If n is **odd** and $a_n > 0$: left end $\downarrow$, right end $\uparrow$
- If n is **odd** and $a_n < 0$: left end $\uparrow$, right end $\downarrow$

A polynomial of degree n has at most $n - 1$ **turning points** and at most n real roots.

Example 1: Factor Theorem and Full Factorisation

Let $P(x) = x^3 - 3x^2 - 4x + 12$.

- Show that $(x - 3)$ is a factor of $P(x)$.
- Hence express $P(x)$ as a product of linear factors.
- Solve $P(x) = 0$.

Solution:

- $P(3) = 27 - 27 - 12 + 12 = 0$. By the factor theorem, $(x - 3)$ is a factor.
- Divide $P(x)$ by $(x - 3)$:

$$P(x) = (x - 3)(x^2 + 0x - 4) = (x - 3)(x^2 - 4) = (x - 3)(x - 2)(x + 2)$$

- $P(x) = 0 \implies (x - 3)(x - 2)(x + 2) = 0 \implies x = 3, x = 2, x = -2$.

Example 2: Finding an Unknown Coefficient

The polynomial $P(x) = 2x^3 + ax^2 - 5x + 3$ leaves a remainder of -1 when divided by $(x - 2)$, and $(x + 1)$ is a factor. Find a and the value of the other factor.

Solution:

Using the remainder theorem for $(x - 2)$:

$$\begin{aligned} P(2) &= 2(8) + a(4) - 5(2) + 3 = 16 + 4a - 10 + 3 = 9 + 4a = -1 \\ 4a &= -10 \implies a = -\frac{5}{2} \end{aligned}$$

Check the factor theorem for $(x + 1)$:

$$P(-1) = 2(-1) + \left(-\frac{5}{2}\right)(1) - 5(-1) + 3 = -2 - \frac{5}{2} + 5 + 3 = \frac{9}{2} \neq 0$$

This gives $P(-1) = 9/2 \neq 0$, meaning there is an inconsistency. The problem may require revisiting. However, if we proceed with $a = -5/2$, we perform polynomial long division of $P(x) = 2x^3 - \frac{5}{2}x^2 - 5x + 3$ by $(x + 1)$:

Using synthetic division with root -1 : coefficients are $2, -\frac{5}{2}, -5, 3$.

$$\text{Bring down } 2. \quad 2(-1) + (-\frac{5}{2}) = -\frac{9}{2}.$$

$$(-\frac{9}{2})(-1) + (-5) = \frac{9}{2} - 5 = -\frac{1}{2}.$$

$$(-\frac{1}{2})(-1) + 3 = \frac{7}{2} \neq 0.$$

Indeed $(x + 1)$ is not a factor with $a = -5/2$. This highlights that both conditions may not be simultaneously satisfiable in a single polynomial of the given form, which is itself a useful IB exercise in recognizing contradiction.

2.6 Rational Functions

Definition 2.6.1 (Rational Function). A rational function is a ratio of two polynomials:

$$f(x) = \frac{P(x)}{Q(x)}, \quad Q(x) \neq 0$$

Asymptotes of Rational Functions

Vertical asymptotes: Set $Q(x) = 0$. For each root $x = a$ of $Q(x)$ that does *not* cancel with $P(x)$, the line $x = a$ is a vertical asymptote.

Horizontal asymptotes (as $x \rightarrow \pm\infty$): Compare degrees $n = \deg P$ and $m = \deg Q$.

- $n < m$: horizontal asymptote at $y = 0$
- $n = m$: horizontal asymptote at $y = \frac{a_n}{b_m}$ (ratio of leading coefficients)
- $n > m$: no horizontal asymptote (possible oblique asymptote)

Oblique (slant) asymptote: If $\deg P = \deg Q + 1$, perform polynomial long division:

$$\frac{P(x)}{Q(x)} = \text{quotient} + \frac{\text{remainder}}{Q(x)}$$

As $x \rightarrow \pm\infty$, $\frac{\text{remainder}}{Q(x)} \rightarrow 0$, so $y = \text{quotient}$ is the oblique asymptote.

IB Tip: Sketching Rational Functions Systematically

Follow this order:

1. Find the domain (where $Q(x) \neq 0$).
2. Find vertical asymptotes.
3. Find horizontal/oblique asymptotes.
4. Find x - and y -intercepts.
5. Test the sign of $f(x)$ in each region between asymptotes and intercepts.
6. Sketch, showing behaviour near asymptotes.

Example 1: Sketching a Rational Function with Oblique Asymptote

Sketch the graph of $f(x) = \frac{x^2 + x - 6}{x - 2}$, showing all asymptotes and intercepts.

Solution:

Factor: $x^2 + x - 6 = (x + 3)(x - 2)$, so

$$f(x) = \frac{(x + 3)(x - 2)}{x - 2} = x + 3 \quad \text{for } x \neq 2$$

Domain: $x \neq 2$, i.e. $(-\infty, 2) \cup (2, \infty)$.

Vertical asymptote: Technically, $x = 2$ makes the denominator zero, but since the factor cancels, $x = 2$ is a *hole* (removable discontinuity), not an asymptote. The point is $(2, 5)$ (found by substituting or taking the limit).

Oblique asymptote: $y = x + 3$ (the simplified function itself).

x-intercept: $x + 3 = 0 \implies x = -3$, so $(-3, 0)$.

y-intercept: $f(0) = -3$, so $(0, -3)$.

The graph is the line $y = x + 3$ with a hole at $(2, 5)$.

Example 2: Full Rational Function Analysis

For $f(x) = \frac{2x^2 - x - 1}{x^2 - 4}$, find all asymptotes, intercepts, and sketch.

Solution:

Domain: $x^2 - 4 \neq 0 \implies x \neq \pm 2$.

Vertical asymptotes: Since $x^2 - 4 = (x - 2)(x + 2)$ does not factor out of the numerator, we have vertical asymptotes at $x = -2$ and $x = 2$.

Horizontal asymptote: $\deg P = \deg Q = 2$, so

$$y = \frac{2}{1} = 2$$

x-intercepts: $2x^2 - x - 1 = 0 \implies (2x + 1)(x - 1) = 0 \implies x = -\frac{1}{2}$ or $x = 1$.

y-intercept: $f(0) = \frac{-1}{-4} = \frac{1}{4}$.

Sign analysis: Test points in the intervals $(-\infty, -2)$, $(-2, -1/2)$, $(-1/2, 1)$, $(1, 2)$, $(2, \infty)$.

The graph approaches $y = 2$ as $x \rightarrow \pm\infty$, has vertical asymptotes at $x = \pm 2$, and passes through the three intercepts found above.

2.7 Reciprocal Functions

Definition 2.7.1 (Reciprocal Function). *The reciprocal function is*

$$f(x) = \frac{1}{x}, \quad x \neq 0$$

Properties of $f(x) = \frac{1}{x}$

- Domain: $x \in \mathbb{R} \setminus \{0\}$
- Range: $y \in \mathbb{R} \setminus \{0\}$
- Vertical asymptote: $x = 0$ (y -axis)

- Horizontal asymptote: $y = 0$ (x -axis)
- The function is **odd**: $f(-x) = -f(x)$

Theorem 2.7.1 (Self-Inverse Property). *The function $f(x) = \frac{1}{x}$ is its own inverse:*

$$f^{-1}(x) = f(x) = \frac{1}{x}$$

Proof. Set $y = \frac{1}{x}$. Swap x and y : $x = \frac{1}{y}$. Solving: $y = \frac{1}{x}$.

Therefore $f^{-1}(x) = \frac{1}{x} = f(x)$, so f is self-inverse. □

Properties of $f(x) = \frac{1}{x^2}$

- Domain: $x \in \mathbb{R} \setminus \{0\}$
- Range: $y \in (0, \infty)$
- Vertical asymptote: $x = 0$
- Horizontal asymptote: $y = 0$
- The function is **even**: $f(-x) = f(x)$
- The graph lies entirely above the x -axis with a “double branch” shape

Example 1: Transformations of $f(x) = \frac{1}{x}$

Describe the transformation and sketch $g(x) = \frac{2}{x-3} + 1$.

Solution:

Starting from $f(x) = \frac{1}{x}$:

1. $\frac{1}{x-3}$: translation of 3 units to the **right**
2. $\frac{2}{x-3}$: vertical stretch by factor 2
3. $\frac{2}{x-3} + 1$: translation of 1 unit **up**

Resulting features:

- Vertical asymptote: $x = 3$
- Horizontal asymptote: $y = 1$
- Domain: $x \neq 3$
- Range: $y \neq 1$

Example 2: Sketching $f(x) = \frac{1}{x^2}$ with Transformations

Sketch $h(x) = -\frac{1}{(x+1)^2} + 4$.

Solution:

Starting from $\frac{1}{x^2}$:

1. $\frac{1}{(x+1)^2}$: shift 1 unit **left** (vertical asymptote at $x = -1$)
2. $-\frac{1}{(x+1)^2}$: reflection in the x -axis (graph now below the x -axis)
3. $-\frac{1}{(x+1)^2} + 4$: shift 4 units **up**

Features:

- Vertical asymptote: $x = -1$
- Horizontal asymptote: $y = 4$
- Range: $y < 4$ (the branches approach $y = 4$ from below)
- y -intercept: $h(0) = -1 + 4 = 3$, so $(0, 3)$

2.8 Exponential and Logarithmic Functions

2.8.1 Exponential Functions

Definition 2.8.1 (Exponential Function). *An exponential function has the form*

$$f(x) = a^x, \quad a > 0, a \neq 1$$

The number $e \approx 2.71828$ is the natural base.

Properties of Exponential Functions

For $f(x) = a^x$ with $a > 0$, $a \neq 1$:

- Domain: $\mathbb{R}$; Range: $(0, \infty)$
- y -intercept: $(0, 1)$ always
- Horizontal asymptote: $y = 0$
- $a > 1$: strictly increasing (growth)
- $0 < a < 1$: strictly decreasing (decay)
- $a^m \times a^n = a^{m+n}$, $\frac{a^m}{a^n} = a^{m-n}$, $(a^m)^n = a^{mn}$

2.8.2 Logarithmic Functions

Definition 2.8.2 (Logarithmic Function). *The logarithmic function is the inverse of the exponential function:*

$$y = \log_a x \iff a^y = x, \quad a > 0, a \neq 1$$

The natural logarithm is $\ln x = \log_e x$.

Properties of Logarithms

$\log_a(xy) = \log_a x + \log_a y$	(product rule)
$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$	(quotient rule)
$\log_a(x^n) = n \log_a x$	(power rule)
$\log_a a = 1$	for any base a
$\log_a 1 = 0$	for any base a
$a^{\log_a x} = x$	(inverse property)
$\log_a(a^x) = x$	(inverse property)
$\log_a x = \frac{\ln x}{\ln a}$	(change of base)

Theorem 2.8.1 (Inverse Relationship). *The functions $f(x) = a^x$ and $g(x) = \log_a x$ are inverses of each other:*

$$\log_a(a^x) = x \quad \text{and} \quad a^{\log_a x} = x$$

Their graphs are reflections of each other in the line $y = x$.

Solving Exponential and Logarithmic Equations

Exponential form to logarithmic:

$$a^x = b \implies x = \log_a b$$

This requires $a > 0$, $a \neq 1$, $b > 0$.

Same base:

$$a^{f(x)} = a^{g(x)} \implies f(x) = g(x)$$

Taking logarithms:

$$a^x = b \implies x \ln a = \ln b \implies x = \frac{\ln b}{\ln a}$$

Exponential Growth and Decay Models

Growth:

$$P(t) = P_0 \cdot a^t \quad \text{where } a > 1$$

or equivalently

$$P(t) = P_0 e^{kt} \quad \text{where } k > 0$$

Decay:

$$P(t) = P_0 \cdot a^t \quad \text{where } 0 < a < 1$$

or equivalently

$$P(t) = P_0 e^{-kt} \quad \text{where } k > 0$$

Half-life: If $P(t) = P_0 \cdot a^t$ and the half-life is $t_{1/2}$, then

$$a = \left(\frac{1}{2}\right)^{1/t_{1/2}}$$

Example 1: Solving an Exponential Equation

Solve $3^{2x-1} = 7^{x+1}$. Give your answer in terms of logarithms.

Solution:

Take natural logarithms of both sides:

$$\begin{aligned}(2x - 1) \ln 3 &= (x + 1) \ln 7 \\ 2x \ln 3 - \ln 3 &= x \ln 7 + \ln 7 \\ 2x \ln 3 - x \ln 7 &= \ln 7 + \ln 3 \\ x(2 \ln 3 - \ln 7) &= \ln 7 + \ln 3 \\ x &= \frac{\ln 7 + \ln 3}{2 \ln 3 - \ln 7} = \frac{\ln 21}{\ln 9 - \ln 7} = \frac{\ln 21}{\ln(9/7)}\end{aligned}$$

Example 2: Exponential Decay and Half-Life

A radioactive substance decays according to $M(t) = M_0 e^{-0.05t}$, where $M(t)$ is the mass in grams after t years. Find the half-life.

Solution:

Half-life is when $M(t) = \frac{M_0}{2}$:

$$\begin{aligned}M_0 e^{-0.05t} &= \frac{M_0}{2} \\ e^{-0.05t} &= \frac{1}{2} \\ -0.05t &= \ln\left(\frac{1}{2}\right) = -\ln 2 \\ t &= \frac{\ln 2}{0.05} = 20 \ln 2 \approx 13.86 \text{ years}\end{aligned}$$

Example 3: Logarithmic Equation

Solve $\log_2(x + 3) + \log_2(x - 1) = 5$.

Solution:

Use the product rule:

$$\log_2((x + 3)(x - 1)) = 5$$

Convert to exponential form:

$$(x + 3)(x - 1) = 2^5 = 32$$

Expand:

$$\begin{aligned}x^2 + 2x - 3 &= 32 \implies x^2 + 2x - 35 = 0 \\ (x + 7)(x - 5) &= 0 \implies x = -7 \text{ or } x = 5\end{aligned}$$

Check against domain: $x + 3 > 0$ and $x - 1 > 0$ require $x > 1$. Therefore $x = -7$ is rejected. The solution is $x = 5$.

2.9 Absolute Value Functions

Definition 2.9.1 (Absolute Value).

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Transformations Involving Absolute Values

$y = |f(x)|$: Take the graph of $y = f(x)$ and *reflect any portion below the x -axis* above it. Parts already above or on the x -axis remain unchanged.

$y = f(|x|)$: Take the portion of $y = f(x)$ for $x \geq 0$ and *reflect it across the y -axis* to the left side. The result is an even function (symmetric about the y -axis).

Solving Absolute Value Equations and Inequalities

Equation $|f(x)| = g(x)$ (where $g(x) \geq 0$):

1. Solve $f(x) = g(x)$.
2. Solve $f(x) = -g(x)$.
3. Check all solutions in the original equation.

Inequality $|f(x)| > g(x)$:

1. Solve $f(x) > g(x)$.
2. Solve $f(x) < -g(x)$.
3. The solution set is the union.

Inequality $|f(x)| < g(x)$:

1. Solve $f(x) < g(x)$.
2. Solve $f(x) > -g(x)$.
3. The solution set is the intersection.

IB Tip: Graphical Method for Absolute Values

IB examiners often award full marks for solving absolute value problems **graphically**. Sketch both sides, and the solutions correspond to intersection points. For inequalities, identify which regions satisfy the condition visually. Always state “from the graph” to justify the graphical method.

Example 1: Solving $|f(x)| = g(x)$

Solve $|2x - 3| = x + 2$.

Solution:

Case 1: $2x - 3 = x + 2 \implies x = 5$.

Check: $|2(5) - 3| = |7| = 7$ and $5 + 2 = 7$. Valid.

Case 2: $2x - 3 = -(x + 2) \implies 2x - 3 = -x - 2 \implies 3x = 1 \implies x = \frac{1}{3}$.

Check: $|2/3 - 3| = |-7/3| = 7/3$ and $1/3 + 2 = 7/3$. Valid.

Since $x + 2 \geq 0$ requires $x \geq -2$, both solutions satisfy this.

Solution: $x = \frac{1}{3}$ or $x = 5$.

Example 2: Solving $|f(x)| > g(x)$

Solve $|x^2 - 4| > 5$.

Solution:

This gives two cases:

Case 1: $x^2 - 4 > 5 \implies x^2 > 9 \implies x > 3$ or $x < -3$.

Case 2: $x^2 - 4 < -5 \implies x^2 < -1$, which has no real solutions.

Solution: $x \in (-\infty, -3) \cup (3, \infty)$.

Example 3: Graph of $y = |f(x)|$ vs. $y = f(|x|)$

Let $f(x) = x^2 - 4x + 3$. Describe the key differences between $y = |f(x)|$ and $y = f(|x|)$.

Solution:

First, $f(x) = (x-1)(x-3)$. The roots are $x = 1$ and $x = 3$. The vertex is at $(2, -1)$.

For $y = |f(x)|$: The portion of the parabola below the x -axis (between $x = 1$ and $x = 3$) is reflected above. The vertex moves from $(2, -1)$ to $(2, 1)$. The graph is V-shaped at the bottom with “cusps” at $(1, 0)$ and $(3, 0)$.

For $y = f(|x|)$: For $x \geq 0$, the graph is unchanged. For $x < 0$, we use $f(-x) = (-x)^2 - 4(-x) + 3 = x^2 + 4x + 3 = (x+1)(x+3)$. The left branch has roots at $x = -1$ and $x = -3$ with vertex at $(-2, -1)$. The result is symmetric about the y -axis only if f is an even function, which it is not here.

2.10 Trigonometric Functions (Graphs)

Key Features of Basic Trigonometric Functions

$y = \sin x$:

- Domain: $\mathbb{R}$; Range: $[-1, 1]$
- Period: 2π ; Amplitude: 1
- x -intercepts: $x = n\pi, n \in \mathbb{Z}$
- y -intercept: $(0, 0)$

$y = \cos x$:

- Domain: $\mathbb{R}$; Range: $[-1, 1]$
- Period: 2π ; Amplitude: 1
- x -intercepts: $x = \frac{\pi}{2} + n\pi, n \in \mathbb{Z}$
- y -intercept: $(0, 1)$

$y = \tan x$:

- Domain: $x \neq \frac{\pi}{2} + n\pi, n \in \mathbb{Z}$; Range: $\mathbb{R}$
- Period: π ; No amplitude (unbounded)

- Vertical asymptotes: $x = \frac{\pi}{2} + n\pi$
- x -intercepts: $x = n\pi, n \in \mathbb{Z}$
- y -intercept: $(0, 0)$

General Trigonometric Function

$$f(x) = a \sin(bx + c) + d \quad \text{or} \quad f(x) = a \cos(bx + c) + d$$

- $|a| = \mathbf{amplitude}$ (vertical stretch from centre line)
- $\frac{2\pi}{|b|} = \mathbf{period}$
- $-\frac{c}{b} = \mathbf{phase\ shift}$ (horizontal translation)
- $d = \mathbf{vertical\ shift}$ (centre line is $y = d$)
- Range: $[d - |a|, d + |a|]$

For tangent: $f(x) = a \tan(bx + c) + d$ has period $\frac{\pi}{|b|}$ and range $\mathbb{R}$.

IB Tip: Phase Shift Confusion

A common IB error is confusing the phase shift. For $f(x) = a \sin(bx + c) + d$:

- The phase shift is $-\frac{c}{b}$, NOT $-c$.
- If written as $f(x) = a \sin\left(b\left(x - \frac{-c}{b}\right)\right) + d$, the shift is clearer.
- The IB may write $f(x) = a \sin(b(x - h)) + d$ directly, where h is the phase shift.

Example 1: Identifying Trigonometric Parameters

For $f(x) = 3 \sin(2x - \pi) + 1$, find the amplitude, period, phase shift, and range.

Solution:

Comparing with $f(x) = a \sin(bx + c) + d$:

- $a = 3$, so amplitude $= |3| = 3$
- $b = 2$, so period $= \frac{2\pi}{2} = \pi$
- $c = -\pi$, so phase shift $= -\frac{(-\pi)}{2} = \frac{\pi}{2}$ (shift right by $\pi/2$)
- $d = 1$, so range $= [1 - 3, 1 + 3] = [-2, 4]$

Example 2: Finding a Trigonometric Function from Features

A sinusoidal function has a maximum value of 7 at $x = \frac{\pi}{4}$, a minimum value of 1, and a period of π . Find its equation in the form $f(x) = a \sin(bx + c) + d$.

Solution:

Amplitude: $|a| = \frac{7-1}{2} = 3$. Since the function has a maximum at $x = \pi/4$ and sine normally has its first maximum at $x = \pi/2$, we can use $a = 3$.

Vertical shift: $d = \frac{7+1}{2} = 4$.

Period: $\frac{2\pi}{|b|} = \pi \implies b = 2$.

Phase shift: The first maximum of sin is at $bx + c = \pi/2$. We know a maximum occurs at $x = \pi/4$:

$$2\left(\frac{\pi}{4}\right) + c = \frac{\pi}{2} \implies \frac{\pi}{2} + c = \frac{\pi}{2} \implies c = 0$$

Equation:

$$f(x) = 3 \sin(2x) + 4$$

Verification: $f(\pi/4) = 3 \sin(\pi/2) + 4 = 3(1) + 4 = 7$. ✓

2.11 Transformations of Graphs

Summary of Transformations

Given $y = f(x)$:

Translations:

- $y = f(x) + b$: translate **up** by b (down if $b < 0$)
- $y = f(x - a)$: translate **right** by a (left if $a < 0$)

Reflections:

- $y = -f(x)$: reflect in the x -axis
- $y = f(-x)$: reflect in the y -axis

Stretches:

- $y = p f(x)$: vertical stretch by factor $|p|$ (compression if $0 < |p| < 1$)
- $y = f(qx)$: horizontal stretch by factor $\frac{1}{|q|}$ (compression if $|q| > 1$)

Combined:

- $y = -f(x)$ combined with $y = f(-x)$: reflect in **both** axes (equivalent to 180° rotation about origin)

Critical IB Concept: Order of Transformations

When applying multiple transformations to $y = f(x)$, the **order matters**. For the general form

$$y = a f(b(x - h)) + k$$

the recommended order of operations (applied to the graph) is:

1. **Horizontal stretch** by factor $\frac{1}{|b|}$ about the y -axis.
2. **Horizontal translation** by h units.

3. **Vertical stretch** by factor $|a|$ about the x -axis.
4. **Reflection** in x -axis if $a < 0$.
5. **Vertical translation** by k units.

Key rule: Horizontal transformations (stretch/translate) are applied *before* vertical ones. Within horizontal: stretch first, then translate. Within vertical: stretch first, then translate.

Why Order Matters — Worked Demonstration

Consider $f(x) = x^2$ and the transformation to $y = (2x - 4)^2$.

Method A: $y = (2x)^2 = 4x^2$ (horizontal compression by $1/2$), then translate right by 2: $y = 4(x - 2)^2 = 4(x^2 - 4x + 4) = 4x^2 - 16x + 16$.

Method B: $y = (x - 4)^2$ (translate right by 4), then compress horizontally by $1/2$: Replace x with $2x$: $y = (2x - 4)^2 = 4x^2 - 16x + 16$.

Both give the same result *only because we correctly identified the form*. A common mistake is reading $(2x - 4)$ as “compress then translate by 4” (wrong). The correct reading is:

$$y = f(2(x - 2)) = (2(x - 2))^2$$

which is: compress by $1/2$ *then* translate right by 2.

IB Tip: Transformation Sequence Questions

The IB frequently asks: “The graph of $y = f(x)$ is transformed to $y = 2f(3x - 6) + 5$. Describe the sequence of transformations.”

Answer strategy: Always factor the inside first:

$$y = 2f(3(x - 2)) + 5$$

Now read off: horizontal stretch factor $1/3$, translate right 2, vertical stretch factor 2, translate up 5.

Example 1: Full Transformation Description

The graph of $y = f(x)$ is shown. (Assume $f(x)$ passes through $(0, 0)$, $(2, 4)$, $(4, 0)$.)

Describe the transformation that maps $y = f(x)$ onto $y = -\frac{1}{2}f(2x + 4)$, and find the images of the three given points.

Solution:

First, rewrite: $y = -\frac{1}{2}f(2(x + 2))$.

Sequence of transformations:

1. Horizontal stretch by factor $\frac{1}{2}$ (since $q = 2$, factor is $1/2$)
2. Horizontal translation 2 units to the **left** (since $x + 2 = x - (-2)$)
3. Vertical compression by factor $\frac{1}{2}$
4. Reflection in the x -axis (negative sign)

Images of points:

Start with $(0, 0)$: After horizontal stretch: $(0, 0)$. After shift left: $(-2, 0)$. After vertical compression and reflection: $(-2, 0)$.

Start with $(2, 4)$: After horizontal stretch: $(1, 4)$. After shift left: $(-1, 4)$. After vertical compression: $(-1, 2)$. After reflection: $(-1, -2)$.
 Start with $(4, 0)$: After horizontal stretch: $(2, 0)$. After shift left: $(0, 0)$. After vertical compression and reflection: $(0, 0)$.

Example 2: Finding the Equation After Transformations

The function $f(x) = \sqrt{x}$ is transformed by: a vertical stretch by factor 3, a reflection in the y -axis, and a translation of 1 unit down. Find the equation of the transformed function and its domain.

Solution:

Apply transformations in order:

1. Vertical stretch by 3: $y = 3\sqrt{x}$
2. Reflection in y -axis: $y = 3\sqrt{-x}$
3. Translation down by 1: $y = 3\sqrt{-x} - 1$

Domain: We need $-x \geq 0 \implies x \leq 0$.

Range: $3\sqrt{-x} \geq 0$, so $3\sqrt{-x} - 1 \geq -1$. Range = $[-1, \infty)$.

2.12 Inequalities

Sign Analysis Method for Inequalities

To solve $f(x) \geq 0$ (or $> 0, \leq 0, < 0$):

1. Find all **critical values**: roots of $f(x) = 0$ and points where f is undefined.
2. Arrange critical values on a number line, dividing $\mathbb{R}$ into intervals.
3. Test the sign of $f(x)$ in each interval (use any convenient test point).
4. Select intervals satisfying the inequality.
5. For $\geq$ or $\leq$, include endpoints where f is defined.

2.12.1 Quadratic Inequalities

Example 1: Quadratic Inequality

Solve $x^2 - 5x + 6 \leq 0$.

Solution:

Factor: $x^2 - 5x + 6 = (x - 2)(x - 3)$.

Critical values: $x = 2$ and $x = 3$.

Sign analysis:

- $x < 2$: test $x = 0$: $(0 - 2)(0 - 3) = 6 > 0$
- $2 < x < 3$: test $x = 2.5$: $(2.5 - 2)(2.5 - 3) = (0.5)(-0.5) < 0$
- $x > 3$: test $x = 4$: $(4 - 2)(4 - 3) = 2 > 0$

We want $f(x) \leq 0$, so the solution includes where f is negative and where $f = 0$.

$$x \in [2, 3]$$

Example 2: Quadratic Inequality — Strict

Solve $-2x^2 + 3x + 5 > 0$.

Solution:

Factor: $-2x^2 + 3x + 5 = -(2x^2 - 3x - 5) = -(2x - 5)(x + 1)$.

Critical values: $x = \frac{5}{2}$ and $x = -1$.

Sign analysis of $-2x^2 + 3x + 5$:

- $x < -1$: test $x = -2$: $-2(4) + 3(-2) + 5 = -8 - 6 + 5 = -9 < 0$
- $-1 < x < \frac{5}{2}$: test $x = 0$: $-2(0) + 3(0) + 5 = 5 > 0$
- $x > \frac{5}{2}$: test $x = 3$: $-2(9) + 3(3) + 5 = -18 + 9 + 5 = -4 < 0$

Since the inequality is strict ($>$), exclude endpoints:

$$x \in \left(-1, \frac{5}{2}\right)$$

2.12.2 Rational Inequalities

Example 3: Rational Inequality

Solve $\frac{x+1}{x-2} \geq 0$.

Solution:

Critical values: numerator zero at $x = -1$; denominator zero (undefined) at $x = 2$.

Sign analysis of $\frac{x+1}{x-2}$:

- $x < -1$: test $x = -2$: $\frac{-2+1}{-2-2} = \frac{-1}{-4} > 0$
- $-1 < x < 2$: test $x = 0$: $\frac{0+1}{0-2} = \frac{1}{-2} < 0$
- $x > 2$: test $x = 3$: $\frac{3+1}{3-2} = \frac{4}{1} > 0$

We need ≥ 0 . Include $x = -1$ (where the expression equals 0), but exclude $x = 2$ (where it is undefined):

$$x \in (-\infty, -1] \cup (2, \infty)$$

Example 4: Complex Rational Inequality

Solve $\frac{x^2 - x - 6}{x^2 - 1} < 0$.

Solution:

Factor numerator and denominator:

$$\frac{x^2 - x - 6}{x^2 - 1} = \frac{(x - 3)(x + 2)}{(x - 1)(x + 1)}$$

Critical values: $x = -2$, $x = -1$, $x = 1$, $x = 3$.

Sign analysis:

- $x < -2$: test $x = -3$: $\frac{(-6)(-1)}{(-4)(-2)} = \frac{6}{8} > 0$
- $-2 < x < -1$: test $x = -1.5$: $\frac{(-4.5)(0.5)}{(-2.5)(-0.5)} = \frac{-2.25}{1.25} < 0$
- $-1 < x < 1$: test $x = 0$: $\frac{(-3)(2)}{(-1)(1)} = \frac{-6}{-1} > 0$
- $1 < x < 3$: test $x = 2$: $\frac{(-1)(4)}{(1)(3)} = \frac{-4}{3} < 0$
- $x > 3$: test $x = 4$: $\frac{(1)(6)}{(3)(5)} = \frac{6}{15} > 0$

We need < 0 , so the expression is negative in $(-2, -1)$ and $(1, 3)$. Since it is a strict inequality, exclude all critical values:

$$x \in (-2, -1) \cup (1, 3)$$

Example 5: Solving $f(x) \geq g(x)$ with Quadratics

Solve $x^2 + 2x - 3 \geq 2x + 1$.

Solution:

Rearrange to bring everything to one side:

$$x^2 + 2x - 3 - 2x - 1 \geq 0 \implies x^2 - 4 \geq 0$$

Factor: $(x - 2)(x + 2) \geq 0$.

Sign analysis:

- $x < -2$: test $x = -3$: $9 - 4 = 5 > 0$
- $-2 < x < 2$: test $x = 0$: $0 - 4 = -4 < 0$
- $x > 2$: test $x = 3$: $9 - 4 = 5 > 0$

Since ≥ 0 (non-strict), include endpoints:

$$x \in (-\infty, -2] \cup [2, \infty)$$

Graphical check: $y = x^2 - 4$ is a parabola with vertex at $(0, -4)$ and roots at $x = \pm 2$. It is above the x -axis for $x \leq -2$ and $x \geq 2$.

IB Exam Tip: “Determine” vs “Find” for Inequalities

“Find” the solution set of $f(x) > 0$: you may use any valid method (algebraic or graphical) and just state the answer.

“Determine” the solution set: IB examiners expect to see **working**—usually the sign analysis table or a clearly labelled graph. For “determine” questions, simply writing the answer without justification will lose marks.

“Hence” or “hence or otherwise”: if the previous part set up a factorisation or graph, use it. “Hence or otherwise” means you *can* use any method, but the intended path uses prior work.

Common Pitfalls in Inequalities

1. **Multiplying by a negative flips the sign:** If you multiply or divide both sides by a negative quantity, reverse the inequality.
2. **Squaring both sides:** Only valid when both sides are known to be non-negative. Otherwise, consider cases.
3. **Rational inequalities:** Never multiply through by the denominator without considering its sign. Use the sign analysis method instead.
4. **Endpoints:** For strict inequalities ($<$, $>$), always exclude critical values where the expression equals zero or is undefined.

Chapter 3

Geometry and Trigonometry

This chapter brings together the geometry of three-dimensional space, the algebra of vectors, and the full toolkit of trigonometry. Vectors provide the bridge between algebra and geometry, while trigonometric identities and equations are among the most frequently tested topics on the IB exam.

3.1 3D Geometry

The IB syllabus requires fluency with volume and surface-area formulas for common solids, as well as the ability to work with points, distances, and angles in three-dimensional space.

3.1.1 Volume Formulas

Volume of Common Solids

$$\text{Sphere: } V = \frac{4}{3}\pi r^3$$

$$\text{Right circular cone: } V = \frac{1}{3}\pi r^2 h$$

$$\text{Pyramid: } V = \frac{1}{3}Bh$$

where r is the radius, h is the perpendicular height, and B is the area of the base of the pyramid.

3.1.2 Surface Area Formulas

Surface Area of Common Solids

$$\text{Sphere: } A = 4\pi r^2$$

$$\text{Right circular cone (total): } A = \pi r l + \pi r^2 = \pi r(l + r)$$

$$\text{Right circular cone (lateral only): } A = \pi r l$$

where $l = \sqrt{r^2 + h^2}$ is the slant height of the cone.

3.1.3 Hemisphere Formulas

Hemisphere Volume and Surface Area

$$\text{Volume of a hemisphere: } V = \frac{2}{3}\pi r^3$$

$$\text{Curved surface area: } A = 2\pi r^2$$

$$\text{Total surface area (including base): } A = 3\pi r^2$$

Remembering Hemisphere Formulas

A hemisphere is simply half a sphere. The curved surface area is half of $4\pi r^2$, giving $2\pi r^2$. The total surface area adds the circular base πr^2 , yielding $3\pi r^2$. The volume is half of $\frac{4}{3}\pi r^3$, giving $\frac{2}{3}\pi r^3$.

3.1.4 Distance in 3D

Definition 3.1.1. The distance between two points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ in three-dimensional space is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

This is a direct extension of the 2D distance formula (Pythagorean theorem), applied twice.

3.1.5 Midpoint in 3D

Midpoint Formula in 3D

The midpoint M of the line segment joining $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ is

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right).$$

3.1.6 Angle Between a Line and a Plane

Angle Between a Line and a Plane

If a line has direction vector $\mathbf{d}$ and a plane has normal vector $\mathbf{n}$, then the angle α between the line and the plane satisfies

$$\sin \alpha = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}| |\mathbf{n}|}.$$

Equivalently, if θ is the angle between $\mathbf{d}$ and $\mathbf{n}$, then $\alpha = 90^\circ - \theta$, so $\alpha + \theta = 90^\circ$.

The angle between a line and a plane is the complement of the angle between the line and the plane's normal vector.

Example: Distance and Midpoint in 3D

Problem. Find the distance and midpoint between $A(1, -2, 3)$ and $B(4, 2, -1)$.

Solution.

$$d = \sqrt{(4 - 1)^2 + (2 - (-2))^2 + (-1 - 3)^2} = \sqrt{9 + 16 + 16} = \sqrt{41}.$$

The midpoint is

$$M = \left(\frac{1+4}{2}, \frac{-2+2}{2}, \frac{3+(-1)}{2} \right) = \left(\frac{5}{2}, 0, 1 \right).$$

Example: Volume of a Composite Solid

Problem. A solid consists of a hemisphere of radius 5 cm sitting on top of a right circular cone of base radius 5 cm and height 12 cm. Find the total volume.

Solution.

$$V_{\text{hemisphere}} = \frac{2}{3}\pi(5)^3 = \frac{250\pi}{3},$$

$$V_{\text{cone}} = \frac{1}{3}\pi(5)^2(12) = 100\pi,$$

$$V_{\text{total}} = \frac{250\pi}{3} + 100\pi = \frac{250\pi + 300\pi}{3} = \frac{550\pi}{3} \approx 576 \text{ cm}^3.$$

3.2 Vectors: Foundations

Vectors are quantities that have both *magnitude* and *direction*. They are the fundamental language for describing geometry in two and three dimensions.

3.2.1 Position and Displacement Vectors

Definition 3.2.1. A **position vector** $\mathbf{a}$ describes the position of a point A relative to a fixed origin O . We write $\mathbf{a} = \overrightarrow{OA}$.

A **displacement vector** (or free vector) $\overrightarrow{AB}$ describes the displacement from point A to point B :

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a}.$$

3.2.2 Base Vectors

Definition 3.2.2. In three dimensions, the standard base vectors are

$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

They point along the positive x -, y -, and z -axes respectively, each with magnitude 1.

3.2.3 Component Form

Component Form of a Vector

A vector $\mathbf{v}$ with components v_1, v_2, v_3 can be written as

$$\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}.$$

3.2.4 Magnitude

Magnitude of a Vector

$$|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}.$$

For a 2D vector $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j}$: $|\mathbf{v}| = \sqrt{v_1^2 + v_2^2}.$

3.2.5 Unit Vectors

Definition 3.2.3. A **unit vector** has magnitude 1. The unit vector in the direction of $\mathbf{v}$ is

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{|\mathbf{v}|}.$$

3.2.6 Vector Addition and Scalar Multiplication

Vector Operations

Addition: If $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$, then

$$\mathbf{a} + \mathbf{b} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{pmatrix}.$$

Scalar multiplication: For $k \in \mathbb{R}$,

$$k\mathbf{a} = \begin{pmatrix} ka_1 \\ ka_2 \\ ka_3 \end{pmatrix}.$$

3.2.7 Parallel Vectors

Definition 3.2.4. Two vectors $\mathbf{a}$ and $\mathbf{b}$ are **parallel** if and only if one is a scalar multiple of the other:

$$\mathbf{a} = k\mathbf{b} \quad \text{for some } k \in \mathbb{R}, k \neq 0.$$

If $k > 0$ the vectors point in the same direction; if $k < 0$ they point in opposite directions.

Checking Parallelism

To check if two vectors are parallel, verify that the ratios of corresponding components are equal:

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = k.$$

Watch out for zero components: if $\mathbf{b} = \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix}$, then $\mathbf{a}$ must also have $a_1 = 0$ to be parallel.

Example: Vector Operations

Problem. Given $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = -\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$, find:

- (i) $3\mathbf{a} - 2\mathbf{b}$
- (ii) $|\mathbf{a} - \mathbf{b}|$
- (iii) The unit vector in the direction of $\mathbf{a}$

Solution.

$$(i) \quad 3\mathbf{a} - 2\mathbf{b} = \begin{pmatrix} 6 \\ -3 \\ 9 \end{pmatrix} - \begin{pmatrix} -2 \\ 8 \\ -4 \end{pmatrix} = \begin{pmatrix} 8 \\ -11 \\ 13 \end{pmatrix} = 8\mathbf{i} - 11\mathbf{j} + 13\mathbf{k}.$$

$$(ii) \quad \mathbf{a} - \mathbf{b} = \begin{pmatrix} 3 \\ -5 \\ 5 \end{pmatrix}, \text{ so } |\mathbf{a} - \mathbf{b}| = \sqrt{9 + 25 + 25} = \sqrt{59}.$$

$$(iii) \quad |\mathbf{a}| = \sqrt{4 + 1 + 9} = \sqrt{14}, \text{ so } \hat{\mathbf{a}} = \frac{1}{\sqrt{14}} \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}.$$

Example: Displacement Vectors and Parallelism

Problem. Points A, B, C have position vectors $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\mathbf{b} = 3\mathbf{i} - \mathbf{j} + 3\mathbf{k}$, $\mathbf{c} = 5\mathbf{i} - 4\mathbf{j} + 7\mathbf{k}$. Show that A, B, C are collinear.

Solution.

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix},$$

$$\overrightarrow{BC} = \mathbf{c} - \mathbf{b} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}.$$

Since $\overrightarrow{AB} = \overrightarrow{BC}$, the vectors are parallel (with $k = 1$) and share the point B , so A, B, C are collinear. In fact, B is the midpoint of AC .

3.3 Dot Product

The dot product (also called the scalar product) is an operation that takes two vectors and returns a scalar. It encodes information about the angle between vectors and is essential for work involving projections, perpendicularity, and distances.

3.3.1 Algebraic Definition

Definition 3.3.1. The *dot product* of $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ is

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3.$$

3.3.2 Geometric Definition

Definition 3.3.2. The *geometric definition* of the dot product is

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta,$$

where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$ ($0 \leq \theta \leq \pi$).

3.3.3 Equivalence of the Two Definitions

Proof: Algebraic Definition Equals Geometric Definition

We prove that $a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}| |\mathbf{b}| \cos \theta$ using the cosine rule.

Place the tails of $\mathbf{a}$ and $\mathbf{b}$ at the origin. Let $\mathbf{c} = \mathbf{a} - \mathbf{b}$ be the vector from the tip of $\mathbf{b}$ to the tip of $\mathbf{a}$. The triangle formed by $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ has angle θ between $\mathbf{a}$ and $\mathbf{b}$.

By the cosine rule on this triangle:

$$|\mathbf{c}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}| |\mathbf{b}| \cos \theta.$$

But $\mathbf{c} = \mathbf{a} - \mathbf{b}$, so

$$|\mathbf{c}|^2 = (a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2.$$

Expanding:

$$\begin{aligned} |\mathbf{c}|^2 &= (a_1^2 - 2a_1b_1 + b_1^2) + (a_2^2 - 2a_2b_2 + b_2^2) + (a_3^2 - 2a_3b_3 + b_3^2) \\ &= (a_1^2 + a_2^2 + a_3^2) + (b_1^2 + b_2^2 + b_3^2) - 2(a_1b_1 + a_2b_2 + a_3b_3) \\ &= |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2(\mathbf{a} \cdot \mathbf{b}). \end{aligned}$$

Equating the two expressions for $|\mathbf{c}|^2$:

$$|\mathbf{a}|^2 + |\mathbf{b}|^2 - 2(\mathbf{a} \cdot \mathbf{b}) = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}| |\mathbf{b}| \cos \theta.$$

Cancelling $|\mathbf{a}|^2 + |\mathbf{b}|^2$ and dividing by -2 :

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta.$$

3.3.4 Angle Between Vectors

Angle Between Two Vectors

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}, \quad \theta = \arccos \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} \right).$$

3.3.5 Perpendicular Vectors

Theorem 3.3.1. Two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular (orthogonal) if and only if

$$\mathbf{a} \cdot \mathbf{b} = 0.$$

Proof. If $\mathbf{a} \perp \mathbf{b}$, then $\theta = 90^\circ$, so $\cos \theta = 0$ and $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cdot 0 = 0$.

Conversely, if $\mathbf{a} \cdot \mathbf{b} = 0$ and both vectors are non-zero, then $|\mathbf{a}| |\mathbf{b}| \cos \theta = 0$, so $\cos \theta = 0$, giving $\theta = 90^\circ$. $\square$

3.3.6 Properties of the Dot Product

Theorem 3.3.2 (Properties of the Dot Product). Let $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ be vectors and $k \in \mathbb{R}$.

- (i) **Commutative:** $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
- (ii) **Distributive:** $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$
- (iii) **Scalar compatibility:** $(k\mathbf{a}) \cdot \mathbf{b} = k(\mathbf{a} \cdot \mathbf{b})$
- (iv) **NOT associative:** In general, $(\mathbf{a} \cdot \mathbf{b})\mathbf{c} \neq \mathbf{a} \cdot (\mathbf{b} \cdot \mathbf{c})$ since the left side is a vector and the right side is a scalar.
- (v) $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$

Why the Dot Product Is Not Associative

The expression $\mathbf{a} \cdot (\mathbf{b} \cdot \mathbf{c})$ is not even defined: $\mathbf{b} \cdot \mathbf{c}$ is a scalar, and you cannot take the dot product of a vector with a scalar. The expression $(\mathbf{a} \cdot \mathbf{b})\mathbf{c}$ means scaling $\mathbf{c}$ by the scalar $\mathbf{a} \cdot \mathbf{b}$, which is a valid (but different) operation.

Example: Finding the Angle Between Vectors

Problem. Find the angle between $\mathbf{a} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\mathbf{b} = -\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$.

Solution.

$$\mathbf{a} \cdot \mathbf{b} = (3)(-1) + (1)(2) + (-2)(4) = -3 + 2 - 8 = -9,$$

$$|\mathbf{a}| = \sqrt{9 + 1 + 4} = \sqrt{14},$$

$$|\mathbf{b}| = \sqrt{1 + 4 + 16} = \sqrt{21},$$

$$\cos \theta = \frac{-9}{\sqrt{14}\sqrt{21}} = \frac{-9}{\sqrt{294}} = \frac{-9}{7\sqrt{6}} \approx -0.526,$$

$$\theta \approx 121.7^\circ.$$

Example: Perpendicularity Check

Problem. Find the value of p such that $\mathbf{a} = p\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} - p\mathbf{j} + 5\mathbf{k}$ are perpendicular.

Solution. For perpendicularity: $\mathbf{a} \cdot \mathbf{b} = 0$:

$$3p + 2(-p) + (-1)(5) = 0 \implies 3p - 2p - 5 = 0 \implies p = 5.$$

3.4 Vector Equation of a Line

3.4.1 Vector Form

Vector Equation of a Line (2D and 3D)

The line passing through point A (position vector $\mathbf{a}$) with direction vector $\mathbf{b}$ is

$$\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}, \quad \lambda \in \mathbb{R}.$$

Here $\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ is the position vector of a general point on the line, and λ is a scalar parameter.

3.4.2 Parametric Form

Parametric Equations of a Line

If $\mathbf{a} = \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$, then the parametric form is

$$x = x_0 + \lambda a, \quad y = y_0 + \lambda b, \quad z = z_0 + \lambda c.$$

3.4.3 Cartesian Form

Cartesian (Symmetric) Equations of a Line

If none of a, b, c is zero, we can eliminate λ to obtain:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

If a direction component is zero, the corresponding numerator equals zero. For example, if $a = 0$ and $b, c \neq 0$:

$$x = x_0, \quad \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

3.4.4 Angle Between Two Lines

The angle between two lines with direction vectors $\mathbf{d}_1$ and $\mathbf{d}_2$ is found using the dot product:

$$\cos \theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1| |\mathbf{d}_2|}.$$

The acute angle is $\theta = \arccos \left(\frac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{|\mathbf{d}_1| |\mathbf{d}_2|} \right)$.

3.4.5 Intersection of Two Lines

To find the intersection of two lines, set their parametric equations equal and solve for the parameters. If a consistent solution exists for both parameters, the lines intersect at the resulting point.

Example: Finding the Equation of a Line

Problem. Find the vector equation of the line through $A(1, 2, -1)$ and $B(3, 0, 4)$.

Solution. The direction vector is $\overrightarrow{AB} = \begin{pmatrix} 2 \\ -2 \\ 5 \end{pmatrix}$. Using point A :

$$\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -2 \\ 5 \end{pmatrix}, \quad \lambda \in \mathbb{R}.$$

Parametric form: $x = 1 + 2\lambda, y = 2 - 2\lambda, z = -1 + 5\lambda$.

Cartesian form: $\frac{x - 1}{2} = \frac{y - 2}{-2} = \frac{z + 1}{5}$.

Example: Intersection of Two Lines

Problem. Find the intersection of $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$.

Solution. Setting the parametric forms equal:

$$\begin{aligned} 1 + \lambda &= 2 - \mu, \\ 0 - \lambda &= 1 + 2\mu, \\ 2 + \lambda &= -1 + \mu. \end{aligned}$$

From the first equation: $\lambda = 1 - \mu$. Substituting into the second: $-(1 - \mu) = 1 + 2\mu \implies -1 + \mu = 1 + 2\mu \implies \mu = -2$.

Then $\lambda = 1 - (-2) = 3$.

Check in the third equation: $2 + 3 = 5$ and $-1 + (-2) = -3$. Since $5 \neq -3$, the lines do *not* intersect (they are skew).

3.5 Lines in Space

In three dimensions, two lines can be related in four fundamentally different ways.

Definition 3.5.1 (Classification of Two Lines in 3D). (i) **Coincident:** The lines are the same line. Their direction vectors are parallel, and they share a point.

(ii) **Parallel:** The direction vectors are parallel, but the lines do not share any point.

(iii) **Intersecting:** The lines meet at exactly one point.

(iv) **Skew:** The lines are not parallel and do not intersect (they lie in different planes).

Distinguishing Cases for Two Lines

Given lines $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{d}_1$ and $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{d}_2$:

Step 1: Check if $\mathbf{d}_1 = k\mathbf{d}_2$ (parallel direction vectors).

- If **yes**: the lines are either coincident or parallel. Check if $\mathbf{a}_2 - \mathbf{a}_1$ is parallel to $\mathbf{d}_1$ (coincident) or not (parallel).
- If **no**: the lines are either intersecting or skew. Solve the system of three equations in λ and μ . If a consistent solution exists, they intersect; otherwise, they are skew.

IB Exam Strategy: Line Classification

1. Always check parallelism *first* by seeing if the direction vectors are scalar multiples.
1. If not parallel, attempt to solve the three equations simultaneously.
1. If the solution satisfies *all three* equations, the lines intersect.
1. If you get a contradiction, the lines are skew.

Example: Classifying Two Lines

Problem. Classify the relationship between $l_1 : \mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix}$.

Solution. Direction vectors: $\mathbf{d}_1 = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$, $\mathbf{d}_2 = \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix}$.

Check parallelism: $\frac{2}{-1} = -2$, but $\frac{1}{2} = 0.5 \neq -2$. Not parallel.

Set parametric equations equal:

$$\begin{aligned} 2\lambda &= 1 - \mu, \\ 1 + \lambda &= 3 + 2\mu, \\ -1 + 3\lambda &= 1. \end{aligned}$$

From (iii): $3\lambda = 2 \implies \lambda = \frac{2}{3}$.

From (i): $\mu = 1 - 2 \cdot \frac{2}{3} = 1 - \frac{4}{3} = -\frac{1}{3}$.

Check in (ii): LHS = $1 + \frac{2}{3} = \frac{5}{3}$, RHS = $3 + 2(-\frac{1}{3}) = 3 - \frac{2}{3} = \frac{7}{3}$.

Since $\frac{5}{3} \neq \frac{7}{3}$, the lines are **skew**.

3.6 Cross Product

The cross product (also called the vector product) is an operation on two vectors that produces another vector, in contrast to the dot product which produces a scalar. It is defined only in three dimensions.

3.6.1 Definition via Determinant

Definition 3.6.1. The **cross product** of $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}$ is

$$\mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = (v_2w_3 - v_3w_2)\mathbf{i} - (v_1w_3 - v_3w_1)\mathbf{j} + (v_1w_2 - v_2w_1)\mathbf{k}.$$

Computing the Cross Product

A useful mnemonic: for each component, “cover” the corresponding row and column, then compute the 2×2 determinant of the remaining entries, alternating signs $+-+$:

$$\begin{aligned} \text{coeff of } \mathbf{i}: & + \begin{vmatrix} v_2 & v_3 \\ w_2 & w_3 \end{vmatrix} = v_2w_3 - v_3w_2, \\ \text{coeff of } \mathbf{j}: & - \begin{vmatrix} v_1 & v_3 \\ w_1 & w_3 \end{vmatrix} = -(v_1w_3 - v_3w_1), \\ \text{coeff of } \mathbf{k}: & + \begin{vmatrix} v_1 & v_2 \\ w_1 & w_2 \end{vmatrix} = v_1w_2 - v_2w_1. \end{aligned}$$

3.6.2 Properties

Theorem 3.6.1 (Properties of the Cross Product). Let $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ be vectors and $k \in \mathbb{R}$.

- (i) **Anti-commutative:** $\mathbf{v} \times \mathbf{w} = -(\mathbf{w} \times \mathbf{v})$
- (ii) **Distributive:** $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$
- (iii) **Scalar compatibility:** $(k\mathbf{v}) \times \mathbf{w} = k(\mathbf{v} \times \mathbf{w})$
- (iv) **NOT associative:** In general, $\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) \neq (\mathbf{u} \times \mathbf{v}) \times \mathbf{w}$
- (v) $\mathbf{v} \times \mathbf{v} = \mathbf{0}$ (the zero vector)
- (vi) $\mathbf{v} \times \mathbf{w} = \mathbf{0}$ if and only if $\mathbf{v}$ and $\mathbf{w}$ are parallel
- (vii) $\mathbf{i} \times \mathbf{j} = \mathbf{k}$, $\mathbf{j} \times \mathbf{k} = \mathbf{i}$, $\mathbf{k} \times \mathbf{i} = \mathbf{j}$

3.6.3 Geometric Interpretation

Theorem 3.6.2. The magnitude of the cross product equals the area of the parallelogram spanned by $\mathbf{v}$ and $\mathbf{w}$:

$$|\mathbf{v} \times \mathbf{w}| = |\mathbf{v}| |\mathbf{w}| \sin \theta,$$

where θ is the angle between $\mathbf{v}$ and $\mathbf{w}$ ($0 \leq \theta \leq \pi$).

Proof: Magnitude of Cross Product Equals Parallelogram Area

Consider the parallelogram spanned by $\mathbf{v}$ and $\mathbf{w}$, with angle θ between them. The base has length $|\mathbf{v}|$ and the perpendicular height is $|\mathbf{w}| \sin \theta$.

The area of the parallelogram is:

$$\text{Area} = \text{base} \times \text{height} = |\mathbf{v}| \cdot |\mathbf{w}| \sin \theta.$$

We now show that $|\mathbf{v} \times \mathbf{w}|^2 = |\mathbf{v}|^2 |\mathbf{w}|^2 \sin^2 \theta$ using the algebraic definition.

Let $\mathbf{v} \times \mathbf{w} = \begin{pmatrix} p \\ q \\ r \end{pmatrix}$ where $p = v_2 w_3 - v_3 w_2$, $q = v_3 w_1 - v_1 w_3$, $r = v_1 w_2 - v_2 w_1$.

Then:

$$\begin{aligned} |\mathbf{v} \times \mathbf{w}|^2 &= p^2 + q^2 + r^2 \\ &= (v_2 w_3 - v_3 w_2)^2 + (v_3 w_1 - v_1 w_3)^2 + (v_1 w_2 - v_2 w_1)^2. \end{aligned}$$

Expanding all three squares and collecting terms (Lagrange's identity):

$$\begin{aligned} |\mathbf{v} \times \mathbf{w}|^2 &= (v_1^2 + v_2^2 + v_3^2)(w_1^2 + w_2^2 + w_3^2) - (v_1 w_1 + v_2 w_2 + v_3 w_3)^2 \\ &= |\mathbf{v}|^2 |\mathbf{w}|^2 - (\mathbf{v} \cdot \mathbf{w})^2 \\ &= |\mathbf{v}|^2 |\mathbf{w}|^2 - |\mathbf{v}|^2 |\mathbf{w}|^2 \cos^2 \theta \\ &= |\mathbf{v}|^2 |\mathbf{w}|^2 (1 - \cos^2 \theta) \\ &= |\mathbf{v}|^2 |\mathbf{w}|^2 \sin^2 \theta. \end{aligned}$$

Taking square roots (since $\sin \theta \geq 0$ for $0 \leq \theta \leq \pi$):

$$|\mathbf{v} \times \mathbf{w}| = |\mathbf{v}| |\mathbf{w}| \sin \theta.$$

3.6.4 Right-Hand Rule

The Right-Hand Rule

The direction of $\mathbf{v} \times \mathbf{w}$ is given by the right-hand rule: point the fingers of your right hand along $\mathbf{v}$, then curl them toward $\mathbf{w}$ through the smaller angle. Your thumb points in the direction of $\mathbf{v} \times \mathbf{w}$.

Consequences:

- $\mathbf{v} \times \mathbf{w}$ is perpendicular to both $\mathbf{v}$ and $\mathbf{w}$.
- $\mathbf{w} \times \mathbf{v}$ points in the opposite direction (anti-commutativity).

Area of a Triangle

Since a triangle has half the area of the parallelogram spanned by two sides, the area of the triangle with sides $\overrightarrow{AB}$ and $\overrightarrow{AC}$ is:

$$\text{Area} = \frac{1}{2} |\overrightarrow{AB} \times \overrightarrow{AC}|.$$

Example: Computing a Cross Product

Problem. Find $\mathbf{a} \times \mathbf{b}$ where $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}$, and find the area of the parallelogram they span.

Solution.

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -1 \\ 3 & 0 & 2 \end{vmatrix} = (4 - 0)\mathbf{i} - (2 - (-3))\mathbf{j} + (0 - 6)\mathbf{k} = 4\mathbf{i} - 5\mathbf{j} - 6\mathbf{k}.$$

$$\text{Area of parallelogram} = |\mathbf{a} \times \mathbf{b}| = \sqrt{16 + 25 + 36} = \sqrt{77}.$$

$$\text{Area of triangle} = \frac{\sqrt{77}}{2}.$$

3.7 Vector Equation of a Plane

3.7.1 Parametric Form of a Plane

Parametric Equation of a Plane

The plane passing through point A (position vector $\mathbf{a}$) and containing two non-parallel direction vectors $\mathbf{b}$ and $\mathbf{c}$ is

$$\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}, \quad \lambda, \mu \in \mathbb{R}.$$

Every point on the plane is obtained by starting at A and moving by scalar multiples of $\mathbf{b}$ and $\mathbf{c}$ independently.

3.7.2 Normal Vector and Normal Form

Definition 3.7.1. A **normal vector** $\mathbf{n}$ to a plane is any vector perpendicular to every vector lying in the plane. If $\mathbf{b}$ and $\mathbf{c}$ are direction vectors in the plane, then

$$\mathbf{n} = \mathbf{b} \times \mathbf{c}$$

is a normal vector to the plane.

Normal (Scalar Product) Equation of a Plane

If a plane has normal vector $\mathbf{n}$ and passes through point A (position vector $\mathbf{a}$), then for any point P (position vector $\mathbf{r}$) on the plane:

$$\overrightarrow{AP} \cdot \mathbf{n} = 0 \implies (\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0 \implies \mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}.$$

3.7.3 Cartesian Equation of a Plane

Derivation: Cartesian Equation from Normal Form

Starting with $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$, let $\mathbf{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$, $\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, and $\mathbf{a} \cdot \mathbf{n} = d$.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} a \\ b \\ c \end{pmatrix} = d,$$

$$ax + by + cz = d.$$

So the Cartesian equation of the plane is $ax + by + cz = d$, where $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is the normal vector.

Cartesian Equation of a Plane

$$ax + by + cz = d,$$

where $\mathbf{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is a normal vector to the plane and d is a constant. The value of d is determined by substituting any known point on the plane.

Finding the Normal Vector

- Given parametric form with direction vectors $\mathbf{b}$ and $\mathbf{c}$: compute $\mathbf{n} = \mathbf{b} \times \mathbf{c}$.
- Given Cartesian form $ax + by + cz = d$: read off $\mathbf{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$.
- Given three points A, B, C : compute $\overrightarrow{AB} \times \overrightarrow{AC}$.

Example: Finding the Equation of a Plane

Problem. Find the Cartesian equation of the plane through $A(1, 0, 2)$, $B(3, -1, 1)$, and $C(0, 2, -1)$.

Solution.

$$\overrightarrow{AB} = \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}, \quad \overrightarrow{AC} = \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}.$$

$$\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & -1 \\ -1 & 2 & -3 \end{vmatrix}$$

$$= (3 - (-2))\mathbf{i} - (-6 - 1)\mathbf{j} + (4 - 1)\mathbf{k} = 5\mathbf{i} + 7\mathbf{j} + 3\mathbf{k}.$$

Using point $A(1, 0, 2)$: $5(1) + 7(0) + 3(2) = 5 + 6 = 11$.

The Cartesian equation is $5x + 7y + 3z = 11$.

Example: Converting Between Forms

Problem. Convert the Cartesian equation $2x - y + 3z = 7$ to vector (parametric) form.

Solution. A normal vector is $\mathbf{n} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$. We need two vectors perpendicular to $\mathbf{n}$. Choose

$$\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \text{ (check: } 2(1) + (-1)(2) + 3(0) = 0 \text{) and } \mathbf{c} = \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} \text{ (check: } 2(0) + (-1)(3) + 3(1) = 0 \text{)}.$$

Find a point on the plane: set $y = 0$, $z = 1$, then $2x + 3 = 7 \implies x = 2$. So $A = (2, 0, 1)$.

Parametric form:

$$\mathbf{r} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix}, \quad \lambda, \mu \in \mathbb{R}.$$

3.8 Intersections and Angles in 3D

3.8.1 Line–Plane Intersection

Method: Line–Plane Intersection

To find the intersection of line $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$ with plane $px + qy + rz = s$:

Step 1. Write the parametric equations: $x = a_1 + \lambda d_1$, $y = a_2 + \lambda d_2$, $z = a_3 + \lambda d_3$.

Step 2. Substitute into the Cartesian equation of the plane.

Step 3. Solve for λ .

Step 4. Substitute λ back to find the point of intersection.

Special cases:

- If $\mathbf{d} \cdot \mathbf{n} \neq 0$: the line intersects the plane at a unique point.
- If $\mathbf{d} \cdot \mathbf{n} = 0$ and the line lies in the plane: infinitely many solutions (line lies in plane).
- If $\mathbf{d} \cdot \mathbf{n} = 0$ and the line does not lie in the plane: no solution (line is parallel to plane).

3.8.2 Intersection of Two Planes

Two planes with normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ intersect in a line (provided they are not parallel). The direction vector of the line of intersection is $\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2$.

To find the full equation of the line of intersection:

1. Compute $\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2$.
2. Set one variable to zero (say $z = 0$) and solve the resulting 2×2 system for x and y to find a point on the line.
3. Write $\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$.

3.8.3 Intersection of Three Planes

Three planes can intersect in:

- (i) A **single point** (unique solution): the three normal vectors are not coplanar, i.e., $\mathbf{n}_1 \cdot (\mathbf{n}_2 \times \mathbf{n}_3) \neq 0$. The system has a unique solution.
- (ii) A **line** (infinitely many solutions): the normals are coplanar but not all parallel, and the system is consistent but with a free parameter.
- (iii) A **plane** (infinitely many solutions): all three planes are the same plane.
- (iv) **No solution**: the system is inconsistent. This occurs when, for example, two parallel planes never meet, or the three planes form a triangular prism.

IB Exam Tip: Three Planes

When solving a 3×3 system to find the intersection of three planes, use row reduction (Gaussian elimination) or substitution. If you find a row that simplifies to $0 = k$ where $k \neq 0$, the system is inconsistent (no solution). If a row becomes $0 = 0$, there are infinitely many solutions.

3.8.4 Angle Between a Line and a Plane

Angle Between a Line and a Plane

If a line has direction vector $\mathbf{d}$ and a plane has normal vector $\mathbf{n}$, let θ be the angle between $\mathbf{d}$ and $\mathbf{n}$, and let α be the angle between the line and the plane. Then

$$\alpha = 90^\circ - \theta, \quad \text{so} \quad \sin \alpha = \cos \theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}| |\mathbf{n}|}.$$

3.8.5 Angle Between Two Planes

Angle Between Two Planes

The angle between two planes with normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ is the same as the angle between their normals:

$$\cos \theta = \frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{|\mathbf{n}_1| |\mathbf{n}_2|}.$$

Example: Line–Plane Intersection

Problem. Find where the line $\mathbf{r} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ intersects the plane $x + 2y - z = 4$.

Solution. Parametric form: $x = 1 + 2\lambda$, $y = -1 + \lambda$, $z = 3 - \lambda$.
Substitute into the plane equation:

$$(1 + 2\lambda) + 2(-1 + \lambda) - (3 - \lambda) = 4.$$

$$1 + 2\lambda - 2 + 2\lambda - 3 + \lambda = 4 \implies 5\lambda - 4 = 4 \implies 5\lambda = 8 \implies \lambda = \frac{8}{5}.$$

The point of intersection:

$$x = 1 + \frac{16}{5} = \frac{21}{5}, \quad y = -1 + \frac{8}{5} = \frac{3}{5}, \quad z = 3 - \frac{8}{5} = \frac{7}{5}.$$

Example: Angle Between Two Planes

Problem. Find the acute angle between the planes $2x - y + 3z = 5$ and $x + 2y - z = 1$.

Solution. Normal vectors: $\mathbf{n}_1 = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$, $\mathbf{n}_2 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$.

$$\mathbf{n}_1 \cdot \mathbf{n}_2 = 2(1) + (-1)(2) + 3(-1) = 2 - 2 - 3 = -3,$$

$$|\mathbf{n}_1| = \sqrt{4 + 1 + 9} = \sqrt{14}, \quad |\mathbf{n}_2| = \sqrt{1 + 4 + 1} = \sqrt{6},$$

$$\cos \theta = \frac{|-3|}{\sqrt{14}\sqrt{6}} = \frac{3}{\sqrt{84}} = \frac{3}{2\sqrt{21}} = \frac{\sqrt{21}}{14} \approx 0.327,$$

$$\theta \approx 70.9^\circ.$$

Example: Line of Intersection of Two Planes

Problem. Find the line of intersection of the planes $x + y + z = 6$ and $2x - y + z = 3$.

Solution.

$$\mathbf{n}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{n}_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix},$$

$$\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{vmatrix} = (1 - (-1))\mathbf{i} - (1 - 2)\mathbf{j} + (-1 - 2)\mathbf{k} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}.$$

To find a point, set $z = 0$:

$$x + y = 6, \quad 2x - y = 3.$$

Adding: $3x = 9 \implies x = 3$, $y = 3$. So $A = (3, 3, 0)$.

The line of intersection is:

$$\mathbf{r} = \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}, \quad \lambda \in \mathbb{R}.$$

3.9 Trigonometric Foundations

3.9.1 Right-Angled Triangle Ratios

SOH-CAH-TOA

For a right-angled triangle with angle θ :

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}},$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}},$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sin \theta}{\cos \theta}.$$

3.9.2 Exact Trigonometric Values

Exact Values for Special Angles

	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

Equivalent values using periodicity and symmetry:

- $120^\circ = 180^\circ - 60^\circ$: $\sin 120^\circ = \frac{\sqrt{3}}{2}$, $\cos 120^\circ = -\frac{1}{2}$, $\tan 120^\circ = -\sqrt{3}$
- $135^\circ = 180^\circ - 45^\circ$: $\sin 135^\circ = \frac{\sqrt{2}}{2}$, $\cos 135^\circ = -\frac{\sqrt{2}}{2}$, $\tan 135^\circ = -1$
- $150^\circ = 180^\circ - 30^\circ$: $\sin 150^\circ = \frac{1}{2}$, $\cos 150^\circ = -\frac{\sqrt{3}}{2}$, $\tan 150^\circ = -\frac{1}{\sqrt{3}}$

Memory Trick: The 45-45-90 and 30-60-90 Triangles

45-45-90 triangle: sides 1, 1, $\sqrt{2}$. Hence $\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$.

30-60-90 triangle: sides 1, $\sqrt{3}$, 2. Opposite 30° is 1, opposite 60° is $\sqrt{3}$, hypotenuse is 2.

3.9.3 Unit Circle Definitions

Definition 3.9.1. On the unit circle (radius 1, centered at the origin), for an angle θ measured counter-clockwise from the positive x -axis, the coordinates of the point on the circle are:

$$(\cos \theta, \sin \theta).$$

Then $\tan \theta = \frac{\sin \theta}{\cos \theta}$ (defined when $\cos \theta \neq 0$).

3.9.4 Radian Measure

Radian Measure

$$\pi \text{ radians} = 180^\circ \implies 1 \text{ radian} = \frac{180^\circ}{\pi} \implies 1^\circ = \frac{\pi}{180} \text{ radians.}$$

Common conversions:

$$\begin{array}{llll} 30^\circ = \frac{\pi}{6}, & 45^\circ = \frac{\pi}{4}, & 60^\circ = \frac{\pi}{3}, & 90^\circ = \frac{\pi}{2}, \\ 120^\circ = \frac{2\pi}{3}, & 135^\circ = \frac{3\pi}{4}, & 150^\circ = \frac{5\pi}{6}, & 180^\circ = \pi. \end{array}$$

3.9.5 Arc Length

Arc Length

For a circle of radius r subtending an angle θ (in radians) at the center:

$$s = r\theta.$$

3.9.6 Sector Area

Area of a Sector

For a circle of radius r and central angle θ (in radians):

$$A = \frac{1}{2}r^2\theta.$$

Derivation: Arc Length and Sector Area

Arc length. The full circumference of a circle of radius r is $2\pi r$. The fraction of the circumference corresponding to angle θ (out of 2π) is $\frac{\theta}{2\pi}$. Therefore:

$$s = \frac{\theta}{2\pi} \cdot 2\pi r = r\theta.$$

Sector area. The full area of a circle is πr^2 . The fraction of the area corresponding to angle θ (out of 2π) is $\frac{\theta}{2\pi}$. Therefore:

$$A = \frac{\theta}{2\pi} \cdot \pi r^2 = \frac{1}{2}r^2\theta.$$

Example: Arc Length and Sector Area

Problem. A sector of a circle with radius 8 cm has an angle of 1.5 radians. Find (a) the arc length and (b) the area of the sector.

Solution.

(a) $s = r\theta = 8 \times 1.5 = 12 \text{ cm.}$

(b) $A = \frac{1}{2}r^2\theta = \frac{1}{2}(64)(1.5) = 48 \text{ cm}^2.$

3.10 Sine and Cosine Rules

The sine and cosine rules generalize right-angled triangle trigonometry to *all* triangles.

3.10.1 The Sine Rule

The Sine Rule

For any triangle with sides a, b, c opposite angles A, B, C respectively:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

Derivation of the Sine Rule

Consider triangle ABC with sides a, b, c . Drop an altitude h from C perpendicular to AB .

Case 1: Both A and B are acute.

In triangle ACD (where D is the foot of the altitude):

$$\sin A = \frac{h}{b} \implies h = b \sin A.$$

In triangle BCD :

$$\sin B = \frac{h}{a} \implies h = a \sin B.$$

Since both equal h :

$$a \sin B = b \sin A \implies \frac{a}{\sin A} = \frac{b}{\sin B}.$$

Similarly, by dropping the altitude from A , we get $\frac{b}{\sin B} = \frac{c}{\sin C}$.

Case 2: One angle is obtuse.

If $B > 90^\circ$, then in triangle BCD we use the supplementary angle:

$$\sin(180^\circ - B) = \frac{h}{a} \implies \sin B = \frac{h}{a}.$$

The result follows identically.

3.10.2 The Cosine Rule

The Cosine Rule

$$c^2 = a^2 + b^2 - 2ab \cos C.$$

Equivalently:

$$a^2 = b^2 + c^2 - 2bc \cos A, \quad b^2 = a^2 + c^2 - 2ac \cos B.$$

Solving for the angle:

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}.$$

Derivation of the Cosine Rule

Consider triangle ABC with sides a, b, c where $c = AB$. Drop altitude h from C to point D on AB . Let $AD = x$ and $DB = c - x$.

In right triangle ACD :

$$h^2 + x^2 = b^2. \quad (1)$$

In right triangle BCD :

$$h^2 + (c - x)^2 = a^2. \quad (2)$$

Expanding (2):

$$h^2 + c^2 - 2cx + x^2 = a^2. \quad (3)$$

Substituting $h^2 + x^2 = b^2$ from (1) into (3):

$$b^2 + c^2 - 2cx = a^2. \quad (4)$$

From right triangle ACD : $\cos A = \frac{x}{b}$, so $x = b \cos A$.

Substituting into (4):

$$a^2 = b^2 + c^2 - 2c(b \cos A) = b^2 + c^2 - 2bc \cos A.$$

3.10.3 Area of a Triangle (Trigonometric)

Area of a Triangle Using Trigonometry

$$\text{Area} = \frac{1}{2}ab \sin C = \frac{1}{2}bc \sin A = \frac{1}{2}ac \sin B.$$

Derivation of the Area Formula

Drop altitude h from C perpendicular to AB . The area is:

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times c \times h.$$

In right triangle ACD : $\sin A = \frac{h}{b}$, so $h = b \sin A$.

Therefore:

$$\text{Area} = \frac{1}{2} \times c \times b \sin A = \frac{1}{2}bc \sin A.$$

3.10.4 The Ambiguous Case of the Sine Rule

The Ambiguous Case (SSA)

When given two sides and a non-included angle (SSA configuration), the sine rule may produce two possible triangles.

Given sides a and b and angle A :

- If $a < b \sin A$: **no triangle** exists.
- If $a = b \sin A$: **one right-angled triangle** exists.
- If $b \sin A < a < b$: **two triangles** exist (the angle B can be acute or obtuse, since $\sin B = \sin(180^\circ - B)$).
- If $a \geq b$: **one triangle** exists (angle B must be acute).

On the IB exam: If using the sine rule gives $\sin B = k$ where $0 < k < 1$, always check whether both $B = \arcsin k$ and $B = 180^\circ - \arcsin k$ produce valid triangles.

IB Exam Tip: When to Use Which Rule

- **Right-angled triangle:** Use SOH–CAH–TOA directly.
- **Given two angles and one side (ASA or AAS):** Use the **sine rule**.
- **Given two sides and the included angle (SAS):** Use the **cosine rule** or **area formula**.
- **Given three sides (SSS):** Use the **cosine rule** to find an angle.
- **Given two sides and a non-included angle (SSA):** Use the **sine rule** but check for the ambiguous case.
- **Finding area:** Use $\frac{1}{2}ab \sin C$ if two sides and the included angle are known, or Heron's formula ($A = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = \frac{a+b+c}{2}$) if three sides are known.

Example: Using the Cosine Rule

Problem. In triangle ABC , $a = 7$, $b = 5$, $c = 3$. Find angle C .

Solution.

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{49 + 25 - 9}{2 \times 7 \times 5} = \frac{65}{70} = \frac{13}{14}.$$

$$C = \arccos\left(\frac{13}{14}\right) \approx 21.8^\circ.$$

Example: Ambiguous Case

Problem. In triangle ABC , $a = 8$, $b = 10$, and $A = 40^\circ$. Find the possible values of B .

Solution. By the sine rule:

$$\sin B = \frac{b \sin A}{a} = \frac{10 \sin 40^\circ}{8} = \frac{10 \times 0.6428}{8} \approx 0.8035.$$

$$B_1 = \arcsin(0.8035) \approx 53.5^\circ, \quad B_2 = 180^\circ - 53.5^\circ = 126.5^\circ.$$

Check: $A + B_1 = 40^\circ + 53.5^\circ = 93.5^\circ < 180^\circ$ (valid). $A + B_2 = 40^\circ + 126.5^\circ = 166.5^\circ < 180^\circ$ (valid).

Both triangles exist. The two possible values of B are approximately 53.5° and 126.5° .

3.11 Trigonometric Identities

Trigonometric identities are equations that hold for all values of the variable (in the domain). They are essential tools for simplifying expressions, solving equations, and proving other identities.

3.11.1 Pythagorean Identity

Identity 3.11.1 (Fundamental Pythagorean Identity).

$$\sin^2 \theta + \cos^2 \theta = 1.$$

Proof. On the unit circle, a point at angle θ has coordinates $(\cos \theta, \sin \theta)$. Since the point lies on the circle $x^2 + y^2 = 1$:

$$\cos^2 \theta + \sin^2 \theta = 1.$$

□

3.11.2 Derived Pythagorean Identities

Derived Identities

$$1 + \tan^2 \theta = \sec^2 \theta,$$

$$1 + \cot^2 \theta = \csc^2 \theta.$$

Proof of Derived Pythagorean Identities

Proof of $1 + \tan^2 \theta = \sec^2 \theta$:

Start with $\sin^2 \theta + \cos^2 \theta = 1$. Divide both sides by $\cos^2 \theta$:

$$\frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta},$$

$$\tan^2 \theta + 1 = \sec^2 \theta.$$

Proof of $1 + \cot^2 \theta = \csc^2 \theta$:

Start with $\sin^2 \theta + \cos^2 \theta = 1$. Divide both sides by $\sin^2 \theta$:

$$\frac{\sin^2 \theta}{\sin^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} = \frac{1}{\sin^2 \theta},$$

$$1 + \cot^2 \theta = \csc^2 \theta.$$

3.11.3 Double Angle Formulas

Double Angle Formulas

$$\sin 2\theta = 2 \sin \theta \cos \theta,$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta,$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}.$$

Derivation of Double Angle Formulas**Derivation of $\sin 2\theta$:**

Using the compound angle formula $\sin(A + B)$ (proved below) with $A = B = \theta$:

$$\sin 2\theta = \sin(\theta + \theta) = \sin \theta \cos \theta + \cos \theta \sin \theta = 2 \sin \theta \cos \theta.$$

Derivation of $\cos 2\theta$:

Using $\cos(A + B)$ with $A = B = \theta$:

$$\cos 2\theta = \cos(\theta + \theta) = \cos \theta \cos \theta - \sin \theta \sin \theta = \cos^2 \theta - \sin^2 \theta.$$

Alternative forms using $\sin^2 \theta + \cos^2 \theta = 1$:

$$\cos 2\theta = \cos^2 \theta - (1 - \cos^2 \theta) = 2 \cos^2 \theta - 1,$$

$$\cos 2\theta = (1 - \sin^2 \theta) - \sin^2 \theta = 1 - 2 \sin^2 \theta.$$

Derivation of $\tan 2\theta$:

Using $\tan(A + B)$ with $A = B = \theta$:

$$\tan 2\theta = \tan(\theta + \theta) = \frac{\tan \theta + \tan \theta}{1 - \tan \theta \cdot \tan \theta} = \frac{2 \tan \theta}{1 - \tan^2 \theta}.$$

Choosing the Right Form of $\cos 2\theta$

- Use $\cos 2\theta = 2 \cos^2 \theta - 1$ when the problem involves $\cos^2 \theta$ or when you want to express $\cos^2 \theta$ in terms of $\cos 2\theta$.
- Use $\cos 2\theta = 1 - 2 \sin^2 \theta$ when the problem involves $\sin^2 \theta$.
- Use $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ when both $\sin \theta$ and $\cos \theta$ appear.

3.11.4 Compound Angle (Addition) Formulas**Compound Angle Formulas**

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B,$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B,$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}.$$

Note: In $\cos(A \pm B)$, the sign on the right is *opposite* to the sign on the left.

Derivation of Compound Angle Formulas

Proof of $\cos(A + B) = \cos A \cos B - \sin A \sin B$:

Consider the unit circle. Let point P be at angle A and point Q at angle $-(A + B)$. The chord length PQ can be computed in two ways using the distance formula.

Alternatively, using rotation matrices or Euler's formula. Here we use a classical geometric proof.

Consider the unit circle. Let $P_1 = (\cos A, \sin A)$, $P_2 = (\cos(-B), \sin(-B)) = (\cos B, -\sin B)$, $P_3 = (\cos(A + B), \sin(A + B))$, and $P_4 = (1, 0)$.

Rotating the entire figure by angle A sends P_4 to P_1 and P_2 to P_3 . Since rotation preserves distances:

$$|P_1 P_2|^2 = |P_3 P_4|^2.$$

Compute $|P_1 P_2|^2$:

$$\begin{aligned} |P_1 P_2|^2 &= (\cos A - \cos B)^2 + (\sin A + \sin B)^2 \\ &= \cos^2 A - 2 \cos A \cos B + \cos^2 B + \sin^2 A + 2 \sin A \sin B + \sin^2 B \\ &= (\cos^2 A + \sin^2 A) + (\cos^2 B + \sin^2 B) - 2 \cos A \cos B + 2 \sin A \sin B \\ &= 2 - 2 \cos A \cos B + 2 \sin A \sin B. \end{aligned}$$

Compute $|P_3 P_4|^2$:

$$\begin{aligned} |P_3 P_4|^2 &= (\cos(A + B) - 1)^2 + \sin^2(A + B) \\ &= \cos^2(A + B) - 2 \cos(A + B) + 1 + \sin^2(A + B) \\ &= 2 - 2 \cos(A + B). \end{aligned}$$

Setting equal:

$$\begin{aligned} 2 - 2 \cos(A + B) &= 2 - 2 \cos A \cos B + 2 \sin A \sin B, \\ \cos(A + B) &= \cos A \cos B - \sin A \sin B. \end{aligned}$$

Proof of $\cos(A - B) = \cos A \cos B + \sin A \sin B$:

Replace B with $-B$ in the formula for $\cos(A + B)$:

$$\cos(A - B) = \cos A \cos(-B) - \sin A \sin(-B) = \cos A \cos B + \sin A \sin B.$$

Proof of $\sin(A + B) = \sin A \cos B + \cos A \sin B$:

Use the cofunction identity $\sin \theta = \cos(90^\circ - \theta)$:

$$\begin{aligned} \sin(A + B) &= \cos(90^\circ - (A + B)) = \cos((90^\circ - A) - B) \\ &= \cos(90^\circ - A) \cos B + \sin(90^\circ - A) \sin B \\ &= \sin A \cos B + \cos A \sin B. \end{aligned}$$

Proof of $\sin(A - B) = \sin A \cos B - \cos A \sin B$:

Replace B with $-B$:

$$\sin(A - B) = \sin A \cos(-B) + \cos A \sin(-B) = \sin A \cos B - \cos A \sin B.$$

Proof of $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$:

$$\tan(A + B) = \frac{\sin(A + B)}{\cos(A + B)} = \frac{\sin A \cos B + \cos A \sin B}{\cos A \cos B - \sin A \sin B}.$$

Divide numerator and denominator by $\cos A \cos B$:

$$\tan(A + B) = \frac{\frac{\sin A}{\cos A} + \frac{\sin B}{\cos B}}{1 - \frac{\sin A}{\cos A} \cdot \frac{\sin B}{\cos B}} = \frac{\tan A + \tan B}{1 - \tan A \tan B}.$$

3.11.5 Reduction Formulas

Reduction (Reference Angle) Formulas

$$\sin(\pi - \theta) = \sin \theta, \quad \cos(\pi - \theta) = -\cos \theta, \quad \tan(\pi - \theta) = -\tan \theta.$$

$$\sin(\pi + \theta) = -\sin \theta, \quad \cos(\pi + \theta) = -\cos \theta, \quad \tan(\pi + \theta) = \tan \theta.$$

$$\sin(2\pi - \theta) = -\sin \theta, \quad \cos(2\pi - \theta) = \cos \theta, \quad \tan(2\pi - \theta) = -\tan \theta.$$

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta, \quad \cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta, \quad \tan\left(\frac{\pi}{2} - \theta\right) = \cot \theta.$$

$$\sin\left(\frac{\pi}{2} + \theta\right) = \cos \theta, \quad \cos\left(\frac{\pi}{2} + \theta\right) = -\sin \theta, \quad \tan\left(\frac{\pi}{2} + \theta\right) = -\cot \theta.$$

ASTC Rule (All, Sine, Tangent, Cosine)

In each quadrant:

- **All** trig functions are positive in the first quadrant ($0 < \theta < \frac{\pi}{2}$).
- **Sine** is positive in the second quadrant ($\frac{\pi}{2} < \theta < \pi$).
- **Tangent** is positive in the third quadrant ($\pi < \theta < \frac{3\pi}{2}$).
- **Cosine** is positive in the fourth quadrant ($\frac{3\pi}{2} < \theta < 2\pi$).

3.11.6 Proof Strategies for Trigonometric Identities

Strategies for Proving Trigonometric Identities

1. **Work on one side:** Transform the more complicated side until it matches the simpler side.
2. **Work on both sides:** Simplify each side independently and show they are equal.
3. **Convert to sine and cosine:** If the identity involves $\tan$, $\sec$, $\csc$, $\cot$, convert everything to $\sin$ and $\cos$.
4. **Look for known identities:** Factor, expand, or substitute using Pythagorean identities, double angle formulas, or compound angle formulas.
5. **Common factorizations:**
 - $1 - 2\sin^2 \theta = \cos 2\theta$
 - $\sin^2 \theta - \cos^2 \theta = -\cos 2\theta$
 - $\sin \theta + \cos \theta = \sqrt{2} \sin\left(\theta + \frac{\pi}{4}\right)$
6. **Never cross-multiply or square both sides** unless you check for extraneous solutions.

Example: Proving a Trigonometric Identity

Problem. Prove that $\frac{1 - \cos 2\theta}{\sin 2\theta} = \tan \theta$.

Proof. Starting with the left-hand side:

$$\begin{aligned}\frac{1 - \cos 2\theta}{\sin 2\theta} &= \frac{1 - (1 - 2\sin^2 \theta)}{2\sin \theta \cos \theta} \\ &= \frac{2\sin^2 \theta}{2\sin \theta \cos \theta} \\ &= \frac{\cancel{2}\sin\cancel{\theta} \cdot \sin \theta}{\cancel{2}\sin\cancel{\theta} \cdot \cos \theta} \\ &= \frac{\sin \theta}{\cos \theta} = \tan \theta.\end{aligned}$$

Example: Using Double Angle and Compound Angle Formulas

Problem. Express $\cos 3\theta$ in terms of $\cos \theta$.

Solution.

$$\begin{aligned}\cos 3\theta &= \cos(2\theta + \theta) \\ &= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta \\ &= (2\cos^2 \theta - 1)\cos \theta - (2\sin \theta \cos \theta)\sin \theta \\ &= 2\cos^3 \theta - \cos \theta - 2\sin^2 \theta \cos \theta \\ &= 2\cos^3 \theta - \cos \theta - 2(1 - \cos^2 \theta)\cos \theta \\ &= 2\cos^3 \theta - \cos \theta - 2\cos \theta + 2\cos^3 \theta \\ &= 4\cos^3 \theta - 3\cos \theta.\end{aligned}$$

This is the triple angle formula for cosine.

3.12 Reciprocal and Inverse Trig Functions

3.12.1 Reciprocal Trig Functions

Definition 3.12.1.

$$\begin{aligned}\sec \theta &= \frac{1}{\cos \theta}, & \cos \theta &\neq 0, \\ \csc \theta &= \frac{1}{\sin \theta}, & \sin \theta &\neq 0, \\ \cot \theta &= \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta}, & \sin \theta &\neq 0.\end{aligned}$$

IB Notation

The IB uses $\csc \theta$ (with the “o”), not $\operatorname{csc} \theta$. Both notations are understood, but use the IB convention on exams.

3.12.2 Inverse Trigonometric Functions

Definition 3.12.2. The *inverse trigonometric functions* (also called *arcfunctions*) are defined as follows:

$$y = \arcsin x \iff \sin y = x, \quad x \in [-1, 1], \quad y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right],$$

$$y = \arccos x \iff \cos y = x, \quad x \in [-1, 1], \quad y \in [0, \pi],$$

$$y = \arctan x \iff \tan y = x, \quad x \in \mathbb{R}, \quad y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Domains and Ranges of Inverse Trig Functions

Function	Domain	Range
$\arcsin x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\arccos x$	$[-1, 1]$	$[0, \pi]$
$\arctan x$	$(-\infty, \infty)$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

3.12.3 Key Properties of Inverse Trig Functions

Properties of Inverse Trig Functions

$$\arcsin(-x) = -\arcsin x,$$

$$\arccos(-x) = \pi - \arccos x,$$

$$\arctan(-x) = -\arctan x.$$

Composition with the original function:

$$\sin(\arcsin x) = x, \quad x \in [-1, 1],$$

$$\cos(\arccos x) = x, \quad x \in [-1, 1],$$

$$\tan(\arctan x) = x, \quad x \in \mathbb{R}.$$

Reverse composition (restricted):

$$\arcsin(\sin x) = x, \quad x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right],$$

$$\arccos(\cos x) = x, \quad x \in [0, \pi],$$

$$\arctan(\tan x) = x, \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Why the Restricted Ranges?

Without restricting the ranges, the inverse trig functions would not be functions (they would fail the vertical line test). For example, $\sin x = \frac{1}{2}$ has infinitely many solutions, but $\arcsin \frac{1}{2} = \frac{\pi}{6}$ is the *unique* value in the range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Key observation: $\arcsin$ and $\arctan$ are odd functions, but $\arccos$ is *not* odd. This is because the range $[0, \pi]$ is not symmetric about zero.

Example: Evaluating Inverse Trig Functions

Problem. Evaluate the following:

(i) $\arcsin\left(\frac{\sqrt{3}}{2}\right)$

(ii) $\arccos\left(-\frac{1}{2}\right)$

(iii) $\arctan(1) + \arctan(-1)$

Solution.

(i) We need $\sin y = \frac{\sqrt{3}}{2}$ with $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. So $y = \frac{\pi}{3}$.

(ii) We need $\cos y = -\frac{1}{2}$ with $y \in [0, \pi]$. So $y = \frac{2\pi}{3}$.

(iii) $\arctan(1) = \frac{\pi}{4}$ and $\arctan(-1) = -\frac{\pi}{4}$. Their sum is 0.

Example: Composing Inverse Trig Functions

Problem. Find $\sin(2 \arccos \frac{3}{5})$.

Solution. Let $\theta = \arccos \frac{3}{5}$, so $\cos \theta = \frac{3}{5}$ and $\theta \in [0, \pi]$. Then $\sin \theta = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$ (positive since $\theta \in [0, \pi]$).

Using the double angle formula:

$$\sin 2\theta = 2 \sin \theta \cos \theta = 2 \cdot \frac{4}{5} \cdot \frac{3}{5} = \frac{24}{25}.$$

3.13 Solving Trigonometric Equations

Solving trigonometric equations is a major part of the IB syllabus. The key is to find *all* solutions using general solution formulas, then restrict to the specified interval if needed.

3.13.1 General Solutions

General Solutions for Trigonometric Equations

If $\sin \theta = k$: $\theta = \arcsin k + 2n\pi$ or $\theta = \pi - \arcsin k + 2n\pi$, $n \in \mathbb{Z}$.

If $\cos \theta = k$: $\theta = \arccos k + 2n\pi$ or $\theta = -\arccos k + 2n\pi$, $n \in \mathbb{Z}$.

If $\tan \theta = k$: $\theta = \arctan k + n\pi$, $n \in \mathbb{Z}$.

Understanding the General Solutions

Sine: $\sin \theta = k$ has two solutions per period (2π), symmetric about $\theta = \frac{\pi}{2}$: if one solution is α , the other is $\pi - \alpha$.

Cosine: $\cos \theta = k$ has two solutions per period (2π), symmetric about $\theta = 0$: if one solution is α , the other is $-\alpha$.

Tangent: $\tan \theta = k$ has one solution per half-period (π), since the period of tangent is π .

3.13.2 Solving in Finite Intervals

Method: Solving in a Given Interval

Step 1. Find the principal value(s) using arcsin, arccos, or arctan.

Step 1. Write down *all* general solutions.

Step 1. Substitute integer values of n to find all solutions within the specified interval.

Step 1. List the solutions in order.

Common intervals on the IB exam: $[0, 2\pi]$, $[0, \pi]$, $[-\pi, \pi]$, $[0^\circ, 360^\circ]$.

3.13.3 Equations Leading to Quadratics

Some trigonometric equations can be transformed into quadratic equations by substitution.

Example: Quadratic in $\sin x$

Problem. Solve $2\sin^2 x - \sin x - 1 = 0$ for $x \in [0, 2\pi]$.

Solution. Let $u = \sin x$. Then $2u^2 - u - 1 = 0$:

$$(2u + 1)(u - 1) = 0 \implies u = -\frac{1}{2} \text{ or } u = 1.$$

Case 1: $\sin x = -\frac{1}{2}$.

$$x = \arcsin\left(-\frac{1}{2}\right) = -\frac{\pi}{6}, \quad \text{or} \quad x = \pi - \left(-\frac{\pi}{6}\right) = \frac{7\pi}{6}.$$

Adding 2π to $-\frac{\pi}{6}$: $x = \frac{11\pi}{6}$.

Solutions in $[0, 2\pi]$: $x = \frac{7\pi}{6}, \frac{11\pi}{6}$.

Case 2: $\sin x = 1$.

$$x = \frac{\pi}{2}.$$

All solutions: $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$.

3.13.4 Using Identities to Transform Equations

Example: Using Double Angle Identities**Problem.** Solve $\cos 2x + \cos x + 1 = 0$ for $x \in [0, 2\pi]$.**Solution.** Replace $\cos 2x$ using the identity $\cos 2x = 2 \cos^2 x - 1$:

$$2 \cos^2 x - 1 + \cos x + 1 = 0,$$

$$2 \cos^2 x + \cos x = 0,$$

$$\cos x(2 \cos x + 1) = 0.$$

Case 1: $\cos x = 0 \implies x = \frac{\pi}{2}$ or $x = \frac{3\pi}{2}$.**Case 2:** $\cos x = -\frac{1}{2} \implies x = \frac{2\pi}{3}$ or $x = \frac{4\pi}{3}$.**All solutions:** $x = \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{2}, \frac{4\pi}{3}$.**Example: Using Pythagorean Identity****Problem.** Solve $\sin^2 x = 3 \cos^2 x$ for $x \in [0, 2\pi]$.**Solution.** Replace $\sin^2 x = 1 - \cos^2 x$:

$$1 - \cos^2 x = 3 \cos^2 x,$$

$$1 = 4 \cos^2 x,$$

$$\cos^2 x = \frac{1}{4},$$

$$\cos x = \pm \frac{1}{2}.$$

$$\cos x = \frac{1}{2} : x = \frac{\pi}{3}, \frac{5\pi}{3}.$$

$$\cos x = -\frac{1}{2} : x = \frac{2\pi}{3}, \frac{4\pi}{3}.$$

All solutions: $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$.**Example: IB-Style Question with Compound Angle****Problem.** Solve $\sin(x + \frac{\pi}{3}) = \frac{1}{2}$ for $x \in [-\pi, \pi]$.**Solution.** Let $u = x + \frac{\pi}{3}$. We need $\sin u = \frac{1}{2}$.General solution: $u = \frac{\pi}{6} + 2n\pi$ or $u = \frac{5\pi}{6} + 2n\pi$.So $x = u - \frac{\pi}{3}$:

$$x = \frac{\pi}{6} - \frac{\pi}{3} + 2n\pi = -\frac{\pi}{6} + 2n\pi,$$

$$x = \frac{5\pi}{6} - \frac{\pi}{3} + 2n\pi = \frac{\pi}{2} + 2n\pi.$$

For $n = 0$: $x = -\frac{\pi}{6}$ and $x = \frac{\pi}{2}$. Both are in $[-\pi, \pi]$.For $n = -1$: $x = -\frac{\pi}{6} - 2\pi = -\frac{13\pi}{6}$ (out of range) and $x = \frac{\pi}{2} - 2\pi = -\frac{3\pi}{2}$ (out of range).For $n = 1$: $x = -\frac{\pi}{6} + 2\pi = \frac{11\pi}{6}$ (out of range) and $x = \frac{\pi}{2} + 2\pi = \frac{5\pi}{2}$ (out of range).**Solutions:** $x = -\frac{\pi}{6}, \frac{\pi}{2}$.**Example: Equation with tan and Multiple Angles****Problem.** Solve $\tan 2x = \sqrt{3}$ for $x \in [0, 2\pi]$.**Solution.** $\tan 2x = \sqrt{3} \implies 2x = \frac{\pi}{3} + n\pi \implies x = \frac{\pi}{6} + \frac{n\pi}{2}$.

For $n = 0$: $x = \frac{\pi}{6}$. For $n = 1$: $x = \frac{\pi}{6} + \frac{\pi}{2} = \frac{2\pi}{3}$.
For $n = 2$: $x = \frac{\pi}{6} + \pi = \frac{7\pi}{6}$. For $n = 3$: $x = \frac{\pi}{6} + \frac{3\pi}{2} = \frac{5\pi}{3}$.
For $n = 4$: $x = \frac{\pi}{6} + 2\pi = \frac{13\pi}{6} > 2\pi$ (stop).
Solutions: $x = \frac{\pi}{6}, \frac{2\pi}{3}, \frac{7\pi}{6}, \frac{5\pi}{3}$.

IB Exam Tip: Common Pitfalls in Trig Equations

- **Forgetting solutions:** Always use the general solution formula. Counting on a diagram alone often leads to missed solutions.
- **Wrong interval:** Read the question carefully. If the interval is in degrees, give answers in degrees. If in radians, give answers in radians.
- **Extraneous solutions:** When squaring both sides or multiplying by a trig function, check that all solutions satisfy the original equation. Squaring can introduce spurious solutions.
- **Domain restrictions:** If the equation involves $\sec x$, remember that $\cos x \neq 0$. If it involves $\tan x$, remember that $\cos x \neq 0$.
- **Double-check with calculator:** After finding solutions, substitute them back into the original equation to verify.

Chapter 4

Statistics and Probability

Statistics and probability form a major component of the IB Mathematics AA HL syllabus, spanning descriptive statistics, correlation, probability theory, discrete and continuous distributions, and the normal distribution. This chapter provides every formula, derivation, and exam technique you need.

4.1 Data Representation

4.1.1 Population vs Sample

Definition 4.1.1. A **population** is the entire set of individuals or items about which information is wanted. A **sample** is a subset of the population that is actually observed.

- **Population parameters** are fixed (but often unknown) quantities, denoted by Greek letters: population mean μ , population variance σ^2 , population standard deviation σ .
- **Sample statistics** are computed from data and used to estimate population parameters, denoted by Latin letters: sample mean $\bar{x}$, sample variance s^2 , sample standard deviation s .
- A **census** collects data from every member of the population.
- A **random sample** gives each member of the population an equal chance of selection.

Definition 4.1.2. **Discrete data** can take only specific (usually integer) values, e.g., number of goals scored, number of students in a class. **Continuous data** can take any value within a range, e.g., height, weight, time, temperature.

Classifying Data in IB Exams

In IB exams, discrete data often arise from *counting* (“the number of...”), while continuous data arise from *measuring* (“the time taken to...”, “the mass of...”). If you can meaningfully take a value between two data points, the data is continuous.

4.1.2 Frequency Distributions

Definition 4.1.3. A **frequency distribution** tabulates data by grouping observations into classes (or using exact values for discrete data) and recording the frequency of each class.

For grouped continuous data, classes are written as intervals, e.g., $10 \leq x < 20$. Important conventions:

- Class boundaries must be continuous with no gaps.
- The **class width** is the difference between the upper and lower class boundaries.

- The **midpoint** (class mark) of a class with boundaries a and b is $\frac{a+b}{2}$.
- **Class width** = $b - a$.

4.1.3 Histograms

A histogram is a graphical representation of a frequency distribution using rectangular bars.

Histogram: Equal Class Widths

When all class widths are equal, the **height of each bar equals the frequency** of that class.

Histogram: Unequal Class Widths

When class widths are unequal, the **height of each bar equals the frequency density**:

$$\text{Frequency density} = \frac{\text{Frequency}}{\text{Class width}}$$

The **area** of each bar equals the class frequency:

$$\text{Area} = \text{Frequency density} \times \text{Class width} = \text{Frequency}$$

Histogram Key Insight

The key principle is: **area represents frequency**, not height. This holds regardless of whether class widths are equal or not. When widths are equal, the height happens to be proportional to frequency, but the area–frequency principle is the fundamental rule.

Worked Example: Histogram with Unequal Widths

The masses (in kg) of 50 people are summarised below:

Mass (kg)	Frequency	Class Width
$40 \leq m < 50$	8	10
$50 \leq m < 60$	15	10
$60 \leq m < 70$	14	10
$70 \leq m < 85$	10	15
$85 \leq m < 100$	3	15

Find the frequency densities.

Solution:

$$\text{Class } 40 \leq m < 50 : \quad \text{FD} = \frac{8}{10} = 0.8$$

$$\text{Class } 50 \leq m < 60 : \quad \text{FD} = \frac{15}{10} = 1.5$$

$$\text{Class } 60 \leq m < 70 : \quad \text{FD} = \frac{14}{10} = 1.4$$

$$\text{Class } 70 \leq m < 85 : \quad \text{FD} = \frac{10}{15} \approx 0.667$$

$$\text{Class } 85 \leq m < 100 : \quad \text{FD} = \frac{3}{15} = 0.2$$

On the histogram, bar heights are these frequency density values and bar widths are the class widths.

4.1.4 Cumulative Frequency Graphs

Definition 4.1.4. A **cumulative frequency (CF)** graph plots the running total of frequencies against the upper class boundary of each group.

Constructing a Cumulative Frequency Graph

1. Create a table with columns: class, frequency, upper class boundary, cumulative frequency.
2. The cumulative frequency for each class is the sum of all frequencies up to and including that class.
3. Plot points at (upper class boundary, cumulative frequency).
4. Connect points with a smooth curve (an S-shaped ogive).

Reading a Cumulative Frequency Graph

- To find the median: read off the value at $\frac{n}{2}$ on the CF axis.
- To find the lower quartile Q_1 : read off at $\frac{n}{4}$.
- To find the upper quartile Q_3 : read off at $\frac{3n}{4}$.
- To find the value corresponding to a given percentile p : read off at $\frac{p}{100} \times n$.

Worked Example: Cumulative Frequency

The weekly wages (in dollars) of 80 workers:

Wage (\$)	Frequency	Cumulative Frequency
$200 \leq w < 300$	12	12
$300 \leq w < 400$	28	40
$400 \leq w < 500$	24	64
$500 \leq w < 600$	11	75
$600 \leq w < 800$	5	80

Estimate the median and interquartile range.

Solution:

$$\text{Median position} = \frac{80}{2} = 40$$

$$Q_1 \text{ position} = \frac{80}{4} = 20$$

$$Q_3 \text{ position} = \frac{3 \times 80}{4} = 60$$

Reading from the CF graph (by linear interpolation within the class):

- Median: at CF = 40, the value is at the upper boundary of the second class, so $Q_2 = \$400$.
- Q_1 : at CF = 20, interpolating in $200 \leq w < 300$: $Q_1 = 200 + \frac{20-0}{12-0} \times 100 = 200 + \frac{20}{12} \times 100 \approx 366.67$.
- Q_3 : at CF = 60, interpolating in $400 \leq w < 500$: $Q_3 = 400 + \frac{60-40}{64-40} \times 100 = 400 + \frac{20}{24} \times 100 \approx 483.33$.
- IQR = $Q_3 - Q_1 \approx 483.33 - 366.67 = 116.67$.

4.1.5 Box-and-Whisker Diagrams

Definition 4.1.5. A **box-and-whisker plot** (box plot) is a standardised way of displaying the five-number summary of a dataset: minimum, Q_1 , median (Q_2), Q_3 , and maximum.

Five-Number Summary and Box Plot

The box plot is constructed as follows:

1. Draw a horizontal (or vertical) number line covering the range of the data.
2. Draw a box from Q_1 to Q_3 (the box spans the IQR).
3. Draw a vertical line inside the box at the median Q_2 .
4. Extend whiskers from Q_1 to the minimum value, and from Q_3 to the maximum value.

Finding Quartiles from Raw Data

For an ordered dataset $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$:

- The **lower quartile** Q_1 is the median of the lower half of the data.
- The **upper quartile** Q_3 is the median of the upper half of the data.
- The **interquartile range** (IQR) is $\text{IQR} = Q_3 - Q_1$.

4.1.6 Outlier Detection

Definition 4.1.6. An **outlier** is a data value that lies an abnormal distance from other values in the dataset. For IB Mathematics, outliers are identified using the $1.5 \times \text{IQR}$ rule.

Outlier Detection Rule

A data point x is considered a **potential outlier** if:

$$x < Q_1 - 1.5 \times \text{IQR} \quad \text{or} \quad x > Q_3 + 1.5 \times \text{IQR}$$

These boundaries are sometimes called the **fences**:

$$\text{Lower fence} = Q_1 - 1.5 \text{IQR}, \quad \text{Upper fence} = Q_3 + 1.5 \text{IQR}$$

Worked Example: Outlier Detection

A dataset has $Q_1 = 25$, $Q_3 = 45$, and values: 5, 18, 22, 27, 30, 32, 35, 40, 48, 55, 72.

Identify any outliers.

Solution:

$$\text{IQR} = 45 - 25 = 20$$

$$\text{Lower fence} = 25 - 1.5 \times 20 = 25 - 30 = -5$$

$$\text{Upper fence} = 45 + 1.5 \times 20 = 45 + 30 = 75$$

Values below -5 or above 75 are outliers. Here, no value lies outside $[-5, 75]$, so there are **no outliers** in this dataset.

IB Exam Tip: Outliers on Box Plots

When a box plot shows a data point plotted separately (beyond a whisker), this indicates an outlier. On IB exams, you may be asked to identify outliers *before* drawing the box plot, in which case the whiskers extend only to the most extreme *non-outlier* values (not the minimum and maximum).

4.2 Measures of Central Tendency and Dispersion

4.2.1 The Mean

Definition 4.2.1. The **mean** (arithmetic average) of n observations $x_1, x_2, \dots, x_n$ is:

Arithmetic Mean

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

For grouped data with class midpoints x_i and frequencies f_i :

Mean of Grouped Data

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

where x_i is the midpoint of the i -th class and f_i is its frequency.

4.2.2 The Median and Mode

Definition 4.2.2. The **median** is the middle value when data are arranged in order. For n data points:

- If n is odd: $\text{median} = x_{\frac{n+1}{2}}$.
- If n is even: $\text{median} = \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}$.

Definition 4.2.3. The **mode** is the most frequently occurring value in the dataset. A dataset can be unimodal, bimodal, multimodal, or have no mode.

Choosing a Measure of Central Tendency

- The **mean** uses all data values but is affected by outliers.
- The **median** is robust to outliers and preferred for skewed distributions.
- The **mode** is the only measure applicable to categorical data.
- For a symmetric distribution: $\text{mean} \approx \text{median} \approx \text{mode}$.
- For right-skewed data: $\text{mode} < \text{median} < \text{mean}$.
- For left-skewed data: $\text{mean} < \text{median} < \text{mode}$.

4.2.3 Range and Interquartile Range

Range and IQR

$$\text{Range} = x_{\max} - x_{\min}$$

$$\text{IQR} = Q_3 - Q_1$$

The IQR is preferred over the range because it is not affected by extreme values.

4.2.4 Variance

Definition 4.2.4. **Variance** measures the average squared deviation from the mean. There are two versions depending on whether the data represent a population or a sample.

Population Variance

When the data represent the **entire population**:

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

Sample Variance (Unbiased Estimator)

When the data are a **sample** drawn from a larger population:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

Derivation: Computational Formula for Population Variance

Starting from the definition:

$$\begin{aligned}\sigma^2 &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= \frac{1}{n} \sum_{i=1}^n (x_i^2 - 2x_i\bar{x} + \bar{x}^2) \\ &= \frac{1}{n} \left[\sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + \sum_{i=1}^n \bar{x}^2 \right]\end{aligned}$$

Since $\sum_{i=1}^n x_i = n\bar{x}$ and $\sum_{i=1}^n \bar{x}^2 = n\bar{x}^2$:

$$\begin{aligned}\sigma^2 &= \frac{1}{n} \left[\sum_{i=1}^n x_i^2 - 2\bar{x}(n\bar{x}) + n\bar{x}^2 \right] \\ &= \frac{1}{n} \left[\sum_{i=1}^n x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 \right] \\ &= \frac{1}{n} \left[\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right] \\ &= \frac{\sum_{i=1}^n x_i^2}{n} - \bar{x}^2\end{aligned}$$

Computational Formula for Variance

$$\sigma^2 = \frac{\sum x_i^2}{n} - \bar{x}^2$$

Similarly, for the sample variance:

$$s^2 = \frac{\sum x_i^2 - n\bar{x}^2}{n - 1}$$

4.2.5 Standard Deviation

Standard Deviation

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}, \quad s = \sqrt{s^2} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

The standard deviation has the same units as the original data, making it more interpretable than variance.

4.2.6 Effect of Linear Transformations

Theorem 4.2.1 (Linear Transformation of Data). *If $Y = aX + b$ where $a, b \in \mathbb{R}$ are constants, then:*

$$\begin{aligned}\bar{y} &= a\bar{x} + b \\ \sigma_Y &= |a| \sigma_X \\ s_Y &= |a| s_X\end{aligned}$$

The standard deviation is scaled by $|a|$ but is unaffected by b (adding a constant shifts data but does not change spread).

Derivation: Transformation Properties

Let $y_i = ax_i + b$ for each observation. Then:

$$\begin{aligned}\bar{y} &= \frac{1}{n} \sum_{i=1}^n y_i = \frac{1}{n} \sum_{i=1}^n (ax_i + b) \\ &= \frac{1}{n} \left(a \sum_{i=1}^n x_i + nb \right) = a \cdot \frac{\sum x_i}{n} + b = a\bar{x} + b\end{aligned}$$

For the variance:

$$\begin{aligned}\sigma_Y^2 &= \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 \\ &= \frac{1}{n} \sum_{i=1}^n [(ax_i + b) - (a\bar{x} + b)]^2 \\ &= \frac{1}{n} \sum_{i=1}^n (ax_i - a\bar{x})^2 \\ &= \frac{1}{n} \sum_{i=1}^n a^2 (x_i - \bar{x})^2 \\ &= a^2 \cdot \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = a^2 \sigma_X^2\end{aligned}$$

Therefore $\sigma_Y = |a| \sigma_X$.

IB Exam Tip: n vs $n - 1$

- Use σ (dividing by n) when the data represent the **entire population**.
- Use s (dividing by $n - 1$) when the data are a **sample** from a larger population.

- **Critical:** On your GDC, the “sample standard deviation” function (usually denoted s_x or σ_{n-1}) divides by $n - 1$. The “population standard deviation” (σ_x or σ_n) divides by n .
- In IB exam questions, *always check the wording*: if the question says “calculate the standard deviation of these 20 marks” without mentioning a larger population, use n . If it says “20 students were selected from a school of 500 to estimate the population standard deviation”, use $n - 1$.
- When in doubt, read the question carefully. IB questions are usually explicit about whether the data is a sample or the population.

Worked Example: Mean, Variance, and Transformations

The marks of 6 students in a test (out of 40) are: 28, 35, 18, 32, 25, 22.

(a) Find the mean and standard deviation.

(b) The teacher decides to convert the marks to a percentage by multiplying by 2.5 and adding 0. Find the new mean and standard deviation.

Solution:

$$\bar{x} = \frac{28 + 35 + 18 + 32 + 25 + 22}{6} = \frac{160}{6} = 26.67$$

Using the computational formula:

$$\begin{aligned}\sum x_i^2 &= 28^2 + 35^2 + 18^2 + 32^2 + 25^2 + 22^2 \\ &= 784 + 1225 + 324 + 1024 + 625 + 484 = 4466 \\ \sigma^2 &= \frac{4466}{6} - 26.67^2 = 744.33 - 711.11 = 33.22 \\ \sigma &= \sqrt{33.22} \approx 5.76\end{aligned}$$

(b) Here $Y = 2.5X + 0$:

$$\begin{aligned}\bar{y} &= 2.5 \times 26.67 = 66.67\% \\ \sigma_Y &= 2.5 \times 5.76 = 14.41\end{aligned}$$

4.3 Correlation and Regression

4.3.1 Scatter Diagrams

Definition 4.3.1. A **scatter diagram** (scatter plot) displays pairs of bivariate data points (x_i, y_i) on a Cartesian plane. It is the first step in investigating a relationship between two variables.

- **Positive correlation:** as x increases, y tends to increase.
- **Negative correlation:** as x increases, y tends to decrease.
- **No correlation:** no apparent linear relationship.
- **Strong vs weak:** describes how tightly clustered points are around a line.

4.3.2 Pearson's Correlation Coefficient

Definition 4.3.2. *Pearson's product-moment correlation coefficient* r measures the strength and direction of the linear association between two variables.

Pearson's Correlation Coefficient

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$$

where:

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - n\bar{x}\bar{y}$$

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2$$

$$S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum y_i^2 - n\bar{y}^2$$

Properties of r

- $-1 \leq r \leq 1$.
- $r = 1$: perfect positive linear correlation.
- $r = -1$: perfect negative linear correlation.
- $r = 0$: no linear correlation (but there may be a non-linear relationship).
- r is dimensionless (unit-free).

Interpreting r in Context

In IB exams, you must interpret r *in context*. For example, do not write “ $r = 0.85$ means strong correlation.” Instead write: “There is a strong positive linear correlation between the temperature and the number of ice creams sold, suggesting that as temperature increases, ice cream sales tend to increase.” Always address both **strength** and **direction**, and note that correlation does *not* imply causation.

4.3.3 Regression Line of y on x

Definition 4.3.3. The **least squares regression line of y on x** is the line $y = ax + b$ that minimises the sum of squared vertical deviations $\sum (y_i - \hat{y}_i)^2$.

Regression Line of y on x

$$y = ax + b$$

where:

$$a = \frac{S_{xy}}{S_{xx}}, \quad b = \bar{y} - a\bar{x}$$

The line always passes through the mean point $(\bar{x}, \bar{y})$.

Derivation: Least Squares Regression Formulas

We wish to find a and b that minimise:

$$S(a, b) = \sum_{i=1}^n (y_i - ax_i - b)^2$$

Substituting $b = \bar{y} - a\bar{x}$ (which minimises S with respect to b when we set $\frac{\partial S}{\partial b} = 0$):

$$\begin{aligned} S(a) &= \sum_{i=1}^n [y_i - ax_i - (\bar{y} - a\bar{x})]^2 \\ &= \sum_{i=1}^n [(y_i - \bar{y}) - a(x_i - \bar{x})]^2 \\ &= \sum_{i=1}^n [(y_i - \bar{y})^2 - 2a(x_i - \bar{x})(y_i - \bar{y}) + a^2(x_i - \bar{x})^2] \\ &= S_{yy} - 2a S_{xy} + a^2 S_{xx} \end{aligned}$$

This is a quadratic in a . The minimum occurs at:

$$\begin{aligned} \frac{dS}{da} &= -2S_{xy} + 2a S_{xx} = 0 \\ a &= \frac{S_{xy}}{S_{xx}} \end{aligned}$$

To verify this gives a minimum: $\frac{d^2 S}{da^2} = 2S_{xx} > 0$ (since $S_{xx} > 0$ unless all x_i are equal).

For b , setting $\frac{\partial S}{\partial b} = 0$:

$$\begin{aligned} \frac{\partial S}{\partial b} &= -2 \sum_{i=1}^n (y_i - ax_i - b) = 0 \\ \sum y_i &= a \sum x_i + nb \\ \bar{y} &= a\bar{x} + b \\ b &= \bar{y} - a\bar{x} \end{aligned}$$

Interpretation of a and b

- b is the y -intercept: the predicted value of y when $x = 0$. However, $x = 0$ may not be within the data range, making b meaningless in some contexts.
- a is the gradient (slope): the predicted *change in y* for a one-unit increase in x .
- In IB exams, always interpret a and b in context with appropriate units.

4.3.4 Regression Line of x on y (HL)

Definition 4.3.4. The **regression line of x on y** minimises the sum of squared horizontal deviations. It is used when we wish to predict x from y .

Regression Line of x on y

$$x = a'y + b'$$

where:

$$a' = \frac{S_{xy}}{S_{yy}}, \quad b' = \bar{x} - a'\bar{y}$$

This can be rearranged to: $y = \frac{1}{a'}x - \frac{b'}{a'}$.

Why Two Regression Lines?

The regression line of y on x minimises vertical residuals (used when predicting y from x).
 The regression line of x on y minimises horizontal residuals (used when predicting x from y).
 The two lines are different unless $|r| = 1$. Both pass through the point $(\bar{x}, \bar{y})$.

IB Exam Tip: Interpolation vs Extrapolation

- **Interpolation:** predicting within the range of observed data. This is reliable.
- **Extrapolation:** predicting outside the range of observed data. This is unreliable and should be treated with caution.
- In IB mark schemes, you may lose marks for making extrapolations without commenting on their unreliability.
- Always check whether a requested prediction falls within the data range before using the regression line.

Worked Example: Correlation and Regression

The data below show the temperature (x , in $^{\circ}\text{C}$) and ice cream sales (y , in hundreds) for 5 days:

x	15	20	25	30	35
y	2.1	3.5	5.0	6.2	8.1

- (a) Find the correlation coefficient r .
 (b) Find the equation of the regression line of y on x .
 (c) Estimate the sales when the temperature is 28°C .

Solution:

$$\bar{x} = \frac{15 + 20 + 25 + 30 + 35}{5} = 25$$

$$\bar{y} = \frac{2.1 + 3.5 + 5.0 + 6.2 + 8.1}{5} = 4.98$$

$$\sum x_i^2 = 225 + 400 + 625 + 900 + 1225 = 3375$$

$$\sum y_i^2 = 4.41 + 12.25 + 25 + 38.44 + 65.61 = 145.71$$

$$\sum x_i y_i = 31.5 + 70 + 125 + 186 + 283.5 = 696$$

$$S_{xx} = 3375 - 5(25)^2 = 3375 - 3125 = 250$$

$$S_{yy} = 145.71 - 5(4.98)^2 = 145.71 - 124.002 = 21.708$$

$$S_{xy} = 696 - 5(25)(4.98) = 696 - 622.5 = 73.5$$

(a)

$$r = \frac{73.5}{\sqrt{250 \times 21.708}} = \frac{73.5}{\sqrt{5427}} = \frac{73.5}{73.67} \approx 0.998$$

(b)

$$a = \frac{S_{xy}}{S_{xx}} = \frac{73.5}{250} = 0.294$$

$$b = \bar{y} - a\bar{x} = 4.98 - 0.294(25) = 4.98 - 7.35 = -2.37$$

So $y = 0.294x - 2.37$ (sales in hundreds).

(c) At $x = 28$:

$$\hat{y} = 0.294(28) - 2.37 = 8.232 - 2.37 = 5.862$$

Estimated sales: 586 ice creams (since y is in hundreds). This is an interpolation, so the estimate is reliable.

4.4 Probability Fundamentals

4.4.1 Sample Space, Events, and Outcomes

Definition 4.4.1. • An **experiment** (or trial) is any process that produces an outcome.

- The **sample space** U (or Ω or S) is the set of all possible outcomes.
- An **event** A is a subset of the sample space: $A \subseteq U$.
- An **outcome** is a single element of the sample space.

4.4.2 Basic Probability

Probability of an Event

For an event A in a sample space U where all outcomes are equally likely:

$$P(A) = \frac{n(A)}{n(U)} = \frac{\text{number of outcomes in } A}{\text{total number of outcomes}}$$

- $0 \leq P(A) \leq 1$ for any event A .
- $P(U) = 1$, $P(\emptyset) = 0$.

4.4.3 Complementary Events

Complement Rule

The complement of event A , denoted A' (or $\bar{A}$), is the event that A does *not* occur:

$$P(A') = 1 - P(A)$$

4.4.4 Combined Events

Addition Rule (General)

For any two events A and B :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Mutually Exclusive Events

Events A and B are **mutually exclusive** (disjoint) if they cannot both occur, i.e., $A \cap B = \emptyset$. In this case:

$$P(A \cap B) = 0 \quad \text{and} \quad P(A \cup B) = P(A) + P(B)$$

4.4.5 Venn Diagrams, Tree Diagrams, and Sample Space Diagrams

Visualising Probability

- **Venn diagrams:** best for showing relationships between 2–3 events, especially combined events and complements.
- **Tree diagrams:** best for sequential experiments with two or more stages. Multiply along branches, add between branches.
- **Sample space diagrams (grids):** best for two-stage experiments where each stage has a small number of equally likely outcomes (e.g., rolling two dice).

Tree Diagram Rules

- The probability on each branch = $P(\text{that outcome} \mid \text{previous outcome})$.
- Multiply probabilities **along** branches to get the probability of a path.
- Add probabilities of mutually exclusive paths to get the total probability of an event.

IB Exam Tips on Probability Questions

- Always check whether events are mutually exclusive before using $P(A \cup B) = P(A) + P(B)$.
- For “at least one” problems, use the complement: $P(\text{at least one}) = 1 - P(\text{none})$.
- For “or” problems, use the addition rule. For “and” problems, use multiplication (or conditional probability).
- Read carefully: “given that” or “knowing that” signals conditional probability (Section 4.5).
- IB exams often combine a Venn diagram question with a follow-up conditional probability question.

Worked Example: Probability with Venn Diagram

In a class of 30 students, 18 play football, 15 play basketball, and 5 play neither. A student is chosen at random.

(a) Draw a Venn diagram and find the probability the student plays both sports.

(b) Find the probability the student plays exactly one sport.

Solution:

(a) Let F = plays football, B = plays basketball.

$$n(F \cup B) = 30 - 5 = 25$$

$$n(F \cup B) = n(F) + n(B) - n(F \cap B)$$

$$25 = 18 + 15 - n(F \cap B)$$

$$n(F \cap B) = 8$$

So: $n(F \cap B) = 8$, $n(F \cap B') = 18 - 8 = 10$, $n(F' \cap B) = 15 - 8 = 7$, $n(F' \cap B') = 5$.

$$P(F \cap B) = \frac{8}{30} = \frac{4}{15}$$

(b)

$$P(\text{exactly one sport}) = \frac{10 + 7}{30} = \frac{17}{30}$$

4.5 Conditional Probability and Independence

4.5.1 Conditional Probability

Definition 4.5.1. The **conditional probability** of A given B , denoted $P(A|B)$, is the probability that A occurs given that B has already occurred.

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad \text{provided } P(B) > 0$$

Understanding Conditional Probability Intuitively

$P(A|B)$ restricts the sample space to only those outcomes where B occurs. It asks: “Among all outcomes where B happens, what fraction also have A happening?”

If using a Venn diagram: $P(A|B) = \frac{n(A \cap B)}{n(B)}$ (provided equally likely outcomes).

4.5.2 Independent Events

Definition 4.5.2. Two events A and B are **independent** if the occurrence of one does not affect the probability of the other:

$$P(A \cap B) = P(A) \times P(B)$$

Equivalently:

$$P(A|B) = P(A)$$

$$P(B|A) = P(B)$$

Theorem 4.5.1 (Independence of A and B'). If A and B are independent events, then A and B' are also independent.

Proof. We need to show that $P(A \cap B') = P(A) \times P(B')$.

From the Venn diagram (or addition rule), since $A = (A \cap B) \cup (A \cap B')$ and these are mutually exclusive:

$$\begin{aligned} P(A) &= P(A \cap B) + P(A \cap B') \\ P(A \cap B') &= P(A) - P(A \cap B) \end{aligned}$$

Since A and B are independent, $P(A \cap B) = P(A) \times P(B)$:

$$\begin{aligned} P(A \cap B') &= P(A) - P(A) \times P(B) \\ &= P(A)[1 - P(B)] \\ &= P(A) \times P(B') \end{aligned}$$

Therefore A and B' are independent. By the same argument, A' and B are independent, and A' and B' are independent. $\square$

Mutually Exclusive vs Independent

- **Mutually exclusive:** $A \cap B = \emptyset$, so $P(A \cap B) = 0$.
- **Independent:** $P(A \cap B) = P(A) \times P(B)$.
- If A and B are both mutually exclusive and $P(A) > 0$ and $P(B) > 0$, they **cannot** be independent (since $P(A) \times P(B) \neq 0$).
- Conversely, independent events with positive probability are **never** mutually exclusive.

- The only way events can be both independent and mutually exclusive is if at least one has probability zero.

Worked Example: Conditional Probability and Independence

A bag contains 5 red and 3 blue marbles. Two marbles are drawn **without replacement**.

(a) Find the probability that both are red.

(b) Find the probability that the second is red given that the first is red.

(c) Are the events “first marble is red” and “second marble is red” independent?

Solution:

Let R_1 = first marble is red, R_2 = second marble is red.

(a)

$$P(R_1 \cap R_2) = P(R_1) \times P(R_2|R_1) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$$

(b)

$$P(R_2|R_1) = \frac{4}{7}$$

(c) For independence, we need $P(R_2|R_1) = P(R_2)$. Since $P(R_2|R_1) = \frac{4}{7}$ but $P(R_2) = \frac{5}{8}$ (by symmetry), $\frac{4}{7} \neq \frac{5}{8}$, so the events are **not independent**. This makes sense: drawing without replacement changes the composition of the bag.

Note: if drawing **with replacement**, the events *would* be independent since $P(R_2|R_1) = \frac{5}{8} = P(R_2)$.

4.6 Bayes' Theorem

4.6.1 Bayes' Theorem Formula

Theorem 4.6.1 (Bayes' Theorem). For events A and B with $P(B) > 0$:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Derivation of Bayes' Theorem

Starting from the definition of conditional probability:

For $P(A|B)$:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

For $P(B|A)$:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \implies P(A \cap B) = P(B|A) \cdot P(A)$$

Substituting:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

4.6.2 Extended (Law of Total Probability) Form

Law of Total Probability

If $A_1, A_2, \dots, A_n$ are mutually exclusive and exhaustive events (i.e., they partition the sample space), then for any event B :

$$P(B) = \sum_{i=1}^n P(B|A_i) \cdot P(A_i)$$

Bayes' Theorem: Extended Form

For a partition A_1, A_2, A_3 of the sample space:

$$P(A_k|B) = \frac{P(B|A_k) \cdot P(A_k)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + P(B|A_3)P(A_3)}$$

For two events A and A' :

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B|A)P(A) + P(B|A')P(A')}$$

4.6.3 Tree Diagram Approach

Tree Diagrams vs Bayes' Formula

For IB exams, the **tree diagram approach** is often simpler and less error-prone than using Bayes' formula directly. The procedure is:

1. Draw a tree diagram with the **prior** probabilities on the first set of branches (e.g., $P(A)$, $P(B)$, $P(C)$).
2. On the second set of branches, write the **conditional** probabilities (e.g., $P(\text{positive}|A)$, $P(\text{positive}|B)$).
3. For “reverse” questions (finding $P(A|\text{positive})$):
 - (a) Find the probability of each path leading to the observed outcome.
 - (b) Sum these to get $P(\text{observed outcome})$.
 - (c) Divide the relevant path probability by the sum.

This is equivalent to Bayes' theorem but less prone to algebraic errors.

Worked Example: Bayes' Theorem (Medical Testing)

A disease affects 1% of a population. A test for the disease has a 95% true positive rate ($P(+|D) = 0.95$) and a 10% false positive rate ($P(+|D') = 0.10$).

A person tests positive. Find the probability they actually have the disease.

Solution using a tree diagram:

First branches (prior): $P(D) = 0.01$, $P(D') = 0.99$.

Second branches (conditional):

$$\begin{aligned} P(+|D) &= 0.95, & P(-|D) &= 0.05 \\ P(+|D') &= 0.10, & P(-|D') &= 0.90 \end{aligned}$$

Path probabilities for testing positive:

$$\begin{aligned} P(D \cap +) &= 0.01 \times 0.95 = 0.0095 \\ P(D' \cap +) &= 0.99 \times 0.10 = 0.0990 \end{aligned}$$

$$P(+) = 0.0095 + 0.0990 = 0.1085$$

$$P(D|+) = \frac{P(D \cap +)}{P(+)} = \frac{0.0095}{0.1085} \approx 0.0876$$

Interpretation: Even with a positive test result, there is only about an 8.8% chance the person has the disease! This is because the disease is rare (low prior probability) and the false positive rate is relatively high compared to the prevalence.

Solution using Bayes' formula:

$$P(D|+) = \frac{P(+|D) P(D)}{P(+|D) P(D) + P(+|D') P(D')} = \frac{0.95 \times 0.01}{0.95 \times 0.01 + 0.10 \times 0.99} = \frac{0.0095}{0.1085} \approx 0.0876$$

4.7 Discrete Random Variables

4.7.1 Probability Distributions

Definition 4.7.1. A **discrete random variable** X is a variable that can take a countable number of values $x_1, x_2, \dots$, each with an associated probability $P(X = x_i) = p_i$.

Requirements for a Probability Distribution

1. $0 \leq p_i \leq 1$ for all i .
2. $\sum_{\text{all } i} p_i = 1$.

A probability distribution is often displayed as a table:

x	x_1	x_2	$\cdots$	x_n
$P(X = x)$	p_1	p_2	$\cdots$	p_n

4.7.2 Expected Value

Definition 4.7.2. The **expected value** (or *expectation* or *mean*) of a discrete random variable X is:

Expected Value

$$E(X) = \sum_i x_i p_i = \sum_{\text{all } x} x \cdot P(X = x)$$

Also denoted μ_X .

Understanding $E(X)$

$E(X)$ is the long-run average value of X if the experiment were repeated many times. It is also called the **mean** of the probability distribution. Note that $E(X)$ need not be one of the possible values of X .

4.7.3 Variance of a Discrete Random Variable

Variance of a Discrete Random Variable

$$\text{Var}(X) = \sum_i (x_i - \mu)^2 p_i$$

where $\mu = E(X)$.

Computational Formula for Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

where:

$$E(X^2) = \sum_i x_i^2 p_i$$

(Derivation: $\text{Var}(X) = E(X^2) - [E(X)]^2$) Starting from the definition of variance:

$$\begin{aligned} \text{Var}(X) &= E[(X - \mu)^2] \\ &= E[X^2 - 2\mu X + \mu^2] \\ &= E(X^2) - 2\mu E(X) + \mu^2 \end{aligned}$$

Since $\mu = E(X)$:

$$\begin{aligned} \text{Var}(X) &= E(X^2) - 2\mu \cdot \mu + \mu^2 \\ &= E(X^2) - 2\mu^2 + \mu^2 \\ &= E(X^2) - \mu^2 \\ &= E(X^2) - [E(X)]^2 \end{aligned}$$

4.7.4 Linear Transformations of Random Variables

Theorem 4.7.1 (Effect of Linear Transformations on $E(X)$ and $\text{Var}(X)$). If $Y = aX + b$ where a, b are constants:

$$\begin{aligned} E(Y) &= a E(X) + b \\ \text{Var}(Y) &= a^2 \text{Var}(X) \end{aligned}$$

The standard deviation transforms as: $\sigma_Y = |a| \sigma_X$.

Derivation: Transformation Properties for Random Variables

Expected value:

$$\begin{aligned} E(Y) &= E(aX + b) = \sum (ax_i + b) p_i \\ &= a \sum x_i p_i + b \sum p_i \\ &= a E(X) + b \cdot 1 = a E(X) + b \end{aligned}$$

Variance:

$$\begin{aligned}
\text{Var}(Y) &= E(Y^2) - [E(Y)]^2 \\
&= E[(aX + b)^2] - [aE(X) + b]^2 \\
&= E[a^2X^2 + 2abX + b^2] - [a^2[E(X)]^2 + 2abE(X) + b^2] \\
&= a^2E(X^2) + 2abE(X) + b^2 - a^2[E(X)]^2 - 2abE(X) - b^2 \\
&= a^2[E(X^2) - [E(X)]^2] \\
&= a^2 \text{Var}(X)
\end{aligned}$$

Adding a Constant Does Not Affect Variance

Adding a constant b shifts the entire distribution but does not change its spread, so $\text{Var}(X + b) = \text{Var}(X)$. Only multiplying by a (stretching or compressing) changes the variance, and it does so by a factor of a^2 (not a).

Worked Example: Discrete Random Variable

A biased die has the probability distribution:

x	1	2	3	4	5	6
$P(X = x)$	0.1	0.2	0.3	0.15	0.15	0.1

(a) Find $E(X)$ and $\text{Var}(X)$.

(b) If $Y = 2X + 3$, find $E(Y)$ and $\text{Var}(Y)$.

Solution:

(a)

$$\begin{aligned}
E(X) &= 1(0.1) + 2(0.2) + 3(0.3) + 4(0.15) + 5(0.15) + 6(0.1) \\
&= 0.1 + 0.4 + 0.9 + 0.6 + 0.75 + 0.6 = 3.35
\end{aligned}$$

$$\begin{aligned}
E(X^2) &= 1(0.1) + 4(0.2) + 9(0.3) + 16(0.15) + 25(0.15) + 36(0.1) \\
&= 0.1 + 0.8 + 2.7 + 2.4 + 3.75 + 3.6 = 13.35
\end{aligned}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 13.35 - 3.35^2 = 13.35 - 11.2225 = 2.1275$$

(b)

$$\begin{aligned}
E(Y) &= 2(3.35) + 3 = 9.7 \\
\text{Var}(Y) &= 2^2 \times 2.1275 = 4 \times 2.1275 = 8.51
\end{aligned}$$

4.8 Binomial Distribution

4.8.1 Definition and Formula

Definition 4.8.1. A random variable X has a **binomial distribution** with parameters n and p , written $X \sim B(n, p)$, if X counts the number of successes in n independent trials, each with probability of success p .

Conditions for a Binomial Distribution

The following four conditions must all be satisfied (often remembered as BINS):

1. **B**inary outcomes: each trial has exactly two outcomes (success/failure).
2. **I**ndependent trials: the outcome of one trial does not affect another.
3. **N** fixed: the number of trials n is fixed in advance.
4. **S**ame probability: the probability of success p is constant for each trial.

Binomial Probability Mass Function

If $X \sim B(n, p)$, then:

$$P(X = r) = \binom{n}{r} p^r (1 - p)^{n-r}, \quad r = 0, 1, 2, \dots, n$$

where $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ is the binomial coefficient.

4.8.2 Mean and Variance**Mean and Variance of $B(n, p)$**

If $X \sim B(n, p)$:

$$\begin{aligned} E(X) &= np \\ \text{Var}(X) &= np(1 - p) \end{aligned}$$

The standard deviation is $\sigma = \sqrt{np(1 - p)}$.

Derivation: Mean of the Binomial Distribution

Let $X \sim B(n, p)$. We can write $X = X_1 + X_2 + \dots + X_n$ where each X_i is an indicator variable for the i -th trial:

$$X_i = \begin{cases} 1 & \text{if trial } i \text{ is a success} \\ 0 & \text{if trial } i \text{ is a failure} \end{cases}$$

Each $X_i \sim B(1, p)$, so $E(X_i) = 1 \cdot p + 0 \cdot (1 - p) = p$.

By linearity of expectation:

$$E(X) = \sum_{i=1}^n E(X_i) = n \cdot p = np$$

Derivation: Variance of the Binomial Distribution

Since the trials are independent, the X_i are independent, so:

$$\text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i)$$

For each $X_i \sim B(1, p)$:

$$\begin{aligned} E(X_i^2) &= 1^2 \cdot p + 0^2 \cdot (1 - p) = p \\ \text{Var}(X_i) &= E(X_i^2) - [E(X_i)]^2 = p - p^2 = p(1 - p) \end{aligned}$$

Therefore:

$$\text{Var}(X) = n \cdot p(1 - p) = np(1 - p)$$

Using Your GDC for Binomial Calculations

- **Individual probability** $P(X = r)$: use the binomial PDF function.
- **Cumulative probability** $P(X \leq r)$: use the binomial CDF function.
- $P(X \geq r)$: calculate as $1 - P(X \leq r - 1)$.
- $P(a \leq X \leq b)$: calculate as $P(X \leq b) - P(X \leq a - 1)$.
- Know the menu path on your specific GDC model. On TI-Nspire: Menu $\rightarrow$ Probability $\rightarrow$ Distributions. On Casio fx-CG50: STAT $\rightarrow$ DIST $\rightarrow$ BINOMIAL.

IB Exam Tip: Identifying Binomial Context

In IB exams, binomial questions often use wording like:

- “A fair coin is tossed 10 times...” ($n = 10, p = 0.5$)
- “65% of people prefer...A random sample of 20 people is selected...” ($n = 20, p = 0.65$)
- “On average, 8% of items are defective. Find the probability that in a batch of 50...” ($n = 50, p = 0.08$)

Always verify the four conditions (BINS) before using the binomial distribution. If the question says “without replacement”, the trials are *not* independent and you should use the hypergeometric distribution (not required for IB, but worth recognising).

Worked Example: Binomial Distribution

A multiple choice test has 10 questions, each with 4 options (only one correct). A student guesses every answer.

(a) Find the probability the student gets exactly 3 correct.

(b) Find the probability the student gets at least 6 correct.

(c) Find the expected number and standard deviation of correct answers.

Solution: Here $X \sim B(10, 0.25)$.

(a)

$$\begin{aligned} P(X = 3) &= \binom{10}{3} (0.25)^3 (0.75)^7 \\ &= 120 \times 0.015625 \times 0.13348 \\ &= 120 \times 0.002087 \\ &\approx 0.2503 \end{aligned}$$

(b)

$$\begin{aligned} P(X \geq 6) &= 1 - P(X \leq 5) \\ &= 1 - \sum_{r=0}^5 \binom{10}{r} (0.25)^r (0.75)^{10-r} \\ &\approx 1 - 0.9803 = 0.0197 \end{aligned}$$

(c)

$$\begin{aligned} E(X) &= 10 \times 0.25 = 2.5 \\ \text{Var}(X) &= 10 \times 0.25 \times 0.75 = 1.875 \\ \sigma &= \sqrt{1.875} \approx 1.369 \end{aligned}$$

4.9 Normal Distribution

4.9.1 The Normal Distribution

Definition 4.9.1. A continuous random variable X has a **normal distribution** with parameters μ (mean) and σ^2 (variance), written $X \sim N(\mu, \sigma^2)$, if its probability density function is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}$$

4.9.2 Properties of the Normal Curve

Bell Curve Properties

- **Symmetric** about the mean μ .
- **Bell-shaped**: maximum at $x = \mu$, tails approach but never reach the x -axis.
- The **inflection points** of the curve are at $x = \mu \pm \sigma$.
- **Total area** under the curve = 1.
- Mean = median = mode = μ .

The 68–95–99.7 Rule (Empirical Rule)

For $X \sim N(\mu, \sigma^2)$:

$$\begin{aligned} P(\mu - \sigma < X < \mu + \sigma) &\approx 0.6827 \\ P(\mu - 2\sigma < X < \mu + 2\sigma) &\approx 0.9545 \\ P(\mu - 3\sigma < X < \mu + 3\sigma) &\approx 0.9973 \end{aligned}$$

The 68–95–99.7 Rule in Exams

This rule is useful for quick estimates and for checking whether GDC answers are reasonable. However, for exact answers in IB exams, always use your GDC's normal distribution functions.

4.9.3 Standardization

Definition 4.9.2. Standardization converts any normal variable to the standard normal $Z \sim N(0, 1)$:

Standardization Formula

$$Z = \frac{X - \mu}{\sigma}, \quad Z \sim N(0, 1)$$

Derivation of Standardization

Let $X \sim N(\mu, \sigma^2)$ and define $Z = \frac{X - \mu}{\sigma}$.

Showing $E(Z) = 0$:

$$E(Z) = E\left(\frac{X - \mu}{\sigma}\right) = \frac{E(X) - \mu}{\sigma} = \frac{\mu - \mu}{\sigma} = 0$$

Showing $\text{Var}(Z) = 1$:

$$\begin{aligned} \text{Var}(Z) &= \text{Var}\left(\frac{X - \mu}{\sigma}\right) = \frac{1}{\sigma^2} \text{Var}(X - \mu) \\ &= \frac{1}{\sigma^2} \text{Var}(X) = \frac{\sigma^2}{\sigma^2} = 1 \end{aligned}$$

Since any linear transformation of a normal variable is also normal, and we have shown Z has mean 0 and variance 1, we conclude $Z \sim N(0, 1)$.

4.9.4 Finding Probabilities

Normal Probabilities

For $X \sim N(\mu, \sigma^2)$, use your GDC to find:

- $P(X < a)$: use normal CDF with lower bound $-\infty$ (or a very small number) and upper bound a .
- $P(a < X < b)$: use normal CDF with lower bound a and upper bound b .
- $P(X > a) = 1 - P(X < a)$.
- By symmetry: $P(X < \mu - a) = P(X > \mu + a)$.

4.9.5 Inverse Normal

Inverse Normal

Given a probability p , the **inverse normal** function finds the value a such that:

$$P(X < a) = p$$

Use the inverse normal function on your GDC with the given μ , σ , and area p .

4.9.6 Normal Approximation to the Binomial

Theorem 4.9.1 (Normal Approximation to the Binomial Distribution). *If $X \sim B(n, p)$ and both $np > 5$ and $n(1 - p) > 5$, then X can be approximated by:*

$$X \approx Y \sim N(np, np(1 - p))$$

Conditions for Normal Approximation

$$np > 5 \quad \text{and} \quad n(1 - p) > 5$$

These ensure the binomial distribution is sufficiently symmetric. If p is close to 0 or 1, a larger n is needed.

Continuity Correction

Since the binomial distribution is discrete and the normal is continuous, we apply a **continuity correction**:

$$\begin{aligned} P(X \leq k) &\approx P(Y < k + 0.5) \\ P(X \geq k) &\approx P(Y > k - 0.5) \\ P(X = k) &\approx P(k - 0.5 < Y < k + 0.5) \\ P(a \leq X \leq b) &\approx P(a - 0.5 < Y < b + 0.5) \end{aligned}$$

Continuity Correction Rule of Thumb

The continuity correction extends by 0.5 in the direction that includes the relevant discrete value(s). For “ $\leq k$ ”, include k by going up to $k + 0.5$. For “ $\geq k$ ”, include k by going down to $k - 0.5$.

IB Exam Tip: When to Use Normal Approximation

- If n is small enough that your GDC can compute exact binomial probabilities (typically $n < 1000$), use the **exact binomial** — it is always more accurate.
- Normal approximation is mainly useful when n is very large or when the question explicitly asks for it.
- Always check the conditions $np > 5$ and $n(1-p) > 5$ *before* applying the approximation.
- Never forget the continuity correction — marks are deducted for omitting it.
- Common IB exam wording: “Using the normal approximation to the binomial distribution, find...” This signals that you must use the approximation.

Worked Example: Normal Distribution

The heights of IB students are normally distributed with mean 165 cm and standard deviation 8 cm. Let $H \sim N(165, 64)$.

(a) Find $P(H > 175)$.

(b) Find $P(160 < H < 172)$.

(c) The top 10% of students are selected for a basketball team. Find the minimum height required.

Solution:

(a) Using the GDC: $P(H > 175) = 1 - P(H < 175) \approx 1 - 0.8944 = 0.1056$.

Alternatively, using standardization:

$$z = \frac{175 - 165}{8} = 1.25$$

$$P(Z > 1.25) = 1 - \Phi(1.25) \approx 1 - 0.8944 = 0.1056$$

(b)

$$\begin{aligned} P(160 < H < 172) &= P\left(\frac{160 - 165}{8} < Z < \frac{172 - 165}{8}\right) \\ &= P(-0.625 < Z < 0.875) \\ &= \Phi(0.875) - \Phi(-0.625) \\ &\approx 0.8092 - 0.2660 = 0.5432 \end{aligned}$$

(c) We need h such that $P(H < h) = 0.90$. Using the inverse normal on the GDC:

$$h \approx 175.25 \text{ cm}$$

So the minimum height is approximately 175.3 cm.

Worked Example: Normal Approximation to Binomial

A fair coin is tossed 200 times. Using the normal approximation, find the probability of getting between 90 and 110 heads (inclusive).

Solution: Let $X \sim B(200, 0.5)$.

Check conditions: $np = 100 > 5$ and $n(1-p) = 100 > 5$. Both satisfied.

Normal approximation: $Y \sim N(100, 50)$ where $\mu = 100$ and $\sigma^2 = 50$, so $\sigma = \sqrt{50} \approx 7.071$.

With continuity correction:

$$\begin{aligned}
 P(90 \leq X \leq 110) &\approx P(89.5 < Y < 110.5) \\
 &= P\left(\frac{89.5 - 100}{7.071} < Z < \frac{110.5 - 100}{7.071}\right) \\
 &= P(-1.485 < Z < 1.485) \\
 &= \Phi(1.485) - \Phi(-1.485) \\
 &\approx 0.9312 - 0.0688 = 0.8624
 \end{aligned}$$

4.10 Continuous Random Variables and PDFs

4.10.1 Probability Density Functions

Definition 4.10.1. A **probability density function** (PDF) $f(x)$ for a continuous random variable X satisfies:

1. $f(x) \geq 0$ for all $x \in \mathbb{R}$.
2. $\int_{-\infty}^{\infty} f(x) \, dx = 1$.

Probability from a PDF

For a continuous random variable X with PDF $f(x)$:

$$P(a \leq X \leq b) = \int_a^b f(x) \, dx$$

Note: for continuous variables, $P(X = a) = 0$ for any single value a , so $P(a \leq X \leq b) = P(a < X < b) = P(a \leq X < b) = P(a < X \leq b)$.

Cumulative Distribution Function (CDF)

The CDF $F(x)$ of a continuous random variable X is:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) \, dt$$

- $F'(x) = f(x)$ (the PDF is the derivative of the CDF).
- $F(-\infty) = 0$, $F(+\infty) = 1$.
- $P(a < X \leq b) = F(b) - F(a)$.

4.10.2 Mode and Median of a Continuous Random Variable

Mode and Median

- **Mode:** the value of x at which $f(x)$ is maximised. Find by setting $f'(x) = 0$ and checking $f''(x) < 0$ (or by inspection for simple PDFs).
- **Median:** the value m such that $P(X \leq m) = 0.5$, i.e., solve $\int_{-\infty}^m f(x) \, dx = 0.5$.

4.10.3 Expected Value and Variance

Expected Value of a Continuous Random Variable

$$E(X) = \int_{-\infty}^{\infty} x f(x) \, dx$$

For a PDF defined on $[a, b]$:

$$E(X) = \int_a^b x f(x) \, dx$$

Variance of a Continuous Random Variable

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) \, dx = E(X^2) - [E(X)]^2$$

where:

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) \, dx$$

Strategy for IB Questions on Continuous Random Variables

Typical IB exam questions on this topic follow a pattern:

1. **Find k :** use the condition $\int f(x) \, dx = 1$ to determine an unknown constant.
2. **Find a probability:** compute a definite integral of $f(x)$.
3. **Find the median:** solve $\int f(x) \, dx = 0.5$ for the upper limit.
4. **Find $E(X)$ and $\text{Var}(X)$:** compute the appropriate integrals.
5. **Find the mode:** maximise $f(x)$ using calculus or inspection.

Worked Example: Continuous Random Variable

A continuous random variable X has PDF:

$$f(x) = \begin{cases} kx(4-x) & 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the value of k .
- (b) Find $P(1 < X < 3)$.
- (c) Find the median of X .
- (d) Find $E(X)$ and $\text{Var}(X)$.

Solution:

- (a) Using $\int_{-\infty}^{\infty} f(x) \, dx = 1$:

$$\begin{aligned} \int_0^4 kx(4-x) \, dx &= 1 \\ k \int_0^4 (4x - x^2) \, dx &= 1 \\ k \left[2x^2 - \frac{x^3}{3} \right]_0^4 &= 1 \\ k \left(32 - \frac{64}{3} \right) &= 1 \\ k \left(\frac{96 - 64}{3} \right) &= 1 \\ k \left(\frac{32}{3} \right) &= 1 \implies k = \frac{3}{32} \end{aligned}$$

(b)

$$\begin{aligned}
 P(1 < X < 3) &= \int_1^3 \frac{3}{32} x(4-x) \, dx \\
 &= \frac{3}{32} \int_1^3 (4x - x^2) \, dx \\
 &= \frac{3}{32} \left[2x^2 - \frac{x^3}{3} \right]_1^3 \\
 &= \frac{3}{32} \left[(18 - 9) - \left(2 - \frac{1}{3} \right) \right] \\
 &= \frac{3}{32} \left[9 - \frac{5}{3} \right] = \frac{3}{32} \times \frac{22}{3} = \frac{22}{32} = \frac{11}{16}
 \end{aligned}$$

(c) Find m such that $\int_0^m \frac{3}{32} x(4-x) \, dx = 0.5$:

$$\begin{aligned}
 \frac{3}{32} \left[2x^2 - \frac{x^3}{3} \right]_0^m &= 0.5 \\
 \frac{3}{32} \left(2m^2 - \frac{m^3}{3} \right) &= \frac{1}{2} \\
 2m^2 - \frac{m^3}{3} &= \frac{16}{3} \\
 6m^2 - m^3 &= 16
 \end{aligned}$$

By trial or GDC, $m \approx 2.30$ (to 3 s.f.). (This cubic has one root in $[0, 4]$, which is the median.)(d) $E(X)$:

$$\begin{aligned}
 E(X) &= \int_0^4 x \cdot \frac{3}{32} x(4-x) \, dx = \frac{3}{32} \int_0^4 (4x^2 - x^3) \, dx \\
 &= \frac{3}{32} \left[\frac{4x^3}{3} - \frac{x^4}{4} \right]_0^4 = \frac{3}{32} \left(\frac{256}{3} - 64 \right) \\
 &= \frac{3}{32} \left(\frac{256 - 192}{3} \right) = \frac{3}{32} \times \frac{64}{3} = 2
 \end{aligned}$$

 $E(X^2)$:

$$\begin{aligned}
 E(X^2) &= \int_0^4 x^2 \cdot \frac{3}{32} x(4-x) \, dx = \frac{3}{32} \int_0^4 (4x^3 - x^4) \, dx \\
 &= \frac{3}{32} \left[x^4 - \frac{x^5}{5} \right]_0^4 = \frac{3}{32} \left(256 - \frac{1024}{5} \right) \\
 &= \frac{3}{32} \left(\frac{1280 - 1024}{5} \right) = \frac{3}{32} \times \frac{256}{5} = \frac{768}{160} = \frac{24}{5} = 4.8
 \end{aligned}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 4.8 - 2^2 = 4.8 - 4 = 0.8$$

Chapter 5

Calculus

Calculus is the single most important topic in IB Mathematics AA HL. It accounts for a substantial proportion of marks across both Paper 1 and Paper 2, and it appears in the Internal Assessment and Extended Essay with great frequency. This chapter provides a comprehensive, rigorous treatment of every calculus topic in the syllabus, complete with proofs, derivations, worked examples, and exam strategies.

5.1 Limits

5.1.1 Intuitive Understanding

The concept of a limit concerns the behavior of a function $f(x)$ as the input x approaches a particular value a . We are not interested in the value of the function *at* $x = a$ (it may not even be defined there), but rather what value $f(x)$ gets arbitrarily close to as x gets arbitrarily close to a .

Definition 5.1.1. We write $\lim_{x \rightarrow a} f(x) = L$ and say “the limit of $f(x)$ as x approaches a equals L ” if we can make the values of $f(x)$ as close to L as we like by taking x sufficiently close to a (but $x \neq a$).

Evaluating a Simple Limit

Evaluate $\lim_{x \rightarrow 2} (3x^2 - 4x + 1)$.

Since the function $f(x) = 3x^2 - 4x + 1$ is a polynomial (and therefore continuous everywhere), we can simply substitute:

$$f(2) = 3(4) - 4(2) + 1 = 12 - 8 + 1 = 5$$

Therefore $\lim_{x \rightarrow 2} (3x^2 - 4x + 1) = 5$.

Limit at a Hole

Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$.

The function is undefined at $x = 3$ (division by zero), so we cannot simply substitute. Instead, we factor and simplify:

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \lim_{x \rightarrow 3} \frac{(x - 3)(x + 3)}{x - 3} = \lim_{x \rightarrow 3} (x + 3) = 6$$

The limit exists and equals 6, even though $f(3)$ is undefined.

5.1.2 One-Sided Limits

Sometimes the behavior of a function differs depending on whether x approaches a from the left or from the right.

Definition 5.1.2. The *right-hand limit* is

$$\lim_{x \rightarrow a^+} f(x) = L$$

if $f(x)$ approaches L as x approaches a from values greater than a .

The *left-hand limit* is

$$\lim_{x \rightarrow a^-} f(x) = L$$

if $f(x)$ approaches L as x approaches a from values less than a .

Existence of a Two-Sided Limit

$$\lim_{x \rightarrow a} f(x) = L \quad \text{if and only if} \quad \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = L$$

If the one-sided limits are not equal, then the (two-sided) limit does not exist.

One-Sided Limits with Piecewise Functions

Let $f(x) = \begin{cases} x^2 & \text{if } x \leq 1 \\ 3x - 2 & \text{if } x > 1 \end{cases}$. Find $\lim_{x \rightarrow 1} f(x)$ if it exists.

Left-hand limit: $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x^2 = 1$.

Right-hand limit: $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (3x - 2) = 1$.

Since both one-sided limits equal 1, we have $\lim_{x \rightarrow 1} f(x) = 1$.

Note that $f(1) = 1^2 = 1$, so f is continuous at $x = 1$ in this case.

5.1.3 Infinite Limits and Vertical Asymptotes

When $f(x)$ grows without bound as x approaches a , we say the limit is infinite.

Infinite Limits

$$\lim_{x \rightarrow a} f(x) = +\infty \quad \text{or} \quad \lim_{x \rightarrow a} f(x) = -\infty$$

This means $f(x)$ increases or decreases without bound. The line $x = a$ is a **vertical asymptote** of f .

Vertical Asymptotes

Find the vertical asymptotes of $f(x) = \frac{1}{(x-2)(x+5)}$.

The function is undefined at $x = 2$ and $x = -5$.

$$\lim_{x \rightarrow 2^+} \frac{1}{(x-2)(x+5)} = +\infty, \quad \lim_{x \rightarrow 2^-} \frac{1}{(x-2)(x+5)} = -\infty$$

$$\lim_{x \rightarrow (-5)^+} \frac{1}{(x-2)(x+5)} = -\infty, \quad \lim_{x \rightarrow (-5)^-} \frac{1}{(x-2)(x+5)} = +\infty$$

Both $x = 2$ and $x = -5$ are vertical asymptotes.

5.1.4 Limits at Infinity and Horizontal Asymptotes

Definition 5.1.3.

$$\lim_{x \rightarrow \infty} f(x) = L$$

means that $f(x)$ approaches L as x grows without bound. If such an L exists, the line $y = L$ is a **horizontal asymptote**.

Key Limit at Infinity

$$\lim_{x \rightarrow \infty} \frac{1}{x^n} = 0 \quad \text{for any } n > 0$$

Similarly, $\lim_{x \rightarrow -\infty} \frac{1}{x^n} = 0$ for even n ; for odd n the limit is 0 as well (the sign is accounted for by the sign of x in the numerator of the original expression).

Finding Horizontal Asymptotes

Find the horizontal asymptotes of $f(x) = \frac{3x^2 - 2x + 1}{5x^2 + 7}$.

Divide numerator and denominator by x^2 :

$$\lim_{x \rightarrow \infty} \frac{3 - \frac{2}{x} + \frac{1}{x^2}}{5 + \frac{7}{x^2}} = \frac{3 - 0 + 0}{5 + 0} = \frac{3}{5}$$

The horizontal asymptote is $y = \frac{3}{5}$.

Strategy for Rational Function Limits at Infinity

For $\lim_{x \rightarrow \infty} \frac{P(x)}{Q(x)}$ where P and Q are polynomials:

- If $\deg P < \deg Q$: the limit is 0.
- If $\deg P = \deg Q$: the limit is $\frac{\text{leading coefficient of } P}{\text{leading coefficient of } Q}$.
- If $\deg P > \deg Q$: the limit is $\pm\infty$.

5.1.5 Special Trigonometric Limit

Fundamental Trigonometric Limit

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \quad (\text{where } \theta \text{ is in radians})$$

Equivalently, $\lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} = 1$.

A direct consequence: $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta} = 0$ and $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = 1$.

Using the Trigonometric Limit

Evaluate $\lim_{x \rightarrow 0} \frac{\sin 5x}{3x}$.

Rewrite to use the standard form:

$$\lim_{x \rightarrow 0} \frac{\sin 5x}{3x} = \lim_{x \rightarrow 0} \frac{5}{3} \cdot \frac{\sin 5x}{5x} = \frac{5}{3} \cdot 1 = \frac{5}{3}$$

Trigonometric Limit with Multiplication by Conjugate

Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$.

Multiply numerator and denominator by $1 + \cos x$:

$$\lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{x^2(1 + \cos x)} = \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2(1 + \cos x)} = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^2 \cdot \frac{1}{1 + \cos x} = 1^2 \cdot \frac{1}{2} = \frac{1}{2}$$

5.1.6 L'Hôpital's Rule

When a limit produces an indeterminate form such as $\frac{0}{0}$ or $\frac{\infty}{\infty}$, L'Hôpital's Rule provides a powerful technique.

Theorem 5.1.1 (L'Hôpital's Rule). *If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$ (or both $= \pm\infty$), and $g'(x) \neq 0$ near a , then*

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

provided the limit on the right exists (or is $\pm\infty$).

Applicable Indeterminate Forms

L'Hôpital's Rule applies directly to:

$$\frac{0}{0} \quad \text{and} \quad \frac{\infty}{\infty}$$

For other forms ($0 \cdot \infty$, $\infty - \infty$, 0^0 , 1^∞ , ∞^0), rewrite the expression as a quotient first.

Repeated Application of L'Hôpital's Rule

Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x - \frac{x^2}{2}}{x^3}$.

First attempt (0/0 form):

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x - \frac{x^2}{2}}{x^3} \xrightarrow{\text{L'H}} \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{3x^2}$$

Still 0/0, apply again:

$$\xrightarrow{\text{L'H}} \lim_{x \rightarrow 0} \frac{e^x - 1}{6x}$$

Still 0/0, apply once more:

$$\xrightarrow{\text{L'H}} \lim_{x \rightarrow 0} \frac{e^x}{6} = \frac{1}{6}$$

This took **three** applications of L'Hôpital's Rule.

L'Hôpital with ∞/∞

Evaluate $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$.

This is ∞/∞ . Applying L'Hôpital's Rule:

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

IB Exam Tip: L'Hôpital's Rule vs. Algebraic Manipulation

On IB exams, L'Hôpital's Rule is a valid technique. However, consider:

- **Algebraic manipulation** (factoring, rationalizing, simplifying) is often faster for simple $0/0$ forms and is always acceptable.
- **L'Hôpital's Rule** is essential when algebraic methods are extremely cumbersome or when dealing with exponential/logarithmic limits.
- **Warning:** Always verify the indeterminate form before applying L'Hôpital. Applying it to a non-indeterminate form will give a wrong answer.
- **For Paper 1** (no calculator), L'Hôpital is often the best approach for limits involving e^x , $\ln x$, or trigonometric functions.

5.2 Differentiation: First Principles

5.2.1 The Definition of the Derivative

Definition 5.2.1. The *derivative* of f at x is defined as

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

provided this limit exists. If it exists, we say f is **differentiable** at x .

The quantity $\frac{f(x+h) - f(x)}{h}$ is the slope of the secant line through $(x, f(x))$ and $(x+h, f(x+h))$. As $h \rightarrow 0$, this secant approaches the tangent line, so the derivative gives the slope of the tangent line at the point $(x, f(x))$.

An equivalent form (sometimes more convenient) is:

$$f'(x) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

5.2.2 Deriving the Power Rule from First Principles

Derivative of $f(x) = x^n$ for positive integer n

We prove $\frac{d}{dx}[x^n] = nx^{n-1}$ by induction on n .

Base case ($n = 1$):

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1 = 1 \cdot x^0 \checkmark$$

Inductive step: Assume $\frac{d}{dx}[x^k] = kx^{k-1}$ for some $k \geq 1$. We show $\frac{d}{dx}[x^{k+1}] = (k+1)x^k$.

Let $f(x) = x^{k+1} = x \cdot x^k$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^{k+1} - x^{k+1}}{h}$$

Using the identity $(x+h)^{k+1} - x^{k+1} = h[(x+h)^k + (x+h)^{k-1}x + \cdots + (x+h)x^{k-1} + x^k]$ (factorisation of $a^{k+1} - b^{k+1}$):

$$f'(x) = \lim_{h \rightarrow 0} [(x+h)^k + (x+h)^{k-1}x + \cdots + (x+h)x^{k-1} + x^k]$$

There are $k + 1$ terms, each tending to x^k as $h \rightarrow 0$:

$$f'(x) = (k + 1)x^k \checkmark$$

Alternatively, using the product rule (once we have proved it): $\frac{d}{dx}[x \cdot x^k] = 1 \cdot x^k + x \cdot kx^{k-1} = (k + 1)x^k$.

Conclusion: By mathematical induction, $\frac{d}{dx}[x^n] = nx^{n-1}$ for all positive integers n .

5.2.3 Derivative of $\sin x$ from First Principles

Derivative of $f(x) = \sin x$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sin(x + h) - \sin x}{h}$$

Using the angle addition formula $\sin(x + h) = \sin x \cos h + \cos x \sin h$:

$$= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin x(\cos h - 1) + \cos x \sin h}{h}$$

$$= \lim_{h \rightarrow 0} \left[\sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h} \right]$$

Using the standard limits $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$ and $\lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0$:

$$= \sin x \cdot 0 + \cos x \cdot 1 = \cos x$$

Therefore $\frac{d}{dx}[\sin x] = \cos x$.

IB Exam Tip: First Principles on Paper 1

First principles differentiation questions are **common on Paper 1** (no calculator). Typical formats:

- “Find $f'(x)$ from first principles” for a given function.
- “Use the definition of the derivative to show that ...”
- The key is to set up the limit carefully, simplify the numerator (factor, expand, or use trigonometric identities), and then evaluate the limit.
- Common functions tested: x^n for small n , $\sqrt{x}$ (write as $x^{1/2}$), $\frac{1}{x}$, $\sin x$, $\cos x$.

First Principles: $f(x) = \frac{1}{x}$

Find $f'(x)$ from first principles for $f(x) = \frac{1}{x}$, $x \neq 0$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \lim_{h \rightarrow 0} \frac{\frac{x - (x+h)}{x(x+h)}}{h} = \lim_{h \rightarrow 0} \frac{-h}{hx(x+h)} = \lim_{h \rightarrow 0} \frac{-1}{x(x+h)} = -\frac{1}{x^2}$$

This confirms $\frac{d}{dx}[x^{-1}] = -x^{-2}$, consistent with the power rule.

5.3 Differentiation Rules and Proofs

5.3.1 Basic Rules

Power Rule

$$\frac{d}{dx}[x^n] = nx^{n-1} \quad \text{for all } n \in \mathbb{R}$$

This holds for any real exponent n : positive integers, negative integers, fractions, and irrationals.

Constant Multiple Rule

$$\frac{d}{dx}[cf(x)] = c \cdot f'(x)$$

where c is a constant. The derivative of a constant multiple is the constant times the derivative.

Sum and Difference Rule

$$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$$

The derivative of a sum (or difference) is the sum (or difference) of the derivatives. This extends to any finite number of terms.

5.3.2 Proof of the Product Rule

Proof of the Product Rule

Theorem: If f and g are differentiable at x , then

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$$

Proof (using first principles):

Let $h(x) = f(x)g(x)$. Then:

$$h'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x)g(x + \Delta x) - f(x)g(x)}{\Delta x}$$

We use the trick of adding and subtracting $f(x + \Delta x)g(x)$ in the numerator:

$$= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x)g(x + \Delta x) - f(x + \Delta x)g(x) + f(x + \Delta x)g(x) - f(x)g(x)}{\Delta x}$$

$$\begin{aligned}
&= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x)[g(x + \Delta x) - g(x)] + g(x)[f(x + \Delta x) - f(x)]}{\Delta x} \\
&= \lim_{\Delta x \rightarrow 0} f(x + \Delta x) \cdot \frac{g(x + \Delta x) - g(x)}{\Delta x} + g(x) \cdot \frac{f(x + \Delta x) - f(x)}{\Delta x}
\end{aligned}$$

Since f is differentiable, it is continuous, so $\lim_{\Delta x \rightarrow 0} f(x + \Delta x) = f(x)$:

$$= f(x) \cdot g'(x) + g(x) \cdot f'(x)$$

5.3.3 Proof of the Quotient Rule

Proof of the Quotient Rule

Theorem: If f and g are differentiable at x and $g(x) \neq 0$, then

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Proof: Write $h(x) = \frac{f(x)}{g(x)}$, so $f(x) = h(x) \cdot g(x)$. Applying the product rule:

$$f'(x) = h'(x)g(x) + h(x)g'(x)$$

Solving for $h'(x)$:

$$\begin{aligned}
h'(x)g(x) &= f'(x) - h(x)g'(x) \\
h'(x) &= \frac{f'(x)}{g(x)} - \frac{h(x)g'(x)}{g(x)} = \frac{f'(x)}{g(x)} - \frac{f(x)g'(x)}{[g(x)]^2}
\end{aligned}$$

Combining over a common denominator $[g(x)]^2$:

$$h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

5.3.4 Proof of the Chain Rule

Proof of the Chain Rule

Theorem: If g is differentiable at x and f is differentiable at $g(x)$, then

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$$

Proof (using first principles):

Let $h(x) = f(g(x))$. Define $\Delta u = g(x + \Delta x) - g(x)$, so $g(x + \Delta x) = g(x) + \Delta u$.

$$h'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(g(x + \Delta x)) - f(g(x))}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(g(x) + \Delta u) - f(g(x))}{\Delta x}$$

Multiply and divide by Δu (when $\Delta u \neq 0$):

$$= \lim_{\Delta x \rightarrow 0} \frac{f(g(x) + \Delta u) - f(g(x))}{\Delta u} \cdot \frac{\Delta u}{\Delta x}$$

As $\Delta x \rightarrow 0$, we have $\Delta u \rightarrow 0$ (since g is continuous). Therefore:

$$= \lim_{\Delta u \rightarrow 0} \frac{f(g(x) + \Delta u) - f(g(x))}{\Delta u} \cdot \lim_{\Delta x \rightarrow 0} \frac{g(x + \Delta x) - g(x)}{\Delta x}$$

$$= f'(g(x)) \cdot g'(x)$$

In Leibniz notation: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$, where $y = f(u)$ and $u = g(x)$.

Mnemonic for Differentiation Rules

- **Product Rule:** “ d -first times second, plus first times d -second.”
- **Quotient Rule:** “ d -top times bottom, minus top times d -bottom, over bottom squared.” (“Low d -High minus High d -Low, over the square of what’s below.”)
- **Chain Rule:** Differentiate the outside, leave the inside alone, then multiply by the derivative of the inside.

5.4 Derivatives of Elementary Functions

5.4.1 Complete Derivative Table

Derivatives of Elementary Functions

$f(x)$	$f'(x)$	Condition
x^n	nx^{n-1}	$n \in \mathbb{R}$
$\sin x$	$\cos x$	
$\cos x$	$-\sin x$	
$\tan x$	$\sec^2 x$	$x \neq \frac{\pi}{2} + k\pi$
$\sec x$	$\sec x \tan x$	$x \neq \frac{\pi}{2} + k\pi$
$\csc x$	$-\csc x \cot x$	$x \neq k\pi$
$\cot x$	$-\csc^2 x$	$x \neq k\pi$
e^x	e^x	
a^x	$a^x \ln a$	$a > 0, a \neq 1$
$\ln x$	$\frac{1}{x}$	$x > 0$
$\log_a x$	$\frac{1}{x \ln a}$	$x > 0$
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$	$ x < 1$
$\arccos x$	$\frac{-1}{\sqrt{1-x^2}}$	$ x < 1$
$\arctan x$	$\frac{1}{1+x^2}$	

5.4.2 Derivations

Derivative of $\cos x$ from First Principles

$$\frac{d}{dx}[\cos x] = \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$$

Using $\cos(x+h) = \cos x \cos h - \sin x \sin h$:

$$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} = \lim_{h \rightarrow 0} \frac{\cos x(\cos h - 1) - \sin x \sin h}{h} \\ &= \cos x \cdot \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} - \sin x \cdot \lim_{h \rightarrow 0} \frac{\sin h}{h} = \cos x \cdot 0 - \sin x \cdot 1 = -\sin x \end{aligned}$$

Derivative of $\tan x$ using the Quotient Rule

$\tan x = \frac{\sin x}{\cos x}$, so by the quotient rule:

$$\frac{d}{dx}[\tan x] = \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$$

Derivative of $\sec x$

$\sec x = \frac{1}{\cos x} = (\cos x)^{-1}$. By the chain rule:

$$\frac{d}{dx}[\sec x] = -(\cos x)^{-2} \cdot (-\sin x) = \frac{\sin x}{\cos^2 x} = \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} = \sec x \tan x$$

Derivative of $\csc x$

$\csc x = (\sin x)^{-1}$. By the chain rule:

$$\frac{d}{dx}[\csc x] = -(\sin x)^{-2} \cdot \cos x = -\frac{\cos x}{\sin^2 x} = -\frac{1}{\sin x} \cdot \frac{\cos x}{\sin x} = -\csc x \cot x$$

Derivative of $\cot x$

$\cot x = \frac{\cos x}{\sin x}$. By the quotient rule:

$$\frac{d}{dx}[\cot x] = \frac{(-\sin x)(\sin x) - (\cos x)(\cos x)}{\sin^2 x} = \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} = \frac{-1}{\sin^2 x} = -\csc^2 x$$

Derivative of $\arcsin x$ using Implicit Differentiation

Let $y = \arcsin x$, so $x = \sin y$ where $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.

Differentiate both sides with respect to x :

$$1 = \cos y \cdot \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{1}{\cos y}$$

Since $x = \sin y$, we have $\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$ (positive because $\cos y \geq 0$ for $y \in [-\pi/2, \pi/2]$):

$$\frac{d}{dx}[\arcsin x] = \frac{1}{\sqrt{1 - x^2}}$$

Derivative of $\arccos x$

Let $y = \arccos x$, so $x = \cos y$ where $0 \leq y \leq \pi$.

$$1 = -\sin y \cdot \frac{dy}{dx} \implies \frac{dy}{dx} = -\frac{1}{\sin y} = -\frac{1}{\sqrt{1-x^2}}$$

Here $\sin y \geq 0$ on $[0, \pi]$.

Alternatively, note that $\arccos x = \frac{\pi}{2} - \arcsin x$, so $\frac{d}{dx}[\arccos x] = -\frac{1}{\sqrt{1-x^2}}$.

Derivative of $\arctan x$

Let $y = \arctan x$, so $x = \tan y$ where $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

$$1 = \sec^2 y \cdot \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{\sec^2 y} = \cos^2 y$$

Using the identity $\cos^2 y = \frac{1}{1 + \tan^2 y}$:

$$\frac{d}{dx}[\arctan x] = \frac{1}{1+x^2}$$

Derivative of a^x

Let $y = a^x$. Take $\ln$ of both sides: $\ln y = x \ln a$.

Differentiate implicitly with respect to x :

$$\frac{1}{y} \cdot \frac{dy}{dx} = \ln a \implies \frac{dy}{dx} = y \ln a = a^x \ln a$$

Derivative of $\log_a x$

Let $y = \log_a x$, so $a^y = x$. Take $\ln$ of both sides: $y \ln a = \ln x$, giving $y = \frac{\ln x}{\ln a}$.

$$\frac{dy}{dx} = \frac{1}{\ln a} \cdot \frac{1}{x} = \frac{1}{x \ln a}$$

5.5 Implicit Differentiation

5.5.1 Concept

When y is defined implicitly as a function of x (e.g., $x^2 + y^2 = r^2$), it may be difficult or impossible to solve for y explicitly. **Implicit differentiation** allows us to find $\frac{dy}{dx}$ by differentiating both sides of the equation with respect to x , treating y as a function of x .

Key Idea of Implicit Differentiation

When differentiating a term involving y with respect to x , apply the chain rule:

$$\frac{d}{dx}[y^n] = ny^{n-1} \frac{dy}{dx}$$

Similarly:

$$\begin{aligned}\frac{d}{dx}[\sin y] &= \cos y \cdot \frac{dy}{dx} \\ \frac{d}{dx}[e^y] &= e^y \cdot \frac{dy}{dx} \\ \frac{d}{dx}[\ln y] &= \frac{1}{y} \cdot \frac{dy}{dx}\end{aligned}$$

Circle: $x^2 + y^2 = 25$

Find $\frac{dy}{dx}$ for the circle $x^2 + y^2 = 25$.

Differentiate both sides with respect to x :

$$2x + 2y \frac{dy}{dx} = 0$$

$$2y \frac{dy}{dx} = -2x \implies \frac{dy}{dx} = -\frac{x}{y}$$

Find $\frac{dy}{dx}$ at the point $(3, 4)$: $\frac{dy}{dx} = -\frac{3}{4}$.

This makes sense geometrically: the tangent to the circle at $(3, 4)$ has slope $-\frac{3}{4}$ (it is perpendicular to the radius, which has slope $\frac{4}{3}$).

Ellipse: $\frac{x^2}{9} + \frac{y^2}{4} = 1$

Find $\frac{dy}{dx}$ for $\frac{x^2}{9} + \frac{y^2}{4} = 1$.

Differentiate:

$$\frac{2x}{9} + \frac{2y}{4} \cdot \frac{dy}{dx} = 0 \implies \frac{2x}{9} + \frac{y}{2} \cdot \frac{dy}{dx} = 0$$

$$\frac{y}{2} \cdot \frac{dy}{dx} = -\frac{2x}{9} \implies \frac{dy}{dx} = -\frac{4x}{9y}$$

5.5.2 Second Derivatives Implicitly

To find $\frac{d^2y}{dx^2}$, differentiate $\frac{dy}{dx}$ with respect to x , remembering that $\frac{dy}{dx}$ may contain y (which is a function of x).

Second Derivative: $x^2 + y^2 = 25$

We found $\frac{dy}{dx} = -\frac{x}{y}$. Now find $\frac{d^2y}{dx^2}$.

Using the quotient rule (remembering y depends on x):

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{(-1)(y) - (-x)\frac{dy}{dx}}{y^2} = \frac{-y + x \cdot \left(-\frac{x}{y}\right)}{y^2} = \frac{-y - \frac{x^2}{y}}{y^2} \\ &= \frac{\frac{-y^2 - x^2}{y}}{y^2} = \frac{-(x^2 + y^2)}{y^3} = \frac{-25}{y^3}\end{aligned}$$

$$\text{At } (3, 4): \frac{d^2y}{dx^2} = \frac{-25}{64}.$$

5.6 Logarithmic Differentiation

5.6.1 When to Use Logarithmic Differentiation

Logarithmic differentiation is useful in two situations:

1. When the function is of the form $y = f(x)^{g(x)}$ (variable in both base and exponent).
2. When the function is a product or quotient of many factors (taking $\ln$ converts products to sums and quotients to differences).

Method of Logarithmic Differentiation

1. Take natural logarithms of both sides: $\ln y = \ln[f(x)^{g(x)}] = g(x) \ln f(x)$.
2. Differentiate both sides with respect to x , using implicit differentiation on the left.
3. Solve for $\frac{dy}{dx}$.

Variable Base and Exponent: $y = x^x$

Find $\frac{dy}{dx}$ for $y = x^x$, $x > 0$.

Take $\ln$ of both sides: $\ln y = x \ln x$.

Differentiate: $\frac{1}{y} \cdot \frac{dy}{dx} = 1 \cdot \ln x + x \cdot \frac{1}{x} = \ln x + 1$.

Therefore:

$$\frac{dy}{dx} = y(\ln x + 1) = x^x(\ln x + 1)$$

Product of Many Factors

Find $\frac{dy}{dx}$ for $y = \frac{x^3 e^{2x} (x+1)^5}{\sqrt{x^2+1}}$.

Take $\ln$:

$$\ln y = 3 \ln x + 2x + 5 \ln(x+1) - \frac{1}{2} \ln(x^2+1)$$

Differentiate:

$$\begin{aligned} \frac{1}{y} \cdot \frac{dy}{dx} &= \frac{3}{x} + 2 + \frac{5}{x+1} - \frac{x}{x^2+1} \\ \frac{dy}{dx} &= \frac{x^3 e^{2x} (x+1)^5}{\sqrt{x^2+1}} \left(\frac{3}{x} + 2 + \frac{5}{x+1} - \frac{x}{x^2+1} \right) \end{aligned}$$

5.7 Applications of Differentiation

5.7.1 Tangents and Normals

Equation of a Tangent Line

The tangent to $y = f(x)$ at $x = x_0$ has equation

$$y - y_0 = f'(x_0)(x - x_0)$$

where $y_0 = f(x_0)$ and $f'(x_0)$ is the gradient (slope) of the tangent.

Equation of a Normal Line

The normal to $y = f(x)$ at $x = x_0$ is perpendicular to the tangent. Its gradient is

$$m_{\text{normal}} = -\frac{1}{f'(x_0)} \quad (\text{provided } f'(x_0) \neq 0)$$

and its equation is

$$y - y_0 = -\frac{1}{f'(x_0)}(x - x_0)$$

If $f'(x_0) = 0$, the tangent is horizontal ($y = y_0$) and the normal is vertical ($x = x_0$).

Tangent and Normal

Find the equations of the tangent and normal to $y = x^3 - 3x + 2$ at $x = 1$.

Point: $(1, 1^3 - 3 + 2) = (1, 0)$.

Gradient: $y' = 3x^2 - 3$, so $y'(1) = 3 - 3 = 0$.

Tangent: Since the gradient is 0, the tangent is horizontal: $y = 0$.

Normal: Since the tangent is horizontal, the normal is vertical: $x = 1$.

5.7.2 Stationary Points

Definition 5.7.1. A **stationary point** (or **critical point**) of f is a point where $f'(x) = 0$.

There are three types of stationary points: **local maxima**, **local minima**, and **horizontal points of inflection**.

First Derivative Test

At a stationary point $x = a$ where $f'(a) = 0$:

- **Local maximum:** f' changes from positive to negative as x increases through a .
- **Local minimum:** f' changes from negative to positive as x increases through a .
- **Horizontal point of inflection:** f' does not change sign (e.g., $f(x) = x^3$ at $x = 0$).

Second Derivative Test

At a stationary point $x = a$ where $f'(a) = 0$:

- $f''(a) > 0 \implies$ **local minimum** (function is concave up).
- $f''(a) < 0 \implies$ **local maximum** (function is concave down).
- $f''(a) = 0 \implies$ the test is **inconclusive**. Use the first derivative test instead.

Classifying Stationary Points

Find and classify all stationary points of $f(x) = x^3 - 6x^2 + 9x + 1$.

$$f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3)$$

Stationary points at $x = 1$ and $x = 3$.

Second derivative: $f''(x) = 6x - 12$.

At $x = 1$: $f''(1) = 6 - 12 = -6 < 0$, so **local maximum**. Value: $f(1) = 1 - 6 + 9 + 1 = 5$.

Point: $(1, 5)$.

At $x = 3$: $f''(3) = 18 - 12 = 6 > 0$, so **local minimum**. Value: $f(3) = 27 - 54 + 27 + 1 = 1$.

Point: $(3, 1)$.

5.7.3 Increasing and Decreasing Functions

Monotonicity

- f is **strictly increasing** on an interval if $f'(x) > 0$ for all x in the interval.
- f is **strictly decreasing** on an interval if $f'(x) < 0$ for all x in the interval.
- f is **constant** on an interval if $f'(x) = 0$ for all x in the interval.

5.7.4 Concavity and Points of Inflection

Definition 5.7.2. • f is **concave up** (convex) on an interval if $f''(x) > 0$ on that interval. The graph curves upward (like a cup $\cup$).

- f is **concave down** (concave) on an interval if $f''(x) < 0$ on that interval. The graph curves downward (like a cap $\cap$).

Definition 5.7.3. A **point of inflection** is a point on the curve where the concavity changes. At a point of inflection $x = a$:

1. $f''(a) = 0$ (or $f''(a)$ does not exist), AND
2. $f''(x)$ changes sign as x passes through a .

Critical IB Distinction: $f''(a) = 0$ is NOT Sufficient

A very common IB exam trap: $f''(a) = 0$ does **not** guarantee a point of inflection.

Counter-example: $f(x) = x^4$. We have $f'(x) = 4x^3$ and $f''(x) = 12x^2$.

- $f''(0) = 0$, but $f''(x) \geq 0$ for all x , so there is **no sign change** in f'' .
- The function is concave up on both sides of $x = 0$.
- $x = 0$ is a local minimum, **not** a point of inflection.

Always check for a sign change in $f''(x)$ to confirm a point of inflection.

Finding Points of Inflection

Find all points of inflection of $f(x) = \frac{x^4}{4} - 2x^3 + 6x^2 - 8x + 3$.

$$f'(x) = x^3 - 6x^2 + 12x - 8 = (x - 2)^3$$

$$f''(x) = 3x^2 - 12x + 12 = 3(x^2 - 4x + 4) = 3(x - 2)^2$$

$f''(x) = 3(x - 2)^2 \geq 0$ for all x , and $f''(x) = 0$ only at $x = 2$.

Since $f''(x)$ does not change sign at $x = 2$ (it is positive on both sides), $x = 2$ is **not** a point of inflection. The function has no points of inflection.

Note: $f'(2) = 0$ and $f''(2) = 0$, so this is a horizontal point of inflection for f' , meaning $x = 2$ is a **stationary point of inflection** of f ? No — it is a stationary point where the second derivative test is inconclusive, but the first derivative test shows $f'(x) = (x - 2)^3$ changes from negative to positive, so $x = 2$ is actually a **local minimum**.

True Point of Inflection

Find the point of inflection of $f(x) = x^3 - 3x^2 + 4$.

$$f'(x) = 3x^2 - 6x, \quad f''(x) = 6x - 6 = 6(x - 1).$$

$f''(x) = 0$ when $x = 1$. Check the sign change:

- For $x < 1$: $f''(x) < 0$ (concave down).
- For $x > 1$: $f''(x) > 0$ (concave up).

Since f'' changes sign, $x = 1$ is a point of inflection. The point is $(1, 1 - 3 + 4) = (1, 2)$.

5.7.5 Optimization Problems

Optimization involves finding the maximum or minimum value of a quantity subject to given constraints.

General Strategy for Optimization

1. **Identify** the quantity to be maximized or minimized. Call it Q .
2. **Write** Q as a function of one variable.
3. **Find** the domain (the feasible region).
4. **Differentiate** Q and set $Q' = 0$ to find stationary points.
5. **Classify** each stationary point (first or second derivative test).
6. **Check** endpoints of the domain if it is closed and bounded.
7. **Answer** the question in context, with appropriate units.

Optimization: Minimizing Surface Area

A closed cylindrical can must hold 500 cm^3 of liquid. Find the dimensions that minimize the surface area.

Let r = radius and h = height.

$$\text{Volume: } V = \pi r^2 h = 500, \text{ so } h = \frac{500}{\pi r^2}.$$

$$\text{Surface area: } S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + \frac{1000}{r}.$$

$$\text{Differentiate: } \frac{dS}{dr} = 4\pi r - \frac{1000}{r^2}.$$

$$\text{Set to zero: } 4\pi r = \frac{1000}{r^2} \implies 4\pi r^3 = 1000 \implies r^3 = \frac{250}{\pi}.$$

$$r = \sqrt[3]{\frac{250}{\pi}} \approx 4.30 \text{ cm}$$

$$h = \frac{500}{\pi r^2} = \frac{500}{\pi \cdot \left(\frac{250}{\pi}\right)^{2/3}} = 2\sqrt[3]{\frac{250}{\pi}} \approx 8.61 \text{ cm}$$

Note: $h = 2r$, so the optimal can has height equal to diameter.

Second derivative: $\frac{d^2S}{dr^2} = 4\pi + \frac{2000}{r^3} > 0$ for all $r > 0$, confirming a minimum.

5.7.6 Related Rates of Change

Definition 5.7.4. Related rates problems involve finding the rate of change of one quantity given the rate of change of a related quantity, using the chain rule:

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

Strategy for Related Rates

1. **Draw** a diagram and identify all variables.
2. **Write** an equation relating the relevant quantities.
3. **Differentiate** both sides with respect to time t (using the chain rule on all spatial variables).
4. **Substitute** the known values (including the rates) and solve for the unknown rate.
5. **Include** units in your answer.

Related Rates: Expanding Circle

A circular oil spill is expanding. When the radius is 50 m, the radius is increasing at a rate of 3 m/s. How fast is the area increasing?

The area is $A = \pi r^2$. Differentiate with respect to t :

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

Substitute $r = 50$ and $\frac{dr}{dt} = 3$:

$$\frac{dA}{dt} = 2\pi(50)(3) = 300\pi \text{ m}^2/\text{s}$$

Related Rates: Ladder Sliding

A 5 m ladder leans against a wall. The base slides away at 1 m/s. How fast is the top descending when the base is 3 m from the wall?

By Pythagoras: $x^2 + y^2 = 25$, where x = distance of base from wall, y = height of top.

Differentiate: $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$.

When $x = 3$: $y = \sqrt{25 - 9} = 4$.

$$2(3)(1) + 2(4) \frac{dy}{dt} = 0 \implies 6 + 8 \frac{dy}{dt} = 0 \implies \frac{dy}{dt} = -\frac{6}{8} = -0.75 \text{ m/s}$$

The top is descending at 0.75 m/s.

IB Exam Tip: “Show That” Questions

When an IB question says “show that,” you must provide **full working** with every step justified. For differentiation problems:

- State which rule you are using.
- Show the intermediate steps clearly.
- Do not skip algebraic simplification steps.
- For “show that $f'(x) = \dots$ ” questions, the answer is given — focus on demonstrating **how** to arrive at it.
- For “find” questions, you need the correct final answer, but showing working is essential for method marks.

5.8 Kinematics

5.8.1 Displacement, Velocity, and Acceleration

Kinematic Relationships

For a particle moving in a straight line:

Displacement: $s(t)$

Velocity: $v(t) = \frac{ds}{dt} = s'(t)$

Acceleration: $a(t) = \frac{dv}{dt} = v'(t) = s''(t)$

Recovery of Displacement from Velocity

$$s(t) = \int v(t) dt + C$$

The constant C is determined by an initial condition (e.g., $s(0) = s_0$).

5.8.2 Total Distance vs. Displacement

Definition 5.8.1. • **Displacement** is the net change in position: $s(t_2) - s(t_1) = \int_{t_1}^{t_2} v(t) dt$.

- **Total distance** travelled is the integral of speed: $\int_{t_1}^{t_2} |v(t)| dt$.
- **Speed** is the magnitude of velocity: $\text{speed}(t) = |v(t)|$.

Total Distance Travelled

$$\text{Total distance} = \int_{t_1}^{t_2} |v(t)| dt$$

Method: Find when $v(t) = 0$ (the particle changes direction). Split the integral at these times and integrate the absolute value on each subinterval.

Distance vs. Displacement

A particle moves with velocity $v(t) = t^2 - 4t + 3$ for $0 \leq t \leq 5$. Find the total distance travelled and the displacement.

Factor: $v(t) = (t - 1)(t - 3)$. The particle changes direction at $t = 1$ and $t = 3$.

Sign of v : positive for $t < 1$, negative for $1 < t < 3$, positive for $t > 3$.

Displacement:

$$s(5) - s(0) = \int_0^5 (t^2 - 4t + 3) dt = \left[\frac{t^3}{3} - 2t^2 + 3t \right]_0^5 = \frac{125}{3} - 50 + 15 = \frac{40}{3}$$

Total distance:

$$\begin{aligned} \int_0^1 v(t) dt &= \left[\frac{t^3}{3} - 2t^2 + 3t \right]_0^1 = \frac{1}{3} - 2 + 3 = \frac{4}{3} \\ \int_1^3 (-v(t)) dt &= - \left[\frac{t^3}{3} - 2t^2 + 3t \right]_1^3 = - \left(9 - 18 + 9 - \frac{4}{3} \right) = \frac{4}{3} \\ \int_3^5 v(t) dt &= \left[\frac{t^3}{3} - 2t^2 + 3t \right]_3^5 = \frac{40}{3} - 0 = \frac{40}{3} - 0 = \frac{125}{3} - 50 + 15 - (9 - 18 + 9) = \frac{40}{3} \end{aligned}$$

$$\text{Total distance} = \frac{4}{3} + \frac{4}{3} + \frac{40}{3} = \frac{48}{3} = 16 \text{ units.}$$

5.9 Curve Sketching

5.9.1 Systematic Approach to Curve Sketching

Complete Curve Sketching Checklist

Follow this systematic approach to sketch any curve:

- Domain:** Find all x -values where f is defined.
- Intercepts:**
 - y -intercept: set $x = 0$.
 - x -intercepts: set $y = 0$ (solve $f(x) = 0$).
- Symmetry:** Check for even/odd symmetry, or periodicity.
- Asymptotes:**
 - Vertical: values where f is undefined (or limits are infinite).
 - Horizontal: $\lim_{x \rightarrow \pm\infty} f(x)$.
 - Oblique/slant: when degree of numerator = degree of denominator + 1.
- First derivative:** Find $f'(x) = 0$ for stationary points, determine intervals of increase/decrease.
- Second derivative:** Find $f''(x) = 0$ for potential inflection points, determine concavity.

7. **Behavior at extremes:** What happens as x approaches domain boundaries?

8. **Sketch:** Plot key points and asymptotes, then connect smoothly.

Comprehensive Curve Sketch

Sketch $f(x) = \frac{x^2}{x^2 - 4}$.

Domain: $x^2 - 4 \neq 0$, so $x \neq \pm 2$. Domain = $\mathbb{R} \setminus \{-2, 2\}$.

Intercepts: y -intercept: $f(0) = 0$. x -intercepts: $x^2 = 0 \implies x = 0$ (origin).

Symmetry: $f(-x) = f(x)$, so even (symmetric about y -axis).

Asymptotes:

- Vertical: $x = -2$ and $x = 2$.
- Horizontal: $\lim_{x \rightarrow \infty} \frac{x^2}{x^2 - 4} = 1$, so $y = 1$.

First derivative:

$$f'(x) = \frac{2x(x^2 - 4) - x^2(2x)}{(x^2 - 4)^2} = \frac{2x^3 - 8x - 2x^3}{(x^2 - 4)^2} = \frac{-8x}{(x^2 - 4)^2}$$

$f'(x) = 0 \implies x = 0$. $f'(x) > 0$ for $x < -2$ and $-2 < x < 0$; $f'(x) < 0$ for $0 < x < 2$ and $x > 2$.

Stationary point: $(0, 0)$. Since f' changes from $+$ to $-$, it is a local maximum.

Second derivative: $f''(0) < 0$ confirms the maximum. Checking for inflection points is also possible but involved.

Sketch: The curve passes through the origin (local max), approaches $y = 1$ as $x \rightarrow \pm\infty$, and has vertical asymptotes at $x = \pm 2$.

5.10 Integration: Anti-Derivatives

5.10.1 The Indefinite Integral

Definition 5.10.1. The *indefinite integral* (or *anti-derivative*) of $f(x)$ is a function $F(x)$ such that $F'(x) = f(x)$. We write

$$\int f(x) \, dx = F(x) + C$$

where C is the *constant of integration*.

5.10.2 Standard Integral Table

Standard Integrals

$f(x)$	$\int f(x) \, dx$
x^n	$\frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
$\frac{1}{x}$	$\ln x + C$
e^x	$e^x + C$
a^x	$\frac{a^x}{\ln a} + C \quad (a > 0, a \neq 1)$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\sec^2 x$	$\tan x + C$
$\csc^2 x$	$-\cot x + C$
$\sec x \tan x$	$\sec x + C$
$\csc x \cot x$	$-\csc x + C$
$\frac{1}{\sqrt{1-x^2}}$	$\arcsin x + C$
$\frac{1}{1+x^2}$	$\arctan x + C$
$\tan x$	$-\ln \cos x + C = \ln \sec x + C$
$\cot x$	$\ln \sin x + C$
$\csc x$	$-\ln \csc x + \cot x + C$
$\sec x$	$\ln \sec x + \tan x + C$

Basic Integration

$$\int (3x^4 - 2x^2 + 5x - 7) \, dx = \frac{3x^5}{5} - \frac{2x^3}{3} + \frac{5x^2}{2} - 7x + C$$

Integrating with Fractional Powers

$$\int \frac{2}{\sqrt{x}} \, dx = \int 2x^{-1/2} \, dx = 2 \cdot \frac{x^{1/2}}{1/2} + C = 4\sqrt{x} + C$$

5.11 Definite Integrals and the Fundamental Theorem

5.11.1 The Definite Integral

Definition 5.11.1. The *definite integral* of f from a to b is

$$\int_a^b f(x) \, dx = F(b) - F(a)$$

where F is any anti-derivative of f .

Theorem 5.11.1 (Fundamental Theorem of Calculus, Part 1). *If f is continuous on $[a, b]$ and $F(x) = \int_a^x f(t) dt$, then $F'(x) = f(x)$ for all $x \in [a, b]$.*

In other words, integration and differentiation are inverse processes: the derivative of an integral gives back the original function.

Theorem 5.11.2 (Fundamental Theorem of Calculus, Part 2). *If f is continuous on $[a, b]$ and F is any anti-derivative of f , then*

$$\int_a^b f(x) dx = F(b) - F(a)$$

This connects the definite integral (area-like concept) with the anti-derivative (algebraic concept).

5.11.2 Properties of Definite Integrals

Properties of Definite Integrals

$$\begin{aligned}\int_a^a f(x) dx &= 0 \\ \int_a^b f(x) dx &= -\int_b^a f(x) dx \quad (\text{Reversal}) \\ \int_a^b [f(x) \pm g(x)] dx &= \int_a^b f(x) dx \pm \int_a^b g(x) dx \quad (\text{Linearity}) \\ \int_a^b c \cdot f(x) dx &= c \int_a^b f(x) dx \quad (\text{Constant multiple}) \\ \int_a^b f(x) dx &= \int_a^c f(x) dx + \int_c^b f(x) dx \quad (\text{Additivity})\end{aligned}$$

Evaluating a Definite Integral

$$\int_1^3 (x^2 + 2x) dx = \left[\frac{x^3}{3} + x^2 \right]_1^3 = \left(\frac{27}{3} + 9 \right) - \left(\frac{1}{3} + 1 \right) = 18 - \frac{4}{3} = \frac{50}{3}$$

Definite Integral with Trigonometric Functions

$$\int_0^{\pi/2} \sin^2 x dx = \int_0^{\pi/2} \frac{1 - \cos 2x}{2} dx = \left[\frac{x}{2} - \frac{\sin 2x}{4} \right]_0^{\pi/2} = \left(\frac{\pi}{4} - 0 \right) - (0 - 0) = \frac{\pi}{4}$$

5.12 Integration Techniques

5.12.1 Substitution (Reverse Chain Rule)

Integration by Substitution

If $u = g(x)$, then $du = g'(x) dx$, and

$$\int f(g(x)) g'(x) dx = \int f(u) du = F(u) + C = F(g(x)) + C$$

For a **definite integral**, change the limits:

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

When to Use Substitution

- Look for a function and its derivative (or a multiple thereof) in the integrand.
- If the integrand has a “composite function” structure, try substituting the inner function.
- For $\int f'(x) \cdot [f(x)]^n dx$, let $u = f(x)$.
- Sometimes a substitution transforms a seemingly complex integral into a standard form.

Substitution: $\int 2xe^{x^2} dx$

Let $u = x^2$, so $du = 2x dx$.

$$\int 2xe^{x^2} dx = \int e^u du = e^u + C = e^{x^2} + C$$

Substitution with Definite Integral

Evaluate $\int_0^1 x\sqrt{1+x^2} dx$.

Let $u = 1 + x^2$, so $du = 2x dx$.

When $x = 0$: $u = 1$. When $x = 1$: $u = 2$.

$$\int_0^1 x\sqrt{1+x^2} dx = \frac{1}{2} \int_1^2 u^{1/2} du = \frac{1}{2} \left[\frac{2u^{3/2}}{3} \right]_1^2 = \frac{1}{3} (2^{3/2} - 1) = \frac{2\sqrt{2} - 1}{3}$$

Trigonometric Substitution

Evaluate $\int \frac{x}{\sqrt{1-x^2}} dx$.

Let $u = 1 - x^2$, so $du = -2x dx$, i.e., $-\frac{1}{2}du = x dx$.

$$\int \frac{x}{\sqrt{1-x^2}} dx = -\frac{1}{2} \int u^{-1/2} du = -u^{1/2} + C = -\sqrt{1-x^2} + C$$

5.12.2 Integration by Parts

Integration by Parts

$$\int u dv = uv - \int v du$$

Equivalently, for definite integrals:

$$\int_a^b u dv = [uv]_a^b - \int_a^b v du$$

The LIATE Rule for Choosing u

When choosing which part of the integrand is u and which is dv , use the **LIATE** priority (highest to lowest):

1. **L** — Logarithmic functions ($\ln x$, $\log_a x$)
2. **I** — Inverse trigonometric functions ($\arcsin x$, $\arctan x$, etc.)
3. **A** — Algebraic functions (polynomials, x^n , rational functions)
4. **T** — Trigonometric functions ($\sin x$, $\cos x$, $\tan x$, etc.)
5. **E** — Exponential functions (e^x , a^x)

Choose u from the highest priority category available. The remaining factor is dv .

Integration by Parts: Polynomial times Exponential

Evaluate $\int xe^x dx$.

By LIATE, let $u = x$ (Algebraic) and $dv = e^x dx$.
Then $du = dx$ and $v = e^x$.

$$\int xe^x dx = xe^x - \int e^x dx = xe^x - e^x + C = e^x(x - 1) + C$$

Integration by Parts: Polynomial times Trigonometric

Evaluate $\int x^2 \sin x dx$.

This requires **repeated** integration by parts.

First application: $u = x^2$, $dv = \sin x dx$, $du = 2x dx$, $v = -\cos x$.

$$\int x^2 \sin x dx = -x^2 \cos x + \int 2x \cos x dx$$

Second application (on $\int 2x \cos x dx$): $u = 2x$, $dv = \cos x dx$, $du = 2 dx$, $v = \sin x$.

$$\int 2x \cos x dx = 2x \sin x - \int 2 \sin x dx = 2x \sin x + 2 \cos x + C$$

Combining:

$$\int x^2 \sin x dx = -x^2 \cos x + 2x \sin x + 2 \cos x + C$$

Integration by Parts: $\int \ln x dx$

Let $u = \ln x$ and $dv = dx$. Then $du = \frac{1}{x} dx$ and $v = x$.

$$\int \ln x dx = x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - \int 1 dx = x \ln x - x + C$$

Cyclic Integration by Parts: $\int e^x \sin x dx$

Let $I = \int e^x \sin x dx$.

First application: $u = \sin x$, $dv = e^x dx$, $du = \cos x dx$, $v = e^x$.

$$I = e^x \sin x - \int e^x \cos x dx$$

Second application: $u = \cos x$, $dv = e^x dx$, $du = -\sin x dx$, $v = e^x$.

$$I = e^x \sin x - \left(e^x \cos x + \int e^x \sin x dx \right) = e^x \sin x - e^x \cos x - I$$

Solving for I :

$$2I = e^x(\sin x - \cos x) \implies I = \frac{e^x(\sin x - \cos x)}{2} + C$$

Tabular Method for Repeated Integration by Parts

When integrating $x^n \cdot f(x)$ where $f(x)$ is e^x , $\sin x$, or $\cos x$ (functions that cycle upon repeated differentiation), use the **tabular method**:

Sign	Differentiate (u)	Integrate (dv)
+	x^n	$f(x)$
−	nx^{n-1}	$\int f(x) dx$
+	$n(n-1)x^{n-2}$	$\iint f(x) dx^2$
$\vdots$	$\vdots$	$\vdots$

Multiply diagonally with alternating signs. Stop when the “Differentiate” column reaches 0 or a repeat of $f(x)$.

5.12.3 Partial Fractions for Integration

When the integrand is a rational function with distinct linear factors in the denominator, partial fractions can decompose it into simpler terms.

Partial Fraction Decomposition

For $\frac{1}{(x-a)(x-b)}$ with $a \neq b$:

$$\frac{1}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$$

where $A = \frac{1}{a-b}$ and $B = \frac{1}{b-a}$.

Integration Using Partial Fractions

Evaluate $\int \frac{1}{(x-1)(x-3)} dx$.

Write: $\frac{1}{(x-1)(x-3)} = \frac{A}{x-1} + \frac{B}{x-3}$.

$$1 = A(x-3) + B(x-1).$$

$$x = 1: 1 = -2A \implies A = -\frac{1}{2}.$$

$$x = 3: 1 = 2B \implies B = \frac{1}{2}.$$

$$\begin{aligned}\int \frac{1}{(x-1)(x-3)} dx &= -\frac{1}{2} \int \frac{1}{x-1} dx + \frac{1}{2} \int \frac{1}{x-3} dx = -\frac{1}{2} \ln|x-1| + \frac{1}{2} \ln|x-3| + C \\ &= \frac{1}{2} \ln \left| \frac{x-3}{x-1} \right| + C\end{aligned}$$

Integration with Repeated Linear Factor

Evaluate $\int \frac{2x+1}{x(x^2+1)} dx$.

$$\frac{2x+1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}.$$

$$2x+1 = A(x^2+1) + (Bx+C)x = (A+B)x^2 + Cx + A.$$

Comparing: $A+B=0$, $C=2$, $A=1$. So $B=-1$.

$$\int \left(\frac{1}{x} + \frac{-x+2}{x^2+1} \right) dx = \ln|x| - \frac{1}{2} \ln(x^2+1) + 2 \arctan x + C$$

5.12.4 Trigonometric Integrals

Reduction Formulas for Powers of Sine and Cosine

Odd power of sin: $\int \sin^{2k+1} x dx = \int \sin^{2k} x \cdot \sin x dx = \int (1 - \cos^2 x)^k \cdot \sin x dx$

Use $u = \cos x$, $du = -\sin x dx$.

Odd power of cos: Similar strategy with $u = \sin x$.

Even powers: Use the half-angle (power-reducing) identities:

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

Integral of $\sin^3 x$

$$\int \sin^3 x dx = \int \sin^2 x \cdot \sin x dx = \int (1 - \cos^2 x) \sin x dx$$

Let $u = \cos x$, $du = -\sin x dx$:

$$= - \int (1 - u^2) du = - \left(u - \frac{u^3}{3} \right) + C = -\cos x + \frac{\cos^3 x}{3} + C$$

Integral of $\cos^2 x$

$$\int \cos^2 x dx = \int \frac{1 + \cos 2x}{2} dx = \frac{x}{2} + \frac{\sin 2x}{4} + C$$

5.13 Applications of Integration

5.13.1 Area Under a Curve

Area Under a Curve (above x -axis)

If $f(x) \geq 0$ on $[a, b]$, the area between the curve $y = f(x)$ and the x -axis from $x = a$ to $x = b$ is

$$A = \int_a^b f(x) \, dx$$

Area Between Curve and x -axis (general case)

When $f(x)$ changes sign on $[a, b]$, the total area is

$$A = \int_a^b |f(x)| \, dx$$

Method: Find where $f(x) = 0$, split the integral at these points, and ensure each sub-integral is positive.

Area Below the x -axis

Find the area enclosed by $y = x^2 - 4$ and the x -axis.

x -intercepts: $x^2 = 4 \implies x = \pm 2$.

On $[-2, 2]$, $x^2 - 4 \leq 0$, so:

$$A = \int_{-2}^2 |x^2 - 4| \, dx = \int_{-2}^2 (4 - x^2) \, dx = \left[4x - \frac{x^3}{3} \right]_{-2}^2 = \left(8 - \frac{8}{3} \right) - \left(-8 + \frac{8}{3} \right) = \frac{32}{3}$$

5.13.2 Area Between Two Curves

Area Between Two Curves

The area between $y = f(x)$ and $y = g(x)$ from $x = a$ to $x = b$ (where $f(x) \geq g(x)$) is

$$A = \int_a^b [f(x) - g(x)] \, dx$$

In general, if the curves cross:

$$A = \int_a^b |f(x) - g(x)| \, dx$$

Area Between Two Curves

Find the area between $y = x^2$ and $y = x + 2$.

Find intersection points: $x^2 = x + 2 \implies x^2 - x - 2 = 0 \implies (x - 2)(x + 1) = 0$, so $x = -1$ and $x = 2$.

On $[-1, 2]$, $x + 2 \geq x^2$ (since the line is above the parabola in this interval):

$$A = \int_{-1}^2 [(x + 2) - x^2] \, dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2 = \left(2 + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) = \frac{9}{2}$$

5.13.3 Area Between a Curve and the y -axis

Area Between Curve and y -axis

The area between the curve $x = g(y)$ and the y -axis from $y = c$ to $y = d$ is

$$A = \int_c^d |g(y)| \, dy$$

This is useful when it is easier to express x as a function of y .

5.13.4 Volume of Revolution

Volume of Revolution about the x -axis (Disk Method)

When the region bounded by $y = f(x)$, the x -axis, $x = a$, and $x = b$ is rotated about the x -axis:

$$V = \pi \int_a^b [f(x)]^2 \, dx$$

Volume of Revolution about the y -axis

Method 1 (using x as a function of y):

$$V = \pi \int_c^d [g(y)]^2 \, dy \quad \text{where } x = g(y)$$

Method 2 (shell method):

$$V = 2\pi \int_a^b x|f(x)| \, dx$$

Derivation of the Disk Formula

Consider rotating the region under $y = f(x)$ from $x = a$ to $x = b$ about the x -axis. Partition $[a, b]$ into subintervals of width Δx . The i -th subinterval $[x_{i-1}, x_i]$ sweeps out a **disk** (or washer) with:

- Outer radius: $R_i = f(x_i^*)$ (where $x_i^* \in [x_{i-1}, x_i]$)
- Thickness: Δx
- Volume of disk: $\pi R_i^2 \Delta x$

The total volume is approximated by the Riemann sum:

$$V \approx \sum_i \pi [f(x_i^*)]^2 \Delta x$$

Taking the limit as $\Delta x \rightarrow 0$ (as the number of subintervals $\rightarrow \infty$):

$$V = \lim_{n \rightarrow \infty} \sum_{i=1}^n \pi [f(x_i^*)]^2 \Delta x = \pi \int_a^b [f(x)]^2 \, dx$$

Volume of Revolution

Find the volume generated when $y = \sqrt{x}$, $x = 0$, $x = 4$ is rotated about the x -axis.

$$V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \int_0^4 x dx = \pi \left[\frac{x^2}{2} \right]_0^4 = \pi \cdot 8 = 8\pi$$

Volume of Revolution about the y -axis

Find the volume when $y = x^2$ ($0 \leq x \leq 2$) is rotated about the y -axis.

Using the shell method:

$$V = 2\pi \int_0^2 x \cdot x^2 dx = 2\pi \int_0^2 x^3 dx = 2\pi \left[\frac{x^4}{4} \right]_0^2 = 2\pi \cdot 4 = 8\pi$$

Using Method 1: $x = \sqrt{y}$, y ranges from 0 to 4.

$$V = \pi \int_0^4 (\sqrt{y})^2 dy = \pi \int_0^4 y dy = \pi \left[\frac{y^2}{2} \right]_0^4 = 8\pi \checkmark$$

5.13.5 Average Value of a Function**Average Value**

The **average (mean) value** of f on $[a, b]$ is

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx$$

Derivation of the Average Value Formula

Partition $[a, b]$ into n equal subintervals of width $\Delta x = \frac{b-a}{n}$. Sample points $x_1^*, x_2^*, \dots, x_n^*$.

The average of the sample values is:

$$\frac{f(x_1^*) + f(x_2^*) + \dots + f(x_n^*)}{n} = \frac{1}{b-a} \sum_{i=1}^n f(x_i^*) \Delta x$$

As $n \rightarrow \infty$, this becomes:

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx$$

Average Value

Find the average value of $f(x) = \sin x$ on $[0, \pi]$.

$$\bar{f} = \frac{1}{\pi - 0} \int_0^\pi \sin x dx = \frac{1}{\pi} [-\cos x]_0^\pi = \frac{1}{\pi} (-\cos \pi + \cos 0) = \frac{1}{\pi} (1 + 1) = \frac{2}{\pi}$$

5.14 Differential Equations

5.14.1 Separation of Variables

Definition 5.14.1. A **separable** differential equation is one that can be written in the form

$$\frac{dy}{dx} = f(x) \cdot g(y)$$

i.e., the right-hand side is a product of a function of x and a function of y .

Method of Separation of Variables

1. Separate: $\frac{1}{g(y)} dy = f(x) dx$
2. Integrate both sides: $\int \frac{1}{g(y)} dy = \int f(x) dx$
3. Solve for y (if possible) and include the constant of integration.
4. If an initial condition is given, use it to find the constant.

Separable Differential Equation

Solve $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 2$.

Separate: $y dy = x dx$.

Integrate: $\int y dy = \int x dx \implies \frac{y^2}{2} = \frac{x^2}{2} + C$.

Using $y(0) = 2$: $\frac{4}{2} = 0 + C \implies C = 2$.

So $\frac{y^2}{2} = \frac{x^2}{2} + 2 \implies y^2 = x^2 + 4 \implies y = \sqrt{x^2 + 4}$ (taking positive root since $y(0) = 2 > 0$).

Separable DE with Exponential

Solve $\frac{dy}{dx} = xy$, given $y(0) = 5$.

Separate: $\frac{1}{y} dy = x dx$.

Integrate: $\ln|y| = \frac{x^2}{2} + C$.

$y(0) = 5$: $\ln 5 = 0 + C \implies C = \ln 5$.

$\ln|y| = \frac{x^2}{2} + \ln 5 \implies \ln\left|\frac{y}{5}\right| = \frac{x^2}{2} \implies y = 5e^{x^2/2}$.

5.14.2 Exponential Growth and Decay

Exponential Growth and Decay

If $\frac{dN}{dt} = kN$ (rate of change proportional to the current amount), then

$$N(t) = N_0 e^{kt}$$

where $N_0 = N(0)$ is the initial amount and k is the constant of proportionality:

- $k > 0$: exponential **growth**
- $k < 0$: exponential **decay**

The **half-life** is $t_{1/2} = \frac{\ln 2}{|k|}$.

Derivation of the Exponential Growth/Decay Formula

Given $\frac{dN}{dt} = kN$, separate variables:

$$\frac{1}{N} dN = k dt$$

Integrate both sides:

$$\ln |N| = kt + C$$

Exponentiate:

$$|N| = e^{kt+C} = e^C \cdot e^{kt}$$

Let $N_0 = \pm e^C$ (sign determined by the sign of N):

$$N(t) = N_0 e^{kt}$$

At $t = 0$: $N(0) = N_0 e^0 = N_0$, confirming N_0 is the initial value.

Half-Life Problem

The half-life of carbon-14 is approximately 5730 years. A fossil contains 30% of the original carbon-14. How old is the fossil?

$$k = -\frac{\ln 2}{5730}$$

$N(t) = N_0 e^{kt}$. When $N(t) = 0.3 N_0$:

$$0.3 = e^{-t \ln 2 / 5730} \implies \ln 0.3 = -\frac{t \ln 2}{5730}$$

$$t = \frac{5730 \cdot \ln 0.3}{-\ln 2} = \frac{5730 \cdot \ln(10/3)}{\ln 2} \approx 9953 \text{ years}$$

5.14.3 Euler's Method

Euler's Method for Numerical Solutions

Given $\frac{dy}{dx} = f(x, y)$ with initial condition $y(x_0) = y_0$, approximate y at $x_0 + h, x_0 + 2h, \dots$ using:

$$y_{n+1} = y_n + h \cdot f(x_n, y_n)$$

$$x_{n+1} = x_n + h$$

where h is the step size.

Euler's Method

Use Euler's method with $h = 0.5$ to approximate $y(2)$ for $\frac{dy}{dx} = x + y$, $y(1) = 1$.

Step 1: $x_0 = 1$, $y_0 = 1$.

$$y_1 = y_0 + h \cdot f(x_0, y_0) = 1 + 0.5(1 + 1) = 2, \quad x_1 = 1.5$$

Step 2: $x_1 = 1.5$, $y_1 = 2$.

$$y_2 = y_1 + h \cdot f(x_1, y_1) = 2 + 0.5(1.5 + 2) = 2 + 1.75 = 3.75, \quad x_2 = 2$$

So $y(2) \approx 3.75$ by Euler's method with $h = 0.5$.

IB Exam Tip: Identifying the Type of Differential Equation

On IB exams, you may be asked to identify the type of DE or choose the correct method:

- **Separable:** Look for $\frac{dy}{dx} = f(x)g(y)$ or equivalent forms that can be rearranged so all y -terms are with dy and all x -terms are with dx .
- **Not separable:** If $\frac{dy}{dx}$ cannot be factored into a product of a function of x and a function of y , other methods (like Euler's method or integrating factors, though the latter is not in the syllabus) may be needed.
- **Always check** whether an initial condition is given — it allows you to determine C .
- **Domain consideration:** After solving, state the domain on which the solution is valid.

5.15 Maclaurin Series

5.15.1 Definition

Definition 5.15.1. The **Maclaurin series** of a function f is the Taylor series centered at $x = 0$:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$$

Validity and Convergence

- The Maclaurin series represents $f(x)$ only within its **interval of convergence**.
- For the standard IB functions (e^x , $\sin x$, $\cos x$, $\ln(1+x)$, $(1+x)^p$), the series converges for $|x| < 1$ (or for all x in the case of e^x , $\sin x$, $\cos x$).
- Always check convergence at the endpoints if needed.

5.15.2 Standard Maclaurin Expansions

Standard Maclaurin Series

$f(x)$	Maclaurin Series	Convergence
e^x	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$	all x
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$	all x
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$	all x
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$	$-1 < x \leq 1$
$(1+x)^p$	$1 + px + \frac{p(p-1)}{2!}x^2 + \frac{p(p-1)(p-2)}{3!}x^3 + \dots$	$ x < 1$

For $(1+x)^n$ where n is a positive integer, the series terminates (it is the binomial expansion) and converges for all x .

5.15.3 Derivations of Standard Series**Maclaurin Series for e^x**

Let $f(x) = e^x$. Then $f^{(n)}(x) = e^x$ for all n , so $f^{(n)}(0) = 1$ for all n .

$$e^x = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

This series converges for all $x \in \mathbb{R}$ (by the ratio test, $\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/(n+1)!}{x^n/n!} \right| = \lim_{n \rightarrow \infty} \frac{|x|}{n+1} = 0 < 1$).

Maclaurin Series for $\sin x$

Let $f(x) = \sin x$. Compute derivatives at $x = 0$:

n	$f^{(n)}(x)$	$f^{(n)}(0)$
0	$\sin x$	0
1	$\cos x$	1
2	$-\sin x$	0
3	$-\cos x$	-1
4	$\sin x$	0

The pattern repeats with period 4. Only odd-order terms survive:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

Converges for all $x \in \mathbb{R}$.

Maclaurin Series for $\cos x$

Let $f(x) = \cos x$. Compute derivatives at $x = 0$:

n	$f^{(n)}(x)$	$f^{(n)}(0)$
0	$\cos x$	1
1	$-\sin x$	0
2	$-\cos x$	-1
3	$\sin x$	0
4	$\cos x$	1

Only even-order terms survive:

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$$

Alternative derivation: Since $\frac{d}{dx}[\sin x] = \cos x$, differentiate the series for $\sin x$ term by term:

$$\cos x = \frac{d}{dx} \left[x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \right] = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

Maclaurin Series for $\ln(1+x)$

Let $f(x) = \ln(1+x)$. Then $f(0) = 0$.

$$f'(x) = \frac{1}{1+x} = (1+x)^{-1}$$

$$f''(x) = -(1+x)^{-2}, \quad f'''(x) = 2(1+x)^{-3}, \quad f^{(4)}(x) = -6(1+x)^{-4}$$

In general: $f^{(n)}(x) = (-1)^{n-1}(n-1)!(1+x)^{-n}$ for $n \geq 1$.

So $f^{(n)}(0) = (-1)^{n-1}(n-1)!$ for $n \geq 1$.

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}(n-1)!}{n!} x^n = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

Converges for $-1 < x \leq 1$.

Alternative derivation: Since $\frac{d}{dx}[\ln(1+x)] = \frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots$ for $|x| < 1$, integrate term by term:

$$\ln(1+x) = \int (1 - x + x^2 - x^3 + \cdots) dx = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

5.15.4 Using Substitution in Known Series**Substitution in Maclaurin Series**

Replace x with an expression to obtain new series. For example, in the series for e^x :

- Replace x with $-x$: $e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots$

- Replace x with x^2 : $e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \dots$
- Replace x with $2x$: $e^{2x} = 1 + 2x + 2x^2 + \frac{4x^3}{3} + \dots$

Series for e^{x^2}

Find the Maclaurin series for e^{x^2} up to the term in x^6 .

Replace x with x^2 in the series for e^x :

$$e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \dots = 1 + x^2 + \frac{x^4}{2} + \frac{x^6}{6} + \dots$$

Series for $\ln(1 - x^2)$

Replace x with $(-x^2)$ in the series for $\ln(1 + x)$:

$$\ln(1 - x^2) = (-x^2) - \frac{(-x^2)^2}{2} + \frac{(-x^2)^3}{3} - \dots = -x^2 - \frac{x^4}{2} - \frac{x^6}{3} - \dots$$

valid for $|x| < 1$.

5.15.5 Products of Series

Product of Two Maclaurin Series

Find the Maclaurin series of $e^x \cos x$ up to x^4 .

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots$$

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + \dots$$

Multiply and collect terms up to x^4 :

$$\begin{aligned} e^x \cos x &= \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}\right) \left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right) + \dots \\ &= 1 \left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right) + x \left(1 - \frac{x^2}{2}\right) + \frac{x^2}{2}(1) + \frac{x^3}{6}(1) + \frac{x^4}{24}(1) + \dots \\ &= 1 - \frac{x^2}{2} + \frac{x^4}{24} + x - \frac{x^3}{2} + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots \\ &= 1 + x + \frac{x^4}{24} + \frac{x^4}{24} - \frac{x^3}{2} + \frac{x^3}{6} + \dots \\ &= 1 + x - \frac{x^3}{3} + \frac{x^4}{12} + \dots \end{aligned}$$

Note: the x^2 terms cancel ($-\frac{1}{2} + \frac{1}{2} = 0$).

5.15.6 Differentiating and Integrating Series

Term-by-Term Differentiation and Integration

If $f(x) = \sum_{n=0}^{\infty} a_n x^n$ for $|x| < R$, then:

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \quad (\text{same radius of convergence})$$

$$\int_0^x f(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1} \quad (\text{same radius of convergence})$$

Series from Integration

Find the Maclaurin series for $\arctan x$.

We know $\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \dots$ for $|x| < 1$.

Since $\frac{d}{dx}[\arctan x] = \frac{1}{1+x^2}$, integrate term by term:

$$\arctan x = \int_0^x (1 - t^2 + t^4 - t^6 + \dots) dt = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

valid for $|x| \leq 1$ (converges at the endpoints by the alternating series test).

5.15.7 Series from Differential Equations

Finding a Series from a Differential Equation

Find the first three non-zero terms of the Maclaurin series for the solution to $\frac{dy}{dx} = y + x$, with $y(0) = 1$.

Method 1: Solve the DE directly.

This is a linear DE: $y' - y = x$. The integrating factor is e^{-x} .

$e^{-x}y' - e^{-x}y = xe^{-x}$, so $\frac{d}{dx}[e^{-x}y] = xe^{-x}$.

Integrating (by parts): $e^{-x}y = -xe^{-x} - e^{-x} + C$, so $y = -x - 1 + Ce^x$.

$y(0) = 1$: $1 = -0 - 1 + C \implies C = 2$.

$$y = 2e^x - x - 1 = 2 \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots \right) - x - 1 = 1 + x + x^2 + \frac{x^3}{3} + \dots$$

Method 2: Use successive differentiation.

$y(0) = 1$.

From the DE at $x = 0$: $y'(0) = y(0) + 0 = 1$.

Differentiate the DE: $y'' = y' + 1$. At $x = 0$: $y''(0) = 1 + 1 = 2$.

Differentiate again: $y''' = y''$. At $x = 0$: $y'''(0) = 2$.

$$y = 1 + x + \frac{2}{2!}x^2 + \frac{2}{3!}x^3 + \dots = 1 + x + x^2 + \frac{x^3}{3} + \dots$$

5.15.8 General Binomial Expansion

General Binomial Theorem

For any real p and $|x| < 1$:

$$(1+x)^p = \sum_{n=0}^{\infty} \binom{p}{n} x^n = 1 + px + \frac{p(p-1)}{2!} x^2 + \frac{p(p-1)(p-2)}{3!} x^3 + \dots$$

where $\binom{p}{n} = \frac{p(p-1)(p-2)\cdots(p-n+1)}{n!}$.

When p is a positive integer, the series terminates at $n = p$ (finite sum).

Binomial Expansion for Non-Integer Exponent

Find the Maclaurin series of $\frac{1}{\sqrt{1-x}} = (1-x)^{-1/2}$ up to x^3 .

$$\begin{aligned} (1-x)^{-1/2} &= 1 + \left(-\frac{1}{2}\right)(-x) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2!}(-x)^2 + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{3!}(-x)^3 + \dots \\ &= 1 + \frac{x}{2} + \frac{3x^2}{8} + \frac{5x^3}{16} + \dots \end{aligned}$$

Note: $\frac{1}{\sqrt{1-x^2}}$ is obtained by replacing x with x^2 :

$$\frac{1}{\sqrt{1-x^2}} = 1 + \frac{x^2}{2} + \frac{3x^4}{8} + \frac{5x^6}{16} + \dots$$

This is useful because $\frac{d}{dx}[\arcsin x] = \frac{1}{\sqrt{1-x^2}}$, so integrating gives:

$$\arcsin x = x + \frac{x^3}{6} + \frac{3x^5}{40} + \dots$$

IB Exam Tip: Maclaurin Series Strategy

- **Know the standard series by heart.** The IB data booklet provides some, but fluency saves time.
- **IB questions often ask for** the series “up to the term in x^n ” — write out enough terms and collect like powers.
- **Common IB tasks:**
 - Find the Maclaurin series of a composite function (use substitution in a known series).
 - Find the Maclaurin series of a product (multiply two known series).
 - Find the Maclaurin series of a quotient or a function defined by a differential equation.
 - Use a Maclaurin series to evaluate a limit (e.g., $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$).
 - Use a Maclaurin series to approximate a definite integral.
- **Using series for limits:** Substitute the series and simplify — often the result is immediate.

- **Using series for integrals:** Integrate the series term by term, then evaluate.
- **Error bounds** are sometimes asked: the remainder after n terms for an alternating series is bounded by the first omitted term.

Using Maclaurin Series to Evaluate a Limit

Evaluate $\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{x^5}$.

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$$

$$\sin x - x + \frac{x^3}{6} = \left(x - \frac{x^3}{6} + \frac{x^5}{120} - \dots \right) - x + \frac{x^3}{6} = \frac{x^5}{120} - \dots$$

$$\lim_{x \rightarrow 0} \frac{\frac{x^5}{120} - \dots}{x^5} = \frac{1}{120}$$

Using Maclaurin Series to Approximate an Integral

Use the Maclaurin series for e^{-x^2} to approximate $\int_0^{0.5} e^{-x^2} dx$.

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2} - \frac{x^6}{6} + \dots$$

Integrate term by term:

$$\begin{aligned} \int_0^{0.5} e^{-x^2} dx &= \left[x - \frac{x^3}{3} + \frac{x^5}{10} - \frac{x^7}{42} + \dots \right]_0^{0.5} \\ &= 0.5 - \frac{0.125}{3} + \frac{0.03125}{10} - \frac{0.0078125}{42} + \dots \\ &\approx 0.5 - 0.04167 + 0.003125 - 0.000186 \approx 0.46127 \end{aligned}$$

Chapter 6

IB Exam Mastery

This chapter is your practical guide to maximising marks on the IB Mathematics AA HL examination. It covers the precise meaning of every command term you will encounter, how markschemes are structured, paper-specific strategies, proof-writing technique, common mark-losing errors, calculator skills, and time management. Master this chapter and you will approach every exam paper with confidence and precision.

6.1 Understanding IB Command Terms

Every IB Mathematics question begins with a **command term** that tells you exactly what the examiner expects. Misunderstanding a command term is one of the most reliable ways to lose marks that you otherwise deserve. Below is a complete reference for every command term that appears on AA HL papers.

6.1.1 Hence

Command Term: Hence

When a question says **Hence**, you **must** use the result you obtained in the immediately preceding part of the question. You are *not* permitted to use an alternative method, even if it is valid.

Example: Hence

Question. (a) Find $\frac{d}{dx}(x^2e^x)$. (b) **Hence** evaluate $\int_0^1 2xe^x + x^2e^x dx$.

Part (a): $\frac{d}{dx}(x^2e^x) = 2xe^x + x^2e^x$.

Part (b) — correct approach:

$$\int_0^1 (2xe^x + x^2e^x) dx = [x^2e^x]_0^1 = e - 0 = e.$$

This uses the result from (a) by the Fundamental Theorem of Calculus.

What loses marks: Integrating by parts from scratch, or using any method that does not reference the derivative found in part (a). The examiner provided part (a) specifically so you would use it. If you ignore it, you lose all method marks for part (b), even if your final answer is correct.

6.1.2 Hence or Otherwise

Command Term: Hence or Otherwise

Hence or otherwise means you *may* use the previous result, but you are also free to use any other valid mathematical method. Both approaches can earn full marks. However, using the “hence” route is almost always faster and less error-prone.

Example: Hence or Otherwise

Question. (a) Show that $\sin 2\theta = 2 \sin \theta \cos \theta$. (b) **Hence or otherwise**, solve $\sin 2\theta = \sin \theta$ for $0 \leq \theta \leq 2\pi$.

Using “hence”: Substitute from (a): $2 \sin \theta \cos \theta = \sin \theta$, so $\sin \theta(2 \cos \theta - 1) = 0$, giving $\theta = 0, \pi, 2\pi, \frac{\pi}{3}, \frac{5\pi}{3}$.

Using “otherwise”: Rewrite $\sin 2\theta - \sin \theta = 0$ and use the compound-angle formula independently. This is valid but repeats working already done in (a).

6.1.3 Show That

Command Term: Show That

Show that means the answer is already given to you. You must present a complete, logical chain of working that *leads to* the stated result. The final answer itself carries **no marks** — all marks are for the working. Omitting steps, even obvious ones, will cost marks.

Example: Show That

Question. Show that $\frac{x^2 - 9}{x^2 + 6x + 9} = \frac{x - 3}{x + 3}$, for $x \neq -3$.

Full-credit working:

$$\begin{aligned} \frac{x^2 - 9}{x^2 + 6x + 9} &= \frac{(x - 3)(x + 3)}{(x + 3)(x + 3)} && \text{(factorizing numerator and denominator)} \\ &= \frac{(x - 3)(x + 3)}{(x + 3)^2} && \text{(rewriting denominator)} \\ &= \frac{x - 3}{x + 3} && \text{(cancelling } x + 3 \neq 0). \end{aligned}$$

What loses marks: Writing only the answer $\frac{x - 3}{x + 3}$ with no working. Writing “by factorisation” without showing the factors. Skipping the condition $x \neq -3$ when it is relevant to the domain.

6.1.4 Find

Command Term: Find

Find means you must calculate or determine the answer. Unlike “show that”, the **final answer carries marks**. You still need to show sufficient working, but the examiner is primarily interested in whether you arrive at the correct result.

Example: Find

Question. Find the y -intercept of the graph of $f(x) = \frac{2x-1}{x+3}$.

Solution: The y -intercept occurs at $x = 0$:

$$f(0) = \frac{2(0)-1}{0+3} = -\frac{1}{3}.$$

The y -intercept is $\left(0, -\frac{1}{3}\right)$.

What loses marks: Writing only $-1/3$ without stating the coordinate pair. Arithmetic errors in substitution.

6.1.5 Determine**Command Term: Determine**

Determine is a stronger form of “find”. It requires not just a calculation but also **reasoning, justification, or a decision**. You may need to consider multiple cases, test conditions, or explain why a particular result holds. The reasoning itself attracts marks.

Example: Determine

Question. Determine the number of solutions to $x^3 - 3x + 1 = 0$.

Solution: Let $f(x) = x^3 - 3x + 1$. Then $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$. The critical points are at $x = -1$ and $x = 1$. Evaluating:

$$f(-1) = 3 > 0, \quad f(1) = -1 < 0, \quad \lim_{x \rightarrow -\infty} f(x) = -\infty, \quad \lim_{x \rightarrow +\infty} f(x) = +\infty.$$

Since f changes sign once between $(-\infty, -1)$, once between $(-1, 1)$, and once between $(1, \infty)$, by the Intermediate Value Theorem there are **three** real solutions.

What loses marks: Simply stating “three solutions” without justification. Using the GDC to find approximate roots without explaining the reasoning process.

6.1.6 Prove**Command Term: Prove**

Prove demands a rigorous, complete, and logically sound argument. Every step must be justified. You cannot use the result you are trying to prove at any point in your working (circular reasoning). The level of rigour expected is higher than for “show that”.

Example: Prove

Question. Prove that the product of two odd integers is odd.

Solution: Let $a = 2m + 1$ and $b = 2n + 1$, where $m, n \in \mathbb{Z}$ (since a and b are odd). Then:

$$\begin{aligned} ab &= (2m+1)(2n+1) \\ &= 4mn + 2m + 2n + 1 \\ &= 2(2mn + m + n) + 1. \end{aligned}$$

Since $2mn + m + n \in \mathbb{Z}$, the product ab is of the form $2k + 1$ for some integer k , and is therefore odd.

What loses marks: Giving a single numerical example (e.g., $3 \times 5 = 15$, which is odd). Not defining m and n as integers. Skipping algebraic expansion steps.

6.1.7 Solve

Command Term: Solve

Solve means find *all* values that satisfy the given equation or inequality under the stated conditions. You must consider the domain, check for extraneous solutions, and present the complete solution set.

Example: Solve

Question. Solve $\log_2(x) + \log_2(x - 2) = 3$.

Solution:

$$\begin{aligned}\log_2(x(x - 2)) &= 3 \\ x(x - 2) &= 2^3 = 8 \\ x^2 - 2x - 8 &= 0 \\ (x - 4)(x + 2) &= 0 \\ x &= 4 \quad \text{or} \quad x = -2.\end{aligned}$$

Checking: $x = -2$ is invalid because $\log_2(-2)$ is undefined. Therefore $x = 4$ is the only solution.

What loses marks: Failing to reject $x = -2$. Not stating the domain requirement $x > 2$. Missing one of the potential solutions before checking validity.

6.1.8 Verify

Command Term: Verify

Verify means substitute the given value(s) into the given expression or equation and **confirm** that the stated result holds. This is *not* the same as proving — you are simply checking a claim, not establishing it from first principles. Verification typically carries fewer marks than proof.

Example: Verify

Question. Verify that $x = 3$ is a solution of $x^3 - 6x^2 + 11x - 6 = 0$.

Solution:

$$\begin{aligned}(3)^3 - 6(3)^2 + 11(3) - 6 &= 27 - 54 + 33 - 6 \\ &= 0. \quad \checkmark\end{aligned}$$

Since the left-hand side equals zero when $x = 3$, the value $x = 3$ is verified as a solution.

What loses marks: Attempting to solve the equation from scratch (while not wrong, it wastes time and is not what was asked). Making an arithmetic error in the substitution.

6.1.9 Deduce

Command Term: Deduce

Deduce means derive a conclusion from the information or results already given or established in previous parts. You must show a clear logical chain from the given information to your conclusion. You should *not* introduce new, unexplained assumptions.

Example: Deduce

Question. (a) Find $f'(x)$ where $f(x) = x \ln x - x$. (b) **Deduce** that $f(x)$ has a minimum at $x = 1$.

Solution: From (a), $f'(x) = \ln x + 1 - 1 = \ln x$.

Setting $f'(x) = 0$: $\ln x = 0$, so $x = 1$.

For $0 < x < 1$: $\ln x < 0$, so $f'(x) < 0$ (function decreasing).

For $x > 1$: $\ln x > 0$, so $f'(x) > 0$ (function increasing).

Since f' changes from negative to positive at $x = 1$, by the first derivative test, f has a **minimum** at $x = 1$.

What loses marks: Not connecting the result of part (a) to part (b). Simply stating “ $x = 1$ is the minimum” without using the derivative found in (a).

6.1.10 Write Down

Command Term: Write Down

Write down means state the answer directly, with minimal or no working. This usually applies when the answer can be read off, recalled from memory, or obtained by direct substitution. However, you should still show brief working if there is any doubt — examiners can award *no* marks for an incorrect answer with no working, even on “write down” questions.

Example: Write Down

Question. Write down the value of $\log_2 8$.

Solution: $\log_2 8 = 3$, since $2^3 = 8$.

What loses marks: Writing an incorrect value. If the question is less trivial, writing nothing because you are unsure — always attempt an answer.

6.1.11 Sketch

Command Term: Sketch

Sketch means draw a graph showing the **general shape** and **key features** (intercepts, asymptotes, maxima, minima, end behaviour). The graph does **not** need to be to scale, but all key features must be clearly labelled and correctly positioned relative to each other.

Example: Sketch

Question. Sketch the graph of $f(x) = \frac{1}{x-2} + 1$.

Key features to show:

- Vertical asymptote at $x = 2$

- Horizontal asymptote at $y = 1$
- y -intercept at $\left(0, \frac{1}{2}\right)$
- x -intercept at $(1, 0)$
- Correct general shape (rectangular hyperbola, shifted)
- Behaviour on each side of the asymptote

What loses marks: Drawing the graph to scale but missing key features. Forgetting to label asymptotes. Drawing the wrong branch orientation. Not indicating intercepts.

6.1.12 Draw

Command Term: Draw

Draw means produce an **accurate, to-scale** diagram using a ruler and compass (on Paper 1) or using your GDC (on Paper 2). Unlike “sketch”, the relative positions and proportions must be correct. This command term is rare in AA HL but may appear in geometry questions.

6.1.13 State

Command Term: State

State means give a specific name, value, theorem, or expression. No working is required, but the answer must be precise. This is similar to “write down” but often used when the answer is a definition, a theorem name, or a specific mathematical object.

Example: State

Question. State the Intermediate Value Theorem.

Solution: If f is continuous on the closed interval $[a, b]$ and k is any number between $f(a)$ and $f(b)$, then there exists at least one $c \in (a, b)$ such that $f(c) = k$.

What loses marks: Giving an informal or imprecise statement. Missing the condition of continuity. Confusing it with a different theorem.

6.1.14 Explain

Command Term: Explain

Explain means give a mathematical reason or justification for a statement or result. You must use mathematical language and reasoning, not informal English. The explanation itself carries marks.

Example: Explain

Question. Explain why $f(x) = x^2 + 1$ has no real zeros.

Solution: For all $x \in \mathbb{R}$, $x^2 \geq 0$, so $x^2 + 1 \geq 1 > 0$. Since $f(x) > 0$ for all real x , the equation $x^2 + 1 = 0$ has no real solutions.

What loses marks: Writing “because the parabola doesn’t cross the x -axis” without mathematical justification. Saying “you can’t square root a negative number” without connecting it to the specific function.

6.1.15 Justify

Command Term: Justify

Justify is stronger than “explain”. It requires you to provide **mathematical evidence** — calculations, theorems, or logical deductions — that definitively support your answer. Simply giving a reason is not enough; the evidence must be sufficient to convince a sceptical reader.

Example: Justify

Question. The roots of $x^2 - 5x + 6 = 0$ are α and β . Justify that $\alpha + \beta = 5$ without solving the equation.

Solution: By Vieta’s formulas, for a quadratic $ax^2 + bx + c = 0$ with roots α and β :

$$\alpha + \beta = -\frac{b}{a} = -\frac{(-5)}{1} = 5.$$

What loses marks: Solving the equation to find $\alpha = 2$ and $\beta = 3$, then adding them. While the numerical answer is correct, the question explicitly asked for justification *without solving*, so this approach does not earn full credit.

6.2 Markscheme Structure and Method Marks

Understanding how the markscheme works is essential for maximising your score. The IB uses a precise system of letter-coded marks.

6.2.1 Types of Marks

Mark Types in IB Markschemes

M — Method marks. Awarded for selecting and applying the correct method or approach. If you use the wrong method, you typically lose all M marks for that part, even if your algebra is flawless.

A — Answer marks. Awarded for obtaining the correct answer *following correct working*. If you make an arithmetic error but used the right method, you may still earn M marks but lose the A mark.

R — Reasoning marks. Awarded for correct mathematical reasoning, justification, or explanation. These appear in “explain”, “justify”, “prove”, and “determine” questions.

C — Communication marks. Awarded for clear, logical, and well-organised presentation of your working. This includes using correct notation, labelling steps, and writing in coherent sentences where required.

6.2.2 How Marks Are Distributed

Multi-part questions in IB Mathematics typically follow a predictable pattern:

Typical Multi-Part Question Structure

Part (a) — Foundation (easy marks): Tests a basic skill or recall. Usually 1–3 marks. This part often establishes a result used later.

Part (b) — Application (moderate): Uses the result from part (a) or applies a concept in a straightforward way. Usually 3–5 marks.

Part (c) — Extension (challenging): Combines concepts, requires deeper reasoning, or extends the problem. Usually 4–8 marks. This is where grade boundaries are most affected.

Example: Mark Distribution

Question (7 marks total).

- (a) Find $f'(x)$ where $f(x) = xe^{-x}$. [2 marks: M1, A1]
 (b) Hence find the coordinates of the stationary point of f . [3 marks: M1, A1, A1]
 (c) Determine whether this stationary point is a maximum or a minimum. [2 marks: R1, R1]

In part (c), both marks are for reasoning: you must use the second derivative *or* a sign test and clearly state your conclusion with justification.

6.2.3 Follow-Through Marks

Follow-Through (FT) Marks — When They Apply

Follow-through marks allow you to earn credit in later parts of a question even if you made an error in an earlier part. The examiner follows your **incorrect** answer forward and awards marks for correct *method* applied to your wrong value.

Example: If you incorrectly find $f'(x) = 2xe^{-x} - xe^{-x}$ in part (a) (sign error), but then correctly set this equal to zero, solve for x , and find the y -coordinate in part (b), you can still earn **M marks** in part (b) for the correct method. You would lose the **A marks** because the final numerical answer is wrong.

Follow-Through Marks — When They Do NOT Apply

Follow-through marks are **NOT** awarded in these situations:

- The error makes the subsequent working *trivially simple* (e.g., you get $x = 0$ by mistake and the rest of the question collapses).
- The question uses the command term “hence” and your part (a) answer is wrong — you cannot “follow through” a method you were supposed to use from (a).
- The error is in a “show that” or “prove” question, where the result is already given and you are demonstrating the method.
- The question is essentially a new start, not dependent on the previous part.
- Your working contains a **conceptual** error (not just an arithmetic slip). For example, integrating instead of differentiating is a conceptual error, not a follow-through situation.

6.3 Paper 1: Non-Calculator Strategies

6.3.1 Paper 1 Format

Paper 1 Overview

- **Duration:** 120 minutes
- **Calculator:** NOT permitted
- **Short questions:** Section A (approximately 60 marks)
- **Extended questions:** Section B (approximately 60 marks)
- **Total:** 120 marks
- **Weighting:** 30% of total IB grade

6.3.2 Essential Non-Calculator Skills

The following algebraic skills must be **fluent** — you should be able to perform them quickly and accurately without hesitation.

Algebraic Fluency Checklist

- Expanding brackets: $(ax + b)(cx + d)$, $(a + b)^n$ for small n
- Factorising quadratics: $ax^2 + bx + c$ by inspection
- Completing the square: $ax^2 + bx + c = a(x - h)^2 + k$
- Simplifying algebraic fractions
- Working with surds: $\sqrt{a} + \sqrt{b}$, rationalising denominators
- Laws of indices including negative and fractional powers
- Logarithm laws: $\log(ab)$, $\log(a/b)$, $\log(a^b)$
- Exact trigonometric values: $\sin(\pi/6)$, $\cos(\pi/4)$, $\tan(\pi/3)$, etc.
- Solving simultaneous equations (substitution and elimination)
- Partial fraction decomposition (for integration)

6.3.3 Speed Techniques

Factorising Quadratics Quickly

For $x^2 + bx + c$: find two numbers that *multiply* to c and *add* to b .

Speed Factorisation

$x^2 - 7x + 12$: factors of 12 that sum to -7 are -3 and -4 . Therefore $x^2 - 7x + 12 = (x - 3)(x - 4)$.

$6x^2 + x - 2$: multiply $6 \times (-2) = -12$. Factors of -12 that sum to 1 are 4 and -3 . Split the middle term:

$$6x^2 + 4x - 3x - 2 = 2x(3x + 2) - 1(3x + 2) = (2x - 1)(3x + 2).$$

Completing the Square Mentally

Mental Completing the Square

For $x^2 + bx$: add and subtract $\left(\frac{b}{2}\right)^2$.

$$x^2 + 6x = (x + 3)^2 - 9$$

$$x^2 - 10x = (x - 5)^2 - 25$$

$$x^2 + 5x = \left(x + \frac{5}{2}\right)^2 - \frac{25}{4}$$

For $ax^2 + bx + c$: factor out a first, then complete the square inside.

Recognising Special Forms

Forms to Recognise Instantly

$a^2 - b^2 = (a - b)(a + b)$	(difference of squares)
$a^2 + 2ab + b^2 = (a + b)^2$	(perfect square)
$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$	(sum of cubes)
$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$	(difference of cubes)
$a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + ab^{n-2} + b^{n-1})$	

Efficient Simplification

Efficient Simplification

Simplify $\frac{\sqrt{3} + 1}{\sqrt{3} - 1}$.

Multiply numerator and denominator by $\sqrt{3} + 1$:

$$\frac{(\sqrt{3} + 1)^2}{(\sqrt{3})^2 - 1^2} = \frac{3 + 2\sqrt{3} + 1}{3 - 1} = \frac{4 + 2\sqrt{3}}{2} = 2 + \sqrt{3}.$$

6.3.4 Common Paper 1 Mistakes

Top Paper 1 Errors

- Sign errors in expansion.** Expanding $-(2x - 3)$ as $-2x - 3$ instead of $-2x + 3$. Always use brackets and distribute carefully.
- Forgetting domain restrictions.** When simplifying $\frac{x^2 - 4}{x - 2} = x + 2$, forgetting to state $x \neq 2$. When taking square roots, forgetting that $\sqrt{x^2} = |x|$, not x .
- Not showing enough working for proof questions.** Proof questions on Paper 1 require *every* step. Skipping algebraic steps like “by inspection” or “obviously” will lose marks.
- Arithmetic errors.** Without a calculator, errors in basic arithmetic ($7 \times 8 = 54$, $\sqrt{144} = 11$) are common. Double-check all calculations. Write intermediate steps — do not attempt multi-step arithmetic mentally.
- Incorrect exact values.** Confusing $\sin(\pi/3) = \frac{\sqrt{3}}{2}$ with $\sin(\pi/6) = \frac{1}{2}$. Memorise the exact value table thoroughly.

6. Losing the constant of integration. Writing $\int 2x \, dx = x^2$ instead of $x^2 + C$.

6.3.5 Exact Values Reference

Exact Trigonometric Values for Paper 1

	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

6.4 Paper 2: GDC (Calculator) Strategies

6.4.1 Paper 2 Format

Paper 2 Overview

- **Duration:** 120 minutes
- **Calculator:** Permitted (GDC required)
- **Short questions:** Section A (approximately 60 marks)
- **Extended questions:** Section B (approximately 60 marks)
- **Total:** 120 marks
- **Weighting:** 30% of total IB grade

6.4.2 When to Use the GDC vs. Algebraic Methods

GDC Decision Guide

Use the GDC when:

- The question says “find” or “determine” and the algebra is tedious
- You need numerical solutions to equations that cannot be solved algebraically
- You need to evaluate definite integrals (especially with no elementary antiderivative)
- You need statistical calculations: mean, standard deviation, binomial/normal probabilities
- You need to find intersections, maxima, or minima graphically
- You need matrix operations (multiplication, inverse, determinant)

Use algebraic methods when:

- The question says “show that” or “prove” (the GDC alone is insufficient)
- The question says “hence” and expects you to use a previous algebraic result
- Exact answers are required (the GDC gives decimal approximations)
- The working is straightforward and algebraic is faster
- You are doing proof or derivation (GDC cannot do this)

6.4.3 GDC Techniques

Solving Equations Numerically

Solving Equations on the GDC

Question. Find the solution to $e^x = x + 3$.

Method: Use the solver or find the intersection of $y = e^x$ and $y = x + 3$.

On TI-84: Press **MATH** → **0:Solver**, enter $e^X - X - 3 = 0$, provide an initial guess, and solve. Alternatively, graph both functions and use **2nd** → **CALC** → **5:intersect**.

Result: $x \approx 1.50524$ (to 5 s.f.)

What to write: “Using the GDC solver, $x \approx 1.51$ (3 s.f.)”

Finding Intersections

Finding Intersections

Graph both functions, then use the intersection tool. On most GDCs:

1. Enter both functions in the $Y =$ editor
2. Press **GRAPH**
3. Use the **CALC** or **TRACE** menu
4. Select “Intersect” and specify both curves
5. Move the cursor near the intersection point and confirm

Always verify the intersection visually before accepting the GDC result.

Calculating Definite Integrals

Numerical Integration

Question. Evaluate $\int_0^2 e^{-x^2} dx$.

This integral has no elementary antiderivative. Use the GDC:

On TI-84: **MATH** → **9:fnInt**(, enter e^{-X^2} , X , 0, 2.

Result: $\int_0^2 e^{-x^2} dx \approx 0.88208$ (5 s.f.)

What to write: State the integral, write “using GDC”, then give the answer with appropriate precision.

Statistical Calculations

GDC Statistical Functions

Binomial distribution:

- $\text{binompdf}(n, p, x)$ — probability of exactly x successes
- $\text{binomcdf}(n, p, x)$ — probability of at most x successes ($X \leq x$)
- For $P(X \geq x)$: use $1 - \text{binomcdf}(n, p, x - 1)$
- For $P(a \leq X \leq b)$: use $\text{binomcdf}(n, p, b) - \text{binomcdf}(n, p, a - 1)$

Normal distribution:

- $\text{normalcdf}(a, b, \mu, \sigma) = P(a < X < b)$
- $\text{invNorm}(\text{area}, \mu, \sigma)$ — inverse normal
- For $P(X < a)$: use $\text{normalcdf}(-10^{99}, a, \mu, \sigma)$

Matrix Operations**Matrix Operations on the GDC**

Most GDCs support matrix entry, multiplication, inversion, and determinant calculation. On the TI-84:

- $2\text{nd} \rightarrow \text{MATRIX}$ to access the matrix editor
- Define matrices $[A]$, $[B]$, etc. with appropriate dimensions
- Use $[A]^{-1}$ for the inverse
- Use $\det([A])$ for the determinant
- Use $[A] \times [B]$ for multiplication

Tip: Always store your matrices in named variables rather than retyping them.

6.4.4 Common GDC Mistakes**GDC Errors That Lose Marks**

- 1. Radian vs. degree mode.** This is the single most common GDC error. Before every exam, check your angle mode. If the question involves π or trigonometric functions in calculus, you are almost certainly in radian mode. Set your GDC to radian mode at the start of the exam.
- 2. Rounding too early.** Store intermediate GDC values in variables (e.g., $\text{STO} \rightarrow X$) rather than writing down a rounded value and re-entering it. Rounding to 3 significant figures mid-calculation can change your final answer.
- 3. Wrong window settings.** If you cannot see a graph, check your window. Use ZoomFit or manually set appropriate ranges. For trigonometric functions, ensure the x -range covers at least $[0, 2\pi]$ in radian mode.
- 4. Misinterpreting GDC output.** The GDC gives *decimal approximations*. If the question asks for an exact answer, the GDC result alone is insufficient. Also, the GDC may give multiple solutions — ensure you identify the correct one for the given domain.
- 5. Not writing down GDC input/output.** The markscheme often awards marks for stating what you entered into the GDC and what it returned. Write: “Using the GDC solver on $\cos x = 0.3x$, $x \approx 1.23$ (3 s.f.)”

Golden Rule for Paper 2

Always write down what you entered into the GDC and what it gave you. The examiner must be able to follow your method. A correct answer with no working earns zero marks if the method is not evident.

6.4.5 When the GDC Cannot Help

Situations Where the GDC Is Insufficient

- **“Show that” questions:** The GDC can *verify* a result numerically, but verification is not a demonstration. You must show algebraic working.
- **“Prove” questions:** The GDC cannot construct a proof. You must provide a rigorous logical argument.
- **“Hence” questions:** If the question directs you to use a specific previous result, the GDC approach may not earn method marks.
- **Exact form answers:** When the answer must be in exact form (e.g., $\ln 3$, $\pi/4$, $\sqrt{5}$), the GDC’s decimal output is not acceptable as the final answer, though it can help you check your algebraic result.
- **Derivation questions:** If you need to derive a formula or expression, the GDC cannot do this for you.
- **Paper 2 Section B proof parts:** Extended proof questions within Paper 2 still require full algebraic proof.

6.5 Paper 3: Extended Response and Investigation

6.5.1 Paper 3 Format

Paper 3 Overview

- **Duration:** 75 minutes
- **Calculator:** Permitted (GDC required)
- **Format:** 2–3 extended-response questions (investigation style)
- **Total:** typically 55 marks
- **Weighting:** 20% of total IB grade
- **Nature:** Unseen contexts, combining multiple topics, requiring investigation and generalisation

6.5.2 Nature of Paper 3

Paper 3 is fundamentally different from Papers 1 and 2. Instead of testing individual topics in isolation, it presents **unfamiliar mathematical contexts** that require you to:

- **Investigate** patterns and relationships
- **Generalise** from specific examples
- **Prove** results you have conjectured
- **Combine** techniques from different areas of the syllabus

The mathematics itself is within the syllabus, but the *context* and *presentation* will be unfamiliar. This tests your ability to **think mathematically**, not just recall formulas.

6.5.3 Strategies for Paper 3

Paper 3 Strategy: Read First, Then Act

Step 1: Read the entire question. Before writing anything, read the whole question (all parts, a through f or however many there are). This gives you the overall picture: where the question is going, what results you will need, and what the final goal is. Many students start part (a) without realising that part (c) gives a crucial hint.

Step 2: Identify the mathematical structure. Look past the unfamiliar context. Is this really about arithmetic sequences? Binomial coefficients? Complex numbers on the unit circle? Once you recognise the underlying mathematics, the problem becomes manageable.

Step 3: Work through the early parts carefully. Early parts are designed to guide you. They often provide results or patterns you will need later. Do not rush them.

Step 4: Generalise from specific cases. If the question asks you to compute specific values and then “generalise” or “conjecture”, look for the pattern. Express it in terms of n (or whatever variable is used).

Step 5: Test your conjecture. Before attempting a proof, verify your conjecture with one more specific case that you have not yet used. If it holds, you have more confidence in your proof attempt.

Step 6: Prove your conjecture rigorously. Use the techniques from Section 6.6. State what is given, state what is to be proved, provide clear logical steps, and match your conclusion to the conjecture.

Step 7: If stuck, use the GDC. For investigation questions, generating numerical examples with the GDC can reveal patterns you might otherwise miss.

6.5.4 Common Paper 3 Themes

Frequent Paper 3 Themes

- 1. Extending familiar concepts:** Applying a technique you know (e.g., induction) to a new type of statement (e.g., a divisibility result you have not seen before).
- 2. Combining topics:** A question might require calculus *and* complex numbers, or probability *and* series, or vectors *and* trigonometry. Look for connections.
- 3. Algebraic manipulation with parameters:** Instead of numerical coefficients, you may work with parameters (a, b, k, α). Keep track of conditions on these parameters (e.g., $k > 0$, $a \neq 1$).
- 4. Proof in unfamiliar contexts:** You may be asked to prove a result about a number-theoretic pattern, a geometric property, or a combinatorial identity that you have not encountered before. The proof techniques (induction, contradiction, direct) are the same; only the context is new.
- 5. Investigating families of functions:** You might be asked to explore the graphs of $f_n(x) = x^n e^{-x}$ for different values of n , find patterns in their stationary points, and prove a general result.

6.5.5 Time Management for Paper 3

Paper 3 Time Allocation

With 55 marks in 75 minutes, you have approximately **1.36 minutes per mark**, slightly more generous than Papers 1 and 2. However, the questions are longer and more complex.

Recommended approach:

- **First 5 minutes:** Read both (or all three) questions completely. Decide which to attempt first.
- **Next 30 minutes:** Work on your stronger question. Aim to complete it fully.
- **Next 30 minutes:** Work on the second question. Complete as much as possible.
- **Last 10 minutes:** Review your work, check calculations, ensure all parts are attempted.

Important: On Paper 3, partial marks are extremely valuable. Even if you cannot complete a proof, writing out the correct approach, the base case, or the first few steps of algebraic manipulation will earn marks. **Never leave a part blank.**

6.6 How to Write Proofs for Full Credit

Proof questions are among the most discriminating on the IB exam. They test not only your mathematical knowledge but also your ability to construct a clear, rigorous, and logically sound argument. This section provides a comprehensive guide.

6.6.1 Structure of a Good Proof

Anatomy of a Complete Proof

A well-structured proof follows these steps:

Step 1: State what is given. Begin by clearly identifying the hypotheses or given conditions. Use “Given...” or “Let...” or “Suppose...”

Step 2: State what is to be proved. Restate the proposition or theorem. This shows the examiner you understand the goal.

Step 3: Present logical steps with connectives. Each step should follow from the previous one using a logical connective: **therefore, since, because, hence, but, so, thus, we have, it follows that**. Never leave the examiner to guess how you got from one line to the next.

Step 4: Conclude. End with a statement that matches what was to be proved. Use “Therefore...” or “Hence, as required.” or “This completes the proof.”

6.6.2 Common Proof Mistakes That Lose Marks

Proof Errors Checklist

1. **Circular reasoning.** Starting with what you need to prove and working backwards, then presenting it as if it were forward reasoning.

Example of circular reasoning:

$$\begin{aligned}x^2 + 2x + 1 &= 0 \\(x + 1)^2 &= 0 \\x + 1 &= 0 \\x &= -1\end{aligned}$$

“Therefore $x = -1$ is a root of $x^2 + 2x + 1 = 0$.”

The problem: You assumed the equation has a root and derived $x = -1$. To *show* $x = -1$ is a root, you must *substitute* $x = -1$ and verify it satisfies the equation.

2. Missing cases. For example, proving a result for $n \geq 0$ without checking $n = 0$. Assuming $x > 0$ without justification. Not considering both even and odd cases for integer proofs.

3. Vague language. Writing “it’s obvious” or “clearly” or “we can see”. Every claim must be justified mathematically. If it is truly obvious, write a one-line justification (“since $x^2 \geq 0$ for all $x \in \mathbb{R}$ ”).

4. Skipping algebraic steps. Even if a step seems simple to you, write it out. The markscheme allocates marks to individual steps. Missing a line of algebra may cost you a method mark.

5. Not defining variables. Writing “Let $k = n + 1$ ” without stating what n is. Always define every variable before using it.

6.6.3 Proof by Induction: Complete Checklist

Mathematical Induction Checklist

To earn full marks on a proof by induction, ensure every one of the following is present:

- 1. Basis step.** Clearly stated and verified. Write: “For $n = 1$: LHS = ... = ..., RHS = ... = Since LHS = RHS, the statement is true for $n = 1$.”
- 2. Inductive hypothesis.** Clearly stated using the correct notation. Write: “**Assume** the statement is true for $n = k$, where $k \in \mathbb{N}$, $k \geq 1$. That is, assume ...” (write the full statement with k).
- 3. Inductive step.** Must *start* from the statement for $n = k + 1$ (the LHS for $n = k + 1$) and use the hypothesis to transform it. Write: “Consider the statement for $n = k + 1$.”
- 4. Logical connection.** Use the word “therefore” or “hence” at the point where the hypothesis is applied. Show the examiner exactly where the assumption is used.
- 5. Conclusion.** State that since both the basis step and the inductive step hold, “by the Principle of Mathematical Induction, the statement is true for all $n \in \mathbb{N}$, $n \geq 1$.”

Worked Example: Proof by Induction — Full Marks

Question (6 marks). Prove by mathematical induction that

$$\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}, \quad \text{for all } n \in \mathbb{N}.$$

Solution:

Basis step: For $n = 1$:

$$\text{LHS} = 1^2 = 1, \quad \text{RHS} = \frac{1(2)(3)}{6} = 1. \quad \text{LHS} = \text{RHS}. \quad \checkmark$$

Inductive hypothesis: Assume the statement is true for $n = k$, where $k \in \mathbb{N}$. That is, assume

$$\sum_{r=1}^k r^2 = \frac{k(k+1)(2k+1)}{6}. \quad (\text{IH})$$

Inductive step: Consider the statement for $n = k + 1$:

$$\begin{aligned} \text{LHS} &= \sum_{r=1}^{k+1} r^2 = \sum_{r=1}^k r^2 + (k+1)^2 \\ &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 && (\text{by the inductive hypothesis (IH)}) \\ &= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \\ &= \frac{(k+1)[k(2k+1) + 6(k+1)]}{6} \\ &= \frac{(k+1)(2k^2 + k + 6k + 6)}{6} \\ &= \frac{(k+1)(2k^2 + 7k + 6)}{6} \\ &= \frac{(k+1)(k+2)(2k+3)}{6} \\ &= \frac{(k+1)((k+1)+1)(2(k+1)+1)}{6}. \end{aligned}$$

This is the statement for $n = k + 1$. **Therefore**, if the statement is true for $n = k$, it is also true for $n = k + 1$.

Conclusion: Since the statement is true for $n = 1$ (basis step) and the inductive step holds, by the Principle of Mathematical Induction, the statement is true for all $n \in \mathbb{N}$.

Markscheme annotation: M1 (basis step correct), M1 (hypothesis stated), M1 (starts $n = k + 1$ from sum), M1 (uses hypothesis), A1 (algebra to final form), R1 (conclusion with PMI reference).

6.6.4 Proof by Contradiction: Complete Checklist

Proof by Contradiction Checklist

To earn full marks on a proof by contradiction:

- 1. State the assumption.** Begin: “Assume, for contradiction, that ...” or “Suppose, to the contrary, that ...” State the *negation* of what you want to prove.
- 2. Derive a contradiction.** Using logical deduction from your assumption (and any given conditions), arrive at a statement that is clearly false or contradicts a known fact.
- 3. State what contradicts what.** Explicitly identify the contradiction: “This contradicts the fact that...” or “But this is impossible because...”
- 4. Conclude.** “Therefore, our assumption must be false, and hence [original statement] is true.”

Worked Example: Proof by Contradiction — Full Marks

Question (5 marks). Prove by contradiction that $\sqrt{2}$ is irrational.

Solution:

Assumption: Assume, for contradiction, that $\sqrt{2}$ is rational. Then $\sqrt{2}$ can be written as $\sqrt{2} = \frac{a}{b}$, where $a, b \in \mathbb{Z}$, $b \neq 0$, and $\gcd(a, b) = 1$ (i.e., the fraction is in its lowest terms).

Derivation:

$$\begin{aligned}\sqrt{2} &= \frac{a}{b} \\ 2 &= \frac{a^2}{b^2} && \text{(squaring both sides)} \\ a^2 &= 2b^2.\end{aligned}$$

Since $a^2 = 2b^2$ and b^2 is an integer, a^2 is even, and therefore a is even. (If a were odd, then $a = 2m + 1$ and $a^2 = 4m^2 + 4m + 1$ would be odd, a contradiction.)

So let $a = 2k$ for some $k \in \mathbb{Z}$. Substituting:

$$\begin{aligned}(2k)^2 &= 2b^2 \\ 4k^2 &= 2b^2 \\ b^2 &= 2k^2.\end{aligned}$$

By the same reasoning, since $b^2 = 2k^2$, b^2 is even, and therefore b is even.

Contradiction: We have shown that both a and b are even. But this contradicts our initial assumption that $\gcd(a, b) = 1$.

Conclusion: Therefore, our assumption that $\sqrt{2}$ is rational must be false. Hence, $\sqrt{2}$ is irrational.

Markscheme annotation: M1 (assumes rational, defines a/b in lowest terms), M1 (deduces a is even), M1 (deduces b is even), R1 (identifies contradiction with $\gcd(a, b) = 1$), R1 (concludes $\sqrt{2}$ is irrational).

6.7 Common Mistakes That Lose Marks

This section catalogues the most frequent errors organised by topic. Use it as a checklist when reviewing your practice papers.

6.7.1 Algebra

Common Algebra Mistakes

1. Sign errors in expansion.

$$-(2x - 3) \neq -2x - 3 \quad (\text{wrong}); \quad -(2x - 3) = -2x + 3 \quad (\text{correct})$$

Prevention: Always write brackets and distribute the negative sign term by term.

2. Dropping terms in long expressions. When expanding $(x+1)^5$, omitting the x^3 term or writing the wrong coefficient.

Prevention: Use Pascal's triangle or the binomial theorem systematically. Write all terms before simplifying.

3. Not distributing correctly.

$$3(x+2)^2 \neq 3x^2 + 12 \quad (\text{wrong}); \quad 3(x+2)^2 = 3(x^2 + 4x + 4) = 3x^2 + 12x + 12 \quad (\text{correct})$$

Prevention: Expand the brackets *first*, then multiply by the coefficient.

4. Cancelling incorrectly.

$$\frac{x+3}{x+5} \neq \frac{3}{5} \quad (\text{wrong!})$$

Prevention: Only cancel *factors*, never *terms*. You can cancel $(x+3)$ only if it appears as a factor in both numerator and denominator.

5. Squaring negatives.

$$(-x)^2 \neq -x^2 \quad (\text{wrong}); \quad (-x)^2 = x^2 \quad (\text{correct})$$

6.7.2 Functions

Common Function Mistakes

1. Forgetting domain restrictions. The domain of $f(x) = \ln(x-1)$ is $x > 1$, not $\mathbb{R}$. The domain of $f(x) = \sqrt{x^2 - 4}$ is $|x| \geq 2$. The domain of $f(x) = \frac{1}{x^2 - 9}$ is $x \neq \pm 3$.

2. Wrong inverse function. To find f^{-1} , swap x and y and solve for y . A common error is to simply replace $f(x)$ with $f^{-1}(x)$ without swapping. Also, always state the domain and range of the inverse.

3. Confusing the range of f with the range of f^{-1} . The range of f^{-1} equals the domain of f , not the range of f .

4. Composition order. $f \circ g$ means “ f of g ”, i.e., first apply g , then apply f . Writing $f(g(x))$ as $g(f(x))$ is a common error.

5. Even/odd function confusion. f is even if $f(-x) = f(x)$; f is odd if $f(-x) = -f(x)$. A function can be neither. Checking only one value of x is not sufficient.

6.7.3 Calculus

Common Calculus Mistakes

1. Forgetting $+C$ in indefinite integration.

$$\int 2x \, dx = x^2 + C \quad (\text{correct, always include } C)$$

This is one of the most common errors in all of IB Mathematics.

2. Chain rule errors.

$$\frac{d}{dx}[\sin(3x)] \neq \cos(3x) \quad (\text{wrong}); \quad \frac{d}{dx}[\sin(3x)] = 3 \cos(3x) \quad (\text{correct})$$

Prevention: Always identify the “inner function” and multiply by its derivative.

3. Product rule vs. chain rule confusion.

$$\frac{d}{dx}[x^2 \sin x] \neq 2x \cos x \quad (\text{this applies the chain rule, not the product rule})$$

$$\text{Correct: } \frac{d}{dx}[x^2 \sin x] = 2x \sin x + x^2 \cos x.$$

4. Wrong limits of integration. When finding the area between two curves, using the wrong x -values for the limits. When integrating by substitution, forgetting to change the limits.

5. Integration by parts — wrong choice of u and dv . Choosing u as the more complicated part leads to a harder integral. Remember LIATE (Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential) as a guide for choosing u .

6. Confusing definite and indefinite integrals. Writing $+C$ in a definite integral (incorrect) or forgetting $+C$ in an indefinite integral (incorrect).

7. Second derivative test errors. Computing $f''(x)$ incorrectly or evaluating it at the wrong point. Also, $f''(x_0) = 0$ is *inconclusive*, not “inflection point”.

6.7.4 Trigonometry

Common Trigonometry Mistakes

1. Degree/radian mix-up. Using degrees in a calculus context (where radians are required) or vice versa. The derivative of $\sin x$ is $\cos x$ *only in radians*.

2. Wrong quadrant for inverse trig.

$$\arcsin\left(-\frac{1}{2}\right) \neq \frac{7\pi}{6} \quad (\text{wrong}); \quad \arcsin\left(-\frac{1}{2}\right) = -\frac{\pi}{6} \quad (\text{correct})$$

Remember the ranges: $\arcsin \in [-\pi/2, \pi/2]$, $\arccos \in [0, \pi]$, $\arctan \in (-\pi/2, \pi/2)$.

3. Trigonometric identity errors.

$$\sin(A + B) \neq \sin A + \sin B \quad (\text{wrong})$$

$$\cos 2\theta \neq 2 \cos \theta \quad (\text{wrong}); \quad \cos 2\theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta \quad (\text{correct})$$

4. Missing solutions in trigonometric equations. When solving $\sin \theta = \frac{1}{2}$ for $\theta \in [0, 2\pi]$, giving only $\theta = \pi/6$ and forgetting $\theta = 5\pi/6$.

5. Ambiguous case in non-right triangles. When using the sine rule, forgetting that there may be two possible angles (the ambiguous case). Always check whether the supplement also satisfies the conditions.

6.7.5 Probability

Common Probability Mistakes

- 1. Not reading the question carefully.** Confusing “at least” ($\geq$) with “at most” ($\leq$). Confusing “exactly 3” with “at least 3”. Misreading the distribution parameters.
- 2. Wrong distribution.** Using the binomial distribution when the trials are *not* independent (e.g., sampling without replacement — use hypergeometric or normal approximation). Using Poisson when the binomial is appropriate.
- 3. Conditional probability errors.** Confusing $P(A | B)$ with $P(B | A)$. Forgetting that $P(A \cap B) = P(A) \cdot P(B | A)$, not $P(A) \cdot P(B)$ (unless A and B are independent).
- 4. Normal distribution — wrong standardisation.**

$$Z = \frac{X - \mu}{\sigma} \quad (\text{correct}); \quad Z = \frac{X - \sigma}{\mu} \quad (\text{common error — swapped})$$

- 5. Forgetting that probabilities must sum to 1.** When constructing a probability distribution, the sum of all probabilities must equal 1. Always verify this.

6.7.6 Vectors

Common Vector Mistakes

- 1. Confusing position and displacement vectors.** A position vector is measured from the origin. A displacement (or direction) vector is the difference between two position vectors. They are not interchangeable in all formulas.

- 2. Wrong angle formula.**

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} \quad (\text{correct})$$

A common error is to forget to divide by the magnitudes, or to use the cross product magnitude instead.

- 3. Scalar triple product sign errors.** The volume of the parallelepiped is $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$. Forgetting the absolute value means you might report a negative volume.

- 4. Vector equation of a line — confusing point and direction.** The vector equation $\mathbf{r} = \mathbf{a} + t\mathbf{d}$ requires $\mathbf{a}$ to be a point on the line and $\mathbf{d}$ to be the direction vector. Swapping these gives a different line.

- 5. Cross product — wrong order.** $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$. The order matters for the direction of the resulting vector.

6.7.7 Complex Numbers

Common Complex Number Mistakes

1. Wrong argument quadrant. When finding $\arg(z)$, always sketch the complex number to identify the correct quadrant. The GDC function $\angle(z)$ or manual calculation $\arctan(y/x)$ may give the wrong angle if z is in the second or third quadrant.

Prevention: Use $\arg(z) = \arctan(y/x)$ only for the first quadrant. For other quadrants, add or subtract π as appropriate.

2. De Moivre's theorem errors.

$$[r \operatorname{cis} \theta]^n = r^n \operatorname{cis}(n\theta) \quad (\text{correct})$$

Common error: writing $\operatorname{cis}(\theta)^n = \operatorname{cis}(n\theta)$ without the r^n factor, or writing $r^n \operatorname{cis}(\theta^n)$.

3. Roots of unity — missing roots. An n th root equation has exactly n distinct roots (counting multiplicity). Forgetting to list all n roots loses marks.

4. Complex conjugate errors. $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ (correct), but $\overline{z_1 z_2} \neq \bar{z}_1 \cdot \bar{z}_2$ (wrong); the correct formula is $\overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2$.

5. Modulus-argument form to Cartesian — arithmetic errors. When converting $z = r \operatorname{cis} \theta$ to $a + bi$, errors in evaluating $r \cos \theta$ or $r \sin \theta$.

6.7.8 Statistics

Common Statistics Mistakes

1. Using n instead of $n - 1$ for sample variance.

$$s_n^2 = \frac{\sum (x_i - \bar{x})^2}{n} \quad (\text{population variance})$$

$$s_{n-1}^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} \quad (\text{unbiased estimate of population variance})$$

The IB almost always uses s_{n-1}^2 (the unbiased estimate). Check which formula the question expects.

2. Wrong formula for standard deviation. The variance is the *square* of the standard deviation, not the standard deviation itself. The standard deviation is the square root of the variance. Do not confuse the two.

3. Interpreting the p -value incorrectly. The p -value is the probability of observing a test statistic at least as extreme as the one computed, *assuming H_0 is true*. It is *not* the probability that H_0 is true.

4. Type I and Type II error confusion. Type I error: rejecting H_0 when H_0 is true (false positive). Type II error: failing to reject H_0 when H_0 is false (false negative).

5. Confidence interval — wrong critical value. Using $z = 1.96$ when the population standard deviation is unknown and the sample size is small. In this case, the t -distribution should be used (though the IB syllabus focuses primarily on the normal distribution for confidence intervals).

6.7.9 General Errors

Common General Mistakes

- 1. Not reading the question.** Answering a different question from the one asked. For example, finding the x -intercept when asked for the y -intercept.
- 2. Not answering what was asked.** Finding the derivative when asked to find the *equation of the tangent*. Computing the integral when asked to find the *area* (which may require taking absolute values or subtracting curves).
- 3. Units.** Forgetting to include units in the final answer when the question specifies them (e.g., metres, seconds, kilograms). Mixing units (e.g., km/h vs. m/s).
- 4. Rounding.** Rounding to the wrong number of significant figures or decimal places. The question specifies the precision; follow it exactly. If not specified, give 3 significant figures unless the answer is exact.
- 5. Not attempting a question.** A blank answer earns zero marks. Even a few lines of correct working can earn 1–2 marks. Always attempt every part of every question.
- 6. Illegible handwriting.** If the examiner cannot read your answer, they cannot award marks. Write clearly and organise your working neatly.
- 7. Crossing out correct work.** If you write a correct solution and then cross it out to try a different (incorrect) method, the examiner will mark the visible (incorrect) solution. Clearly indicate which solution you want marked.

6.8 Calculator Techniques Reference

This section provides a practical reference for the most important GDC operations on the two most common calculator models: the TI-84 series and the TI-Nspire series.

6.8.1 TI-84 Series Tips

TI-84 Essential Operations

Solving equations:

- MATH $\rightarrow$ 0:Solver — enter equation, set bounds, solve
- Alternatively: graph the function and use 2nd $\rightarrow$ CALC $\rightarrow$ 2:zero

Finding intersections:

- Graph both functions, then 2nd $\rightarrow$ CALC $\rightarrow$ 5:intersect
- Select first curve, second curve, and guess

Numerical derivative:

- MATH $\rightarrow$ 8:nDeriv(— nDeriv(expression, variable, value)
- Example: nDeriv($X^2 \cdot e^X$, X, 1) gives $f'(1)$ for $f(x) = x^2 e^x$
- **Warning:** nDeriv uses a symmetric difference approximation and can be inaccurate near sharp turns or discontinuities

Definite integrals:

- MATH $\rightarrow$ 9:fnInt(— fnInt(expression, variable, lower, upper)
- Example: fnInt(X^2 , X, 0, 3) gives $\int_0^3 x^2 dx$
- Alternatively: graph and use 2nd $\rightarrow$ CALC $\rightarrow$ 7: $\int f(x)dx$

Statistical distributions:

- 2nd $\rightarrow$ VARS $\rightarrow$ DISTR
- A:binompdf(n, p, x) — $P(X = x)$
- B:binomcdf(n, p, x) — $P(X \leq x)$
- 2:normalcdf(a, b, μ, σ) — $P(a < X < b)$
- 3:invNorm(area, μ, σ) — inverse normal

Storing values:

- After a calculation, press STO $\rightarrow$ ALPHA and choose a letter
- Example: 1.2345 STO $\rightarrow$ ALPHA A stores 1.2345 as A

6.8.2 TI-Nspire Tips

TI-Nspire Essential Operations**Solving equations:**

- In a Calculator page: menu $\rightarrow$ **Algebra** $\rightarrow$ **Solve** (or type solve())
- Example: solve($e^x = x + 3, x$)
- For numerical solving when algebraic solve fails: nSolve($e^x = x + 3, x$)

Finding intersections:

- Graph both functions on a Graphs page
- menu $\rightarrow$ **Analyze Graph** $\rightarrow$ **Intersection**
- Click on both curves and set the lower/upper bounds

Numerical derivative:

- Type nDeriv(expression, variable, point)
- Example: nDeriv($x^2 \cdot e^x, x, 1$)

Definite integrals:

- Type the integral template: CTRL + I (or use the template from the palette)
- Fill in the integrand, variable, and limits

Statistical distributions:

- menu $\rightarrow$ **Probability** $\rightarrow$ **Distributions**
- **Binomial Pdf** and **Binomial Cdf**
- **Normal Cdf** and **Inverse Normal**

Defining functions and variables:

- Define a function: Define $f(x) = x^2 \cdot e^x$
- Store a value: 1.2345 $\rightarrow$ a
- Use these in subsequent calculations for precision

6.8.3 Storing Values for Precision

The Golden Rule of GDC Precision

Never round intermediate values. Store exact GDC outputs in variables and use those variables in subsequent calculations.

Example: If your GDC gives $x = 1.386294361$ as the solution to an equation, do NOT write $x = 1.386$ and then use 1.386 in the next part. Instead:

1. Compute x on the GDC
2. Store it: $\text{STO} \rightarrow X$
3. In the next part, use X directly: $X^2 + 3 \cdot X$

This preserves the full precision of the GDC and prevents accumulation of rounding errors.

On Paper 2: Write “Let $a = 1.39$ (3 s.f.)” but use the stored full-precision value in your GDC for further calculations. Round *only* your final answer to the required precision.

6.8.4 Using Tables on the GDC

Table Feature for Pattern Recognition

The table feature (TI-84: $2\text{nd} \rightarrow \text{GRAPH}$; TI-Nspire: $\text{CTRL} + T$ on a Graphs page) is invaluable for:

- Generating sequences and spotting patterns (useful for Paper 3 investigations)
- Evaluating a function at many points quickly
- Checking your algebraic answers numerically
- Finding approximate roots by looking for sign changes
- Verifying that a function satisfies a given condition at specific points

Tip: Set the table start value and step size appropriately. For trigonometric functions in radians, a step of $\pi/12$ or $\pi/6$ is often useful.

6.8.5 Common GDC Models and Their Quirks

GDC Model Quirks

TI-84 Plus CE:

- Battery can drain during long exams — start with a full charge
- The nDeriv function can give inaccurate results near cusps or discontinuities
- Press MODE to check angle units and numerical format
- The $\text{STAT} \rightarrow \text{EDIT}$ screen is easily overwritten — be careful when entering data

TI-84 Plus (non-CE):

- Uses AAA batteries — bring spares
- Slower processor; complex graphs take longer to render
- Same functionality as CE for mathematical operations

TI-Nspire CX (non-CAS):

- Larger screen and better resolution — easier to read graphs
- The nSolve function is more robust than the TI-84 solver
- Can have multiple pages open simultaneously (Calculator, Graphs, Lists)
- Press $\text{CTRL} \rightarrow \text{DOC}$ to switch between pages
- **Important:** Ensure you have the non-CAS version; CAS calculators are *not* permitted

TI-Nspire CX II (non-CAS):

- Same features as CX but faster
- Improved keyboard layout
- De-interlacing feature improves graph readability

All models:

- Press CLEAR (TI-84) or ESC (Nspire) to return to the home screen
- **Reset your calculator before the exam** to clear stored variables and programs (unless programs are permitted)
- Familiarise yourself with *your specific model* well before the exam

6.9 Time Management and Exam Strategy

Effective time management is the difference between a grade 6 and a grade 7. This section provides a practical framework.

6.9.1 Papers 1 and 2: The One-Minute-Per-Mark Rule

Time Budget: Papers 1 and 2

Paper 1: 120 marks in 120 minutes = **1 minute per mark.**

Paper 2: 120 marks in 120 minutes = **1 minute per mark.**

Paper 3: 55 marks in 75 minutes $\approx$ **1.36 minutes per mark.**

Practical Time Allocation for Papers 1 and 2

Section A (short questions, approximately 60 marks):

- Target: 50–55 minutes
- This gives you 5–10 minutes of buffer for harder short questions
- If a 2-mark question takes more than 3 minutes, move on and return later

Section B (extended questions, approximately 60 marks):

- Target: 55–60 minutes
- Read the entire section before starting
- Begin with the question you find most approachable
- Within each question, attempt every part — partial marks add up

Review time:

- Reserve 5–10 minutes at the end
- Check that all parts have been attempted
- Verify answers using alternative methods or the GDC (Paper 2)
- Ensure all answers are rounded to the required precision

6.9.2 When to Skip and Return

The Skip Strategy

Skip a question temporarily if:

- You have spent more than 1.5 minutes per mark without progress
- You cannot see how to start the question at all
- You feel confused or anxious — move on to regain composure

Before skipping, write down:

- Any relevant formula you can identify
- A diagram if applicable
- The first step of the method, even if you cannot continue

This ensures you earn partial marks and gives you a starting point when you return.

When you return:

- Re-read the question carefully — you may notice something you missed
- Look at later parts of the question for hints (later parts often require earlier results, which tells you what approach to use)
- Try a different method from the one you attempted first

6.9.3 How to Check Your Work Efficiently

Efficient Checking Strategies

- 1. Substitute your answer back.** If you solved an equation, substitute your solution back into the original equation. If you found a derivative, check it against the GDC's numerical derivative.
- 2. Use the GDC to verify (Paper 2).** For algebraic results, use the GDC to check numerically. For example, if you found that the stationary point is at $x = \ln 3$, verify that $f'(\ln 3) \approx 0$ on the GDC.
- 3. Check dimensions and units.** If the answer is a length, is it positive? If it is a probability, is it between 0 and 1? If it is an area, does it have square units?
- 4. Check reasonableness.** Does the magnitude of the answer make sense? A probability of 2.3 or a length of -5 metres is clearly wrong.
- 5. Check against your estimate.** Before solving, make a rough estimate. If your exact answer is orders of magnitude different from your estimate, investigate.
- 6. Re-read the question.** The most common source of errors is not the mathematics but the reading. Re-read the question to confirm you answered what was asked.

6.9.4 Partial Marks Strategies

Maximising Partial Marks When Stuck

When you cannot solve a problem completely, use these strategies to earn every possible mark:

- 1. Write the relevant formula.** If you know the formula but cannot complete the calculation, write the formula with the correct values substituted. This typically earns 1 mark.
- 2. Draw a diagram.** For geometry, vectors, or probability questions, a clearly labelled diagram can earn marks even without calculations.
- 3. State what you know.** If a question asks you to prove something and you cannot complete the proof, write the given conditions and the first few logical steps. The basis step of an induction proof, for example, typically earns 1–2 marks on its own.
- 4. Attempt every part.** Even if part (a) is wrong, attempt part (b) using your (wrong) answer from (a). You may earn follow-through method marks.
- 5. Show your working, always.** A correct answer with no working earns full marks if the answer is correct — but if there is any error, no working means zero marks. Working provides a safety net.
- 6. Use the answer space fully.** If you are unsure, write multiple approaches. The examiner will mark the best one. Clearly label different attempts so the examiner can follow your reasoning.

6.9.5 Final Exam Day Checklist

Pre-Exam Checklist

The night before:

- Ensure your GDC is fully charged (or has fresh batteries)
- Reset your GDC to clear stored variables (or check your school's policy)
- Set your GDC to radian mode
- Pack: GDC, spare batteries, pens, pencils, ruler, protractor, compass, eraser
- Review the command terms (Section 6.1) and common mistakes (Section 6.7)
- Do NOT stay up late cramming — sleep is essential for mathematical reasoning

In the exam room:

- Write your candidate number on every page
- Read the entire paper before starting (especially Paper 3)
- Start with questions you are confident about to build momentum
- Keep track of time — check the clock every 15–20 minutes
- Show all working, even for “write down” questions
- Round final answers to the specified precision
- If you finish early, use the time to check every answer

Chapter 7

Ultra-Condensed Formula Sheet

*Every formula from the entire IB Mathematics AA HL course,
compressed into a single rapid-reference appendix.
This sheet alone is sufficient for full-course revision.*

7.1 Number and Algebra – Quick Reference

Scientific Notation

$a = m \times 10^n$, $1 \leq |m| < 10$, $n \in \mathbb{Z}$.
Significant figures: count from the first non-zero digit.

Arithmetic Sequences & Series

Term: $u_n = u_1 + (n-1)d$
Sum: $S_n = \frac{n}{2}(u_1 + u_n) = \frac{n}{2}[2u_1 + (n-1)d]$

Geometric Sequences & Series

Term: $u_n = u_1 r^{n-1}$
Sum (finite): $S_n = \frac{u_1(r^n - 1)}{r - 1} = \frac{u_1(1 - r^n)}{1 - r}$
Sum (infinite): $S_\infty = \frac{u_1}{1 - r}$, $|r| < 1$

Compound Interest & Finance

Compound interest: $A = P(1 + r)^n$
Compounding k times/yr: $A = P\left(1 + \frac{r}{k}\right)^{kn}$
Continuous: $A = Pe^{rt}$
Depreciation: $A = P(1 - r)^n$

Exponent Laws

$a^m \cdot a^n = a^{m+n}$ $(a^m)^n = a^{mn}$
 $\frac{a^m}{a^n} = a^{m-n}$ $(ab)^n = a^n b^n$
 $a^0 = 1$ $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
 $a^{-n} = \frac{1}{a^n}$ $a^{1/n} = \sqrt[n]{a}$

Logarithm Laws

$\log_a(xy) = \log_a x + \log_a y$ $\log_a(x/y) = \log_a x - \log_a y$
 $\log_a(x^n) = n \log_a x$ $\log_a a = 1$, $\log_a 1 = 0$
Change of base: $\log_a b = \frac{\log_c b}{\log_c a} = \frac{\ln b}{\ln a}$
Inverse: $a^{\log_a x} = x$, $\log_a(a^x) = x$

Binomial Theorem

Expansion: $(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$

General term: $T_{r+1} = \binom{n}{r} a^{n-r} b^r$

Binomial coefficient: $\binom{n}{r} = \frac{n!}{r!(n-r)!} = {}^nC_r$

Pascal identity: $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$

Counting Principles

Fundamental: $n(A \times B) = n(A) \cdot n(B)$

Permutations: ${}_nP_r = P(n, r) = \frac{n!}{(n-r)!}$

Combinations: $\binom{n}{r} = C(n, r) = \frac{n!}{r!(n-r)!}$

Relation: $\binom{n}{r} = \frac{{}_nP_r}{r!}$

Circular permutations: $(n-1)!$

Permutations with repetition: $\frac{n!}{n_1! n_2! \cdots n_k!}$

Complex Numbers – Forms

Rectangular: $z = a + bi$, $a = \operatorname{Re}(z)$,
 $b = \operatorname{Im}(z)$

Modulus: $|z| = \sqrt{a^2 + b^2}$

Argument: $\arg(z) = \arctan\left(\frac{b}{a}\right)$, adjusted for quadrant

Polar: $z = r(\cos \theta + i \sin \theta) = r \operatorname{cis} \theta$

Exponential: $z = r e^{i\theta}$

Complex Numbers – Operations

Conjugate: $\bar{z} = a - bi$; $z\bar{z} = |z|^2 = a^2 + b^2$

Division: $\frac{z_1}{z_2} = \frac{z_1 \bar{z}_2}{z_2 \bar{z}_2} = \frac{z_1 \bar{z}_2}{|z_2|^2}$

Multiplication (polar): $z_1 z_2 = r_1 r_2 \operatorname{cis}(\theta_1 + \theta_2)$

Division (polar): $\frac{z_1}{z_2} = \frac{r_1}{r_2} \operatorname{cis}(\theta_1 - \theta_2)$

De Moivre: $z^n = r^n \operatorname{cis}(n\theta) = r^n e^{in\theta}$

Complex Roots

n th roots of unity ($z^n = 1$):

$$z_k = \operatorname{cis}\left(\frac{2k\pi}{n}\right), \quad k = 0, 1, 2, \dots, n-1$$

n th roots of any complex number ($z^n = w = R e^{i\alpha}$):

$$z_k = R^{1/n} \operatorname{cis}\left(\frac{\alpha + 2k\pi}{n}\right), \quad k = 0, 1, \dots, n-1$$

Roots are equally spaced on a circle of radius $R^{1/n}$.

Proof by Mathematical Induction

Step 1: Prove the statement is true for $n = 1$ (base case).

Step 2: Assume true for $n = k$ (inductive hypothesis).

Step 3: Prove true for $n = k + 1$ using the hypothesis.

Step 4: Conclude the statement holds for all $n \in \mathbb{Z}^+$.

Partial Fractions

Case 1 – Distinct linear factors: $\frac{px + q}{(ax + b)(cx + d)} = \frac{A}{ax + b} + \frac{B}{cx + d}$

Case 2 – Repeated linear factor: $\frac{px + q}{(ax + b)^2(cx + d)} = \frac{A}{ax + b} + \frac{B}{(ax + b)^2} + \frac{C}{cx + d}$

Case 3 – Irreducible quadratic: $\frac{px^2 + qx + r}{(ax + b)(x^2 + cx + d)} = \frac{A}{ax + b} + \frac{Bx + C}{x^2 + cx + d}, \quad (c^2 - 4d < 0)$

Absolute Value & Inequalities

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$|x| \leq a \iff -a \leq x \leq a, \quad |x| \geq a \iff x \leq -a \text{ or } x \geq a$$

$$|ab| = |a||b|, \quad |a + b| \leq |a| + |b| \quad (\text{triangle inequality})$$

Quadratic inequality $ax^2 + bx + c > 0$: roots of $= 0$ divide real line into intervals; test one point per interval. If $a > 0$: positive outside the roots (if real). If $\Delta < 0$ and $a > 0$: always positive.

Systems of Linear Equations

2 equations, 2 unknowns: substitution or elimination.

Cramer's rule: For $Ax + By = E, Cx + Dy = F$:

$$x = \frac{\begin{vmatrix} E & B \\ F & D \end{vmatrix}}{\begin{vmatrix} A & B \\ C & D \end{vmatrix}}, \quad y = \frac{\begin{vmatrix} A & E \\ C & F \end{vmatrix}}{\begin{vmatrix} A & B \\ C & D \end{vmatrix}}$$

System has unique solution iff determinant $\neq 0$ (i.e. lines are not parallel).

7.2 Functions – Quick Reference

Linear Function

$$y = mx + c, \quad \text{gradient } m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Point-gradient: } y - y_1 = m(x - x_1)$$

$$x\text{-intercept: set } y = 0; \quad y\text{-intercept: } c$$

$$\text{Parallel: } m_1 = m_2; \quad \text{Perpendicular: } m_1 m_2 = -1$$

Quadratic Function – Three Forms

$$\text{General: } f(x) = ax^2 + bx + c$$

$$\text{Vertex: } f(x) = a(x - h)^2 + k, \quad \text{vertex } (h, k)$$

$$\text{Factored: } f(x) = a(x - p)(x - q), \quad \text{roots } p, q$$

$$\text{Axis of symmetry: } x = -\frac{b}{2a}$$

$$\text{Discriminant: } \Delta = b^2 - 4ac$$

$\Delta > 0$: two real roots; $\Delta = 0$: one repeated root; $\Delta < 0$: no real roots

$$\text{Sum/product of roots: } \alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

Polynomial Functions

General degree n : $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

Remainder theorem: $f(x) = (x - c)Q(x) + R \implies R = f(c)$

Factor theorem: $(x - c)$ is a factor of $f(x) \iff f(c) = 0$

Fundamental theorem of algebra: A polynomial of degree n has exactly n roots in $\mathbb{C}$ (counting multiplicity).

Vieta's Formulas – Cubic

$$ax^3 + bx^2 + cx + d = a(x - \alpha)(x - \beta)(x - \gamma):$$

$$\alpha + \beta + \gamma = -\frac{b}{a}, \quad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}, \quad \alpha\beta\gamma = -\frac{d}{a}$$

Rational Functions – Asymptotes

Vertical asymptote: set denominator = 0 (after cancelling common factors).

Horizontal asymptote ($y = \frac{p(x)}{q(x)}$):

- $\deg(p) < \deg(q)$: $y = 0$
- $\deg(p) = \deg(q)$: $y = \frac{\text{leading coefficient of } p}{\text{leading coefficient of } q}$
- $\deg(p) > \deg(q)$: oblique/slant asymptote (polynomial long division)

Oblique asymptote: divide numerator by denominator; the quotient (ignoring remainder) gives the asymptote.

Function Transformations

Transformation	Effect on $y = f(x)$	Order
Vertical translation	$y = f(x) + k$	Shift up k
Horizontal translation	$y = f(x - h)$	Shift right h
Vertical stretch	$y = a f(x)$	Stretch by $ a $ from x -axis
Reflection in x -axis	$y = -f(x)$	$a = -1$ case
Horizontal stretch	$y = f(bx)$	Stretch by $\frac{1}{ b }$ from y -axis
Reflection in y -axis	$y = f(-x)$	$b = -1$ case
Combined	$y = a f(b(x - h)) + k$	Apply in reverse: h, b, a, k

Inverse Functions

Condition: f must be one-to-one (passes horizontal line test).

Procedure: Write $y = f(x)$, swap x and y , solve for y .

Domain/range swap:

$$\text{Domain of } f^{-1} = \text{Range of } f, \quad \text{Range of } f^{-1} = \text{Domain of } f$$

Key identities: $f^{-1}(f(x)) = x, \quad f(f^{-1}(x)) = x$

Reflection: $y = f^{-1}(x)$ is the reflection of $y = f(x)$ in $y = x$.

Derivative: $(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$

Composite Functions

Definition: $(f \circ g)(x) = f(g(x))$

Domain of $f \circ g$: all x in domain of g such that $g(x)$ is in domain of f .

Key caution: $f \circ g \neq g \circ f$ in general.

Inverse of composite: $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$

Exponential & Logarithmic Functions

Exponential: $f(x) = a^x$, $a > 0$, $a \neq 1$. Always positive. y -intercept: $(0, 1)$. Asymptote: $y = 0$.

Natural exponential: $f(x) = e^x$. Gradient at any point $= e^x$.

Logarithmic: $f(x) = \log_a x$, inverse of a^x . x -intercept: $(1, 0)$. Asymptote: $x = 0$.

$$\log_a x = y \iff a^y = x$$

$$\log_a x > 0 \iff x > 1; \quad \log_a x < 0 \iff 0 < x < 1 \quad (\text{for } a > 1).$$

Modulus Function

$$f(x) = |x|: \quad f(x) = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Graph properties: V-shape, vertex at origin, never negative.

Equation $|f(x)| = a$: solve $f(x) = a$ or $f(x) = -a$.

Equation $|f(x)| = |g(x)|$: solve $f(x) = g(x)$ or $f(x) = -g(x)$.

Inequality $|f(x)| < a$ ($a > 0$): $-a < f(x) < a$.

7.3 Geometry and Trigonometry – Quick Reference**3D Geometry – Volume and Surface Area**

Solid	Volume	Surface Area
Cuboid	lwh	$2(lw + lh + wh)$
Cube (side s)	s^3	$6s^2$
Cylinder	$\pi r^2 h$	$2\pi r(r + h)$
Cone	$\frac{1}{3}\pi r^2 h$	$\pi r(r + l)$, $l = \sqrt{r^2 + h^2}$
Sphere	$\frac{4}{3}\pi r^3$	$4\pi r^2$
Hemisphere	$\frac{2}{3}\pi r^3$	$3\pi r^2$
Pyramid	$\frac{1}{3}Bh$	$B + \text{sum of lateral faces}$

Distance and Midpoint

2D Distance: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

3D Distance: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

Midpoint (2D): $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

Midpoint (3D): $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$

Vectors – Operations

Equality: $\mathbf{a} = \mathbf{b} \iff$ same magnitude and direction.

Magnitude: $|\mathbf{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$

Unit vector: $\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$

Addition/Subtraction: $\mathbf{a} \pm \mathbf{b} = \langle a_1 \pm b_1, a_2 \pm b_2, a_3 \pm b_3 \rangle$

Scalar multiplication: $k\mathbf{a} = \langle ka_1, ka_2, ka_3 \rangle$, $|k\mathbf{a}| = |k| |\mathbf{a}|$

Dot Product and Cross Product**Dot product:**

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}| |\mathbf{b}| \cos \theta$$

$$\mathbf{a} \cdot \mathbf{b} = 0 \iff \mathbf{a} \perp \mathbf{b}. \quad \cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}. \quad \text{Projection: } \text{proj}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2} \mathbf{b}$$

Cross product (3D):

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle$$

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}| |\mathbf{b}| \sin \theta. \quad \mathbf{a} \times \mathbf{b} = \mathbf{0} \iff \mathbf{a} \parallel \mathbf{b}.$$

$$\text{Area of parallelogram: } |\mathbf{a} \times \mathbf{b}|. \quad \text{Area of triangle: } \frac{1}{2} |\mathbf{a} \times \mathbf{b}|.$$

Lines in 3D**Vector form:** $\mathbf{r} = \mathbf{a} + t\mathbf{d}, \quad t \in \mathbb{R}$ **Parametric form:**

$$x = x_0 + ta,$$

$$y = y_0 + tb,$$

$$z = z_0 + tc$$

$$\text{Cartesian (symmetric) form: } \frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

Two lines intersect if the system of parametric equations has a consistent solution.

$$\text{Skew lines: not parallel, no intersection.} \quad \text{Angle between lines: } \cos \theta = \frac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{|\mathbf{d}_1| |\mathbf{d}_2|}$$

$$\text{Shortest distance (point to line): } d = \frac{|\overrightarrow{AP} \times \mathbf{d}|}{|\mathbf{d}|}, \quad A \text{ on line, } P \text{ external point.}$$

$$\text{Shortest distance (skew lines): } d = \frac{|(\mathbf{a}_2 - \mathbf{a}_1) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|}{|\mathbf{d}_1 \times \mathbf{d}_2|}$$

Planes in 3D**Vector equation:** $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}, \quad \mathbf{n} = \langle A, B, C \rangle$ (normal vector)**Cartesian equation:** $Ax + By + Cz = D$ **Parametric:** $\mathbf{r} = \mathbf{a} + s\mathbf{u} + t\mathbf{v}, \quad s, t \in \mathbb{R}$

$$\text{Angle between planes: } \cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1| |\mathbf{n}_2|}$$

$$\text{Angle between line and plane: } \sin \theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}| |\mathbf{n}|}$$

$$\text{Distance (point to plane): } d = \frac{|Ax_0 + By_0 + Cz_0 - D|}{\sqrt{A^2 + B^2 + C^2}}$$

Line-plane intersection: substitute parametric line into plane equation, solve for t .**Exact Trigonometric Values**

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undef.	0

2D Coordinate Geometry**Distance:** $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ **Midpoint:** $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$ **Gradient:** $m = \frac{y_2 - y_1}{x_2 - x_1}$ **Perpendicular bisector:** line through midpoint, gradient $= -\frac{1}{m}$ (where m is gradient of segment).**Area of triangle (coordinates):** $\Delta = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$ **Trigonometric Identities – Pythagorean**

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

Double Angle Identities

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Compound Angle Identities

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

Reciprocal Trig Functions & Identities

$$\csc \theta = \frac{1}{\sin \theta}, \quad \sec \theta = \frac{1}{\cos \theta}, \quad \cot \theta = \frac{\cos \theta}{\sin \theta} = \frac{1}{\tan \theta}$$

Reduction formulas ($n \in \mathbb{Z}$):

$$\begin{array}{lll} \sin(\pi - \theta) = \sin \theta & \cos(\pi - \theta) = -\cos \theta & \tan(\pi - \theta) = -\tan \theta \\ \sin(\pi + \theta) = -\sin \theta & \cos(\pi + \theta) = -\cos \theta & \tan(\pi + \theta) = \tan \theta \\ \sin(\frac{\pi}{2} - \theta) = \cos \theta & \cos(\frac{\pi}{2} - \theta) = \sin \theta & \tan(\frac{\pi}{2} - \theta) = \cot \theta \end{array}$$
Sine Rule, Cosine Rule, Area Formula**Sine rule:** $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$ **Cosine rule:** $c^2 = a^2 + b^2 - 2ab \cos C$, $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$ **Area of triangle:** $\Delta = \frac{1}{2} ab \sin C = \frac{1}{2} bc \sin A = \frac{1}{2} ca \sin B$ **Heron's formula:** $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$, $s = \frac{a+b+c}{2}$

General Solutions of Trigonometric Equations

$$\sin \theta = k \quad (k \neq \pm 1) \implies \theta = \arcsin(k) + 2n\pi \quad \text{or} \quad \pi - \arcsin(k) + 2n\pi$$

$$\sin \theta = 1 \implies \theta = \frac{\pi}{2} + 2n\pi$$

$$\sin \theta = -1 \implies \theta = \frac{3\pi}{2} + 2n\pi$$

$$\cos \theta = k \quad (k \neq \pm 1) \implies \theta = \pm \arccos(k) + 2n\pi$$

$$\cos \theta = 1 \implies \theta = 2n\pi$$

$$\cos \theta = -1 \implies \theta = \pi + 2n\pi$$

$$\tan \theta = k \implies \theta = \arctan(k) + n\pi$$

where $n \in \mathbb{Z}$ in all cases.

7.4 Statistics and Probability – Quick Reference

Measures of Central Tendency & Spread

Mean (ungrouped): $\bar{x} = \frac{\sum x_i}{n}$

Mean (grouped): $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$

Variance (population): $\sigma^2 = \frac{\sum (x_i - \mu)^2}{n} = \frac{\sum x_i^2}{n} - \mu^2$

Variance (sample): $s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{\sum x_i^2 - n\bar{x}^2}{n-1}$

Standard deviation: $\sigma = \sqrt{\sigma^2}, \quad s = \sqrt{s^2}$

Interquartile range: $\text{IQR} = Q_3 - Q_1$

Outliers: below $Q_1 - 1.5 \text{IQR}$ or above $Q_3 + 1.5 \text{IQR}$

Linear Transformations of Data

For $Y = aX + b$:

$$\bar{y} = a\bar{x} + b$$

$$s_Y = |a| s_X$$

$$\sigma_Y = |a| \sigma_X$$

Regression & Correlation

PMCC: $r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}} = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$

Regression line of y on x : $y - \bar{y} = \frac{S_{xy}}{S_{xx}}(x - \bar{x})$

Regression line of x on y : $x - \bar{x} = \frac{S_{xy}}{S_{yy}}(y - \bar{y})$

Coefficient of determination: $r^2 = \frac{\text{explained variation}}{\text{total variation}}$

Probability Axioms and Rules

$$0 \leq P(A) \leq 1$$

$$P(A') = 1 - P(A)$$

Mutually excl.: $P(A \cup B) = P(A) + P(B)$

$$P(\emptyset) = 0, \quad P(\Omega) = 1$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Indep.: $P(A \cap B) = P(A)P(B)$

Conditional Probability & Bayes' Theorem

Conditional: $P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$

Multiplication rule: $P(A \cap B) = P(A | B) P(B) = P(B | A) P(A)$

Independent check: $A \perp B \iff P(A | B) = P(A)$

Total probability: $P(A) = \sum_{i=1}^n P(A | B_i) P(B_i), \quad \{B_i\} \text{ partition of } \Omega$

Bayes' theorem: $P(B_k | A) = \frac{P(A | B_k) P(B_k)}{\sum_{i=1}^n P(A | B_i) P(B_i)}$

Discrete Random Variables

Expected value: $E(X) = \mu = \sum x_i p_i$

Variance: $\text{Var}(X) = \sigma^2 = E(X^2) - [E(X)]^2 = \sum x_i^2 p_i - \mu^2$

Standard deviation: $\sigma = \sqrt{\text{Var}(X)}$

Linear transformation: $E(aX + b) = aE(X) + b, \quad \text{Var}(aX + b) = a^2 \text{Var}(X)$

Sum of independent r.v.s: $E(X + Y) = E(X) + E(Y), \quad \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

Properties: $E(a) = a, \quad E(X + a) = E(X) + a, \quad \text{Var}(a) = 0, \quad \text{Var}(X + a) = \text{Var}(X)$

Binomial Distribution $X \sim B(n, p)$

PMF: $P(X = r) = \binom{n}{r} p^r (1 - p)^{n-r}, \quad r = 0, 1, \dots, n$

Mean: $E(X) = np$

Variance: $\text{Var}(X) = np(1 - p)$

Standard deviation: $\sigma = \sqrt{np(1 - p)}$

Conditions: fixed n , independent trials, two outcomes, constant p .

Poisson Distribution $X \sim \text{Po}(\lambda)$

PMF: $P(X = r) = \frac{\lambda^r e^{-\lambda}}{r!}, \quad r = 0, 1, 2, \dots$

Mean: $E(X) = \lambda$

Variance: $\text{Var}(X) = \lambda$

Conditions: events occur independently at a constant average rate in a fixed interval.

Grouped Data and Cumulative Frequency

Estimated mean (grouped): $\bar{x} \approx \frac{\sum f_i m_i}{\sum f_i}, \quad m_i = \text{class midpoint}$

Cumulative frequency polygon (ogive): plot cumulative frequency vs. upper class boundary. Median/quartiles from ogive.

Histogram: frequency density = $\frac{f}{\text{class width}}$; area of bar = frequency.

Normal Distribution $X \sim N(\mu, \sigma^2)$

PDF: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

Standardization: $Z = \frac{X - \mu}{\sigma}, \quad Z \sim N(0, 1)$

68–95–99.7 rule:

$$P(\mu - \sigma < X < \mu + \sigma) \approx 0.6827$$

$$P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.9545$$

$$P(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.9973$$

Inverse: given $P(X < x_0) = p$, find z_0 from $N(0, 1)$ table, then $x_0 = \mu + z_0\sigma$.

7.5 Calculus – Quick Reference

Limits – Key Results

L'Hôpital's rule: If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ yields $\frac{0}{0}$ or $\frac{\infty}{\infty}$:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Special limits:

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1, \quad \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta} = 0, \quad \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

Limit laws: $\lim(f + g) = \lim f + \lim g$, $\lim(cf) = c \lim f$, $\lim(fg) = \lim f \cdot \lim g$,
 $\lim(f/g) = \frac{\lim f}{\lim g}$ (if $\lim g \neq 0$).

Derivative from First Principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Complete Derivative Table

Function $f(x)$	Derivative $f'(x)$
c (constant)	0
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
a^x ($a > 0$)	$a^x \ln a$
$\ln x$	$\frac{1}{x}$
$\log_a x$	$\frac{1}{x \ln a}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\csc x$	$-\csc x \cot x$
$\sec x$	$\sec x \tan x$
$\cot x$	$-\csc^2 x$
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$
$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$
$\arctan x$	$\frac{1}{1+x^2}$
$ x $	$\frac{x}{ x } = \operatorname{sgn}(x)$

Differentiation Rules

Linearity: $(af + bg)' = af' + bg'$

Product rule: $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$

Quotient rule: $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$

Chain rule: $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$

Implicit & Logarithmic Differentiation

Implicit: Differentiate both sides w.r.t. x ; treat y as a function of x so $\frac{d}{dx}[y^n] = ny^{n-1}\frac{dy}{dx}$.

Logarithmic differentiation (for $y = [f(x)]^{g(x)}$):

1. Take $\ln$ of both sides: $\ln y = g(x) \ln f(x)$
2. Differentiate implicitly: $\frac{y'}{y} = g'(x) \ln f(x) + \frac{g(x) f'(x)}{f(x)}$
3. Solve: $y' = y \left[g'(x) \ln f(x) + \frac{g(x) f'(x)}{f(x)} \right]$

Second Derivative & Stationary Points

Second derivative test:

- $f''(x_0) > 0$: local minimum
- $f''(x_0) < 0$: local maximum
- $f''(x_0) = 0$: test inconclusive (use first derivative sign chart)

Point of inflexion: $f''(x_0) = 0$ and f'' changes sign at x_0 .

Complete Integral Table

Integrand	Antiderivative
$\int x^n dx$	$\frac{x^{n+1}}{n+1} + C, \quad n \neq -1$
$\int \frac{1}{x} dx$	$\ln x + C$
$\int e^{kx} dx$	$\frac{1}{k}e^{kx} + C$
$\int a^x dx$	$\frac{a^x}{\ln a} + C$
$\int \sin(kx) dx$	$-\frac{1}{k}\cos(kx) + C$
$\int \cos(kx) dx$	$\frac{1}{k}\sin(kx) + C$
$\int \sec^2(kx) dx$	$\frac{1}{k}\tan(kx) + C$
$\int \csc^2(kx) dx$	$-\frac{1}{k}\cot(kx) + C$
$\int \frac{1}{\sqrt{a^2 - x^2}} dx$	$\arcsin\left(\frac{x}{a}\right) + C$
$\int \frac{1}{a^2 + x^2} dx$	$\frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$
$\int \frac{1}{x^2 - a^2} dx$	$\frac{1}{2a}\ln\left \frac{x-a}{x+a}\right + C$
$\int \tan x dx$	$-\ln \cos x + C$
$\int \cot x dx$	$\ln \sin x + C$

Integration Techniques

Substitution: Let $u = g(x)$, $du = g'(x) dx$. Rewrite integral entirely in terms of u .

Integration by parts:

$$\int u dv = uv - \int v du$$

Use LIATE priority for choosing u : **L**ogarithmic, **I**nverse trig, **A**lgebraic, **T**rigonometric, **E**xponential.

Definite integration by parts:

$$\int_a^b u dv = [uv]_a^b - \int_a^b v du$$

Partial fractions: decompose rational function, then integrate term by term using

$$\int \frac{1}{ax + b} dx = \frac{1}{a} \ln|ax + b|.$$

Fundamental Theorem of Calculus**FTC Part 1:** $\frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x)$ **FTC Part 2:** $\int_a^b f(x) dx = F(b) - F(a), \quad F'(x) = f(x)$ **Properties:**

$$\int_a^a f(x) dx = 0 \qquad \int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$\int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx \qquad \int_a^b k f(x) dx = k \int_a^b f(x) dx$$

Area Formulas**Area under curve** ($f(x) \geq 0$):

$$A = \int_a^b f(x) dx$$

Area between two curves ($f(x) \geq g(x)$ on $[a, b]$):

$$A = \int_a^b [f(x) - g(x)] dx$$

Area between curve and y-axis ($f(y) \geq 0$):

$$A = \int_c^d f(y) dy$$

Signed area: If f crosses the x -axis, integrate $|f(x)|$ or split the integral.**Volume of Revolution****About the x -axis:**

$$V = \pi \int_a^b [f(x)]^2 dx$$

About the y -axis:

$$V = \pi \int_c^d [f(y)]^2 dy$$

About a horizontal line $y = k$:

$$V = \pi \int_a^b [f(x) - k]^2 dx$$

About a vertical line $x = h$:

$$V = \pi \int_c^d [f(y) - h]^2 dy$$

Average Value of a Function

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx$$

Kinematics – Calculus-Based

Position: $s(t)$; **Velocity:** $v(t) = s'(t) = \frac{ds}{dt}$; **Acceleration:** $a(t) = v'(t) = s''(t)$

Displacement: $\Delta s = \int_{t_1}^{t_2} v(t) dt$

Total distance: $D = \int_{t_1}^{t_2} |v(t)| dt$

Speed: $|v(t)|$

Differential Equations – Separation of Variables

If $\frac{dy}{dx} = f(x)g(y)$, separate and integrate:

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$

General solution contains an arbitrary constant C .

Particular solution found using an initial/boundary condition.

Growth/decay: $\frac{dN}{dt} = kN \implies N(t) = N_0 e^{kt}$ ($k > 0$: growth, $k < 0$: decay)

Newton's law of cooling: $\frac{dT}{dt} = -k(T - T_{\text{env}})$

Logistic: $\frac{dP}{dt} = kP \left(1 - \frac{P}{L}\right)$

Maclaurin Series

General formula: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$

Function	Maclaurin Series	Interval of Convergence
e^x	$\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$	all $x \in \mathbb{R}$
$\sin x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$	all $x \in \mathbb{R}$
$\cos x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$	all $x \in \mathbb{R}$
$\ln(1+x)$	$\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$	$-1 < x \leq 1$
$\frac{1}{1-x}$	$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots$	$ x < 1$
$(1+x)^k$	$\sum_{n=0}^{\infty} \binom{k}{n} x^n = 1 + kx + \frac{k(k-1)}{2!}x^2 + \dots$	$ x < 1$

Maclaurin Series – Operations

Differentiation: Differentiate the series term by term (within radius of convergence).

Integration: Integrate the series term by term (within radius of convergence).

Substitution: Replace x with an expression, e.g. $e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \dots$

Multiplication: Multiply two series and collect like powers of x .

Standard limits from series:

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

Absolute and Conditional Convergence

Ratio test: $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$.

$L < 1$: converges; $L > 1$: diverges; $L = 1$: inconclusive.

Alternating series test: If $a_n \rightarrow 0$ and a_n is decreasing, then $\sum (-1)^n a_n$ converges.

Absolute convergence: $\sum |a_n|$ converges $\implies \sum a_n$ converges.

Conditional convergence: $\sum a_n$ converges but $\sum |a_n|$ diverges.

Error bound (alternating): $|R_N| \leq |a_{N+1}|$

Comparison test: If $0 \leq a_n \leq b_n$ and $\sum b_n$ converges, then $\sum a_n$ converges. If $\sum a_n$ diverges, then $\sum b_n$ diverges.

Limit comparison test: If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$ with $0 < L < \infty$, then $\sum a_n$ and $\sum b_n$ both converge or both diverge.

Integral test: If f is positive, continuous, and decreasing for $x \geq 1$, then $\sum a_n$ and $\int_1^\infty f(x) dx$ both converge or both diverge.

Continuity and Differentiability

Continuity at $x = a$: $\lim_{x \rightarrow a} f(x) = f(a)$

Differentiability at $x = a$: $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ exists

Key fact: Differentiable $\implies$ Continuous. Continuous $\nRightarrow$ Differentiable.

Intermediate value theorem (IVT): If f continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$, then $f(c) = 0$ for some $c \in (a, b)$.

Eigenvalues and Eigenvectors

For an $n \times n$ matrix A :

Eigenvalue equation: $A\mathbf{v} = \lambda\mathbf{v}$, $\mathbf{v} \neq \mathbf{0}$

Characteristic equation: $\det(A - \lambda I) = 0$

Finding eigenvectors: For each λ , solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$

Diagonalization: $A = PDP^{-1}$, where D is the diagonal matrix of eigenvalues and P has corresponding eigenvectors as columns.

Trace: $\text{tr}(A) = \sum \lambda_i = \sum a_{ii}$

Determinant: $\det(A) = \prod \lambda_i$

Matrix Operations

Addition: $(A + B)_{ij} = a_{ij} + b_{ij}$ (same dimensions)

Scalar multiplication: $(kA)_{ij} = k a_{ij}$

Matrix multiplication: $(AB)_{ij} = \sum_k a_{ik} b_{kj}$; A is $m \times n$, B is $n \times p$, AB is $m \times p$.

Identity: $AI = IA = A$

Inverse: $AA^{-1} = A^{-1}A = I$. $(AB)^{-1} = B^{-1}A^{-1}$.

Determinant of 2×2 : $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Determinant of 3×3 : expand by cofactors along any row/column.

Inverse of 2×2 : $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

$$S = \int_a^b f(x) dx$$

$$z = r \cdot \text{cis}(\theta)$$

INTERNATIONAL BACCALAUREATE

$$P(A|B) = P(A \cap B) / P(B)$$

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$${}^nC_r = n! / (r!(n-r)!)$$

$$V = \pi \cdot \int_a^b [f(x)]^2 dx$$

$$S_{\infty} = u_1 / (1 - r)$$

$$\Delta = b^2 - 4ac$$

$$a^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r$$

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Chapter 1

Number and Algebra

1.1 Scientific Notation

Definition 1.1.1 (Scientific Notation / Standard Form). A positive real number is written in *scientific notation* (also called *standard form*) as

$$a \times 10^k,$$

where $1 \leq a < 10$ and $k \in \mathbb{Z}$. For negative numbers we write $-a \times 10^k$ with the same conditions on a and k .

Scientific Notation Conversion Rules

- **Large numbers** (≥ 1): count how many places the decimal point moves *left* to get a ; this is k .

$$345,000 = 3.45 \times 10^5$$

- **Small numbers** ($0 < x < 1$): count how many places the decimal point moves *right*; this is $-k$.

$$0.0072 = 7.2 \times 10^{-3}$$

- **Significant figures**: the value of a carries the required precision.

$$0.00506237 \approx 5.06 \times 10^{-3} \quad (3 \text{ s.f.})$$

Calculator vs. Non-Calculator Approach

- **Paper 1 (non-calculator)**: You must be comfortable converting by hand. Move the decimal point, count the shifts. Multiplication and division of numbers in scientific notation use the laws of indices: multiply the a -values and add the powers.
- **Paper 2 (calculator)**: Use the GDC. However, you must still write your answer in correct scientific notation on the paper. On the TI-Nspire, press **Ctrl + Enter** to evaluate, and the calculator will display the result.
- **IB warning**: never write “ 34.5×10^4 ” as your final answer. The coefficient *must* satisfy $1 \leq a < 10$.

Worked Example: Scientific Notation Multiplication

Question. The speed of light is approximately 3.00×10^8 m/s. How far does light travel in one year (a light-year), in metres? Give your answer in scientific notation to 3 significant figures.

Solution.

$$\begin{aligned}
 \text{Distance} &= c \times t \\
 &= (3.00 \times 10^8) \times (365.25 \times 24 \times 3600) \\
 &= 3.00 \times 10^8 \times 31,557,600 \\
 &= 3.00 \times 10^8 \times 3.15576 \times 10^7 \\
 &= 9.46728 \times 10^{15} \\
 &\approx 9.47 \times 10^{15} \text{ m} \quad (3 \text{ s.f.})
 \end{aligned}$$

Worked Example: Converting Very Small Numbers

Question. The charge of an electron is approximately 0.000,000,000,000,000,16 coulombs. Express this in scientific notation.

Solution.

$$0.000,000,000,000,000,16 = 1.6 \times 10^{-19} \text{ C}$$

The decimal point moved 19 places to the right, so $k = -19$.

1.2 Arithmetic Sequences and Series

Definition 1.2.1 (Arithmetic Sequence). An **arithmetic sequence** (or *arithmetic progression*) is a sequence in which each term differs from the previous term by a constant called the **common difference** d . That is,

$$u_{n+1} - u_n = d \quad \text{for all } n \geq 1.$$

The sequence takes the form $u_1, u_1 + d, u_1 + 2d, u_1 + 3d, \dots$

Arithmetic Sequence: n th Term

$$u_n = u_1 + (n - 1)d$$

where u_1 is the first term, d is the common difference, and $n \in \mathbb{N}$.

Derivation of the n th Term Formula

Starting from the recursive definition $u_{n+1} = u_n + d$:

$$\begin{aligned}
u_1 &= u_1 \\
u_2 &= u_1 + d \\
u_3 &= u_2 + d = u_1 + 2d \\
u_4 &= u_3 + d = u_1 + 3d \\
&\vdots \\
u_n &= u_1 + \underbrace{d + d + \cdots + d}_{n-1 \text{ times}} \\
&= u_1 + (n-1)d
\end{aligned}$$

We see that the coefficient of d is always one less than the index n .

Arithmetic Series: Sum of First n Terms

$$S_n = \frac{n}{2}(2u_1 + (n-1)d) \quad (1.1)$$

$$= \frac{n}{2}(u_1 + u_n) \quad (1.2)$$

Derivation of the Sum Formula (Pairing Method)

Write out S_n forwards and backwards, then add:

$$\begin{aligned}
S_n &= u_1 + (u_1 + d) + (u_1 + 2d) + \cdots + (u_1 + (n-1)d) \\
S_n &= u_n + (u_n - d) + (u_n - 2d) + \cdots + (u_n - (n-1)d)
\end{aligned}$$

Adding the two rows term by term:

$$\begin{aligned}
2S_n &= (u_1 + u_n) + (u_1 + u_n) + \cdots + (u_1 + u_n) \\
&= n(u_1 + u_n)
\end{aligned}$$

Therefore

$$S_n = \frac{n}{2}(u_1 + u_n).$$

Substituting $u_n = u_1 + (n-1)d$ into the second factor gives the alternative form:

$$S_n = \frac{n}{2}(2u_1 + (n-1)d).$$

Useful Sigma Notation Manipulations

Let $\sum$ denote summation. The following properties are frequently tested on IB exams:

$$\sum_{r=1}^n k = kn \quad (\text{constant out of the sum})$$

$$\sum_{r=1}^n (ar^2 + br + c) = a \sum_{r=1}^n r^2 + b \sum_{r=1}^n r + c \sum_{r=1}^n 1 \quad (\text{linearity})$$

$$\sum_{r=1}^n f(r) = \sum_{r=1}^m f(r) + \sum_{r=m+1}^n f(r) \quad (\text{splitting the sum})$$

$$\sum_{r=k+1}^n f(r) = \sum_{r=1}^n f(r) - \sum_{r=1}^k f(r) \quad (\text{shifting/lowering the index})$$

Sigma Notation Exam Strategy

Common IB trick: You may be asked to find $\sum_{r=21}^{50} u_r$ for an arithmetic sequence. Use the splitting property:

$$\sum_{r=21}^{50} u_r = \sum_{r=1}^{50} u_r - \sum_{r=1}^{20} u_r = S_{50} - S_{20}.$$

Worked Example: Arithmetic Series with Sigma Notation

Question. An arithmetic sequence has first term 5 and common difference 3. Find $\sum_{r=16}^{40} u_r$.

Solution.

$$\begin{aligned} \sum_{r=16}^{40} u_r &= \sum_{r=1}^{40} u_r - \sum_{r=1}^{15} u_r \\ S_{40} &= \frac{40}{2}(2(5) + (40-1)(3)) = 20(10 + 117) = 20 \times 127 = 2540 \\ S_{15} &= \frac{15}{2}(2(5) + (15-1)(3)) = \frac{15}{2}(10 + 42) = \frac{15 \times 52}{2} = 390 \\ \sum_{r=16}^{40} u_r &= 2540 - 390 = 2150 \end{aligned}$$

Worked Example: Using Both Forms of S_n

Question. The sum of the first 20 terms of an arithmetic sequence is 440, and the sum of the first 40 terms is 1700. Find u_1 and d .

Solution. Using $S_n = \frac{n}{2}(2u_1 + (n-1)d)$:

$$S_{20} = \frac{20}{2}(2u_1 + 19d) = 440 \implies 2u_1 + 19d = 44 \quad (1.3)$$

$$S_{40} = \frac{40}{2}(2u_1 + 39d) = 1700 \implies 2u_1 + 39d = 85 \quad (1.4)$$

Subtracting (1.3) from (1.4):

$$20d = 41 \implies d = 2.05.$$

Substituting back: $2u_1 + 19(2.05) = 44$, so $2u_1 + 38.95 = 44$, giving $2u_1 = 5.05$, hence $u_1 = 2.525$.

1.3 Geometric Sequences and Series

Definition 1.3.1 (Geometric Sequence). A **geometric sequence** (or *geometric progression*) is a sequence in which the ratio of consecutive terms is constant. This constant is called the **common ratio** r :

$$\frac{u_{n+1}}{u_n} = r \quad \text{for all } n \geq 1.$$

Geometric Sequence: n th Term

$$u_n = u_1 r^{n-1}$$

Geometric Series: Sum of First n Terms ($r \neq 1$)

$$S_n = \frac{u_1(r^n - 1)}{r - 1} = \frac{u_1(1 - r^n)}{1 - r}$$

Both forms are equivalent; choose the one with the positive denominator to minimise sign errors.

Derivation of the Geometric Sum Formula

Write out S_n and multiply by r :

$$\begin{aligned} S_n &= u_1 + u_1 r + u_1 r^2 + \cdots + u_1 r^{n-1} \\ rS_n &= \quad u_1 r + u_1 r^2 + \cdots + u_1 r^{n-1} + u_1 r^n \end{aligned}$$

Subtracting the second line from the first (most terms cancel):

$$\begin{aligned} S_n - rS_n &= u_1 - u_1 r^n \\ S_n(1 - r) &= u_1(1 - r^n) \\ S_n &= \frac{u_1(1 - r^n)}{1 - r} \end{aligned}$$

Infinite Geometric Series

If $|r| < 1$, then $r^n \rightarrow 0$ as $n \rightarrow \infty$, and the sum to infinity is

$$S_\infty = \frac{u_1}{1 - r}$$

The series **converges** if and only if $|r| < 1$. If $|r| \geq 1$, the series **diverges** (the partial sums grow without bound or oscillate).

Proof of Convergence: $r^n \rightarrow 0$ when $|r| < 1$ **Case 1:** $0 < r < 1$.Write $r = \frac{1}{1+h}$ for some $h > 0$. Then by the binomial theorem (or Bernoulli's inequality):

$$r^n = \frac{1}{(1+h)^n} \leq \frac{1}{1+nh}.$$

As $n \rightarrow \infty$, the denominator $1+nh \rightarrow \infty$, so $r^n \rightarrow 0$.**Case 2:** $-1 < r < 0$.Then $|r| < 1$, so $|r^n| = |r|^n \rightarrow 0$ by Case 1, which implies $r^n \rightarrow 0$.**Divergence when $|r| \geq 1$:**

- If $r = 1$: $S_n = nu_1 \rightarrow \pm\infty$ (unless $u_1 = 0$).
- If $r = -1$: S_n oscillates between u_1 and 0, never settling.
- If $r > 1$: $r^n \rightarrow \infty$, so $|S_n| \rightarrow \infty$.
- If $r < -1$: $|r^n| \rightarrow \infty$ with alternating sign, so S_n diverges by oscillation with increasing magnitude.

IB Exam Tip: Checking Convergence

Always state the condition $|r| < 1$ before using S_∞ . If the question asks “find the sum to infinity” and $|r| \geq 1$, you must write “the series diverges; no finite sum exists.” Marks are awarded for recognising convergence.

Worked Example: Geometric Series with Sum to Infinity**Question.** A geometric sequence has first term 8 and second term -4 . Find the sum to infinity.**Solution.**

$$r = \frac{u_2}{u_1} = \frac{-4}{8} = -0.5.$$

Since $|r| = 0.5 < 1$, the series converges.

$$S_\infty = \frac{u_1}{1-r} = \frac{8}{1-(-0.5)} = \frac{8}{1.5} = \frac{16}{3}.$$

Worked Example: Finding n in a Geometric Sum**Question.** The sum of the first n terms of a geometric sequence with $u_1 = 3$ and $r = 2$ is 3069. Find n .**Solution.**

$$S_n = \frac{3(2^n - 1)}{2 - 1} = 3069$$

$$2^n - 1 = 1029$$

$$2^n = 1028.$$

Since $2^{10} = 1024 \neq 1028$, we check: $n = \frac{\ln 1028}{\ln 2} \approx 10.003$. For integer n this does not work, so we recheck. Actually:

$$\frac{3(2^n - 1)}{1} = 3069 \implies 2^n = 1029 + 1 = 1030? \quad \text{No.}$$

Let us redo: $3(2^n - 1) = 3069$, so $2^n - 1 = 1023$, thus $2^n = 1024 = 2^{10}$, giving $n = 10$.

Worked Example: Mixed Arithmetic–Geometric

Question. Three numbers a, b, c form both an arithmetic sequence and a geometric sequence. Prove that $a = b = c$.

Solution. Since a, b, c are arithmetic: $2b = a + c$.

Since a, b, c are geometric: $b^2 = ac$.

From the arithmetic condition, $a + c = 2b$. Squaring:

$$(a + c)^2 = 4b^2 \implies a^2 + 2ac + c^2 = 4ac \implies a^2 - 2ac + c^2 = 0 \implies (a - c)^2 = 0.$$

Hence $a = c$, and substituting back, $2b = 2a$, so $b = a$. Therefore $a = b = c$.

1.4 Financial Mathematics

Compound Interest

When a principal P is invested at an annual interest rate of r (as a decimal), compounded n times per year for t years, the accumulated amount is

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

For annual compounding ($n = 1$):

$$A = P(1 + r)^t$$

Compound Interest as a Geometric Sequence

Starting with principal P and annual rate r (compounded yearly):

$$\text{After 1 year: } A_1 = P + Pr = P(1 + r)$$

$$\text{After 2 years: } A_2 = A_1(1 + r) = P(1 + r)^2$$

$$\text{After 3 years: } A_3 = A_2(1 + r) = P(1 + r)^3$$

$$\vdots$$

$$\text{After } t \text{ years: } A_t = P(1 + r)^t$$

This is a geometric sequence with first term $P(1 + r)$ and common ratio $(1 + r)$, but evaluated at the t -th term starting from P . Each year, the amount is multiplied by the factor $(1 + r)$.

Annuity: Present Value

An **annuity** is a series of equal periodic payments FV (future value of each payment). The **present value** PV of an annuity paying $\$PMT$ per period for n periods at periodic rate i is

$$PV = \frac{PMT}{i} (1 - (1 + i)^{-n})$$

Annuity: Future Value

The **future value** (accumulated amount) is

$$FV = \frac{PMT}{i} ((1 + i)^n - 1)$$

Derivation of the Annuity Formulas

Consider an annuity with payment PMT at the *end* of each period, interest rate i per period, for n periods.

Future Value derivation. The first payment earns interest for $n - 1$ periods, the second for $n - 2$ periods, etc.:

$$FV = PMT(1 + i)^{n-1} + PMT(1 + i)^{n-2} + \cdots + PMT(1 + i) + PMT.$$

This is a geometric series with first term PMT and common ratio $(1 + i)$:

$$FV = PMT \cdot \frac{(1 + i)^n - 1}{(1 + i) - 1} = \frac{PMT}{i} ((1 + i)^n - 1).$$

Present Value derivation. Discount each payment back to time 0:

$$PV = \frac{PMT}{1 + i} + \frac{PMT}{(1 + i)^2} + \cdots + \frac{PMT}{(1 + i)^n}.$$

This is a geometric series with first term $\frac{PMT}{1 + i}$ and common ratio $\frac{1}{1 + i}$:

$$PV = \frac{PMT}{1 + i} \cdot \frac{1 - \left(\frac{1}{1 + i}\right)^n}{1 - \frac{1}{1 + i}} = \frac{PMT}{1 + i} \cdot \frac{1 - (1 + i)^{-n}}{\frac{i}{1 + i}} = \frac{PMT}{i} (1 - (1 + i)^{-n}).$$

Loan Amortization: Monthly Payment

For a loan of principal L at annual rate r compounded monthly over N months, the monthly payment PMT is

$$PMT = \frac{L \cdot i}{1 - (1 + i)^{-N}}$$

where $i = \frac{r}{12}$ is the monthly interest rate.

Derivation of the Monthly Payment Formula

Setting $PV = L$ (the loan amount equals the present value of all payments):

$$L = \frac{PMT}{i} (1 - (1 + i)^{-N}).$$

Solving for PMT :

$$PMT = \frac{L \cdot i}{1 - (1 + i)^{-N}}.$$

Depreciation

Straight-line depreciation:

$$V_t = V_0 - dt$$

where V_0 is the initial value and d is the depreciation per period.

Compound (reducing-balance) depreciation:

$$V_t = V_0(1 - r)^t$$

where r is the depreciation rate per period. This is identical in form to compound interest but with a decreasing factor $(1 - r) < 1$.

IB Exam Rounding Rules for Financial Mathematics

- **NEVER round intermediate steps.** Store exact values on your GDC and round only the final answer.
- If the question asks for the answer “to the nearest dollar” or “to 2 decimal places,” apply rounding *only* to the final answer.
- Common trap: calculating the monthly payment, rounding it, then using the rounded value to find total interest paid. This gives a wrong answer.
- The IB markscheme will accept answers within a specified tolerance (typically the last digit may differ by 1 due to rounding).

Worked Example: Loan Repayment

Question. A mortgage of \$250,000 is taken out at an annual interest rate of 4.8%, compounded monthly, to be repaid over 25 years. Find the monthly payment and the total interest paid.

Solution.

$$L = 250000, \quad r = 0.048, \quad i = \frac{0.048}{12} = 0.004, \quad N = 25 \times 12 = 300.$$

$$PMT = \frac{250000 \times 0.004}{1 - (1.004)^{-300}} = \frac{1000}{1 - 0.30097\dots} = \frac{1000}{0.69903\dots} \approx \$1430.21$$

Total amount repaid: $300 \times 1430.21 = \$429,063$.

Total interest: $429,063 - 250,000 = \$179,063$.

Worked Example: Compound Depreciation

Question. A car is purchased for \$35,000 and depreciates at 15% per year. Find its value after 6 years and determine when its value falls below \$10,000.

Solution.

$$V_6 = 35000(1 - 0.15)^6 = 35000(0.85)^6 \approx 35000 \times 0.3771 \approx \$13,200.$$

For $V_t < 10000$:

$$\begin{aligned}
 35000(0.85)^t &< 10000 \\
 (0.85)^t &< \frac{2}{7} \\
 t \ln(0.85) &< \ln(2/7) \\
 t &> \frac{\ln(2/7)}{\ln(0.85)} \approx \frac{-1.2528}{-0.1625} \approx 7.71.
 \end{aligned}$$

So the value falls below \$10,000 after **8 complete years**.

1.5 Exponents and Logarithms

1.5.1 Exponent Laws

Fundamental Laws of Exponents

For real base $a > 0$ and real exponents m, n :

$$a^m \times a^n = a^{m+n} \quad (1.5)$$

$$\frac{a^m}{a^n} = a^{m-n} \quad (1.6)$$

$$(a^m)^n = a^{mn} \quad (1.7)$$

$$a^{-n} = \frac{1}{a^n} \quad (1.8)$$

$$a^{1/n} = \sqrt[n]{a} \quad (1.9)$$

$$a^{m/n} = (\sqrt[n]{a})^m = \sqrt[n]{a^m} \quad (1.10)$$

$$a^0 = 1 \quad (a \neq 0) \quad (1.11)$$

Rational Exponents Explained

We define $a^{1/n} = \sqrt[n]{a}$ for $a > 0$ and $n \in \mathbb{N}$. Then:

Fractional exponents: $a^{m/n} = (a^{1/n})^m$.

Verification. Using law (1.7):

$$(a^{1/n})^m = a^{m/n} = a^{(m)(1/n)}.$$

We can also verify by raising both sides of $a^{1/n} = \sqrt[n]{a}$ to the power n :

$$(a^{1/n})^n = a^{n/n} = a^1 = a = (\sqrt[n]{a})^n. \quad \checkmark$$

1.5.2 Logarithm Laws

Definition 1.5.1 (Logarithm). For $a > 0$, $a \neq 1$, the logarithm base a is defined by

$$\log_a x = y \iff a^y = x.$$

The function $\log_a x$ is the inverse of a^x .

Laws of Logarithms

For $a > 0$, $a \neq 1$, and $x, y > 0$:

$$\log_a(xy) = \log_a x + \log_a y \quad (1.12)$$

$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y \quad (1.13)$$

$$\log_a(x^m) = m \log_a x \quad (1.14)$$

$$\log_a 1 = 0 \quad (1.15)$$

$$\log_a a = 1 \quad (1.16)$$

Derivation of Logarithm Laws from Exponent Laws

Product law: Let $\log_a x = m$ and $\log_a y = n$. Then $a^m = x$ and $a^n = y$.

$$xy = a^m \cdot a^n = a^{m+n} \quad (\text{by (1.5)})$$

So $\log_a(xy) = m + n = \log_a x + \log_a y$.

Quotient law: Similarly, $\frac{x}{y} = \frac{a^m}{a^n} = a^{m-n}$ (by (1.6)).

So $\log_a(x/y) = m - n = \log_a x - \log_a y$.

Power law: Let $\log_a x = m$, so $a^m = x$. Then

$$x^k = (a^m)^k = a^{mk} \quad (\text{by (1.7)}).$$

So $\log_a(x^k) = mk = k \log_a x$.

Change of Base Formula

$$\log_b x = \frac{\log_a x}{\log_a b}$$

In practice, use natural logarithms or base 10:

$$\log_b x = \frac{\ln x}{\ln b} = \frac{\log_{10} x}{\log_{10} b}.$$

Proof of the Change of Base Formula

Let $y = \log_b x$. By definition, $b^y = x$.

Take $\log_a$ of both sides:

$$\log_a(b^y) = \log_a x.$$

Using the power law (1.14):

$$y \log_a b = \log_a x.$$

Therefore

$$y = \frac{\log_a x}{\log_a b},$$

which is $\log_b x = \frac{\log_a x}{\log_a b}$.

1.5.3 Solving Exponential and Logarithmic Equations

Strategy for Exponential Equations

1. Isolate the exponential term.
2. Take $\ln$ (or $\log$) of both sides.
3. Use the power law to bring the exponent down.
4. Solve the resulting linear (or polynomial) equation.
5. **Always check** that solutions satisfy the domain of the original equation.

Strategy for Logarithmic Equations

1. Combine all log terms using log laws (if on the same side).
2. Or exponentiate both sides to remove the logarithm.
3. Solve the resulting equation.
4. **Check all solutions in the original equation** — extraneous solutions can arise from squaring or from the domain restriction $\log_a x$ requires $x > 0$.

Worked Example: Exponential Equation

Question. Solve $3^{2x-1} = 5^{x+2}$. Give your answer to 3 significant figures.

Solution.

$$\begin{aligned}
 3^{2x-1} &= 5^{x+2} \\
 \ln(3^{2x-1}) &= \ln(5^{x+2}) \\
 (2x-1)\ln 3 &= (x+2)\ln 5 \\
 2x\ln 3 - \ln 3 &= x\ln 5 + 2\ln 5 \\
 2x\ln 3 - x\ln 5 &= \ln 3 + 2\ln 5 \\
 x(2\ln 3 - \ln 5) &= \ln 3 + 2\ln 5 \\
 x &= \frac{\ln 3 + 2\ln 5}{2\ln 3 - \ln 5} = \frac{\ln 3 + \ln 25}{\ln 9 - \ln 5} = \frac{\ln 75}{\ln(9/5)} \approx \frac{4.317}{0.5878} \approx 7.34 \quad (3 \text{ s.f.})
 \end{aligned}$$

Worked Example: Logarithmic Equation

Question. Solve $\log_2(x+3) + \log_2(x-1) = 4$.

Solution. Domain: $x+3 > 0$ and $x-1 > 0$, so $x > 1$.

$$\begin{aligned}
 \log_2((x+3)(x-1)) &= 4 \\
 (x+3)(x-1) &= 2^4 = 16 \\
 x^2 + 2x - 3 &= 16 \\
 x^2 + 2x - 19 &= 0 \\
 x &= \frac{-2 \pm \sqrt{4+76}}{2} = \frac{-2 \pm \sqrt{80}}{2} = -1 \pm 2\sqrt{5}.
 \end{aligned}$$

Since $x > 1$, we take $x = -1 + 2\sqrt{5} \approx 3.47$. The other root $-1 - 2\sqrt{5} \approx -5.47$ is rejected (fails the domain condition).

Worked Example: Hidden Quadratic in Exponent

Question. Solve $4^x + 2^{x+1} - 3 = 0$.

Solution. Note that $4^x = (2^2)^x = 2^{2x} = (2^x)^2$. Let $u = 2^x$:

$$\begin{aligned}u^2 + 2u - 3 &= 0 \\(u + 3)(u - 1) &= 0.\end{aligned}$$

Since $u = 2^x > 0$, we reject $u = -3$. So $u = 1$, giving $2^x = 1$, hence $x = 0$.

1.6 Proof

1.6.1 Direct Proof

Structure of a Direct Proof

1. **State** what you are going to prove clearly.
2. Start from known facts, definitions, or given conditions.
3. Use logical deductions (algebra, known theorems) to reach the conclusion.
4. End with a clear concluding statement such as “hence proved” or “as required.”

Worked Example: Direct Proof

Question. Prove that the product of two odd integers is odd.

Solution. Let $a = 2m + 1$ and $b = 2n + 1$ be two odd integers, where $m, n \in \mathbb{Z}$.
Then

$$ab = (2m + 1)(2n + 1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1.$$

Since $2mn + m + n \in \mathbb{Z}$, the product ab is of the form $2k + 1$ with $k \in \mathbb{Z}$, which is odd. ■

1.6.2 Proof by Contradiction

Structure of Proof by Contradiction

1. **Assume** the negation of the statement to be proved.
2. **Derive** a logical contradiction (something absurd or false).
3. **Conclude** that the assumption must be false, so the original statement is true.

Worked Example: Proof by Contradiction

Question. Prove that $\sqrt{2}$ is irrational.

Solution. Assume, for contradiction, that $\sqrt{2}$ is rational. Then

$$\sqrt{2} = \frac{p}{q},$$

where $p, q \in \mathbb{Z}^+$, $\gcd(p, q) = 1$ (the fraction is in lowest terms).

Squaring: $2 = \frac{p^2}{q^2}$, so $p^2 = 2q^2$.

This means p^2 is even, so p is even (since the square of an odd number is odd). Write $p = 2k$. Then $(2k)^2 = 2q^2$, so $4k^2 = 2q^2$, giving $q^2 = 2k^2$.

This means q^2 is even, so q is even.

But if both p and q are even, then $\gcd(p, q) \geq 2$, contradicting $\gcd(p, q) = 1$.

Therefore the assumption is false: $\sqrt{2}$ is irrational. ■

1.6.3 Counterexamples

How to Find and Present Counterexamples

- A **counterexample** is a single specific case that disproves a general statement.
- To disprove “for all x , $P(x)$,” find one x_0 such that $P(x_0)$ is false.
- **Present clearly:** state the proposed claim, give the counterexample, show the computation, and conclude.
- **IB warning:** a counterexample must be explicit. “There exists some value” is not enough — you must state the value.

Worked Example: Counterexample

Question. A student claims: “If n^2 is divisible by 4, then n is divisible by 4.” Disprove this statement.

Solution. Consider $n = 6$. Then $n^2 = 36$, and 36 is divisible by 4 (since $36/4 = 9$). However, 6 is *not* divisible by 4 (since $6/4 = 1.5$). This is a counterexample, so the claim is false.

1.6.4 Mathematical Induction

Structure of Proof by Mathematical Induction

To prove a statement $P(n)$ for all integers $n \geq n_0$:

Basis step. Verify $P(n_0)$ is true (usually $n_0 = 1$).

Inductive hypothesis.

Assume $P(k)$ is true for some arbitrary $k \geq n_0$.

Inductive step.

Using the hypothesis, prove $P(k + 1)$ is true.

Conclusion. By the principle of mathematical induction, $P(n)$ is true for all $n \geq n_0$.

IB Mark Allocation for Proofs

- **M1** (Method): Correctly setting up the proof structure (stating basis step, hypothesis, etc.).
- **A1** (Answer): Correct algebraic manipulation within each step.
- **R1** (Reasoning): The concluding statement “by mathematical induction” or “hence proved for all n .”

- **Common loss of marks:** Forgetting to state the inductive hypothesis explicitly; forgetting the concluding statement; algebraic errors in the inductive step.

Worked Example: Induction — Sum Formula

Question. Prove by induction that $\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \in \mathbb{N}$.

Solution.

Basis step ($n = 1$): $\sum_{r=1}^1 r^2 = 1$, and $\frac{1 \times 2 \times 3}{6} = 1$. ✓

Inductive hypothesis: Assume for some $k \geq 1$:

$$\sum_{r=1}^k r^2 = \frac{k(k+1)(2k+1)}{6}. \quad (\text{IH})$$

Inductive step: We must show

$$\sum_{r=1}^{k+1} r^2 = \frac{(k+1)(k+2)(2k+3)}{6}.$$

$$\begin{aligned} \sum_{r=1}^{k+1} r^2 &= \sum_{r=1}^k r^2 + (k+1)^2 \\ &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \quad (\text{by IH}) \\ &= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \\ &= \frac{(k+1)[k(2k+1) + 6(k+1)]}{6} \\ &= \frac{(k+1)(2k^2 + k + 6k + 6)}{6} \\ &= \frac{(k+1)(2k^2 + 7k + 6)}{6}. \end{aligned}$$

Factorising the quadratic: $2k^2 + 7k + 6 = (k+2)(2k+3)$. Therefore:

$$\sum_{r=1}^{k+1} r^2 = \frac{(k+1)(k+2)(2k+3)}{6}. \quad \checkmark$$

Conclusion. By the principle of mathematical induction, the formula holds for all $n \in \mathbb{N}$. ■

Worked Example: Induction — Divisibility

Question. Prove by induction that $3^{2n} - 1$ is divisible by 8 for all $n \in \mathbb{N}$.

Solution.

Basis step ($n = 1$): $3^2 - 1 = 9 - 1 = 8$, which is divisible by 8. ✓

Inductive hypothesis: Assume $3^{2k} - 1 = 8m$ for some integer m .

Inductive step: Consider $n = k + 1$:

$$\begin{aligned}
 3^{2(k+1)} - 1 &= 3^{2k+2} - 1 = 9 \cdot 3^{2k} - 1 \\
 &= 9 \cdot 3^{2k} - 9 + 8 \\
 &= 9(3^{2k} - 1) + 8 \\
 &= 9 \cdot 8m + 8 \quad (\text{by IH}) \\
 &= 8(9m + 1).
 \end{aligned}$$

Since $9m + 1 \in \mathbb{Z}$, $3^{2(k+1)} - 1$ is divisible by 8. ✓

Conclusion. By mathematical induction, $3^{2n} - 1$ is divisible by 8 for all $n \in \mathbb{N}$. ■

1.7 Binomial Theorem

Definition 1.7.1 (Binomial Coefficient). For non-negative integers n and r with $0 \leq r \leq n$:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

This counts the number of ways to choose r objects from n distinct objects.

The Binomial Theorem (Positive Integer n)

$$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$$

Expanding:

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \cdots + \binom{n}{n-1} a b^{n-1} + b^n.$$

General term:

$$T_{r+1} = \binom{n}{r} a^{n-r} b^r$$

is the term containing b^r (or equivalently, the $(r + 1)$ -th term).

Proof of the Binomial Theorem by Induction

Claim: $(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$ for all $n \in \mathbb{N}_0$.

Basis step ($n = 0$): $(a + b)^0 = 1$ and $\binom{0}{0} a^0 b^0 = 1$. ✓

Basis step ($n = 1$): $(a + b)^1 = a + b$ and $\binom{1}{0} a^1 b^0 + \binom{1}{1} a^0 b^1 = a + b$. ✓

Inductive hypothesis: Assume for some $k \geq 1$:

$$(a + b)^k = \sum_{r=0}^k \binom{k}{r} a^{k-r} b^r. \quad (\text{IH})$$

Inductive step:

$$\begin{aligned}
 (a+b)^{k+1} &= (a+b)(a+b)^k \\
 &= (a+b) \sum_{r=0}^k \binom{k}{r} a^{k-r} b^r \\
 &= \sum_{r=0}^k \binom{k}{r} a^{k+1-r} b^r + \sum_{r=0}^k \binom{k}{r} a^{k-r} b^{r+1}.
 \end{aligned}$$

Shift the index in the second sum (let $s = r + 1$, so $r = s - 1$):

$$\begin{aligned}
 &= \sum_{r=0}^k \binom{k}{r} a^{k+1-r} b^r + \sum_{s=1}^{k+1} \binom{k}{s-1} a^{k+1-s} b^s \\
 &= a^{k+1} + \sum_{r=1}^k \binom{k}{r} a^{k+1-r} b^r + \sum_{s=1}^k \binom{k}{s-1} a^{k+1-s} b^s + b^{k+1} \\
 &= a^{k+1} + \sum_{r=1}^k \left[\binom{k}{r} + \binom{k}{r-1} \right] a^{k+1-r} b^r + b^{k+1}.
 \end{aligned}$$

Using Pascal's identity $\binom{k}{r} + \binom{k}{r-1} = \binom{k+1}{r}$:

$$(a+b)^{k+1} = \binom{k+1}{0} a^{k+1} + \sum_{r=1}^k \binom{k+1}{r} a^{k+1-r} b^r + \binom{k+1}{k+1} b^{k+1} = \sum_{r=0}^{k+1} \binom{k+1}{r} a^{k+1-r} b^r. \quad \checkmark$$

Conclusion. By induction, the binomial theorem holds for all $n \in \mathbb{N}_0$. ■

Pascal's Triangle Connection

- Row n of Pascal's triangle gives the coefficients of $(a+b)^n$.
- The identity $\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$ is why adjacent entries in Pascal's triangle sum to the entry below.
- For small n (up to $n = 5$ or 6), writing out Pascal's triangle is faster than computing $\binom{n}{r}$ individually.

Extension: Binomial Expansion for Rational n

For $n \in \mathbb{Q}$ (not necessarily a positive integer) and $|x| < 1$:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \frac{n(n-1)(n-2)}{3!} x^3 + \cdots = \sum_{r=0}^{\infty} \binom{n}{r} x^r$$

where the generalised binomial coefficient is

$$\binom{n}{r} = \frac{n(n-1)(n-2) \cdots (n-r+1)}{r!}.$$

Validity: The expansion converges when $|x| < 1$. When n is a positive integer, it reduces to the finite expansion (the series terminates after $r = n$).

Binomial Coefficient Properties

$$\binom{n}{r} = \binom{n}{n-r} \quad (\text{Symmetry})$$

$$\sum_{r=0}^n \binom{n}{r} = 2^n \quad (\text{Sum of row})$$

$$\sum_{r=0}^n (-1)^r \binom{n}{r} = 0 \quad (\text{Alternating sum})$$

Proof of Sum of Row = 2^n

Set $a = 1$, $b = 1$ in the binomial theorem:

$$(1 + 1)^n = 2^n = \sum_{r=0}^n \binom{n}{r} 1^{n-r} 1^r = \sum_{r=0}^n \binom{n}{r}.$$

Similarly, for the alternating sum, set $a = 1$, $b = -1$:

$$(1 - 1)^n = 0 = \sum_{r=0}^n \binom{n}{r} 1^{n-r} (-1)^r = \sum_{r=0}^n (-1)^r \binom{n}{r}.$$

Worked Example: Finding a Specific Term

Question. Find the coefficient of x^3 in the expansion of $(2 - 3x)^7$.

Solution. The general term is

$$T_{r+1} = \binom{7}{r} (2)^{7-r} (-3x)^r = \binom{7}{r} 2^{7-r} (-3)^r x^r.$$

We need $r = 3$:

$$\text{Coefficient} = \binom{7}{3} 2^4 (-3)^3 = 35 \times 16 \times (-27) = -15120.$$

Worked Example: Infinite Binomial Expansion

Question. Find the coefficient of x^3 in the expansion of $\frac{1}{(1 - 2x)^3}$ up to and including the term in x^3 .

Solution. Write $\frac{1}{(1 - 2x)^3} = (1 - 2x)^{-3}$. Using the generalised binomial theorem with $n = -3$ and x replaced by $-2x$ (valid when $|-2x| < 1$, i.e. $|x| < 0.5$):

$$\begin{aligned} (1 + y)^{-3} &= 1 + (-3)y + \frac{(-3)(-4)}{2!} y^2 + \frac{(-3)(-4)(-5)}{3!} y^3 + \dots \\ &= 1 - 3y + 6y^2 - 10y^3 + \dots \end{aligned}$$

Substituting $y = -2x$:

$$\begin{aligned} (1 - 2x)^{-3} &= 1 - 3(-2x) + 6(-2x)^2 - 10(-2x)^3 + \dots \\ &= 1 + 6x + 24x^2 + 80x^3 + \dots \end{aligned}$$

The coefficient of x^3 is **80**.

Worked Example: Binomial Approximation

Question. Use the binomial expansion to approximate $\sqrt{1.02}$ to 5 decimal places.

Solution. $\sqrt{1.02} = (1.02)^{1/2} = (1 + 0.02)^{1/2}$.

$$\begin{aligned}(1+x)^{1/2} &= 1 + \frac{1}{2}x + \frac{\frac{1}{2} \cdot (-\frac{1}{2})}{2!}x^2 + \frac{\frac{1}{2} \cdot (-\frac{1}{2}) \cdot (-\frac{3}{2})}{3!}x^3 \\ &= 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \dots\end{aligned}$$

With $x = 0.02$:

$$\begin{aligned}(1.02)^{1/2} &\approx 1 + 0.01 - \frac{0.0004}{8} + \frac{0.000008}{16} \\ &= 1 + 0.01 - 0.00005 + 0.0000005 \\ &= 1.0099505 \\ &\approx 1.00995.\end{aligned}$$

1.8 Counting

Definition 1.8.1 (Factorial).

$$n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1, \quad n \in \mathbb{N}.$$

By convention, $0! = 1$.

Permutations

The number of ways to arrange r objects chosen from n distinct objects, **where order matters**:

$${}^n P_r = \frac{n!}{(n-r)!}$$

Equivalently written as $P(n, r)$.

Combinations

The number of ways to choose r objects from n distinct objects, **where order does not matter**:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Equivalently written as ${}^n C_r$ or $C(n, r)$.

Derivation of Permutation and Combination Formulas

Permutations. We have n distinct objects and r positions to fill.

- Position 1: n choices.
- Position 2: $n - 1$ remaining choices.
- Position 3: $n - 2$ remaining choices.
- ...
- Position r : $n - (r - 1) = n - r + 1$ remaining choices.

By the multiplication principle:

$${}^nP_r = n(n-1)(n-2)\cdots(n-r+1) = \frac{n!}{(n-r)!}.$$

Combinations. Each combination of r objects can be arranged in $r!$ different orders (permutations). Therefore:

$${}^nP_r = \binom{n}{r} \times r!$$

Solving:

$$\binom{n}{r} = \frac{{}^nP_r}{r!} = \frac{n!}{r!(n-r)!}.$$

Circular Arrangements

The number of distinct ways to arrange n distinct objects in a circle is

$$(n-1)!$$

Reason: Fix one person's position to break the rotational symmetry. The remaining $(n-1)$ people can be arranged in $(n-1)!$ ways.

With reflection equivalence (e.g. a necklace that can be flipped):

$$\frac{(n-1)!}{2} \quad \text{for } n \geq 3.$$

Arrangements with Restrictions

- **Items must be together:** Treat the group as a single “block.” If k items must sit together, treat them as one unit, giving $(n-k+1)!$ arrangements of the blocks, then multiply by $k!$ for internal arrangements of the block.

$$\text{Total} = (n-k+1)! \times k!$$

- **Items must NOT be together:** Use the complement rule.

$$(\text{Items not together}) = \text{Total arrangements} - (\text{Items together})$$

For two specific people not sitting together in a line of n : $n! - (n-1)! \times 2!$.

- **Items in specific positions:** Place the restricted items first, then arrange the rest.

IB Exam Strategies for Counting Problems

- **Identify** whether order matters (permutation) or not (combination).
- **Draw** a simple diagram or list small cases to verify your reasoning.
- **Look for** “at least” or “at most” phrasing — these often require summing cases or using complements.
- **Check** your answer makes sense: ${}^nC_r \leq {}^nP_r \leq n^r$.
- **Common pitfall:** confusing permutation and combination. Ask yourself: “if I swap two elements, is it a different arrangement?”

- For complex multi-step problems, use the **multiplication principle**: if step 1 has a options and step 2 has b options, the total is $a \times b$.

Worked Example: Committee Selection

Question. A committee of 5 people is to be selected from 7 men and 6 women. How many committees can be formed if the committee must contain at least 2 women?

Solution. Count by cases (number of women on the committee):

$$2 \text{ women, 3 men: } \binom{6}{2} \times \binom{7}{3} = 15 \times 35 = 525$$

$$3 \text{ women, 2 men: } \binom{6}{3} \times \binom{7}{2} = 20 \times 21 = 420$$

$$4 \text{ women, 1 man: } \binom{6}{4} \times \binom{7}{1} = 15 \times 7 = 105$$

$$5 \text{ women, 0 men: } \binom{6}{5} \times \binom{7}{0} = 6 \times 1 = 6$$

$$\text{Total} = 525 + 420 + 105 + 6 = 1056.$$

Alternative (complement method):

$$\text{Total: } \binom{13}{5} = 1287$$

$$0 \text{ women: } \binom{7}{5} = 21$$

$$1 \text{ woman: } \binom{6}{1} \binom{7}{4} = 6 \times 35 = 210$$

$$\text{Answer} = 1287 - 21 - 210 = 1056. \quad \checkmark$$

Worked Example: Circular Arrangement with Restrictions

Question. In how many ways can 5 boys and 3 girls sit around a circular table if two particular girls, Anna and Bella, must not sit next to each other?

Solution. Total circular arrangements of 8 people: $(8 - 1)! = 7! = 5040$.

Arrangements where Anna and Bella ARE together: Treat them as a block. We have 7 units (the block + 6 others). Circular arrangements: $(7 - 1)! = 6! = 720$. The block has $2! = 2$ internal orderings. So: $720 \times 2 = 1440$.

Answer:

$$5040 - 1440 = 3600.$$

Worked Example: Seating with Restrictions

Question. How many ways can the letters of the word ARRANGE be arranged?

Solution. The word ARRANGE has 7 letters with repetitions: A appears twice, R appears twice.

$$\text{Arrangements} = \frac{7!}{2! \times 2!} = \frac{5040}{4} = 1260.$$

1.9 Partial Fractions

Partial fractions decompose a rational function $\frac{P(x)}{Q(x)}$ (where $\deg P < \deg Q$) into a sum of simpler fractions. This technique is essential for integration.

Partial Fractions: Distinct Linear Factors

For

$$\frac{P(x)}{(x-a)(x-b)(x-c)},$$

write

$$\frac{P(x)}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}.$$

Partial Fractions: Repeated Linear Factors

For

$$\frac{P(x)}{(x-a)^2(x-b)},$$

write

$$\frac{P(x)}{(x-a)^2(x-b)} = \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}.$$

Important: Include *both* $\frac{A}{x-a}$ and $\frac{B}{(x-a)^2}$ — a common mistake is to omit the first term.

Partial Fractions: Irreducible Quadratic Factor

If $x^2 + bx + c$ has no real roots (discriminant < 0), it is irreducible. For

$$\frac{P(x)}{(x-a)(x^2+bx+c)},$$

write

$$\frac{P(x)}{(x-a)(x^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}.$$

Note: The numerator over the quadratic is a *linear* expression $Bx + C$, not just a constant.

Method for Finding Coefficients

1. **Multiply through** by the denominator to clear fractions.
2. **Substitute convenient values** of x (the roots of the denominator) to find some coefficients immediately.
3. **Equate coefficients** of like powers of x to set up simultaneous equations for the remaining coefficients.
4. **Check** your answer by recombining the partial fractions.

Connection to Integration

Partial fractions are primarily used in IB Math AA HL to integrate rational functions. For example:

$$\int \frac{1}{(x-1)(x+2)} dx = \int \frac{1/3}{x-1} dx + \int \frac{-1/3}{x+2} dx = \frac{1}{3} \ln|x-1| - \frac{1}{3} \ln|x+2| + C.$$

The **logarithm integration rule** $\int \frac{1}{ax+b} dx = \frac{1}{a} \ln |ax+b| + C$ applies directly to each partial fraction with a linear denominator.

Worked Example: Distinct Linear Factors

Question. Express $\frac{5x+7}{(x+1)(x-2)}$ in partial fractions.

Solution. Write

$$\frac{5x+7}{(x+1)(x-2)} = \frac{A}{x+1} + \frac{B}{x-2}.$$

Multiplying through by $(x+1)(x-2)$:

$$5x+7 = A(x-2) + B(x+1).$$

Substitute $x = -1$: $5(-1) + 7 = A(-3) + B(0)$, so $2 = -3A$, giving $A = -\frac{2}{3}$.

Substitute $x = 2$: $5(2) + 7 = A(0) + B(3)$, so $17 = 3B$, giving $B = \frac{17}{3}$.

$$\frac{5x+7}{(x+1)(x-2)} = \frac{-2/3}{x+1} + \frac{17/3}{x-2}.$$

Worked Example: Repeated Factors

Question. Express $\frac{2x^2+3x+1}{(x-1)^2(x+1)}$ in partial fractions.

Solution. Write

$$\frac{2x^2+3x+1}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1}.$$

Multiplying through by $(x-1)^2(x+1)$:

$$2x^2+3x+1 = A(x-1)(x+1) + B(x+1) + C(x-1)^2.$$

Substitute $x = 1$: $2 + 3 + 1 = A(0) + B(2) + C(0)$, so $B = 3$.

Substitute $x = -1$: $2 - 3 + 1 = A(0) + B(0) + C(4)$, so $0 = 4C$, giving $C = 0$.

Equate coefficients of x^2 : $2 = A + C = A + 0$, so $A = 2$.

$$\frac{2x^2+3x+1}{(x-1)^2(x+1)} = \frac{2}{x-1} + \frac{3}{(x-1)^2}.$$

Worked Example: Improper Fraction — Long Division First

Question. Express $\frac{x^3+1}{x^2-1}$ in partial fractions.

Solution. Since the degree of the numerator (3) exceeds the degree of the denominator (2), first perform polynomial long division:

$$\frac{x^3+1}{x^2-1} = x + \frac{x+1}{x^2-1} = x + \frac{x+1}{(x-1)(x+1)} = x + \frac{1}{x-1}.$$

Note that $\frac{x+1}{(x-1)(x+1)} = \frac{1}{x-1}$ (after cancellation, valid for $x \neq -1$).

1.10 Complex Numbers

1.10.1 Definitions and Basic Operations

Definition 1.10.1 (Imaginary Unit). The **imaginary unit** i is defined by $i^2 = -1$. A **complex number** is a number of the form

$$z = a + bi,$$

where $a, b \in \mathbb{R}$. Here $a = \operatorname{Re}(z)$ is the real part and $b = \operatorname{Im}(z)$ is the imaginary part.

Definition 1.10.2 (Complex Conjugate). The **complex conjugate** of $z = a + bi$ is

$$\bar{z} = a - bi.$$

Definition 1.10.3 (Modulus). The **modulus** (absolute value) of $z = a + bi$ is

$$|z| = \sqrt{a^2 + b^2}.$$

Definition 1.10.4 (Argument). The **argument** of $z = a + bi$ is the angle $\theta = \arg(z)$ measured from the positive real axis to the line joining the origin to z , where

$$\theta = \arctan\left(\frac{b}{a}\right).$$

Quadrant awareness is essential. Use the signs of a and b to determine the correct quadrant:

- $a > 0, b > 0$: first quadrant, $\theta \in (0, \pi/2)$.
- $a < 0, b > 0$: second quadrant, $\theta \in (\pi/2, \pi)$.
- $a < 0, b < 0$: third quadrant, $\theta \in (-\pi, -\pi/2)$.
- $a > 0, b < 0$: fourth quadrant, $\theta \in (-\pi/2, 0)$.

The **principal argument** $\operatorname{Arg}(z)$ satisfies $-\pi < \operatorname{Arg}(z) \leq \pi$.

Complex Conjugate Properties

$$\begin{aligned} z\bar{z} &= |z|^2 = a^2 + b^2 \\ \overline{z+w} &= \bar{z} + \bar{w} \\ \overline{zw} &= \bar{z}\bar{w} \\ z + \bar{z} &= 2\operatorname{Re}(z) \\ z - \bar{z} &= 2i\operatorname{Im}(z) \end{aligned}$$

1.10.2 The Complex Plane (Argand Diagram)

Working with the Argand Diagram

- The horizontal axis represents $\operatorname{Re}(z)$ and the vertical axis represents $\operatorname{Im}(z)$.
- The modulus $|z|$ is the distance from the origin to the point representing z .
- The argument $\arg(z)$ is the angle from the positive real axis, measured anticlockwise (positive) or clockwise (negative).

- Geometric interpretation: $|z - w|$ is the distance between the points representing z and w on the Argand diagram.
- $|z - a| = r$ represents a circle with centre a and radius r .
- $\arg(z - a) = \theta$ represents a half-line from a at angle θ .

1.10.3 Polar and Exponential Forms

Polar Form

Any non-zero complex number $z = a + bi$ can be written as

$$z = r(\cos \theta + i \sin \theta) = r \operatorname{cis} \theta$$

where $r = |z| = \sqrt{a^2 + b^2}$ and $\theta = \arg(z)$.

Euler's Formula and Exponential Form

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Therefore the **Euler form** (exponential form) of z is

$$z = r e^{i\theta}$$

Connection: Euler's Formula via Taylor Series

Euler's formula can be understood by comparing the Taylor series expansions:

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \\ \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \end{aligned}$$

Substituting $x = i\theta$ into the series for e^x :

$$\begin{aligned} e^{i\theta} &= 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \cdots \\ &= 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \frac{i\theta^5}{5!} - \cdots \\ &= \underbrace{\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right)}_{=\cos \theta} + i \underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right)}_{=\sin \theta} \\ &= \cos \theta + i \sin \theta. \end{aligned}$$

This gives the celebrated identity $e^{i\theta} = \cos \theta + i \sin \theta$. Setting $\theta = \pi$ yields Euler's identity: $e^{i\pi} + 1 = 0$.

Key Identity: Euler's Identity

$$e^{i\pi} + 1 = 0$$

This connects five fundamental constants: e , i , π , 1 , and 0 .

1.10.4 De Moivre's Theorem

Theorem 1.10.1 (De Moivre's Theorem). *For any integer n and any complex number $z = r \operatorname{cis} \theta$:*

$$z^n = r^n \operatorname{cis}(n\theta) = r^n (\cos(n\theta) + i \sin(n\theta))$$

Equivalently, in Euler form: $(r e^{i\theta})^n = r^n e^{in\theta}$.

Proof of De Moivre's Theorem by Induction. Claim: For $z = r \operatorname{cis} \theta$ and all $n \in \mathbb{N}$, $z^n = r^n \operatorname{cis}(n\theta)$.

Basis step ($n = 1$): $z^1 = r \operatorname{cis} \theta = r^1 \operatorname{cis}(1 \cdot \theta)$. ✓

Inductive hypothesis: Assume $z^k = r^k \operatorname{cis}(k\theta)$ for some $k \geq 1$.

Inductive step:

$$\begin{aligned} z^{k+1} &= z^k \cdot z \\ &= r^k \operatorname{cis}(k\theta) \cdot r \operatorname{cis} \theta \quad (\text{by IH}) \\ &= r^{k+1} \operatorname{cis}(k\theta) \operatorname{cis}(\theta). \end{aligned}$$

Using the compound angle formulas:

$$\begin{aligned} \operatorname{cis}(k\theta) \operatorname{cis}(\theta) &= (\cos(k\theta) + i \sin(k\theta)) (\cos \theta + i \sin \theta) \\ &= \cos(k\theta) \cos \theta - \sin(k\theta) \sin \theta \\ &\quad + i(\sin(k\theta) \cos \theta + \cos(k\theta) \sin \theta) \\ &= \cos((k+1)\theta) + i \sin((k+1)\theta) \\ &= \operatorname{cis}((k+1)\theta). \end{aligned}$$

Therefore $z^{k+1} = r^{k+1} \operatorname{cis}((k+1)\theta)$. ✓

Conclusion. By mathematical induction, De Moivre's theorem holds for all $n \in \mathbb{N}$. The extension to negative integers follows since $z^{-n} = \frac{1}{z^n}$. □

1.10.5 Roots of Complex Numbers

Theorem 1.10.2 (Roots of a Complex Number). *The equation $z^n = w$, where $w = R \operatorname{cis} \alpha \neq 0$ and $n \in \mathbb{N}$, has exactly n distinct solutions:*

$$z_k = R^{1/n} \operatorname{cis}\left(\frac{\alpha + 2k\pi}{n}\right), \quad k = 0, 1, 2, \dots, n-1$$

*These n roots are **equally spaced** around a circle of radius $R^{1/n}$ centred at the origin, with consecutive roots separated by an angle of $\frac{2\pi}{n}$.*

Derivation of the Roots Formula

Let $z = r \operatorname{cis} \theta$ and $w = R \operatorname{cis} \alpha$. We want $z^n = w$.

By De Moivre's theorem: $r^n \operatorname{cis}(n\theta) = R \operatorname{cis} \alpha$.

Equating moduli: $r^n = R$, so $r = R^{1/n}$.

Equating arguments: $n\theta = \alpha + 2k\pi$ for $k \in \mathbb{Z}$ (since the argument is periodic with period 2π).

Therefore

$$\theta = \frac{\alpha + 2k\pi}{n}.$$

The distinct values are obtained for $k = 0, 1, 2, \dots, n-1$. For $k = n$, we get

$$\theta = \frac{\alpha + 2n\pi}{n} = \frac{\alpha}{n} + 2\pi,$$

which gives the same complex number as $k = 0$. So there are exactly n distinct roots.

Sum of Roots of Unity

The n th roots of unity (solutions to $z^n = 1$) are $\operatorname{cis}\left(\frac{2k\pi}{n}\right)$ for $k = 0, 1, \dots, n-1$. Their sum is:

$$\sum_{k=0}^{n-1} \operatorname{cis}\left(\frac{2k\pi}{n}\right) = 0.$$

This is because they form a geometric series with ratio $\operatorname{cis}(2\pi/n) \neq 1$, and their sum equals $\frac{1 - r^n}{1 - r} = \frac{1 - 1}{1 - r} = 0$.

Worked Example: Converting to Polar Form

Question. Express $z = -1 + \sqrt{3}i$ in polar and exponential form.

Solution.

$$r = |z| = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{1 + 3} = 2.$$

$$\theta = \arctan\left(\frac{\sqrt{3}}{-1}\right).$$

Since z is in the second quadrant ($a < 0$, $b > 0$): $\theta = \pi - \arctan(\sqrt{3}) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$.

$$z = 2 \operatorname{cis}\left(\frac{2\pi}{3}\right) = 2e^{2\pi i/3}.$$

Worked Example: De Moivre for Large Powers

Question. Find $(1 + i)^8$.

Solution. First convert $1 + i$ to polar form:

$$r = \sqrt{1^2 + 1^2} = \sqrt{2}, \quad \theta = \arctan(1) = \frac{\pi}{4}.$$

$$(1 + i)^8 = (\sqrt{2} \operatorname{cis}(\pi/4))^8 = (\sqrt{2})^8 \operatorname{cis}(8 \cdot \pi/4) = 16 \operatorname{cis}(2\pi) = 16(\cos 2\pi + i \sin 2\pi) = 16.$$

Worked Example: Finding Cube Roots

Question. Find all cube roots of $z = -8i$.

Solution. Express $w = -8i$ in polar form:

$$R = |-8i| = 8, \quad \alpha = \arg(-8i) = -\frac{\pi}{2}.$$

The cube roots are

$$z_k = 8^{1/3} \operatorname{cis}\left(\frac{-\pi/2 + 2k\pi}{3}\right) = 2 \operatorname{cis}\left(\frac{-\pi + 4k\pi}{6}\right),$$

for $k = 0, 1, 2$:

$$k = 0: \quad z_0 = 2 \operatorname{cis}\left(-\frac{\pi}{6}\right) = 2 \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = \sqrt{3} - i.$$

$$k = 1: \quad z_1 = 2 \operatorname{cis}\left(\frac{3\pi}{6}\right) = 2 \operatorname{cis}\left(\frac{\pi}{2}\right) = 2i.$$

$$k = 2: \quad z_2 = 2 \operatorname{cis}\left(\frac{7\pi}{6}\right) = 2 \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = -\sqrt{3} - i.$$

These three roots are equally spaced at 120° intervals on a circle of radius 2.

Worked Example: Geometric Locus on the Argand Diagram

Question. Sketch the locus of z satisfying $|z - 2 - i| = 3$ and find the maximum value of $\arg(z)$ for points on this locus.

Solution. The equation $|z - (2 + i)| = 3$ represents a circle with centre $2 + i$ (the point $(2, 1)$ on the Argand diagram) and radius 3.

The maximum argument occurs at the point on the circle where the tangent from the origin touches the circle in the upper half-plane. The argument of the centre is $\alpha = \arctan(1/2)$. The angle between the line from the origin to the centre and the tangent is $\beta = \arcsin(3/\sqrt{5})$, where $\sqrt{5}$ is the distance from the origin to the centre.

$$\max \arg(z) = \alpha + \beta = \arctan\left(\frac{1}{2}\right) + \arcsin\left(\frac{3}{\sqrt{5}}\right).$$

Since $\arcsin(3/\sqrt{5})$ requires $3/\sqrt{5} \leq 1$, i.e. $\sqrt{5} \geq 3$, but $\sqrt{5} \approx 2.236 < 3$. This means the origin lies *inside* the circle. In this case, the maximum argument is found from the tangent from the origin to the circle.

Actually, since the origin is at distance $\sqrt{2^2 + 1^2} = \sqrt{5} \approx 2.236$ from the centre, and the radius is $3 > \sqrt{5}$, the origin lies *inside* the circle. The maximum argument is achieved at the point where a line from the origin is tangent to the circle. Using the relationship:

$$\max \arg(z) = \arctan\left(\frac{1}{2}\right) + \arccos\left(\frac{\sqrt{5}}{3}\right).$$

IB Exam Tips for Complex Numbers

- **Polar form is essential** for multiplication, division, and powers. Always convert before applying De Moivre.
- **Cartesian form** is best for addition and subtraction.

- When asked for roots, always give *all* n roots and verify they are equally spaced.
- **Show working** for the argument — do not just write the calculator value. Show which quadrant z is in.
- For locus questions, clearly label the centre and radius, and indicate the relevant region.
- When simplifying $\frac{z}{w}$, multiply numerator and denominator by $\bar{w}$ to rationalise: $\frac{z}{w} = \frac{z\bar{w}}{w\bar{w}} = \frac{z\bar{w}}{|w|^2}$.
- **Common error:** confusing the argument of a product: $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$ (not the product of the arguments).

Chapter 2

Functions

Functions form the backbone of the IB Mathematics AA HL syllabus. Nearly every topic—from calculus to probability—relies on a solid understanding of functions. This chapter covers every function type examined at HL, from linear fundamentals through to solving inequalities with absolute values and rational expressions. Every formula is boxed, every key derivation is shown, and every subtopic includes worked examples in the IB exam style.

2.1 Function Fundamentals

Definition 2.1.1 (Function). A function f from a set A to a set B , written $f: A \rightarrow B$, is a rule that assigns to each element $x \in A$ **exactly one** element $f(x) \in B$.

- **Domain:** the set of all valid inputs, A .
- **Codomain:** the set B into which outputs are mapped.
- **Range (image):** the subset of B consisting of all actual outputs:

$$\text{Range}(f) = \{f(x) \mid x \in A\} \subseteq B$$

Definition 2.1.2 (Function Notation). The value of f at x is written $f(x)$. In applied contexts, common notations include:

- $v(t)$ — velocity as a function of time
- $s(t)$ — displacement as a function of time
- $P(n)$ — population as a function of n
- **Composite notation:** $(f \circ g)(x) = f(g(x))$

Vertical Line Test

A relation represented graphically is a function if and only if **every vertical line** intersects the graph **at most once**.

Definition 2.1.3 (One-to-One (Injective) Function). A function f is one-to-one (injective) if

$$f(x_1) = f(x_2) \implies x_1 = x_2$$

*Equivalently: distinct inputs produce distinct outputs. Graphically, **every horizontal line** intersects the graph at most once (horizontal line test).*

Definition 2.1.4 (Onto (Surjective) Function). A function $f: A \rightarrow B$ is onto (surjective) if every element of B is mapped to by at least one element of A :

$$\forall y \in B, \exists x \in A \text{ such that } f(x) = y$$

This means $\text{Range}(f) = B$.

Definition 2.1.5 (Bijective Function). A function that is **both injective and surjective** is called bijective (a one-to-one correspondence).

Example 1: Determining if a Relation is a Function

Let $f = \{(1, 3), (2, 5), (3, 5), (4, 7)\}$ and $g = \{(1, 2), (2, 3), (1, 4)\}$. Determine whether f and g are functions.

Solution:

f : each x -value $\{1, 2, 3, 4\}$ appears exactly once. Therefore f is a function. Since $f(2) = f(3) = 5$ while $2 \neq 3$, f is *not* one-to-one.

g : the input $x = 1$ is mapped to both 2 and 4. Therefore g is *not* a function.

Example 2: Finding Domain and Range

Find the natural domain and range of $f(x) = \sqrt{4 - x^2}$.

Solution:

For the square root to be defined (over $\mathbb{R}$), we require

$$4 - x^2 \geq 0 \implies x^2 \leq 4 \implies -2 \leq x \leq 2$$

Therefore, $\text{Domain}(f) = [-2, 2]$.

Since $x^2 \geq 0$ for all x , we have $4 - x^2 \leq 4$, so $f(x) = \sqrt{4 - x^2} \leq 2$. The minimum value is $f(2) = f(-2) = 0$. Therefore, $\text{Range}(f) = [0, 2]$.

2.2 Inverse and Composite Functions

2.2.1 Inverse Functions

Definition 2.2.1 (Inverse Function). The inverse of a function f , denoted f^{-1} , is the function that “undoes” f :

$$f^{-1}(f(x)) = x \quad \text{for all } x \in \text{Domain}(f)$$

$$f(f^{-1}(y)) = y \quad \text{for all } y \in \text{Range}(f)$$

Theorem 2.2.1 (Existence of Inverse). A function f has an inverse function f^{-1} if and only if f is **one-to-one** (injective). If f is not injective, we must **restrict its domain** to a subset on which it is injective.

Algebraic Method for Finding f^{-1}

Given $y = f(x)$:

1. Replace $f(x)$ with y .
2. **Swap** x and y .
3. Solve for y in terms of x .
4. Replace y with $f^{-1}(x)$.

5. State the domain of f^{-1} (equals the range of f).

Graphical Interpretation

The graph of $y = f^{-1}(x)$ is the **reflection** of the graph of $y = f(x)$ in the line $y = x$.

Key relationships:

- $\text{Domain}(f^{-1}) = \text{Range}(f)$
- $\text{Range}(f^{-1}) = \text{Domain}(f)$
- If (a, b) lies on f , then (b, a) lies on f^{-1}

Example 1: Finding an Inverse Function

Let $f(x) = \frac{2x+3}{x-1}$ for $x > 1$. Find f^{-1} and state its domain.

Solution:

Set $y = \frac{2x+3}{x-1}$.

Swap x and y :

$$x = \frac{2y+3}{y-1}$$

Solve for y :

$$x(y-1) = 2y+3$$

$$xy - x = 2y + 3$$

$$xy - 2y = x + 3$$

$$y(x-2) = x+3$$

$$y = \frac{x+3}{x-2}$$

Therefore $f^{-1}(x) = \frac{x+3}{x-2}$ with domain $x > 2$ (since the range of f is $y > 2$, which can be verified by noting $f(x) = 2 + \frac{5}{x-1}$).

Example 2: Restricting Domain for Invertibility

The function $f(x) = x^2 - 4x + 3$ does not have an inverse on its natural domain. Restrict the domain so that f^{-1} exists.

Solution:

Complete the square:

$$f(x) = (x-2)^2 - 1$$

The vertex is at $x = 2$. For f to be one-to-one, restrict to one side of the vertex:

- Option A: $\text{Domain}(f) = [2, \infty)$, then $\text{Range}(f) = [-1, \infty)$
- Option B: $\text{Domain}(f) = (-\infty, 2]$, then $\text{Range}(f) = [-1, \infty)$

Using Option A, set $y = (x - 2)^2 - 1$, swap and solve:

$$\begin{aligned}x &= (y - 2)^2 - 1 \\x + 1 &= (y - 2)^2 \\\sqrt{x + 1} &= y - 2 \quad (\text{since } y \geq 2) \\y &= 2 + \sqrt{x + 1}\end{aligned}$$

So $f^{-1}(x) = 2 + \sqrt{x + 1}$ with domain $[-1, \infty)$.

2.2.2 Composite Functions

Definition 2.2.2 (Composite Function). *Given functions f and g , the composite $f \circ g$ is defined by*

$$(f \circ g)(x) = f(g(x))$$

This means: first apply g , then apply f to the result.

Domain of a Composite Function

The domain of $(f \circ g)(x)$ is the set of all x in the domain of g such that $g(x)$ is in the domain of f :

$$\text{Domain}(f \circ g) = \{x \in \text{Domain}(g) \mid g(x) \in \text{Domain}(f)\}$$

IB Tip: “Hence” Questions

When a question says “hence find f^{-1} ” after asking you to find $(f \circ g)(x) = x$, it is telling you that $g = f^{-1}$. IB examiners frequently use this structure:

1. Part (a): find $(f \circ g)(x)$ and simplify to x .
2. Part (b): “Hence write down $f^{-1}(x)$.”

The answer to part (b) is simply $g(x)$.

Common IB Error

Composite functions are **not** commutative in general: $(f \circ g)(x) \neq (g \circ f)(x)$. Always check which function is applied first. Reading right to left: $f(g(x))$ means g first, then f .

Example 3: Composite Functions and Domain

Let $f(x) = \sqrt{x - 1}$ and $g(x) = x^2 + 3$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$, stating the domain of each.

Solution:

Part 1: $(f \circ g)(x) = f(g(x)) = f(x^2 + 3) = \sqrt{x^2 + 3 - 1} = \sqrt{x^2 + 2}$.

Domain: we need $g(x) \geq 1$ (the domain condition of f), i.e. $x^2 + 3 \geq 1$, i.e. $x^2 \geq -2$, which is always true. Since $\text{Domain}(g) = \mathbb{R}$, we have $\text{Domain}(f \circ g) = \mathbb{R}$.

Part 2: $(g \circ f)(x) = g(f(x)) = g(\sqrt{x - 1}) = (\sqrt{x - 1})^2 + 3 = x - 1 + 3 = x + 2$.

Domain: we need $x - 1 \geq 0$ (domain of f), so $\text{Domain}(g \circ f) = [1, \infty)$.

Note: The simplified expressions $\sqrt{x^2 + 2}$ and $x + 2$ are different functions, confirming $f \circ g \neq g \circ f$.

Example 4: “Hence” Inverse Question

Let $f(x) = 2x + 1$ and $g(x) = \frac{x-1}{2}$.

(a) Find $(f \circ g)(x)$.

(b) Hence write down $f^{-1}(x)$.

Solution:

(a)

$$(f \circ g)(x) = f\left(\frac{x-1}{2}\right) = 2\left(\frac{x-1}{2}\right) + 1 = (x-1) + 1 = x$$

(b) Since $(f \circ g)(x) = x$, we have $g = f^{-1}$, so

$$f^{-1}(x) = \frac{x-1}{2}$$

2.3 Linear Functions

Definition 2.3.1 (Linear Function). A linear function has the form

$$y = mx + c$$

where m is the **gradient** (slope) and c is the **y-intercept**.

Key Linear Formulas

Gradient (two-point form):

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Midpoint:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Distance:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Parallel lines: $m_1 = m_2$

Perpendicular lines: $m_1 \times m_2 = -1$

Derivation of the Perpendicular Gradient Condition

If line L_1 has gradient m and line L_2 is perpendicular to L_1 , then the angle between them is 90° . Using the tangent of the angle between two lines:

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

Setting $\theta = 90^\circ$, we have $\tan 90^\circ$ undefined, so the denominator must be zero:

$$1 + m_1 m_2 = 0 \implies m_1 m_2 = -1$$

Therefore, if $m_2 = -\frac{1}{m_1}$, the lines are perpendicular.

IB Tip: Coordinate Geometry

In Paper 2, coordinate geometry questions frequently combine the distance formula with gradients. Common question structures:

- “Find the equation of the perpendicular bisector of AB .” — find midpoint of AB , find gradient of AB , use negative reciprocal, substitute midpoint.
- “Show that ABC is a right-angled triangle.” — show two gradients multiply to -1 .

Example 1: Equation of a Perpendicular Bisector

The points $A(1, 3)$ and $B(7, -1)$ are given. Find the equation of the perpendicular bisector of AB .

Solution:

Step 1: Midpoint of AB :

$$M = \left(\frac{1+7}{2}, \frac{3+(-1)}{2} \right) = (4, 1)$$

Step 2: Gradient of AB :

$$m_{AB} = \frac{-1-3}{7-1} = \frac{-4}{6} = -\frac{2}{3}$$

Step 3: Gradient of perpendicular bisector:

$$m_{\perp} = -\frac{1}{m_{AB}} = -\frac{1}{-2/3} = \frac{3}{2}$$

Step 4: Equation through $M(4, 1)$:

$$y - 1 = \frac{3}{2}(x - 4) \implies y = \frac{3}{2}x - 5$$

Example 2: Right-Angled Triangle

Show that the triangle with vertices $A(0, 0)$, $B(4, 2)$, $C(1, 7)$ is right-angled.

Solution:

Compute the gradients:

$$m_{AB} = \frac{2-0}{4-0} = \frac{1}{2}$$

$$m_{AC} = \frac{7-0}{1-0} = 7$$

$$m_{BC} = \frac{7-2}{1-4} = \frac{5}{-3} = -\frac{5}{3}$$

Check products of pairs:

$$m_{AB} \times m_{AC} = \frac{1}{2} \times 7 = \frac{7}{2} \neq -1$$

$$m_{AB} \times m_{BC} = \frac{1}{2} \times \left(-\frac{5}{3}\right) = -\frac{5}{6} \neq -1$$

$$m_{AC} \times m_{BC} = 7 \times \left(-\frac{5}{3}\right) = -\frac{35}{3} \neq -1$$

None of the products equal -1 , so no two sides are perpendicular. Alternatively, use the Pythagorean theorem:

$$AB^2 = 4^2 + 2^2 = 20$$

$$AC^2 = 1^2 + 7^2 = 50$$

$$BC^2 = 3^2 + 5^2 = 34$$

Check: $20 + 34 = 54 \neq 50$. Since $AB^2 + BC^2 \neq AC^2$, the triangle is *not* right-angled at B . Checking $AB^2 + AC^2 = 70 \neq 34$ and $AC^2 + BC^2 = 84 \neq 20$, we confirm $\triangle ABC$ is **not** right-angled.

2.4 Quadratic Functions

Definition 2.4.1 (Quadratic Function). A quadratic function is a polynomial of degree 2. It can be expressed in **three forms**:

Three Forms of a Quadratic

1. Standard (general) form:

$$f(x) = ax^2 + bx + c, \quad a \neq 0$$

2. Factored (intercept) form:

$$f(x) = a(x - p)(x - q)$$

where p and q are the x -intercepts (roots).

3. Vertex form:

$$f(x) = a(x - h)^2 + k$$

where (h, k) is the vertex of the parabola.

2.4.1 Completing the Square

Completing the Square: Full Derivation

Start with $f(x) = ax^2 + bx + c$ where $a \neq 0$.

Step 1: Factor out a from the first two terms:

$$f(x) = a\left(x^2 + \frac{b}{a}x\right) + c$$

Step 2: Take half the coefficient of x and square it:

$$\left(\frac{b/a}{2}\right)^2 = \frac{b^2}{4a^2}$$

Step 3: Add and subtract this inside the brackets:

$$\begin{aligned}
 f(x) &= a \left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} - \frac{b^2}{4a^2} \right) + c \\
 &= a \left(x + \frac{b}{2a} \right)^2 - a \cdot \frac{b^2}{4a^2} + c \\
 &= a \left(x + \frac{b}{2a} \right)^2 - \frac{b^2}{4a} + c \\
 &= a \left(x + \frac{b}{2a} \right)^2 + \frac{4ac - b^2}{4a}
 \end{aligned}$$

Comparing with $f(x) = a(x - h)^2 + k$, we identify:

$$h = -\frac{b}{2a}, \quad k = \frac{4ac - b^2}{4a} = c - \frac{b^2}{4a}$$

2.4.2 The Discriminant

The Discriminant

For $f(x) = ax^2 + bx + c$, the *discriminant* is

$$\Delta = b^2 - 4ac$$

Theorem 2.4.1 (Discriminant Analysis). 1. $\Delta > 0$: Two distinct real roots. The parabola crosses the x -axis at two points.

2. $\Delta = 0$: One repeated real root (a double root). The parabola is tangent to the x -axis at the vertex.

3. $\Delta < 0$: No real roots. The parabola does not intersect the x -axis.

2.4.3 Quadratic Formula

Derivation of the Quadratic Formula

Start with the vertex form derived by completing the square:

$$a \left(x + \frac{b}{2a} \right)^2 = \frac{b^2 - 4ac}{4a}$$

Divide both sides by a :

$$\left(x + \frac{b}{2a} \right)^2 = \frac{b^2 - 4ac}{4a^2}$$

Take square roots:

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

Therefore:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Quadratic Formula

$$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2.4.4 Vieta's Formulas for Quadratics

Theorem 2.4.2 (Vieta's Formulas — Quadratic). If $ax^2 + bx + c = 0$ has roots α and β , then:

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

Proof of Vieta's Formulas

Starting from the factored form:

$$a(x - \alpha)(x - \beta) = 0$$

Expanding:

$$a(x^2 - (\alpha + \beta)x + \alpha\beta) = ax^2 - a(\alpha + \beta)x + a\alpha\beta = 0$$

Comparing with $ax^2 + bx + c = 0$:

$$-a(\alpha + \beta) = b \implies \alpha + \beta = -\frac{b}{a}$$

$$a\alpha\beta = c \implies \alpha\beta = \frac{c}{a}$$

IB Tip: Converting Between Forms

Standard to vertex: Complete the square (see derivation above).

Standard to factored: Factor or use the quadratic formula to find roots, then write $f(x) = a(x - \alpha)(x - \beta)$.

Factored to standard: Expand: $a(x - p)(x - q) = ax^2 - a(p + q)x + apq$.

Factored to vertex: Expand to standard, then complete the square. Or note that $h = \frac{p + q}{2}$ (midpoint of roots).

Vertex to standard: Expand: $a(x - h)^2 + k = ax^2 - 2ahx + (ah^2 + k)$.

Example 1: Discriminant and Nature of Roots

For $f(x) = 2x^2 - 6x + k$, find the values of k for which the equation $f(x) = 0$ has: **(a)** two distinct real roots, **(b)** a repeated root, **(c)** no real roots.

Solution:

$a = 2$, $b = -6$, $c = k$. The discriminant is:

$$\Delta = (-6)^2 - 4(2)(k) = 36 - 8k$$

(a) Two distinct roots: $\Delta > 0 \implies 36 - 8k > 0 \implies k < \frac{36}{8} = \frac{9}{2}$.

(b) Repeated root: $\Delta = 0 \implies 36 - 8k = 0 \implies k = \frac{9}{2}$.

(c) No real roots: $\Delta < 0 \implies 36 - 8k < 0 \implies k > \frac{9}{2}$.

Example 2: Completing the Square and Vertex

Express $f(x) = -3x^2 + 12x - 7$ in vertex form. Hence state the maximum value and the value of x at which it occurs.

Solution:

$$\begin{aligned} f(x) &= -3x^2 + 12x - 7 \\ &= -3(x^2 - 4x) - 7 \\ &= -3(x^2 - 4x + 4 - 4) - 7 \\ &= -3(x - 2)^2 + 12 - 7 \\ &= -3(x - 2)^2 + 5 \end{aligned}$$

Vertex form: $f(x) = -3(x - 2)^2 + 5$. Since $a = -3 < 0$, the parabola opens downward. The maximum value is $k = 5$, occurring at $x = h = 2$.

Example 3: Vieta's Formulas Application

The quadratic equation $2x^2 + 5x - 3 = 0$ has roots α and β . Without solving the equation, find: (a) $\alpha + \beta$, (b) $\alpha\beta$, (c) $\alpha^2 + \beta^2$, (d) $\frac{1}{\alpha} + \frac{1}{\beta}$.

Solution:

By Vieta's formulas: $\alpha + \beta = -\frac{5}{2}$ and $\alpha\beta = -\frac{3}{2}$.

(c)

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(-\frac{5}{2}\right)^2 - 2\left(-\frac{3}{2}\right) = \frac{25}{4} + 3 = \frac{37}{4}$$

(d)

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-5/2}{-3/2} = \frac{5}{3}$$

2.5 Polynomial Functions

Definition 2.5.1 (Polynomial Function). A polynomial function *has the form*

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where $n \in \mathbb{N}_0$ is the **degree**, $a_n \neq 0$ is the **leading coefficient**, and a_0 is the **constant term**.

2.5.1 Factor and Remainder Theorems

Theorem 2.5.1 (Factor Theorem). Let $P(x)$ be a polynomial. Then:

$$P(a) = 0 \iff (x - a) \text{ is a factor of } P(x)$$

Theorem 2.5.2 (Remainder Theorem). When a polynomial $P(x)$ is divided by $(x - a)$, the remainder is $P(a)$.

Proof of the Remainder Theorem. By the polynomial division algorithm, dividing $P(x)$ by $(x - a)$ gives:

$$P(x) = (x - a) \cdot Q(x) + R$$

where $Q(x)$ is the quotient and R is a constant (since $\deg(x - a) = 1$).

Substituting $x = a$:

$$P(a) = (a - a) \cdot Q(a) + R = 0 \cdot Q(a) + R = R$$

Therefore the remainder is $R = P(a)$. □

IB Tip: Using the Remainder Theorem

The remainder theorem is frequently tested in two ways:

1. “Find the remainder when $P(x)$ is divided by $(x - a)$ ” — simply evaluate $P(a)$.
2. “Find the value of k such that $(x - a)$ is a factor” — set $P(a) = 0$ and solve for the unknown.

2.5.2 Vieta's Formulas for Cubics and Quartics

Theorem 2.5.3 (Vieta's Formulas — Cubic). *If $ax^3 + bx^2 + cx + d = 0$ has roots α, β, γ , then:*

$$\begin{aligned}\alpha + \beta + \gamma &= -\frac{b}{a} \\ \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} \\ \alpha\beta\gamma &= -\frac{d}{a}\end{aligned}$$

Theorem 2.5.4 (Vieta's Formulas — Quartic). *If $ax^4 + bx^3 + cx^2 + dx + e = 0$ has roots $\alpha, \beta, \gamma, \delta$, then:*

$$\begin{aligned}\sum \alpha &= -\frac{b}{a} \\ \sum \alpha\beta &= \frac{c}{a} \\ \sum \alpha\beta\gamma &= -\frac{d}{a} \\ \alpha\beta\gamma\delta &= \frac{e}{a}\end{aligned}$$

where the sums are taken over all distinct combinations.

Proof of Vieta's Formulas for a Cubic

Start from the factored form:

$$a(x - \alpha)(x - \beta)(x - \gamma) = 0$$

Expanding step by step:

$$\begin{aligned}(x - \alpha)(x - \beta)(x - \gamma) &= (x^2 - (\alpha + \beta)x + \alpha\beta)(x - \gamma) \\ &= x^3 - \gamma x^2 - (\alpha + \beta)x^2 + (\alpha + \beta)\gamma x + \alpha\beta x - \alpha\beta\gamma \\ &= x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \alpha\gamma + \beta\gamma)x - \alpha\beta\gamma\end{aligned}$$

Multiplying by a and comparing with $ax^3 + bx^2 + cx + d = 0$:

$$-a(\alpha + \beta + \gamma) = b \implies \alpha + \beta + \gamma = -\frac{b}{a}$$

$$a(\alpha\beta + \alpha\gamma + \beta\gamma) = c \implies \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$-a\alpha\beta\gamma = d \implies \alpha\beta\gamma = -\frac{d}{a}$$

The proof for quartics follows the same pattern by expanding $(x - \alpha)(x - \beta)(x - \gamma)(x - \delta)$.

2.5.3 Graph Behaviour of Polynomials

End Behaviour of Polynomials

For $P(x) = a_n x^n + \dots$:

- If n is **even** and $a_n > 0$: both ends go $\uparrow$
- If n is **even** and $a_n < 0$: both ends go $\downarrow$
- If n is **odd** and $a_n > 0$: left end $\downarrow$, right end $\uparrow$
- If n is **odd** and $a_n < 0$: left end $\uparrow$, right end $\downarrow$

A polynomial of degree n has at most $n - 1$ **turning points** and at most n real roots.

Example 1: Factor Theorem and Full Factorisation

Let $P(x) = x^3 - 3x^2 - 4x + 12$.

(a) Show that $(x - 3)$ is a factor of $P(x)$.

(b) Hence express $P(x)$ as a product of linear factors.

(c) Solve $P(x) = 0$.

Solution:

(a) $P(3) = 27 - 27 - 12 + 12 = 0$. By the factor theorem, $(x - 3)$ is a factor.

(b) Divide $P(x)$ by $(x - 3)$:

$$P(x) = (x - 3)(x^2 + 0x - 4) = (x - 3)(x^2 - 4) = (x - 3)(x - 2)(x + 2)$$

(c) $P(x) = 0 \implies (x - 3)(x - 2)(x + 2) = 0 \implies x = 3, x = 2, x = -2$.

Example 2: Finding an Unknown Coefficient

The polynomial $P(x) = 2x^3 + ax^2 - 5x + 3$ leaves a remainder of -1 when divided by $(x - 2)$, and $(x + 1)$ is a factor. Find a and the value of the other factor.

Solution:

Using the remainder theorem for $(x - 2)$:

$$P(2) = 2(8) + a(4) - 5(2) + 3 = 16 + 4a - 10 + 3 = 9 + 4a = -1$$

$$4a = -10 \implies a = -\frac{5}{2}$$

Check the factor theorem for $(x + 1)$:

$$P(-1) = 2(-1) + \left(-\frac{5}{2}\right)(1) - 5(-1) + 3 = -2 - \frac{5}{2} + 5 + 3 = \frac{9}{2} \neq 0$$

This gives $P(-1) = 9/2 \neq 0$, meaning there is an inconsistency. The problem may require revisiting. However, if we proceed with $a = -5/2$, we perform polynomial long division of $P(x) = 2x^3 - \frac{5}{2}x^2 - 5x + 3$ by $(x + 1)$:

Using synthetic division with root -1 : coefficients are $2, -\frac{5}{2}, -5, 3$.

$$\text{Bring down } 2. \quad 2(-1) + (-\frac{5}{2}) = -\frac{9}{2}.$$

$$(-\frac{9}{2})(-1) + (-5) = \frac{9}{2} - 5 = -\frac{1}{2}.$$

$$(-\frac{1}{2})(-1) + 3 = \frac{7}{2} \neq 0.$$

Indeed $(x + 1)$ is not a factor with $a = -5/2$. This highlights that both conditions may not be simultaneously satisfiable in a single polynomial of the given form, which is itself a useful IB exercise in recognizing contradiction.

2.6 Rational Functions

Definition 2.6.1 (Rational Function). A rational function is a ratio of two polynomials:

$$f(x) = \frac{P(x)}{Q(x)}, \quad Q(x) \neq 0$$

Asymptotes of Rational Functions

Vertical asymptotes: Set $Q(x) = 0$. For each root $x = a$ of $Q(x)$ that does *not* cancel with $P(x)$, the line $x = a$ is a vertical asymptote.

Horizontal asymptotes (as $x \rightarrow \pm\infty$): Compare degrees $n = \deg P$ and $m = \deg Q$.

- $n < m$: horizontal asymptote at $y = 0$
- $n = m$: horizontal asymptote at $y = \frac{a_n}{b_m}$ (ratio of leading coefficients)
- $n > m$: no horizontal asymptote (possible oblique asymptote)

Oblique (slant) asymptote: If $\deg P = \deg Q + 1$, perform polynomial long division:

$$\frac{P(x)}{Q(x)} = \text{quotient} + \frac{\text{remainder}}{Q(x)}$$

As $x \rightarrow \pm\infty$, $\frac{\text{remainder}}{Q(x)} \rightarrow 0$, so $y = \text{quotient}$ is the oblique asymptote.

IB Tip: Sketching Rational Functions Systematically

Follow this order:

1. Find the domain (where $Q(x) \neq 0$).
2. Find vertical asymptotes.
3. Find horizontal/oblique asymptotes.
4. Find x - and y -intercepts.
5. Test the sign of $f(x)$ in each region between asymptotes and intercepts.
6. Sketch, showing behaviour near asymptotes.

Example 1: Sketching a Rational Function with Oblique Asymptote

Sketch the graph of $f(x) = \frac{x^2 + x - 6}{x - 2}$, showing all asymptotes and intercepts.

Solution:

Factor: $x^2 + x - 6 = (x + 3)(x - 2)$, so

$$f(x) = \frac{(x + 3)(x - 2)}{x - 2} = x + 3 \quad \text{for } x \neq 2$$

Domain: $x \neq 2$, i.e. $(-\infty, 2) \cup (2, \infty)$.

Vertical asymptote: Technically, $x = 2$ makes the denominator zero, but since the factor cancels, $x = 2$ is a *hole* (removable discontinuity), not an asymptote. The point is $(2, 5)$ (found by substituting or taking the limit).

Oblique asymptote: $y = x + 3$ (the simplified function itself).

x-intercept: $x + 3 = 0 \implies x = -3$, so $(-3, 0)$.

y-intercept: $f(0) = -3$, so $(0, -3)$.

The graph is the line $y = x + 3$ with a hole at $(2, 5)$.

Example 2: Full Rational Function Analysis

For $f(x) = \frac{2x^2 - x - 1}{x^2 - 4}$, find all asymptotes, intercepts, and sketch.

Solution:

Domain: $x^2 - 4 \neq 0 \implies x \neq \pm 2$.

Vertical asymptotes: Since $x^2 - 4 = (x - 2)(x + 2)$ does not factor out of the numerator, we have vertical asymptotes at $x = -2$ and $x = 2$.

Horizontal asymptote: $\deg P = \deg Q = 2$, so

$$y = \frac{2}{1} = 2$$

x-intercepts: $2x^2 - x - 1 = 0 \implies (2x + 1)(x - 1) = 0 \implies x = -\frac{1}{2}$ or $x = 1$.

y-intercept: $f(0) = \frac{-1}{-4} = \frac{1}{4}$.

Sign analysis: Test points in the intervals $(-\infty, -2)$, $(-2, -1/2)$, $(-1/2, 1)$, $(1, 2)$, $(2, \infty)$.

The graph approaches $y = 2$ as $x \rightarrow \pm\infty$, has vertical asymptotes at $x = \pm 2$, and passes through the three intercepts found above.

2.7 Reciprocal Functions

Definition 2.7.1 (Reciprocal Function). *The reciprocal function is*

$$f(x) = \frac{1}{x}, \quad x \neq 0$$

Properties of $f(x) = \frac{1}{x}$

- Domain: $x \in \mathbb{R} \setminus \{0\}$
- Range: $y \in \mathbb{R} \setminus \{0\}$
- Vertical asymptote: $x = 0$ (y -axis)

- Horizontal asymptote: $y = 0$ (x -axis)
- The function is **odd**: $f(-x) = -f(x)$

Theorem 2.7.1 (Self-Inverse Property). *The function $f(x) = \frac{1}{x}$ is its own inverse:*

$$f^{-1}(x) = f(x) = \frac{1}{x}$$

Proof. Set $y = \frac{1}{x}$. Swap x and y : $x = \frac{1}{y}$. Solving: $y = \frac{1}{x}$.

Therefore $f^{-1}(x) = \frac{1}{x} = f(x)$, so f is self-inverse. □

Properties of $f(x) = \frac{1}{x^2}$

- Domain: $x \in \mathbb{R} \setminus \{0\}$
- Range: $y \in (0, \infty)$
- Vertical asymptote: $x = 0$
- Horizontal asymptote: $y = 0$
- The function is **even**: $f(-x) = f(x)$
- The graph lies entirely above the x -axis with a “double branch” shape

Example 1: Transformations of $f(x) = \frac{1}{x}$

Describe the transformation and sketch $g(x) = \frac{2}{x-3} + 1$.

Solution:

Starting from $f(x) = \frac{1}{x}$:

1. $\frac{1}{x-3}$: translation of 3 units to the **right**
2. $\frac{2}{x-3}$: vertical stretch by factor 2
3. $\frac{2}{x-3} + 1$: translation of 1 unit **up**

Resulting features:

- Vertical asymptote: $x = 3$
- Horizontal asymptote: $y = 1$
- Domain: $x \neq 3$
- Range: $y \neq 1$

Example 2: Sketching $f(x) = \frac{1}{x^2}$ with Transformations

Sketch $h(x) = -\frac{1}{(x+1)^2} + 4$.

Solution:

Starting from $\frac{1}{x^2}$:

1. $\frac{1}{(x+1)^2}$: shift 1 unit **left** (vertical asymptote at $x = -1$)
2. $-\frac{1}{(x+1)^2}$: reflection in the x -axis (graph now below the x -axis)
3. $-\frac{1}{(x+1)^2} + 4$: shift 4 units **up**

Features:

- Vertical asymptote: $x = -1$
- Horizontal asymptote: $y = 4$
- Range: $y < 4$ (the branches approach $y = 4$ from below)
- y -intercept: $h(0) = -1 + 4 = 3$, so $(0, 3)$

2.8 Exponential and Logarithmic Functions

2.8.1 Exponential Functions

Definition 2.8.1 (Exponential Function). *An exponential function has the form*

$$f(x) = a^x, \quad a > 0, a \neq 1$$

The number $e \approx 2.71828$ is the natural base.

Properties of Exponential Functions

For $f(x) = a^x$ with $a > 0$, $a \neq 1$:

- Domain: $\mathbb{R}$; Range: $(0, \infty)$
- y -intercept: $(0, 1)$ always
- Horizontal asymptote: $y = 0$
- $a > 1$: strictly increasing (growth)
- $0 < a < 1$: strictly decreasing (decay)
- $a^m \times a^n = a^{m+n}$, $\frac{a^m}{a^n} = a^{m-n}$, $(a^m)^n = a^{mn}$

2.8.2 Logarithmic Functions

Definition 2.8.2 (Logarithmic Function). *The logarithmic function is the inverse of the exponential function:*

$$y = \log_a x \iff a^y = x, \quad a > 0, a \neq 1$$

The natural logarithm is $\ln x = \log_e x$.

Properties of Logarithms

$\log_a(xy) = \log_a x + \log_a y$	(product rule)
$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$	(quotient rule)
$\log_a(x^n) = n \log_a x$	(power rule)
$\log_a a = 1$	for any base a
$\log_a 1 = 0$	for any base a
$a^{\log_a x} = x$	(inverse property)
$\log_a(a^x) = x$	(inverse property)
$\log_a x = \frac{\ln x}{\ln a}$	(change of base)

Theorem 2.8.1 (Inverse Relationship). *The functions $f(x) = a^x$ and $g(x) = \log_a x$ are inverses of each other:*

$$\log_a(a^x) = x \quad \text{and} \quad a^{\log_a x} = x$$

Their graphs are reflections of each other in the line $y = x$.

Solving Exponential and Logarithmic Equations

Exponential form to logarithmic:

$$a^x = b \implies x = \log_a b$$

This requires $a > 0$, $a \neq 1$, $b > 0$.

Same base:

$$a^{f(x)} = a^{g(x)} \implies f(x) = g(x)$$

Taking logarithms:

$$a^x = b \implies x \ln a = \ln b \implies x = \frac{\ln b}{\ln a}$$

Exponential Growth and Decay Models

Growth:

$$P(t) = P_0 \cdot a^t \quad \text{where } a > 1$$

or equivalently

$$P(t) = P_0 e^{kt} \quad \text{where } k > 0$$

Decay:

$$P(t) = P_0 \cdot a^t \quad \text{where } 0 < a < 1$$

or equivalently

$$P(t) = P_0 e^{-kt} \quad \text{where } k > 0$$

Half-life: If $P(t) = P_0 \cdot a^t$ and the half-life is $t_{1/2}$, then

$$a = \left(\frac{1}{2}\right)^{1/t_{1/2}}$$

Example 1: Solving an Exponential Equation

Solve $3^{2x-1} = 7^{x+1}$. Give your answer in terms of logarithms.

Solution:

Take natural logarithms of both sides:

$$\begin{aligned}(2x - 1) \ln 3 &= (x + 1) \ln 7 \\ 2x \ln 3 - \ln 3 &= x \ln 7 + \ln 7 \\ 2x \ln 3 - x \ln 7 &= \ln 7 + \ln 3 \\ x(2 \ln 3 - \ln 7) &= \ln 7 + \ln 3 \\ x &= \frac{\ln 7 + \ln 3}{2 \ln 3 - \ln 7} = \frac{\ln 21}{\ln 9 - \ln 7} = \frac{\ln 21}{\ln(9/7)}\end{aligned}$$

Example 2: Exponential Decay and Half-Life

A radioactive substance decays according to $M(t) = M_0 e^{-0.05t}$, where $M(t)$ is the mass in grams after t years. Find the half-life.

Solution:

Half-life is when $M(t) = \frac{M_0}{2}$:

$$\begin{aligned}M_0 e^{-0.05t} &= \frac{M_0}{2} \\ e^{-0.05t} &= \frac{1}{2} \\ -0.05t &= \ln\left(\frac{1}{2}\right) = -\ln 2 \\ t &= \frac{\ln 2}{0.05} = 20 \ln 2 \approx 13.86 \text{ years}\end{aligned}$$

Example 3: Logarithmic Equation

Solve $\log_2(x + 3) + \log_2(x - 1) = 5$.

Solution:

Use the product rule:

$$\log_2((x + 3)(x - 1)) = 5$$

Convert to exponential form:

$$(x + 3)(x - 1) = 2^5 = 32$$

Expand:

$$\begin{aligned}x^2 + 2x - 3 &= 32 \implies x^2 + 2x - 35 = 0 \\ (x + 7)(x - 5) &= 0 \implies x = -7 \text{ or } x = 5\end{aligned}$$

Check against domain: $x + 3 > 0$ and $x - 1 > 0$ require $x > 1$. Therefore $x = -7$ is rejected. The solution is $x = 5$.

2.9 Absolute Value Functions

Definition 2.9.1 (Absolute Value).

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Transformations Involving Absolute Values

$y = |f(x)|$: Take the graph of $y = f(x)$ and *reflect any portion below the x -axis* above it. Parts already above or on the x -axis remain unchanged.

$y = f(|x|)$: Take the portion of $y = f(x)$ for $x \geq 0$ and *reflect it across the y -axis* to the left side. The result is an even function (symmetric about the y -axis).

Solving Absolute Value Equations and Inequalities

Equation $|f(x)| = g(x)$ (where $g(x) \geq 0$):

1. Solve $f(x) = g(x)$.
2. Solve $f(x) = -g(x)$.
3. Check all solutions in the original equation.

Inequality $|f(x)| > g(x)$:

1. Solve $f(x) > g(x)$.
2. Solve $f(x) < -g(x)$.
3. The solution set is the union.

Inequality $|f(x)| < g(x)$:

1. Solve $f(x) < g(x)$.
2. Solve $f(x) > -g(x)$.
3. The solution set is the intersection.

IB Tip: Graphical Method for Absolute Values

IB examiners often award full marks for solving absolute value problems **graphically**. Sketch both sides, and the solutions correspond to intersection points. For inequalities, identify which regions satisfy the condition visually. Always state “from the graph” to justify the graphical method.

Example 1: Solving $|f(x)| = g(x)$

Solve $|2x - 3| = x + 2$.

Solution:

Case 1: $2x - 3 = x + 2 \implies x = 5$.

Check: $|2(5) - 3| = |7| = 7$ and $5 + 2 = 7$. Valid.

Case 2: $2x - 3 = -(x + 2) \implies 2x - 3 = -x - 2 \implies 3x = 1 \implies x = \frac{1}{3}$.

Check: $|2/3 - 3| = |-7/3| = 7/3$ and $1/3 + 2 = 7/3$. Valid.

Since $x + 2 \geq 0$ requires $x \geq -2$, both solutions satisfy this.

Solution: $x = \frac{1}{3}$ or $x = 5$.

Example 2: Solving $|f(x)| > g(x)$

Solve $|x^2 - 4| > 5$.

Solution:

This gives two cases:

Case 1: $x^2 - 4 > 5 \implies x^2 > 9 \implies x > 3$ or $x < -3$.

Case 2: $x^2 - 4 < -5 \implies x^2 < -1$, which has no real solutions.

Solution: $x \in (-\infty, -3) \cup (3, \infty)$.

Example 3: Graph of $y = |f(x)|$ vs. $y = f(|x|)$

Let $f(x) = x^2 - 4x + 3$. Describe the key differences between $y = |f(x)|$ and $y = f(|x|)$.

Solution:

First, $f(x) = (x-1)(x-3)$. The roots are $x = 1$ and $x = 3$. The vertex is at $(2, -1)$.

For $y = |f(x)|$: The portion of the parabola below the x -axis (between $x = 1$ and $x = 3$) is reflected above. The vertex moves from $(2, -1)$ to $(2, 1)$. The graph is V-shaped at the bottom with “cusps” at $(1, 0)$ and $(3, 0)$.

For $y = f(|x|)$: For $x \geq 0$, the graph is unchanged. For $x < 0$, we use $f(-x) = (-x)^2 - 4(-x) + 3 = x^2 + 4x + 3 = (x+1)(x+3)$. The left branch has roots at $x = -1$ and $x = -3$ with vertex at $(-2, -1)$. The result is symmetric about the y -axis only if f is an even function, which it is not here.

2.10 Trigonometric Functions (Graphs)

Key Features of Basic Trigonometric Functions

$y = \sin x$:

- Domain: $\mathbb{R}$; Range: $[-1, 1]$
- Period: 2π ; Amplitude: 1
- x -intercepts: $x = n\pi, n \in \mathbb{Z}$
- y -intercept: $(0, 0)$

$y = \cos x$:

- Domain: $\mathbb{R}$; Range: $[-1, 1]$
- Period: 2π ; Amplitude: 1
- x -intercepts: $x = \frac{\pi}{2} + n\pi, n \in \mathbb{Z}$
- y -intercept: $(0, 1)$

$y = \tan x$:

- Domain: $x \neq \frac{\pi}{2} + n\pi, n \in \mathbb{Z}$; Range: $\mathbb{R}$
- Period: π ; No amplitude (unbounded)

- Vertical asymptotes: $x = \frac{\pi}{2} + n\pi$
- x -intercepts: $x = n\pi, n \in \mathbb{Z}$
- y -intercept: $(0, 0)$

General Trigonometric Function

$$f(x) = a \sin(bx + c) + d \quad \text{or} \quad f(x) = a \cos(bx + c) + d$$

- $|a| = \mathbf{amplitude}$ (vertical stretch from centre line)
- $\frac{2\pi}{|b|} = \mathbf{period}$
- $-\frac{c}{b} = \mathbf{phase\ shift}$ (horizontal translation)
- $d = \mathbf{vertical\ shift}$ (centre line is $y = d$)
- Range: $[d - |a|, d + |a|]$

For tangent: $f(x) = a \tan(bx + c) + d$ has period $\frac{\pi}{|b|}$ and range $\mathbb{R}$.

IB Tip: Phase Shift Confusion

A common IB error is confusing the phase shift. For $f(x) = a \sin(bx + c) + d$:

- The phase shift is $-\frac{c}{b}$, NOT $-c$.
- If written as $f(x) = a \sin\left(b\left(x - \frac{-c}{b}\right)\right) + d$, the shift is clearer.
- The IB may write $f(x) = a \sin(b(x - h)) + d$ directly, where h is the phase shift.

Example 1: Identifying Trigonometric Parameters

For $f(x) = 3 \sin(2x - \pi) + 1$, find the amplitude, period, phase shift, and range.

Solution:

Comparing with $f(x) = a \sin(bx + c) + d$:

- $a = 3$, so amplitude $= |3| = 3$
- $b = 2$, so period $= \frac{2\pi}{2} = \pi$
- $c = -\pi$, so phase shift $= -\frac{(-\pi)}{2} = \frac{\pi}{2}$ (shift right by $\pi/2$)
- $d = 1$, so range $= [1 - 3, 1 + 3] = [-2, 4]$

Example 2: Finding a Trigonometric Function from Features

A sinusoidal function has a maximum value of 7 at $x = \frac{\pi}{4}$, a minimum value of 1, and a period of π . Find its equation in the form $f(x) = a \sin(bx + c) + d$.

Solution:

Amplitude: $|a| = \frac{7-1}{2} = 3$. Since the function has a maximum at $x = \pi/4$ and sine normally has its first maximum at $x = \pi/2$, we can use $a = 3$.

Vertical shift: $d = \frac{7+1}{2} = 4$.

Period: $\frac{2\pi}{|b|} = \pi \implies b = 2$.

Phase shift: The first maximum of sin is at $bx + c = \pi/2$. We know a maximum occurs at $x = \pi/4$:

$$2\left(\frac{\pi}{4}\right) + c = \frac{\pi}{2} \implies \frac{\pi}{2} + c = \frac{\pi}{2} \implies c = 0$$

Equation:

$$f(x) = 3 \sin(2x) + 4$$

Verification: $f(\pi/4) = 3 \sin(\pi/2) + 4 = 3(1) + 4 = 7$. ✓

2.11 Transformations of Graphs

Summary of Transformations

Given $y = f(x)$:

Translations:

- $y = f(x) + b$: translate **up** by b (down if $b < 0$)
- $y = f(x - a)$: translate **right** by a (left if $a < 0$)

Reflections:

- $y = -f(x)$: reflect in the x -axis
- $y = f(-x)$: reflect in the y -axis

Stretches:

- $y = p f(x)$: vertical stretch by factor $|p|$ (compression if $0 < |p| < 1$)
- $y = f(qx)$: horizontal stretch by factor $\frac{1}{|q|}$ (compression if $|q| > 1$)

Combined:

- $y = -f(x)$ combined with $y = f(-x)$: reflect in **both** axes (equivalent to 180° rotation about origin)

Critical IB Concept: Order of Transformations

When applying multiple transformations to $y = f(x)$, the **order matters**. For the general form

$$y = a f(b(x - h)) + k$$

the recommended order of operations (applied to the graph) is:

1. **Horizontal stretch** by factor $\frac{1}{|b|}$ about the y -axis.
2. **Horizontal translation** by h units.

3. **Vertical stretch** by factor $|a|$ about the x -axis.
4. **Reflection** in x -axis if $a < 0$.
5. **Vertical translation** by k units.

Key rule: Horizontal transformations (stretch/translate) are applied *before* vertical ones. Within horizontal: stretch first, then translate. Within vertical: stretch first, then translate.

Why Order Matters — Worked Demonstration

Consider $f(x) = x^2$ and the transformation to $y = (2x - 4)^2$.

Method A: $y = (2x)^2 = 4x^2$ (horizontal compression by $1/2$), then translate right by 2: $y = 4(x - 2)^2 = 4(x^2 - 4x + 4) = 4x^2 - 16x + 16$.

Method B: $y = (x - 4)^2$ (translate right by 4), then compress horizontally by $1/2$: Replace x with $2x$: $y = (2x - 4)^2 = 4x^2 - 16x + 16$.

Both give the same result *only because we correctly identified the form*. A common mistake is reading $(2x - 4)$ as “compress then translate by 4” (wrong). The correct reading is:

$$y = f(2(x - 2)) = (2(x - 2))^2$$

which is: compress by $1/2$ *then* translate right by 2.

IB Tip: Transformation Sequence Questions

The IB frequently asks: “The graph of $y = f(x)$ is transformed to $y = 2f(3x - 6) + 5$. Describe the sequence of transformations.”

Answer strategy: Always factor the inside first:

$$y = 2f(3(x - 2)) + 5$$

Now read off: horizontal stretch factor $1/3$, translate right 2, vertical stretch factor 2, translate up 5.

Example 1: Full Transformation Description

The graph of $y = f(x)$ is shown. (Assume $f(x)$ passes through $(0, 0)$, $(2, 4)$, $(4, 0)$.)

Describe the transformation that maps $y = f(x)$ onto $y = -\frac{1}{2}f(2x + 4)$, and find the images of the three given points.

Solution:

First, rewrite: $y = -\frac{1}{2}f(2(x + 2))$.

Sequence of transformations:

1. Horizontal stretch by factor $\frac{1}{2}$ (since $q = 2$, factor is $1/2$)
2. Horizontal translation 2 units to the **left** (since $x + 2 = x - (-2)$)
3. Vertical compression by factor $\frac{1}{2}$
4. Reflection in the x -axis (negative sign)

Images of points:

Start with $(0, 0)$: After horizontal stretch: $(0, 0)$. After shift left: $(-2, 0)$. After vertical compression and reflection: $(-2, 0)$.

Start with $(2, 4)$: After horizontal stretch: $(1, 4)$. After shift left: $(-1, 4)$. After vertical compression: $(-1, 2)$. After reflection: $(-1, -2)$.
 Start with $(4, 0)$: After horizontal stretch: $(2, 0)$. After shift left: $(0, 0)$. After vertical compression and reflection: $(0, 0)$.

Example 2: Finding the Equation After Transformations

The function $f(x) = \sqrt{x}$ is transformed by: a vertical stretch by factor 3, a reflection in the y -axis, and a translation of 1 unit down. Find the equation of the transformed function and its domain.

Solution:

Apply transformations in order:

1. Vertical stretch by 3: $y = 3\sqrt{x}$
2. Reflection in y -axis: $y = 3\sqrt{-x}$
3. Translation down by 1: $y = 3\sqrt{-x} - 1$

Domain: We need $-x \geq 0 \implies x \leq 0$.

Range: $3\sqrt{-x} \geq 0$, so $3\sqrt{-x} - 1 \geq -1$. Range = $[-1, \infty)$.

2.12 Inequalities

Sign Analysis Method for Inequalities

To solve $f(x) \geq 0$ (or $> 0, \leq 0, < 0$):

1. Find all **critical values**: roots of $f(x) = 0$ and points where f is undefined.
2. Arrange critical values on a number line, dividing $\mathbb{R}$ into intervals.
3. Test the sign of $f(x)$ in each interval (use any convenient test point).
4. Select intervals satisfying the inequality.
5. For $\geq$ or $\leq$, include endpoints where f is defined.

2.12.1 Quadratic Inequalities

Example 1: Quadratic Inequality

Solve $x^2 - 5x + 6 \leq 0$.

Solution:

Factor: $x^2 - 5x + 6 = (x - 2)(x - 3)$.

Critical values: $x = 2$ and $x = 3$.

Sign analysis:

- $x < 2$: test $x = 0$: $(0 - 2)(0 - 3) = 6 > 0$
- $2 < x < 3$: test $x = 2.5$: $(2.5 - 2)(2.5 - 3) = (0.5)(-0.5) < 0$
- $x > 3$: test $x = 4$: $(4 - 2)(4 - 3) = 2 > 0$

We want $f(x) \leq 0$, so the solution includes where f is negative and where $f = 0$.

$$x \in [2, 3]$$

Example 2: Quadratic Inequality — Strict

Solve $-2x^2 + 3x + 5 > 0$.

Solution:

Factor: $-2x^2 + 3x + 5 = -(2x^2 - 3x - 5) = -(2x - 5)(x + 1)$.

Critical values: $x = \frac{5}{2}$ and $x = -1$.

Sign analysis of $-2x^2 + 3x + 5$:

- $x < -1$: test $x = -2$: $-2(4) + 3(-2) + 5 = -8 - 6 + 5 = -9 < 0$
- $-1 < x < \frac{5}{2}$: test $x = 0$: $-2(0) + 3(0) + 5 = 5 > 0$
- $x > \frac{5}{2}$: test $x = 3$: $-2(9) + 3(3) + 5 = -18 + 9 + 5 = -4 < 0$

Since the inequality is strict ($>$), exclude endpoints:

$$x \in \left(-1, \frac{5}{2}\right)$$

2.12.2 Rational Inequalities

Example 3: Rational Inequality

Solve $\frac{x+1}{x-2} \geq 0$.

Solution:

Critical values: numerator zero at $x = -1$; denominator zero (undefined) at $x = 2$.

Sign analysis of $\frac{x+1}{x-2}$:

- $x < -1$: test $x = -2$: $\frac{-2+1}{-2-2} = \frac{-1}{-4} > 0$
- $-1 < x < 2$: test $x = 0$: $\frac{0+1}{0-2} = \frac{1}{-2} < 0$
- $x > 2$: test $x = 3$: $\frac{3+1}{3-2} = \frac{4}{1} > 0$

We need ≥ 0 . Include $x = -1$ (where the expression equals 0), but exclude $x = 2$ (where it is undefined):

$$x \in (-\infty, -1] \cup (2, \infty)$$

Example 4: Complex Rational Inequality

Solve $\frac{x^2 - x - 6}{x^2 - 1} < 0$.

Solution:

Factor numerator and denominator:

$$\frac{x^2 - x - 6}{x^2 - 1} = \frac{(x - 3)(x + 2)}{(x - 1)(x + 1)}$$

Critical values: $x = -2$, $x = -1$, $x = 1$, $x = 3$.

Sign analysis:

- $x < -2$: test $x = -3$: $\frac{(-6)(-1)}{(-4)(-2)} = \frac{6}{8} > 0$
- $-2 < x < -1$: test $x = -1.5$: $\frac{(-4.5)(0.5)}{(-2.5)(-0.5)} = \frac{-2.25}{1.25} < 0$
- $-1 < x < 1$: test $x = 0$: $\frac{(-3)(2)}{(-1)(1)} = \frac{-6}{-1} > 0$
- $1 < x < 3$: test $x = 2$: $\frac{(-1)(4)}{(1)(3)} = \frac{-4}{3} < 0$
- $x > 3$: test $x = 4$: $\frac{(1)(6)}{(3)(5)} = \frac{6}{15} > 0$

We need < 0 , so the expression is negative in $(-2, -1)$ and $(1, 3)$. Since it is a strict inequality, exclude all critical values:

$$x \in (-2, -1) \cup (1, 3)$$

Example 5: Solving $f(x) \geq g(x)$ with Quadratics

Solve $x^2 + 2x - 3 \geq 2x + 1$.

Solution:

Rearrange to bring everything to one side:

$$x^2 + 2x - 3 - 2x - 1 \geq 0 \implies x^2 - 4 \geq 0$$

Factor: $(x - 2)(x + 2) \geq 0$.

Sign analysis:

- $x < -2$: test $x = -3$: $9 - 4 = 5 > 0$
- $-2 < x < 2$: test $x = 0$: $0 - 4 = -4 < 0$
- $x > 2$: test $x = 3$: $9 - 4 = 5 > 0$

Since ≥ 0 (non-strict), include endpoints:

$$x \in (-\infty, -2] \cup [2, \infty)$$

Graphical check: $y = x^2 - 4$ is a parabola with vertex at $(0, -4)$ and roots at $x = \pm 2$. It is above the x -axis for $x \leq -2$ and $x \geq 2$.

IB Exam Tip: “Determine” vs “Find” for Inequalities

“Find” the solution set of $f(x) > 0$: you may use any valid method (algebraic or graphical) and just state the answer.

“Determine” the solution set: IB examiners expect to see **working**—usually the sign analysis table or a clearly labelled graph. For “determine” questions, simply writing the answer without justification will lose marks.

“Hence” or “hence or otherwise”: if the previous part set up a factorisation or graph, use it. “Hence or otherwise” means you *can* use any method, but the intended path uses prior work.

Common Pitfalls in Inequalities

1. **Multiplying by a negative flips the sign:** If you multiply or divide both sides by a negative quantity, reverse the inequality.
2. **Squaring both sides:** Only valid when both sides are known to be non-negative. Otherwise, consider cases.
3. **Rational inequalities:** Never multiply through by the denominator without considering its sign. Use the sign analysis method instead.
4. **Endpoints:** For strict inequalities ($<$, $>$), always exclude critical values where the expression equals zero or is undefined.

Chapter 3

Geometry and Trigonometry

This chapter brings together the geometry of three-dimensional space, the algebra of vectors, and the full toolkit of trigonometry. Vectors provide the bridge between algebra and geometry, while trigonometric identities and equations are among the most frequently tested topics on the IB exam.

3.1 3D Geometry

The IB syllabus requires fluency with volume and surface-area formulas for common solids, as well as the ability to work with points, distances, and angles in three-dimensional space.

3.1.1 Volume Formulas

Volume of Common Solids

$$\text{Sphere: } V = \frac{4}{3}\pi r^3$$

$$\text{Right circular cone: } V = \frac{1}{3}\pi r^2 h$$

$$\text{Pyramid: } V = \frac{1}{3}Bh$$

where r is the radius, h is the perpendicular height, and B is the area of the base of the pyramid.

3.1.2 Surface Area Formulas

Surface Area of Common Solids

$$\text{Sphere: } A = 4\pi r^2$$

$$\text{Right circular cone (total): } A = \pi r l + \pi r^2 = \pi r(l + r)$$

$$\text{Right circular cone (lateral only): } A = \pi r l$$

where $l = \sqrt{r^2 + h^2}$ is the slant height of the cone.

3.1.3 Hemisphere Formulas

Hemisphere Volume and Surface Area

$$\text{Volume of a hemisphere: } V = \frac{2}{3}\pi r^3$$

$$\text{Curved surface area: } A = 2\pi r^2$$

$$\text{Total surface area (including base): } A = 3\pi r^2$$

Remembering Hemisphere Formulas

A hemisphere is simply half a sphere. The curved surface area is half of $4\pi r^2$, giving $2\pi r^2$. The total surface area adds the circular base πr^2 , yielding $3\pi r^2$. The volume is half of $\frac{4}{3}\pi r^3$, giving $\frac{2}{3}\pi r^3$.

3.1.4 Distance in 3D

Definition 3.1.1. The distance between two points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ in three-dimensional space is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

This is a direct extension of the 2D distance formula (Pythagorean theorem), applied twice.

3.1.5 Midpoint in 3D

Midpoint Formula in 3D

The midpoint M of the line segment joining $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ is

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right).$$

3.1.6 Angle Between a Line and a Plane

Angle Between a Line and a Plane

If a line has direction vector $\mathbf{d}$ and a plane has normal vector $\mathbf{n}$, then the angle α between the line and the plane satisfies

$$\sin \alpha = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}| |\mathbf{n}|}.$$

Equivalently, if θ is the angle between $\mathbf{d}$ and $\mathbf{n}$, then $\alpha = 90^\circ - \theta$, so $\alpha + \theta = 90^\circ$.

The angle between a line and a plane is the complement of the angle between the line and the plane's normal vector.

Example: Distance and Midpoint in 3D

Problem. Find the distance and midpoint between $A(1, -2, 3)$ and $B(4, 2, -1)$.

Solution.

$$d = \sqrt{(4 - 1)^2 + (2 - (-2))^2 + (-1 - 3)^2} = \sqrt{9 + 16 + 16} = \sqrt{41}.$$

The midpoint is

$$M = \left(\frac{1+4}{2}, \frac{-2+2}{2}, \frac{3+(-1)}{2} \right) = \left(\frac{5}{2}, 0, 1 \right).$$

Example: Volume of a Composite Solid

Problem. A solid consists of a hemisphere of radius 5 cm sitting on top of a right circular cone of base radius 5 cm and height 12 cm. Find the total volume.

Solution.

$$V_{\text{hemisphere}} = \frac{2}{3}\pi(5)^3 = \frac{250\pi}{3},$$

$$V_{\text{cone}} = \frac{1}{3}\pi(5)^2(12) = 100\pi,$$

$$V_{\text{total}} = \frac{250\pi}{3} + 100\pi = \frac{250\pi + 300\pi}{3} = \frac{550\pi}{3} \approx 576 \text{ cm}^3.$$

3.2 Vectors: Foundations

Vectors are quantities that have both *magnitude* and *direction*. They are the fundamental language for describing geometry in two and three dimensions.

3.2.1 Position and Displacement Vectors

Definition 3.2.1. A **position vector** $\mathbf{a}$ describes the position of a point A relative to a fixed origin O . We write $\mathbf{a} = \overrightarrow{OA}$.

A **displacement vector** (or free vector) $\overrightarrow{AB}$ describes the displacement from point A to point B :

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a}.$$

3.2.2 Base Vectors

Definition 3.2.2. In three dimensions, the standard base vectors are

$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

They point along the positive x -, y -, and z -axes respectively, each with magnitude 1.

3.2.3 Component Form

Component Form of a Vector

A vector $\mathbf{v}$ with components v_1, v_2, v_3 can be written as

$$\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}.$$

3.2.4 Magnitude

Magnitude of a Vector

$$|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}.$$

For a 2D vector $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j}$: $|\mathbf{v}| = \sqrt{v_1^2 + v_2^2}.$

3.2.5 Unit Vectors

Definition 3.2.3. A **unit vector** has magnitude 1. The unit vector in the direction of $\mathbf{v}$ is

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{|\mathbf{v}|}.$$

3.2.6 Vector Addition and Scalar Multiplication

Vector Operations

Addition: If $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$, then

$$\mathbf{a} + \mathbf{b} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{pmatrix}.$$

Scalar multiplication: For $k \in \mathbb{R}$,

$$k\mathbf{a} = \begin{pmatrix} ka_1 \\ ka_2 \\ ka_3 \end{pmatrix}.$$

3.2.7 Parallel Vectors

Definition 3.2.4. Two vectors $\mathbf{a}$ and $\mathbf{b}$ are **parallel** if and only if one is a scalar multiple of the other:

$$\mathbf{a} = k\mathbf{b} \quad \text{for some } k \in \mathbb{R}, k \neq 0.$$

If $k > 0$ the vectors point in the same direction; if $k < 0$ they point in opposite directions.

Checking Parallelism

To check if two vectors are parallel, verify that the ratios of corresponding components are equal:

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = k.$$

Watch out for zero components: if $\mathbf{b} = \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix}$, then $\mathbf{a}$ must also have $a_1 = 0$ to be parallel.

Example: Vector Operations

Problem. Given $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = -\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$, find:

- (i) $3\mathbf{a} - 2\mathbf{b}$
- (ii) $|\mathbf{a} - \mathbf{b}|$
- (iii) The unit vector in the direction of $\mathbf{a}$

Solution.

$$(i) \quad 3\mathbf{a} - 2\mathbf{b} = \begin{pmatrix} 6 \\ -3 \\ 9 \end{pmatrix} - \begin{pmatrix} -2 \\ 8 \\ -4 \end{pmatrix} = \begin{pmatrix} 8 \\ -11 \\ 13 \end{pmatrix} = 8\mathbf{i} - 11\mathbf{j} + 13\mathbf{k}.$$

$$(ii) \quad \mathbf{a} - \mathbf{b} = \begin{pmatrix} 3 \\ -5 \\ 5 \end{pmatrix}, \text{ so } |\mathbf{a} - \mathbf{b}| = \sqrt{9 + 25 + 25} = \sqrt{59}.$$

$$(iii) \quad |\mathbf{a}| = \sqrt{4 + 1 + 9} = \sqrt{14}, \text{ so } \hat{\mathbf{a}} = \frac{1}{\sqrt{14}} \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}.$$

Example: Displacement Vectors and Parallelism

Problem. Points A, B, C have position vectors $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\mathbf{b} = 3\mathbf{i} - \mathbf{j} + 3\mathbf{k}$, $\mathbf{c} = 5\mathbf{i} - 4\mathbf{j} + 7\mathbf{k}$. Show that A, B, C are collinear.

Solution.

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix},$$

$$\overrightarrow{BC} = \mathbf{c} - \mathbf{b} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}.$$

Since $\overrightarrow{AB} = \overrightarrow{BC}$, the vectors are parallel (with $k = 1$) and share the point B , so A, B, C are collinear. In fact, B is the midpoint of AC .

3.3 Dot Product

The dot product (also called the scalar product) is an operation that takes two vectors and returns a scalar. It encodes information about the angle between vectors and is essential for work involving projections, perpendicularity, and distances.

3.3.1 Algebraic Definition

Definition 3.3.1. The *dot product* of $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ is

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3.$$

3.3.2 Geometric Definition

Definition 3.3.2. The *geometric definition* of the dot product is

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta,$$

where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$ ($0 \leq \theta \leq \pi$).

3.3.3 Equivalence of the Two Definitions

Proof: Algebraic Definition Equals Geometric Definition

We prove that $a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}| |\mathbf{b}| \cos \theta$ using the cosine rule.

Place the tails of $\mathbf{a}$ and $\mathbf{b}$ at the origin. Let $\mathbf{c} = \mathbf{a} - \mathbf{b}$ be the vector from the tip of $\mathbf{b}$ to the tip of $\mathbf{a}$. The triangle formed by $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ has angle θ between $\mathbf{a}$ and $\mathbf{b}$.

By the cosine rule on this triangle:

$$|\mathbf{c}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}| |\mathbf{b}| \cos \theta.$$

But $\mathbf{c} = \mathbf{a} - \mathbf{b}$, so

$$|\mathbf{c}|^2 = (a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2.$$

Expanding:

$$\begin{aligned} |\mathbf{c}|^2 &= (a_1^2 - 2a_1b_1 + b_1^2) + (a_2^2 - 2a_2b_2 + b_2^2) + (a_3^2 - 2a_3b_3 + b_3^2) \\ &= (a_1^2 + a_2^2 + a_3^2) + (b_1^2 + b_2^2 + b_3^2) - 2(a_1b_1 + a_2b_2 + a_3b_3) \\ &= |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2(\mathbf{a} \cdot \mathbf{b}). \end{aligned}$$

Equating the two expressions for $|\mathbf{c}|^2$:

$$|\mathbf{a}|^2 + |\mathbf{b}|^2 - 2(\mathbf{a} \cdot \mathbf{b}) = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}| |\mathbf{b}| \cos \theta.$$

Cancelling $|\mathbf{a}|^2 + |\mathbf{b}|^2$ and dividing by -2 :

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta.$$

3.3.4 Angle Between Vectors

Angle Between Two Vectors

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}, \quad \theta = \arccos \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} \right).$$

3.3.5 Perpendicular Vectors

Theorem 3.3.1. Two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular (orthogonal) if and only if

$$\mathbf{a} \cdot \mathbf{b} = 0.$$

Proof. If $\mathbf{a} \perp \mathbf{b}$, then $\theta = 90^\circ$, so $\cos \theta = 0$ and $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cdot 0 = 0$.

Conversely, if $\mathbf{a} \cdot \mathbf{b} = 0$ and both vectors are non-zero, then $|\mathbf{a}| |\mathbf{b}| \cos \theta = 0$, so $\cos \theta = 0$, giving $\theta = 90^\circ$. $\square$

3.3.6 Properties of the Dot Product

Theorem 3.3.2 (Properties of the Dot Product). Let $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ be vectors and $k \in \mathbb{R}$.

- (i) **Commutative:** $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
- (ii) **Distributive:** $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$
- (iii) **Scalar compatibility:** $(k\mathbf{a}) \cdot \mathbf{b} = k(\mathbf{a} \cdot \mathbf{b})$
- (iv) **NOT associative:** In general, $(\mathbf{a} \cdot \mathbf{b})\mathbf{c} \neq \mathbf{a} \cdot (\mathbf{b} \cdot \mathbf{c})$ since the left side is a vector and the right side is a scalar.
- (v) $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$

Why the Dot Product Is Not Associative

The expression $\mathbf{a} \cdot (\mathbf{b} \cdot \mathbf{c})$ is not even defined: $\mathbf{b} \cdot \mathbf{c}$ is a scalar, and you cannot take the dot product of a vector with a scalar. The expression $(\mathbf{a} \cdot \mathbf{b})\mathbf{c}$ means scaling $\mathbf{c}$ by the scalar $\mathbf{a} \cdot \mathbf{b}$, which is a valid (but different) operation.

Example: Finding the Angle Between Vectors

Problem. Find the angle between $\mathbf{a} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\mathbf{b} = -\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$.

Solution.

$$\mathbf{a} \cdot \mathbf{b} = (3)(-1) + (1)(2) + (-2)(4) = -3 + 2 - 8 = -9,$$

$$|\mathbf{a}| = \sqrt{9 + 1 + 4} = \sqrt{14},$$

$$|\mathbf{b}| = \sqrt{1 + 4 + 16} = \sqrt{21},$$

$$\cos \theta = \frac{-9}{\sqrt{14}\sqrt{21}} = \frac{-9}{\sqrt{294}} = \frac{-9}{7\sqrt{6}} \approx -0.526,$$

$$\theta \approx 121.7^\circ.$$

Example: Perpendicularity Check

Problem. Find the value of p such that $\mathbf{a} = p\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} - p\mathbf{j} + 5\mathbf{k}$ are perpendicular.

Solution. For perpendicularity: $\mathbf{a} \cdot \mathbf{b} = 0$:

$$3p + 2(-p) + (-1)(5) = 0 \implies 3p - 2p - 5 = 0 \implies p = 5.$$

3.4 Vector Equation of a Line

3.4.1 Vector Form

Vector Equation of a Line (2D and 3D)

The line passing through point A (position vector $\mathbf{a}$) with direction vector $\mathbf{b}$ is

$$\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}, \quad \lambda \in \mathbb{R}.$$

Here $\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ is the position vector of a general point on the line, and λ is a scalar parameter.

3.4.2 Parametric Form

Parametric Equations of a Line

If $\mathbf{a} = \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$, then the parametric form is

$$x = x_0 + \lambda a, \quad y = y_0 + \lambda b, \quad z = z_0 + \lambda c.$$

3.4.3 Cartesian Form

Cartesian (Symmetric) Equations of a Line

If none of a, b, c is zero, we can eliminate λ to obtain:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

If a direction component is zero, the corresponding numerator equals zero. For example, if $a = 0$ and $b, c \neq 0$:

$$x = x_0, \quad \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

3.4.4 Angle Between Two Lines

The angle between two lines with direction vectors $\mathbf{d}_1$ and $\mathbf{d}_2$ is found using the dot product:

$$\cos \theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1| |\mathbf{d}_2|}.$$

The acute angle is $\theta = \arccos \left(\frac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{|\mathbf{d}_1| |\mathbf{d}_2|} \right)$.

3.4.5 Intersection of Two Lines

To find the intersection of two lines, set their parametric equations equal and solve for the parameters. If a consistent solution exists for both parameters, the lines intersect at the resulting point.

Example: Finding the Equation of a Line

Problem. Find the vector equation of the line through $A(1, 2, -1)$ and $B(3, 0, 4)$.

Solution. The direction vector is $\overrightarrow{AB} = \begin{pmatrix} 2 \\ -2 \\ 5 \end{pmatrix}$. Using point A :

$$\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -2 \\ 5 \end{pmatrix}, \quad \lambda \in \mathbb{R}.$$

Parametric form: $x = 1 + 2\lambda, y = 2 - 2\lambda, z = -1 + 5\lambda$.

Cartesian form: $\frac{x - 1}{2} = \frac{y - 2}{-2} = \frac{z + 1}{5}$.

Example: Intersection of Two Lines

Problem. Find the intersection of $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$.

Solution. Setting the parametric forms equal:

$$\begin{aligned} 1 + \lambda &= 2 - \mu, \\ 0 - \lambda &= 1 + 2\mu, \\ 2 + \lambda &= -1 + \mu. \end{aligned}$$

From the first equation: $\lambda = 1 - \mu$. Substituting into the second: $-(1 - \mu) = 1 + 2\mu \implies -1 + \mu = 1 + 2\mu \implies \mu = -2$.

Then $\lambda = 1 - (-2) = 3$.

Check in the third equation: $2 + 3 = 5$ and $-1 + (-2) = -3$. Since $5 \neq -3$, the lines do *not* intersect (they are skew).

3.5 Lines in Space

In three dimensions, two lines can be related in four fundamentally different ways.

Definition 3.5.1 (Classification of Two Lines in 3D). (i) **Coincident:** The lines are the same line. Their direction vectors are parallel, and they share a point.

(ii) **Parallel:** The direction vectors are parallel, but the lines do not share any point.

(iii) **Intersecting:** The lines meet at exactly one point.

(iv) **Skew:** The lines are not parallel and do not intersect (they lie in different planes).

Distinguishing Cases for Two Lines

Given lines $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{d}_1$ and $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{d}_2$:

Step 1: Check if $\mathbf{d}_1 = k\mathbf{d}_2$ (parallel direction vectors).

- If **yes**: the lines are either coincident or parallel. Check if $\mathbf{a}_2 - \mathbf{a}_1$ is parallel to $\mathbf{d}_1$ (coincident) or not (parallel).
- If **no**: the lines are either intersecting or skew. Solve the system of three equations in λ and μ . If a consistent solution exists, they intersect; otherwise, they are skew.

IB Exam Strategy: Line Classification

1. Always check parallelism *first* by seeing if the direction vectors are scalar multiples.
1. If not parallel, attempt to solve the three equations simultaneously.
1. If the solution satisfies *all three* equations, the lines intersect.
1. If you get a contradiction, the lines are skew.

Example: Classifying Two Lines

Problem. Classify the relationship between $l_1 : \mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix}$.

Solution. Direction vectors: $\mathbf{d}_1 = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$, $\mathbf{d}_2 = \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix}$.

Check parallelism: $\frac{2}{-1} = -2$, but $\frac{1}{2} = 0.5 \neq -2$. Not parallel.

Set parametric equations equal:

$$\begin{aligned} 2\lambda &= 1 - \mu, \\ 1 + \lambda &= 3 + 2\mu, \\ -1 + 3\lambda &= 1. \end{aligned}$$

From (iii): $3\lambda = 2 \implies \lambda = \frac{2}{3}$.

From (i): $\mu = 1 - 2 \cdot \frac{2}{3} = 1 - \frac{4}{3} = -\frac{1}{3}$.

Check in (ii): LHS = $1 + \frac{2}{3} = \frac{5}{3}$, RHS = $3 + 2(-\frac{1}{3}) = 3 - \frac{2}{3} = \frac{7}{3}$.

Since $\frac{5}{3} \neq \frac{7}{3}$, the lines are **skew**.

3.6 Cross Product

The cross product (also called the vector product) is an operation on two vectors that produces another vector, in contrast to the dot product which produces a scalar. It is defined only in three dimensions.

3.6.1 Definition via Determinant

Definition 3.6.1. The **cross product** of $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}$ is

$$\mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = (v_2w_3 - v_3w_2)\mathbf{i} - (v_1w_3 - v_3w_1)\mathbf{j} + (v_1w_2 - v_2w_1)\mathbf{k}.$$

Computing the Cross Product

A useful mnemonic: for each component, “cover” the corresponding row and column, then compute the 2×2 determinant of the remaining entries, alternating signs $+-+$:

$$\begin{aligned} \text{coeff of } \mathbf{i}: & + \begin{vmatrix} v_2 & v_3 \\ w_2 & w_3 \end{vmatrix} = v_2w_3 - v_3w_2, \\ \text{coeff of } \mathbf{j}: & - \begin{vmatrix} v_1 & v_3 \\ w_1 & w_3 \end{vmatrix} = -(v_1w_3 - v_3w_1), \\ \text{coeff of } \mathbf{k}: & + \begin{vmatrix} v_1 & v_2 \\ w_1 & w_2 \end{vmatrix} = v_1w_2 - v_2w_1. \end{aligned}$$

3.6.2 Properties

Theorem 3.6.1 (Properties of the Cross Product). Let $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ be vectors and $k \in \mathbb{R}$.

- (i) **Anti-commutative:** $\mathbf{v} \times \mathbf{w} = -(\mathbf{w} \times \mathbf{v})$
- (ii) **Distributive:** $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$
- (iii) **Scalar compatibility:** $(k\mathbf{v}) \times \mathbf{w} = k(\mathbf{v} \times \mathbf{w})$
- (iv) **NOT associative:** In general, $\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) \neq (\mathbf{u} \times \mathbf{v}) \times \mathbf{w}$
- (v) $\mathbf{v} \times \mathbf{v} = \mathbf{0}$ (the zero vector)
- (vi) $\mathbf{v} \times \mathbf{w} = \mathbf{0}$ if and only if $\mathbf{v}$ and $\mathbf{w}$ are parallel
- (vii) $\mathbf{i} \times \mathbf{j} = \mathbf{k}$, $\mathbf{j} \times \mathbf{k} = \mathbf{i}$, $\mathbf{k} \times \mathbf{i} = \mathbf{j}$

3.6.3 Geometric Interpretation

Theorem 3.6.2. The magnitude of the cross product equals the area of the parallelogram spanned by $\mathbf{v}$ and $\mathbf{w}$:

$$|\mathbf{v} \times \mathbf{w}| = |\mathbf{v}| |\mathbf{w}| \sin \theta,$$

where θ is the angle between $\mathbf{v}$ and $\mathbf{w}$ ($0 \leq \theta \leq \pi$).

Proof: Magnitude of Cross Product Equals Parallelogram Area

Consider the parallelogram spanned by $\mathbf{v}$ and $\mathbf{w}$, with angle θ between them. The base has length $|\mathbf{v}|$ and the perpendicular height is $|\mathbf{w}| \sin \theta$.

The area of the parallelogram is:

$$\text{Area} = \text{base} \times \text{height} = |\mathbf{v}| \cdot |\mathbf{w}| \sin \theta.$$

We now show that $|\mathbf{v} \times \mathbf{w}|^2 = |\mathbf{v}|^2 |\mathbf{w}|^2 \sin^2 \theta$ using the algebraic definition.

Let $\mathbf{v} \times \mathbf{w} = \begin{pmatrix} p \\ q \\ r \end{pmatrix}$ where $p = v_2 w_3 - v_3 w_2$, $q = v_3 w_1 - v_1 w_3$, $r = v_1 w_2 - v_2 w_1$.

Then:

$$\begin{aligned} |\mathbf{v} \times \mathbf{w}|^2 &= p^2 + q^2 + r^2 \\ &= (v_2 w_3 - v_3 w_2)^2 + (v_3 w_1 - v_1 w_3)^2 + (v_1 w_2 - v_2 w_1)^2. \end{aligned}$$

Expanding all three squares and collecting terms (Lagrange's identity):

$$\begin{aligned} |\mathbf{v} \times \mathbf{w}|^2 &= (v_1^2 + v_2^2 + v_3^2)(w_1^2 + w_2^2 + w_3^2) - (v_1 w_1 + v_2 w_2 + v_3 w_3)^2 \\ &= |\mathbf{v}|^2 |\mathbf{w}|^2 - (\mathbf{v} \cdot \mathbf{w})^2 \\ &= |\mathbf{v}|^2 |\mathbf{w}|^2 - |\mathbf{v}|^2 |\mathbf{w}|^2 \cos^2 \theta \\ &= |\mathbf{v}|^2 |\mathbf{w}|^2 (1 - \cos^2 \theta) \\ &= |\mathbf{v}|^2 |\mathbf{w}|^2 \sin^2 \theta. \end{aligned}$$

Taking square roots (since $\sin \theta \geq 0$ for $0 \leq \theta \leq \pi$):

$$|\mathbf{v} \times \mathbf{w}| = |\mathbf{v}| |\mathbf{w}| \sin \theta.$$

3.6.4 Right-Hand Rule

The Right-Hand Rule

The direction of $\mathbf{v} \times \mathbf{w}$ is given by the right-hand rule: point the fingers of your right hand along $\mathbf{v}$, then curl them toward $\mathbf{w}$ through the smaller angle. Your thumb points in the direction of $\mathbf{v} \times \mathbf{w}$.

Consequences:

- $\mathbf{v} \times \mathbf{w}$ is perpendicular to both $\mathbf{v}$ and $\mathbf{w}$.
- $\mathbf{w} \times \mathbf{v}$ points in the opposite direction (anti-commutativity).

Area of a Triangle

Since a triangle has half the area of the parallelogram spanned by two sides, the area of the triangle with sides $\overrightarrow{AB}$ and $\overrightarrow{AC}$ is:

$$\text{Area} = \frac{1}{2} |\overrightarrow{AB} \times \overrightarrow{AC}|.$$

Example: Computing a Cross Product

Problem. Find $\mathbf{a} \times \mathbf{b}$ where $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}$, and find the area of the parallelogram they span.

Solution.

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -1 \\ 3 & 0 & 2 \end{vmatrix} = (4 - 0)\mathbf{i} - (2 - (-3))\mathbf{j} + (0 - 6)\mathbf{k} = 4\mathbf{i} - 5\mathbf{j} - 6\mathbf{k}.$$

$$\text{Area of parallelogram} = |\mathbf{a} \times \mathbf{b}| = \sqrt{16 + 25 + 36} = \sqrt{77}.$$

$$\text{Area of triangle} = \frac{\sqrt{77}}{2}.$$

3.7 Vector Equation of a Plane

3.7.1 Parametric Form of a Plane

Parametric Equation of a Plane

The plane passing through point A (position vector $\mathbf{a}$) and containing two non-parallel direction vectors $\mathbf{b}$ and $\mathbf{c}$ is

$$\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}, \quad \lambda, \mu \in \mathbb{R}.$$

Every point on the plane is obtained by starting at A and moving by scalar multiples of $\mathbf{b}$ and $\mathbf{c}$ independently.

3.7.2 Normal Vector and Normal Form

Definition 3.7.1. A **normal vector** $\mathbf{n}$ to a plane is any vector perpendicular to every vector lying in the plane. If $\mathbf{b}$ and $\mathbf{c}$ are direction vectors in the plane, then

$$\mathbf{n} = \mathbf{b} \times \mathbf{c}$$

is a normal vector to the plane.

Normal (Scalar Product) Equation of a Plane

If a plane has normal vector $\mathbf{n}$ and passes through point A (position vector $\mathbf{a}$), then for any point P (position vector $\mathbf{r}$) on the plane:

$$\overrightarrow{AP} \cdot \mathbf{n} = 0 \implies (\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0 \implies \mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}.$$

3.7.3 Cartesian Equation of a Plane

Derivation: Cartesian Equation from Normal Form

Starting with $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$, let $\mathbf{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$, $\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, and $\mathbf{a} \cdot \mathbf{n} = d$.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} a \\ b \\ c \end{pmatrix} = d,$$

$$ax + by + cz = d.$$

So the Cartesian equation of the plane is $ax + by + cz = d$, where $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is the normal vector.

Cartesian Equation of a Plane

$$ax + by + cz = d,$$

where $\mathbf{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is a normal vector to the plane and d is a constant. The value of d is determined by substituting any known point on the plane.

Finding the Normal Vector

- Given parametric form with direction vectors $\mathbf{b}$ and $\mathbf{c}$: compute $\mathbf{n} = \mathbf{b} \times \mathbf{c}$.
- Given Cartesian form $ax + by + cz = d$: read off $\mathbf{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$.
- Given three points A, B, C : compute $\overrightarrow{AB} \times \overrightarrow{AC}$.

Example: Finding the Equation of a Plane

Problem. Find the Cartesian equation of the plane through $A(1, 0, 2)$, $B(3, -1, 1)$, and $C(0, 2, -1)$.

Solution.

$$\overrightarrow{AB} = \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}, \quad \overrightarrow{AC} = \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}.$$

$$\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & -1 \\ -1 & 2 & -3 \end{vmatrix}$$

$$= (3 - (-2))\mathbf{i} - (-6 - 1)\mathbf{j} + (4 - 1)\mathbf{k} = 5\mathbf{i} + 7\mathbf{j} + 3\mathbf{k}.$$

Using point $A(1, 0, 2)$: $5(1) + 7(0) + 3(2) = 5 + 6 = 11$.

The Cartesian equation is $5x + 7y + 3z = 11$.

Example: Converting Between Forms

Problem. Convert the Cartesian equation $2x - y + 3z = 7$ to vector (parametric) form.

Solution. A normal vector is $\mathbf{n} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$. We need two vectors perpendicular to $\mathbf{n}$. Choose

$$\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \text{ (check: } 2(1) + (-1)(2) + 3(0) = 0 \text{) and } \mathbf{c} = \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} \text{ (check: } 2(0) + (-1)(3) + 3(1) = 0 \text{)}.$$

Find a point on the plane: set $y = 0$, $z = 1$, then $2x + 3 = 7 \implies x = 2$. So $A = (2, 0, 1)$.

Parametric form:

$$\mathbf{r} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix}, \quad \lambda, \mu \in \mathbb{R}.$$

3.8 Intersections and Angles in 3D

3.8.1 Line–Plane Intersection

Method: Line–Plane Intersection

To find the intersection of line $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$ with plane $px + qy + rz = s$:

Step 1. Write the parametric equations: $x = a_1 + \lambda d_1$, $y = a_2 + \lambda d_2$, $z = a_3 + \lambda d_3$.

Step 2. Substitute into the Cartesian equation of the plane.

Step 3. Solve for λ .

Step 4. Substitute λ back to find the point of intersection.

Special cases:

- If $\mathbf{d} \cdot \mathbf{n} \neq 0$: the line intersects the plane at a unique point.
- If $\mathbf{d} \cdot \mathbf{n} = 0$ and the line lies in the plane: infinitely many solutions (line lies in plane).
- If $\mathbf{d} \cdot \mathbf{n} = 0$ and the line does not lie in the plane: no solution (line is parallel to plane).

3.8.2 Intersection of Two Planes

Two planes with normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ intersect in a line (provided they are not parallel). The direction vector of the line of intersection is $\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2$.

To find the full equation of the line of intersection:

1. Compute $\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2$.
2. Set one variable to zero (say $z = 0$) and solve the resulting 2×2 system for x and y to find a point on the line.
3. Write $\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$.

3.8.3 Intersection of Three Planes

Three planes can intersect in:

- (i) A **single point** (unique solution): the three normal vectors are not coplanar, i.e., $\mathbf{n}_1 \cdot (\mathbf{n}_2 \times \mathbf{n}_3) \neq 0$. The system has a unique solution.
- (ii) A **line** (infinitely many solutions): the normals are coplanar but not all parallel, and the system is consistent but with a free parameter.
- (iii) A **plane** (infinitely many solutions): all three planes are the same plane.
- (iv) **No solution**: the system is inconsistent. This occurs when, for example, two parallel planes never meet, or the three planes form a triangular prism.

IB Exam Tip: Three Planes

When solving a 3×3 system to find the intersection of three planes, use row reduction (Gaussian elimination) or substitution. If you find a row that simplifies to $0 = k$ where $k \neq 0$, the system is inconsistent (no solution). If a row becomes $0 = 0$, there are infinitely many solutions.

3.8.4 Angle Between a Line and a Plane

Angle Between a Line and a Plane

If a line has direction vector $\mathbf{d}$ and a plane has normal vector $\mathbf{n}$, let θ be the angle between $\mathbf{d}$ and $\mathbf{n}$, and let α be the angle between the line and the plane. Then

$$\alpha = 90^\circ - \theta, \quad \text{so} \quad \sin \alpha = \cos \theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}| |\mathbf{n}|}.$$

3.8.5 Angle Between Two Planes

Angle Between Two Planes

The angle between two planes with normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ is the same as the angle between their normals:

$$\cos \theta = \frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{|\mathbf{n}_1| |\mathbf{n}_2|}.$$

Example: Line–Plane Intersection

Problem. Find where the line $\mathbf{r} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ intersects the plane $x + 2y - z = 4$.

Solution. Parametric form: $x = 1 + 2\lambda$, $y = -1 + \lambda$, $z = 3 - \lambda$.
Substitute into the plane equation:

$$(1 + 2\lambda) + 2(-1 + \lambda) - (3 - \lambda) = 4.$$

$$1 + 2\lambda - 2 + 2\lambda - 3 + \lambda = 4 \implies 5\lambda - 4 = 4 \implies 5\lambda = 8 \implies \lambda = \frac{8}{5}.$$

The point of intersection:

$$x = 1 + \frac{16}{5} = \frac{21}{5}, \quad y = -1 + \frac{8}{5} = \frac{3}{5}, \quad z = 3 - \frac{8}{5} = \frac{7}{5}.$$

Example: Angle Between Two Planes

Problem. Find the acute angle between the planes $2x - y + 3z = 5$ and $x + 2y - z = 1$.

Solution. Normal vectors: $\mathbf{n}_1 = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$, $\mathbf{n}_2 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$.

$$\mathbf{n}_1 \cdot \mathbf{n}_2 = 2(1) + (-1)(2) + 3(-1) = 2 - 2 - 3 = -3,$$

$$|\mathbf{n}_1| = \sqrt{4 + 1 + 9} = \sqrt{14}, \quad |\mathbf{n}_2| = \sqrt{1 + 4 + 1} = \sqrt{6},$$

$$\cos \theta = \frac{|-3|}{\sqrt{14}\sqrt{6}} = \frac{3}{\sqrt{84}} = \frac{3}{2\sqrt{21}} = \frac{\sqrt{21}}{14} \approx 0.327,$$

$$\theta \approx 70.9^\circ.$$

Example: Line of Intersection of Two Planes

Problem. Find the line of intersection of the planes $x + y + z = 6$ and $2x - y + z = 3$.

Solution.

$$\mathbf{n}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{n}_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix},$$

$$\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{vmatrix} = (1 - (-1))\mathbf{i} - (1 - 2)\mathbf{j} + (-1 - 2)\mathbf{k} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}.$$

To find a point, set $z = 0$:

$$x + y = 6, \quad 2x - y = 3.$$

Adding: $3x = 9 \implies x = 3$, $y = 3$. So $A = (3, 3, 0)$.

The line of intersection is:

$$\mathbf{r} = \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}, \quad \lambda \in \mathbb{R}.$$

3.9 Trigonometric Foundations

3.9.1 Right-Angled Triangle Ratios

SOH-CAH-TOA

For a right-angled triangle with angle θ :

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}},$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}},$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sin \theta}{\cos \theta}.$$

3.9.2 Exact Trigonometric Values

Exact Values for Special Angles

	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

Equivalent values using periodicity and symmetry:

- $120^\circ = 180^\circ - 60^\circ$: $\sin 120^\circ = \frac{\sqrt{3}}{2}$, $\cos 120^\circ = -\frac{1}{2}$, $\tan 120^\circ = -\sqrt{3}$
- $135^\circ = 180^\circ - 45^\circ$: $\sin 135^\circ = \frac{\sqrt{2}}{2}$, $\cos 135^\circ = -\frac{\sqrt{2}}{2}$, $\tan 135^\circ = -1$
- $150^\circ = 180^\circ - 30^\circ$: $\sin 150^\circ = \frac{1}{2}$, $\cos 150^\circ = -\frac{\sqrt{3}}{2}$, $\tan 150^\circ = -\frac{1}{\sqrt{3}}$

Memory Trick: The 45-45-90 and 30-60-90 Triangles

45-45-90 triangle: sides 1, 1, $\sqrt{2}$. Hence $\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$.

30-60-90 triangle: sides 1, $\sqrt{3}$, 2. Opposite 30° is 1, opposite 60° is $\sqrt{3}$, hypotenuse is 2.

3.9.3 Unit Circle Definitions

Definition 3.9.1. On the unit circle (radius 1, centered at the origin), for an angle θ measured counter-clockwise from the positive x -axis, the coordinates of the point on the circle are:

$$(\cos \theta, \sin \theta).$$

Then $\tan \theta = \frac{\sin \theta}{\cos \theta}$ (defined when $\cos \theta \neq 0$).

3.9.4 Radian Measure

Radian Measure

$$\pi \text{ radians} = 180^\circ \implies 1 \text{ radian} = \frac{180^\circ}{\pi} \implies 1^\circ = \frac{\pi}{180} \text{ radians.}$$

Common conversions:

$$\begin{array}{llll} 30^\circ = \frac{\pi}{6}, & 45^\circ = \frac{\pi}{4}, & 60^\circ = \frac{\pi}{3}, & 90^\circ = \frac{\pi}{2}, \\ 120^\circ = \frac{2\pi}{3}, & 135^\circ = \frac{3\pi}{4}, & 150^\circ = \frac{5\pi}{6}, & 180^\circ = \pi. \end{array}$$

3.9.5 Arc Length

Arc Length

For a circle of radius r subtending an angle θ (in radians) at the center:

$$s = r\theta.$$

3.9.6 Sector Area

Area of a Sector

For a circle of radius r and central angle θ (in radians):

$$A = \frac{1}{2}r^2\theta.$$

Derivation: Arc Length and Sector Area

Arc length. The full circumference of a circle of radius r is $2\pi r$. The fraction of the circumference corresponding to angle θ (out of 2π) is $\frac{\theta}{2\pi}$. Therefore:

$$s = \frac{\theta}{2\pi} \cdot 2\pi r = r\theta.$$

Sector area. The full area of a circle is πr^2 . The fraction of the area corresponding to angle θ (out of 2π) is $\frac{\theta}{2\pi}$. Therefore:

$$A = \frac{\theta}{2\pi} \cdot \pi r^2 = \frac{1}{2}r^2\theta.$$

Example: Arc Length and Sector Area

Problem. A sector of a circle with radius 8 cm has an angle of 1.5 radians. Find (a) the arc length and (b) the area of the sector.

Solution.

(a) $s = r\theta = 8 \times 1.5 = 12 \text{ cm.}$

(b) $A = \frac{1}{2}r^2\theta = \frac{1}{2}(64)(1.5) = 48 \text{ cm}^2.$

3.10 Sine and Cosine Rules

The sine and cosine rules generalize right-angled triangle trigonometry to *all* triangles.

3.10.1 The Sine Rule

The Sine Rule

For any triangle with sides a, b, c opposite angles A, B, C respectively:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

Derivation of the Sine Rule

Consider triangle ABC with sides a, b, c . Drop an altitude h from C perpendicular to AB .

Case 1: Both A and B are acute.

In triangle ACD (where D is the foot of the altitude):

$$\sin A = \frac{h}{b} \implies h = b \sin A.$$

In triangle BCD :

$$\sin B = \frac{h}{a} \implies h = a \sin B.$$

Since both equal h :

$$a \sin B = b \sin A \implies \frac{a}{\sin A} = \frac{b}{\sin B}.$$

Similarly, by dropping the altitude from A , we get $\frac{b}{\sin B} = \frac{c}{\sin C}$.

Case 2: One angle is obtuse.

If $B > 90^\circ$, then in triangle BCD we use the supplementary angle:

$$\sin(180^\circ - B) = \frac{h}{a} \implies \sin B = \frac{h}{a}.$$

The result follows identically.

3.10.2 The Cosine Rule

The Cosine Rule

$$c^2 = a^2 + b^2 - 2ab \cos C.$$

Equivalently:

$$a^2 = b^2 + c^2 - 2bc \cos A, \quad b^2 = a^2 + c^2 - 2ac \cos B.$$

Solving for the angle:

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}.$$

Derivation of the Cosine Rule

Consider triangle ABC with sides a, b, c where $c = AB$. Drop altitude h from C to point D on AB . Let $AD = x$ and $DB = c - x$.

In right triangle ACD :

$$h^2 + x^2 = b^2. \quad (1)$$

In right triangle BCD :

$$h^2 + (c - x)^2 = a^2. \quad (2)$$

Expanding (2):

$$h^2 + c^2 - 2cx + x^2 = a^2. \quad (3)$$

Substituting $h^2 + x^2 = b^2$ from (1) into (3):

$$b^2 + c^2 - 2cx = a^2. \quad (4)$$

From right triangle ACD : $\cos A = \frac{x}{b}$, so $x = b \cos A$.

Substituting into (4):

$$a^2 = b^2 + c^2 - 2c(b \cos A) = b^2 + c^2 - 2bc \cos A.$$

3.10.3 Area of a Triangle (Trigonometric)

Area of a Triangle Using Trigonometry

$$\text{Area} = \frac{1}{2}ab \sin C = \frac{1}{2}bc \sin A = \frac{1}{2}ac \sin B.$$

Derivation of the Area Formula

Drop altitude h from C perpendicular to AB . The area is:

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times c \times h.$$

In right triangle ACD : $\sin A = \frac{h}{b}$, so $h = b \sin A$.

Therefore:

$$\text{Area} = \frac{1}{2} \times c \times b \sin A = \frac{1}{2}bc \sin A.$$

3.10.4 The Ambiguous Case of the Sine Rule

The Ambiguous Case (SSA)

When given two sides and a non-included angle (SSA configuration), the sine rule may produce two possible triangles.

Given sides a and b and angle A :

- If $a < b \sin A$: **no triangle** exists.
- If $a = b \sin A$: **one right-angled triangle** exists.
- If $b \sin A < a < b$: **two triangles** exist (the angle B can be acute or obtuse, since $\sin B = \sin(180^\circ - B)$).
- If $a \geq b$: **one triangle** exists (angle B must be acute).

On the IB exam: If using the sine rule gives $\sin B = k$ where $0 < k < 1$, always check whether both $B = \arcsin k$ and $B = 180^\circ - \arcsin k$ produce valid triangles.

IB Exam Tip: When to Use Which Rule

- **Right-angled triangle:** Use SOH–CAH–TOA directly.
- **Given two angles and one side (ASA or AAS):** Use the **sine rule**.
- **Given two sides and the included angle (SAS):** Use the **cosine rule** or **area formula**.
- **Given three sides (SSS):** Use the **cosine rule** to find an angle.
- **Given two sides and a non-included angle (SSA):** Use the **sine rule** but check for the ambiguous case.
- **Finding area:** Use $\frac{1}{2}ab \sin C$ if two sides and the included angle are known, or Heron's formula ($A = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = \frac{a+b+c}{2}$) if three sides are known.

Example: Using the Cosine Rule

Problem. In triangle ABC , $a = 7$, $b = 5$, $c = 3$. Find angle C .

Solution.

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{49 + 25 - 9}{2 \times 7 \times 5} = \frac{65}{70} = \frac{13}{14}.$$

$$C = \arccos\left(\frac{13}{14}\right) \approx 21.8^\circ.$$

Example: Ambiguous Case

Problem. In triangle ABC , $a = 8$, $b = 10$, and $A = 40^\circ$. Find the possible values of B .

Solution. By the sine rule:

$$\sin B = \frac{b \sin A}{a} = \frac{10 \sin 40^\circ}{8} = \frac{10 \times 0.6428}{8} \approx 0.8035.$$

$$B_1 = \arcsin(0.8035) \approx 53.5^\circ, \quad B_2 = 180^\circ - 53.5^\circ = 126.5^\circ.$$

Check: $A + B_1 = 40^\circ + 53.5^\circ = 93.5^\circ < 180^\circ$ (valid). $A + B_2 = 40^\circ + 126.5^\circ = 166.5^\circ < 180^\circ$ (valid).

Both triangles exist. The two possible values of B are approximately 53.5° and 126.5° .

3.11 Trigonometric Identities

Trigonometric identities are equations that hold for all values of the variable (in the domain). They are essential tools for simplifying expressions, solving equations, and proving other identities.

3.11.1 Pythagorean Identity

Identity 3.11.1 (Fundamental Pythagorean Identity).

$$\sin^2 \theta + \cos^2 \theta = 1.$$

Proof. On the unit circle, a point at angle θ has coordinates $(\cos \theta, \sin \theta)$. Since the point lies on the circle $x^2 + y^2 = 1$:

$$\cos^2 \theta + \sin^2 \theta = 1.$$

□

3.11.2 Derived Pythagorean Identities

Derived Identities

$$1 + \tan^2 \theta = \sec^2 \theta,$$

$$1 + \cot^2 \theta = \csc^2 \theta.$$

Proof of Derived Pythagorean Identities

Proof of $1 + \tan^2 \theta = \sec^2 \theta$:

Start with $\sin^2 \theta + \cos^2 \theta = 1$. Divide both sides by $\cos^2 \theta$:

$$\frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta},$$

$$\tan^2 \theta + 1 = \sec^2 \theta.$$

Proof of $1 + \cot^2 \theta = \csc^2 \theta$:

Start with $\sin^2 \theta + \cos^2 \theta = 1$. Divide both sides by $\sin^2 \theta$:

$$\frac{\sin^2 \theta}{\sin^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} = \frac{1}{\sin^2 \theta},$$

$$1 + \cot^2 \theta = \csc^2 \theta.$$

3.11.3 Double Angle Formulas

Double Angle Formulas

$$\sin 2\theta = 2 \sin \theta \cos \theta,$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta,$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}.$$

Derivation of Double Angle Formulas**Derivation of $\sin 2\theta$:**

Using the compound angle formula $\sin(A + B)$ (proved below) with $A = B = \theta$:

$$\sin 2\theta = \sin(\theta + \theta) = \sin \theta \cos \theta + \cos \theta \sin \theta = 2 \sin \theta \cos \theta.$$

Derivation of $\cos 2\theta$:

Using $\cos(A + B)$ with $A = B = \theta$:

$$\cos 2\theta = \cos(\theta + \theta) = \cos \theta \cos \theta - \sin \theta \sin \theta = \cos^2 \theta - \sin^2 \theta.$$

Alternative forms using $\sin^2 \theta + \cos^2 \theta = 1$:

$$\cos 2\theta = \cos^2 \theta - (1 - \cos^2 \theta) = 2 \cos^2 \theta - 1,$$

$$\cos 2\theta = (1 - \sin^2 \theta) - \sin^2 \theta = 1 - 2 \sin^2 \theta.$$

Derivation of $\tan 2\theta$:

Using $\tan(A + B)$ with $A = B = \theta$:

$$\tan 2\theta = \tan(\theta + \theta) = \frac{\tan \theta + \tan \theta}{1 - \tan \theta \cdot \tan \theta} = \frac{2 \tan \theta}{1 - \tan^2 \theta}.$$

Choosing the Right Form of $\cos 2\theta$

- Use $\cos 2\theta = 2 \cos^2 \theta - 1$ when the problem involves $\cos^2 \theta$ or when you want to express $\cos^2 \theta$ in terms of $\cos 2\theta$.
- Use $\cos 2\theta = 1 - 2 \sin^2 \theta$ when the problem involves $\sin^2 \theta$.
- Use $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ when both $\sin \theta$ and $\cos \theta$ appear.

3.11.4 Compound Angle (Addition) Formulas**Compound Angle Formulas**

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B,$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B,$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}.$$

Note: In $\cos(A \pm B)$, the sign on the right is *opposite* to the sign on the left.

Derivation of Compound Angle Formulas

Proof of $\cos(A + B) = \cos A \cos B - \sin A \sin B$:

Consider the unit circle. Let point P be at angle A and point Q at angle $-(A + B)$. The chord length PQ can be computed in two ways using the distance formula.

Alternatively, using rotation matrices or Euler's formula. Here we use a classical geometric proof.

Consider the unit circle. Let $P_1 = (\cos A, \sin A)$, $P_2 = (\cos(-B), \sin(-B)) = (\cos B, -\sin B)$, $P_3 = (\cos(A + B), \sin(A + B))$, and $P_4 = (1, 0)$.

Rotating the entire figure by angle A sends P_4 to P_1 and P_2 to P_3 . Since rotation preserves distances:

$$|P_1 P_2|^2 = |P_3 P_4|^2.$$

Compute $|P_1 P_2|^2$:

$$\begin{aligned} |P_1 P_2|^2 &= (\cos A - \cos B)^2 + (\sin A + \sin B)^2 \\ &= \cos^2 A - 2 \cos A \cos B + \cos^2 B + \sin^2 A + 2 \sin A \sin B + \sin^2 B \\ &= (\cos^2 A + \sin^2 A) + (\cos^2 B + \sin^2 B) - 2 \cos A \cos B + 2 \sin A \sin B \\ &= 2 - 2 \cos A \cos B + 2 \sin A \sin B. \end{aligned}$$

Compute $|P_3 P_4|^2$:

$$\begin{aligned} |P_3 P_4|^2 &= (\cos(A + B) - 1)^2 + \sin^2(A + B) \\ &= \cos^2(A + B) - 2 \cos(A + B) + 1 + \sin^2(A + B) \\ &= 2 - 2 \cos(A + B). \end{aligned}$$

Setting equal:

$$\begin{aligned} 2 - 2 \cos(A + B) &= 2 - 2 \cos A \cos B + 2 \sin A \sin B, \\ \cos(A + B) &= \cos A \cos B - \sin A \sin B. \end{aligned}$$

Proof of $\cos(A - B) = \cos A \cos B + \sin A \sin B$:

Replace B with $-B$ in the formula for $\cos(A + B)$:

$$\cos(A - B) = \cos A \cos(-B) - \sin A \sin(-B) = \cos A \cos B + \sin A \sin B.$$

Proof of $\sin(A + B) = \sin A \cos B + \cos A \sin B$:

Use the cofunction identity $\sin \theta = \cos(90^\circ - \theta)$:

$$\begin{aligned} \sin(A + B) &= \cos(90^\circ - (A + B)) = \cos((90^\circ - A) - B) \\ &= \cos(90^\circ - A) \cos B + \sin(90^\circ - A) \sin B \\ &= \sin A \cos B + \cos A \sin B. \end{aligned}$$

Proof of $\sin(A - B) = \sin A \cos B - \cos A \sin B$:

Replace B with $-B$:

$$\sin(A - B) = \sin A \cos(-B) + \cos A \sin(-B) = \sin A \cos B - \cos A \sin B.$$

Proof of $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$:

$$\tan(A + B) = \frac{\sin(A + B)}{\cos(A + B)} = \frac{\sin A \cos B + \cos A \sin B}{\cos A \cos B - \sin A \sin B}.$$

Divide numerator and denominator by $\cos A \cos B$:

$$\tan(A + B) = \frac{\frac{\sin A}{\cos A} + \frac{\sin B}{\cos B}}{1 - \frac{\sin A}{\cos A} \cdot \frac{\sin B}{\cos B}} = \frac{\tan A + \tan B}{1 - \tan A \tan B}.$$

3.11.5 Reduction Formulas

Reduction (Reference Angle) Formulas

$\sin(\pi - \theta) = \sin \theta,$	$\cos(\pi - \theta) = -\cos \theta,$	$\tan(\pi - \theta) = -\tan \theta.$
$\sin(\pi + \theta) = -\sin \theta,$	$\cos(\pi + \theta) = -\cos \theta,$	$\tan(\pi + \theta) = \tan \theta.$
$\sin(2\pi - \theta) = -\sin \theta,$	$\cos(2\pi - \theta) = \cos \theta,$	$\tan(2\pi - \theta) = -\tan \theta.$
$\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta,$	$\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta,$	$\tan\left(\frac{\pi}{2} - \theta\right) = \cot \theta.$
$\sin\left(\frac{\pi}{2} + \theta\right) = \cos \theta,$	$\cos\left(\frac{\pi}{2} + \theta\right) = -\sin \theta,$	$\tan\left(\frac{\pi}{2} + \theta\right) = -\cot \theta.$

ASTC Rule (All, Sine, Tangent, Cosine)

In each quadrant:

- **All** trig functions are positive in the first quadrant ($0 < \theta < \frac{\pi}{2}$).
- **Sine** is positive in the second quadrant ($\frac{\pi}{2} < \theta < \pi$).
- **Tangent** is positive in the third quadrant ($\pi < \theta < \frac{3\pi}{2}$).
- **Cosine** is positive in the fourth quadrant ($\frac{3\pi}{2} < \theta < 2\pi$).

3.11.6 Proof Strategies for Trigonometric Identities

Strategies for Proving Trigonometric Identities

1. **Work on one side:** Transform the more complicated side until it matches the simpler side.
2. **Work on both sides:** Simplify each side independently and show they are equal.
3. **Convert to sine and cosine:** If the identity involves $\tan$, $\sec$, $\csc$, $\cot$, convert everything to $\sin$ and $\cos$.
4. **Look for known identities:** Factor, expand, or substitute using Pythagorean identities, double angle formulas, or compound angle formulas.
5. **Common factorizations:**
 - $1 - 2\sin^2 \theta = \cos 2\theta$
 - $\sin^2 \theta - \cos^2 \theta = -\cos 2\theta$
 - $\sin \theta + \cos \theta = \sqrt{2} \sin\left(\theta + \frac{\pi}{4}\right)$
6. **Never cross-multiply or square both sides** unless you check for extraneous solutions.

Example: Proving a Trigonometric Identity

Problem. Prove that $\frac{1 - \cos 2\theta}{\sin 2\theta} = \tan \theta$.

Proof. Starting with the left-hand side:

$$\begin{aligned}\frac{1 - \cos 2\theta}{\sin 2\theta} &= \frac{1 - (1 - 2\sin^2 \theta)}{2\sin \theta \cos \theta} \\ &= \frac{2\sin^2 \theta}{2\sin \theta \cos \theta} \\ &= \frac{\cancel{2}\sin\cancel{\theta} \cdot \sin \theta}{\cancel{2}\sin\cancel{\theta} \cdot \cos \theta} \\ &= \frac{\sin \theta}{\cos \theta} = \tan \theta.\end{aligned}$$

Example: Using Double Angle and Compound Angle Formulas

Problem. Express $\cos 3\theta$ in terms of $\cos \theta$.

Solution.

$$\begin{aligned}\cos 3\theta &= \cos(2\theta + \theta) \\ &= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta \\ &= (2\cos^2 \theta - 1)\cos \theta - (2\sin \theta \cos \theta)\sin \theta \\ &= 2\cos^3 \theta - \cos \theta - 2\sin^2 \theta \cos \theta \\ &= 2\cos^3 \theta - \cos \theta - 2(1 - \cos^2 \theta)\cos \theta \\ &= 2\cos^3 \theta - \cos \theta - 2\cos \theta + 2\cos^3 \theta \\ &= 4\cos^3 \theta - 3\cos \theta.\end{aligned}$$

This is the triple angle formula for cosine.

3.12 Reciprocal and Inverse Trig Functions

3.12.1 Reciprocal Trig Functions

Definition 3.12.1.

$$\begin{aligned}\sec \theta &= \frac{1}{\cos \theta}, & \cos \theta &\neq 0, \\ \csc \theta &= \frac{1}{\sin \theta}, & \sin \theta &\neq 0, \\ \cot \theta &= \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta}, & \sin \theta &\neq 0.\end{aligned}$$

IB Notation

The IB uses $\csc \theta$ (with the “o”), not $\operatorname{csc} \theta$. Both notations are understood, but use the IB convention on exams.

3.12.2 Inverse Trigonometric Functions

Definition 3.12.2. The *inverse trigonometric functions* (also called *arcfunctions*) are defined as follows:

$$y = \arcsin x \iff \sin y = x, \quad x \in [-1, 1], \quad y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right],$$

$$y = \arccos x \iff \cos y = x, \quad x \in [-1, 1], \quad y \in [0, \pi],$$

$$y = \arctan x \iff \tan y = x, \quad x \in \mathbb{R}, \quad y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Domains and Ranges of Inverse Trig Functions

Function	Domain	Range
$\arcsin x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\arccos x$	$[-1, 1]$	$[0, \pi]$
$\arctan x$	$(-\infty, \infty)$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

3.12.3 Key Properties of Inverse Trig Functions

Properties of Inverse Trig Functions

$$\arcsin(-x) = -\arcsin x,$$

$$\arccos(-x) = \pi - \arccos x,$$

$$\arctan(-x) = -\arctan x.$$

Composition with the original function:

$$\sin(\arcsin x) = x, \quad x \in [-1, 1],$$

$$\cos(\arccos x) = x, \quad x \in [-1, 1],$$

$$\tan(\arctan x) = x, \quad x \in \mathbb{R}.$$

Reverse composition (restricted):

$$\arcsin(\sin x) = x, \quad x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right],$$

$$\arccos(\cos x) = x, \quad x \in [0, \pi],$$

$$\arctan(\tan x) = x, \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Why the Restricted Ranges?

Without restricting the ranges, the inverse trig functions would not be functions (they would fail the vertical line test). For example, $\sin x = \frac{1}{2}$ has infinitely many solutions, but $\arcsin \frac{1}{2} = \frac{\pi}{6}$ is the *unique* value in the range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Key observation: $\arcsin$ and $\arctan$ are odd functions, but $\arccos$ is *not* odd. This is because the range $[0, \pi]$ is not symmetric about zero.

Example: Evaluating Inverse Trig Functions

Problem. Evaluate the following:

(i) $\arcsin\left(\frac{\sqrt{3}}{2}\right)$

(ii) $\arccos\left(-\frac{1}{2}\right)$

(iii) $\arctan(1) + \arctan(-1)$

Solution.

(i) We need $\sin y = \frac{\sqrt{3}}{2}$ with $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. So $y = \frac{\pi}{3}$.

(ii) We need $\cos y = -\frac{1}{2}$ with $y \in [0, \pi]$. So $y = \frac{2\pi}{3}$.

(iii) $\arctan(1) = \frac{\pi}{4}$ and $\arctan(-1) = -\frac{\pi}{4}$. Their sum is 0.

Example: Composing Inverse Trig Functions

Problem. Find $\sin(2 \arccos \frac{3}{5})$.

Solution. Let $\theta = \arccos \frac{3}{5}$, so $\cos \theta = \frac{3}{5}$ and $\theta \in [0, \pi]$. Then $\sin \theta = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$ (positive since $\theta \in [0, \pi]$).

Using the double angle formula:

$$\sin 2\theta = 2 \sin \theta \cos \theta = 2 \cdot \frac{4}{5} \cdot \frac{3}{5} = \frac{24}{25}.$$

3.13 Solving Trigonometric Equations

Solving trigonometric equations is a major part of the IB syllabus. The key is to find *all* solutions using general solution formulas, then restrict to the specified interval if needed.

3.13.1 General Solutions

General Solutions for Trigonometric Equations

If $\sin \theta = k$: $\theta = \arcsin k + 2n\pi$ or $\theta = \pi - \arcsin k + 2n\pi$, $n \in \mathbb{Z}$.

If $\cos \theta = k$: $\theta = \arccos k + 2n\pi$ or $\theta = -\arccos k + 2n\pi$, $n \in \mathbb{Z}$.

If $\tan \theta = k$: $\theta = \arctan k + n\pi$, $n \in \mathbb{Z}$.

Understanding the General Solutions

Sine: $\sin \theta = k$ has two solutions per period (2π), symmetric about $\theta = \frac{\pi}{2}$: if one solution is α , the other is $\pi - \alpha$.

Cosine: $\cos \theta = k$ has two solutions per period (2π), symmetric about $\theta = 0$: if one solution is α , the other is $-\alpha$.

Tangent: $\tan \theta = k$ has one solution per half-period (π), since the period of tangent is π .

3.13.2 Solving in Finite Intervals

Method: Solving in a Given Interval

Step 1. Find the principal value(s) using arcsin, arccos, or arctan.

Step 1. Write down *all* general solutions.

Step 1. Substitute integer values of n to find all solutions within the specified interval.

Step 1. List the solutions in order.

Common intervals on the IB exam: $[0, 2\pi]$, $[0, \pi]$, $[-\pi, \pi]$, $[0^\circ, 360^\circ]$.

3.13.3 Equations Leading to Quadratics

Some trigonometric equations can be transformed into quadratic equations by substitution.

Example: Quadratic in $\sin x$

Problem. Solve $2\sin^2 x - \sin x - 1 = 0$ for $x \in [0, 2\pi]$.

Solution. Let $u = \sin x$. Then $2u^2 - u - 1 = 0$:

$$(2u + 1)(u - 1) = 0 \implies u = -\frac{1}{2} \text{ or } u = 1.$$

Case 1: $\sin x = -\frac{1}{2}$.

$$x = \arcsin\left(-\frac{1}{2}\right) = -\frac{\pi}{6}, \quad \text{or} \quad x = \pi - \left(-\frac{\pi}{6}\right) = \frac{7\pi}{6}.$$

Adding 2π to $-\frac{\pi}{6}$: $x = \frac{11\pi}{6}$.

Solutions in $[0, 2\pi]$: $x = \frac{7\pi}{6}, \frac{11\pi}{6}$.

Case 2: $\sin x = 1$.

$$x = \frac{\pi}{2}.$$

All solutions: $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$.

3.13.4 Using Identities to Transform Equations

Example: Using Double Angle Identities**Problem.** Solve $\cos 2x + \cos x + 1 = 0$ for $x \in [0, 2\pi]$.**Solution.** Replace $\cos 2x$ using the identity $\cos 2x = 2\cos^2 x - 1$:

$$2\cos^2 x - 1 + \cos x + 1 = 0,$$

$$2\cos^2 x + \cos x = 0,$$

$$\cos x(2\cos x + 1) = 0.$$

Case 1: $\cos x = 0 \implies x = \frac{\pi}{2}$ or $x = \frac{3\pi}{2}$.**Case 2:** $\cos x = -\frac{1}{2} \implies x = \frac{2\pi}{3}$ or $x = \frac{4\pi}{3}$.**All solutions:** $x = \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{2}, \frac{4\pi}{3}$.**Example: Using Pythagorean Identity****Problem.** Solve $\sin^2 x = 3\cos^2 x$ for $x \in [0, 2\pi]$.**Solution.** Replace $\sin^2 x = 1 - \cos^2 x$:

$$1 - \cos^2 x = 3\cos^2 x,$$

$$1 = 4\cos^2 x,$$

$$\cos^2 x = \frac{1}{4},$$

$$\cos x = \pm \frac{1}{2}.$$

$$\cos x = \frac{1}{2} : x = \frac{\pi}{3}, \frac{5\pi}{3}.$$

$$\cos x = -\frac{1}{2} : x = \frac{2\pi}{3}, \frac{4\pi}{3}.$$

All solutions: $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$.**Example: IB-Style Question with Compound Angle****Problem.** Solve $\sin(x + \frac{\pi}{3}) = \frac{1}{2}$ for $x \in [-\pi, \pi]$.**Solution.** Let $u = x + \frac{\pi}{3}$. We need $\sin u = \frac{1}{2}$.General solution: $u = \frac{\pi}{6} + 2n\pi$ or $u = \frac{5\pi}{6} + 2n\pi$.So $x = u - \frac{\pi}{3}$:

$$x = \frac{\pi}{6} - \frac{\pi}{3} + 2n\pi = -\frac{\pi}{6} + 2n\pi,$$

$$x = \frac{5\pi}{6} - \frac{\pi}{3} + 2n\pi = \frac{\pi}{2} + 2n\pi.$$

For $n = 0$: $x = -\frac{\pi}{6}$ and $x = \frac{\pi}{2}$. Both are in $[-\pi, \pi]$.For $n = -1$: $x = -\frac{\pi}{6} - 2\pi = -\frac{13\pi}{6}$ (out of range) and $x = \frac{\pi}{2} - 2\pi = -\frac{3\pi}{2}$ (out of range).For $n = 1$: $x = -\frac{\pi}{6} + 2\pi = \frac{11\pi}{6}$ (out of range) and $x = \frac{\pi}{2} + 2\pi = \frac{5\pi}{2}$ (out of range).**Solutions:** $x = -\frac{\pi}{6}, \frac{\pi}{2}$.**Example: Equation with tan and Multiple Angles****Problem.** Solve $\tan 2x = \sqrt{3}$ for $x \in [0, 2\pi]$.**Solution.** $\tan 2x = \sqrt{3} \implies 2x = \frac{\pi}{3} + n\pi \implies x = \frac{\pi}{6} + \frac{n\pi}{2}$.

For $n = 0$: $x = \frac{\pi}{6}$. For $n = 1$: $x = \frac{\pi}{6} + \frac{\pi}{2} = \frac{2\pi}{3}$.
For $n = 2$: $x = \frac{\pi}{6} + \pi = \frac{7\pi}{6}$. For $n = 3$: $x = \frac{\pi}{6} + \frac{3\pi}{2} = \frac{5\pi}{3}$.
For $n = 4$: $x = \frac{\pi}{6} + 2\pi = \frac{13\pi}{6} > 2\pi$ (stop).
Solutions: $x = \frac{\pi}{6}, \frac{2\pi}{3}, \frac{7\pi}{6}, \frac{5\pi}{3}$.

IB Exam Tip: Common Pitfalls in Trig Equations

- **Forgetting solutions:** Always use the general solution formula. Counting on a diagram alone often leads to missed solutions.
- **Wrong interval:** Read the question carefully. If the interval is in degrees, give answers in degrees. If in radians, give answers in radians.
- **Extraneous solutions:** When squaring both sides or multiplying by a trig function, check that all solutions satisfy the original equation. Squaring can introduce spurious solutions.
- **Domain restrictions:** If the equation involves $\sec x$, remember that $\cos x \neq 0$. If it involves $\tan x$, remember that $\cos x \neq 0$.
- **Double-check with calculator:** After finding solutions, substitute them back into the original equation to verify.

Chapter 4

Statistics and Probability

Statistics and probability form a major component of the IB Mathematics AA HL syllabus, spanning descriptive statistics, correlation, probability theory, discrete and continuous distributions, and the normal distribution. This chapter provides every formula, derivation, and exam technique you need.

4.1 Data Representation

4.1.1 Population vs Sample

Definition 4.1.1. A **population** is the entire set of individuals or items about which information is wanted. A **sample** is a subset of the population that is actually observed.

- **Population parameters** are fixed (but often unknown) quantities, denoted by Greek letters: population mean μ , population variance σ^2 , population standard deviation σ .
- **Sample statistics** are computed from data and used to estimate population parameters, denoted by Latin letters: sample mean $\bar{x}$, sample variance s^2 , sample standard deviation s .
- A **census** collects data from every member of the population.
- A **random sample** gives each member of the population an equal chance of selection.

Definition 4.1.2. **Discrete data** can take only specific (usually integer) values, e.g., number of goals scored, number of students in a class. **Continuous data** can take any value within a range, e.g., height, weight, time, temperature.

Classifying Data in IB Exams

In IB exams, discrete data often arise from *counting* (“the number of...”), while continuous data arise from *measuring* (“the time taken to...”, “the mass of...”). If you can meaningfully take a value between two data points, the data is continuous.

4.1.2 Frequency Distributions

Definition 4.1.3. A **frequency distribution** tabulates data by grouping observations into classes (or using exact values for discrete data) and recording the frequency of each class.

For grouped continuous data, classes are written as intervals, e.g., $10 \leq x < 20$. Important conventions:

- Class boundaries must be continuous with no gaps.
- The **class width** is the difference between the upper and lower class boundaries.

- The **midpoint** (class mark) of a class with boundaries a and b is $\frac{a+b}{2}$.
- **Class width** = $b - a$.

4.1.3 Histograms

A histogram is a graphical representation of a frequency distribution using rectangular bars.

Histogram: Equal Class Widths

When all class widths are equal, the **height of each bar equals the frequency** of that class.

Histogram: Unequal Class Widths

When class widths are unequal, the **height of each bar equals the frequency density**:

$$\text{Frequency density} = \frac{\text{Frequency}}{\text{Class width}}$$

The **area** of each bar equals the class frequency:

$$\text{Area} = \text{Frequency density} \times \text{Class width} = \text{Frequency}$$

Histogram Key Insight

The key principle is: **area represents frequency**, not height. This holds regardless of whether class widths are equal or not. When widths are equal, the height happens to be proportional to frequency, but the area–frequency principle is the fundamental rule.

Worked Example: Histogram with Unequal Widths

The masses (in kg) of 50 people are summarised below:

Mass (kg)	Frequency	Class Width
$40 \leq m < 50$	8	10
$50 \leq m < 60$	15	10
$60 \leq m < 70$	14	10
$70 \leq m < 85$	10	15
$85 \leq m < 100$	3	15

Find the frequency densities.

Solution:

$$\text{Class } 40 \leq m < 50 : \quad \text{FD} = \frac{8}{10} = 0.8$$

$$\text{Class } 50 \leq m < 60 : \quad \text{FD} = \frac{15}{10} = 1.5$$

$$\text{Class } 60 \leq m < 70 : \quad \text{FD} = \frac{14}{10} = 1.4$$

$$\text{Class } 70 \leq m < 85 : \quad \text{FD} = \frac{10}{15} \approx 0.667$$

$$\text{Class } 85 \leq m < 100 : \quad \text{FD} = \frac{3}{15} = 0.2$$

On the histogram, bar heights are these frequency density values and bar widths are the class widths.

4.1.4 Cumulative Frequency Graphs

Definition 4.1.4. A **cumulative frequency (CF)** graph plots the running total of frequencies against the upper class boundary of each group.

Constructing a Cumulative Frequency Graph

1. Create a table with columns: class, frequency, upper class boundary, cumulative frequency.
2. The cumulative frequency for each class is the sum of all frequencies up to and including that class.
3. Plot points at (upper class boundary, cumulative frequency).
4. Connect points with a smooth curve (an S-shaped ogive).

Reading a Cumulative Frequency Graph

- To find the median: read off the value at $\frac{n}{2}$ on the CF axis.
- To find the lower quartile Q_1 : read off at $\frac{n}{4}$.
- To find the upper quartile Q_3 : read off at $\frac{3n}{4}$.
- To find the value corresponding to a given percentile p : read off at $\frac{p}{100} \times n$.

Worked Example: Cumulative Frequency

The weekly wages (in dollars) of 80 workers:

Wage (\$)	Frequency	Cumulative Frequency
$200 \leq w < 300$	12	12
$300 \leq w < 400$	28	40
$400 \leq w < 500$	24	64
$500 \leq w < 600$	11	75
$600 \leq w < 800$	5	80

Estimate the median and interquartile range.

Solution:

$$\text{Median position} = \frac{80}{2} = 40$$

$$Q_1 \text{ position} = \frac{80}{4} = 20$$

$$Q_3 \text{ position} = \frac{3 \times 80}{4} = 60$$

Reading from the CF graph (by linear interpolation within the class):

- Median: at CF = 40, the value is at the upper boundary of the second class, so $Q_2 = \$400$.
- Q_1 : at CF = 20, interpolating in $200 \leq w < 300$: $Q_1 = 200 + \frac{20-0}{12-0} \times 100 = 200 + \frac{20}{12} \times 100 \approx 366.67$.
- Q_3 : at CF = 60, interpolating in $400 \leq w < 500$: $Q_3 = 400 + \frac{60-40}{64-40} \times 100 = 400 + \frac{20}{24} \times 100 \approx 483.33$.
- IQR = $Q_3 - Q_1 \approx 483.33 - 366.67 = 116.67$.

4.1.5 Box-and-Whisker Diagrams

Definition 4.1.5. A **box-and-whisker plot** (box plot) is a standardised way of displaying the five-number summary of a dataset: minimum, Q_1 , median (Q_2), Q_3 , and maximum.

Five-Number Summary and Box Plot

The box plot is constructed as follows:

1. Draw a horizontal (or vertical) number line covering the range of the data.
2. Draw a box from Q_1 to Q_3 (the box spans the IQR).
3. Draw a vertical line inside the box at the median Q_2 .
4. Extend whiskers from Q_1 to the minimum value, and from Q_3 to the maximum value.

Finding Quartiles from Raw Data

For an ordered dataset $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$:

- The **lower quartile** Q_1 is the median of the lower half of the data.
- The **upper quartile** Q_3 is the median of the upper half of the data.
- The **interquartile range** (IQR) is $\text{IQR} = Q_3 - Q_1$.

4.1.6 Outlier Detection

Definition 4.1.6. An **outlier** is a data value that lies an abnormal distance from other values in the dataset. For IB Mathematics, outliers are identified using the $1.5 \times \text{IQR}$ rule.

Outlier Detection Rule

A data point x is considered a **potential outlier** if:

$$x < Q_1 - 1.5 \times \text{IQR} \quad \text{or} \quad x > Q_3 + 1.5 \times \text{IQR}$$

These boundaries are sometimes called the **fences**:

$$\text{Lower fence} = Q_1 - 1.5 \text{IQR}, \quad \text{Upper fence} = Q_3 + 1.5 \text{IQR}$$

Worked Example: Outlier Detection

A dataset has $Q_1 = 25$, $Q_3 = 45$, and values: 5, 18, 22, 27, 30, 32, 35, 40, 48, 55, 72.

Identify any outliers.

Solution:

$$\text{IQR} = 45 - 25 = 20$$

$$\text{Lower fence} = 25 - 1.5 \times 20 = 25 - 30 = -5$$

$$\text{Upper fence} = 45 + 1.5 \times 20 = 45 + 30 = 75$$

Values below -5 or above 75 are outliers. Here, no value lies outside $[-5, 75]$, so there are **no outliers** in this dataset.

IB Exam Tip: Outliers on Box Plots

When a box plot shows a data point plotted separately (beyond a whisker), this indicates an outlier. On IB exams, you may be asked to identify outliers *before* drawing the box plot, in which case the whiskers extend only to the most extreme *non-outlier* values (not the minimum and maximum).

4.2 Measures of Central Tendency and Dispersion

4.2.1 The Mean

Definition 4.2.1. The **mean** (arithmetic average) of n observations $x_1, x_2, \dots, x_n$ is:

Arithmetic Mean

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

For grouped data with class midpoints x_i and frequencies f_i :

Mean of Grouped Data

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

where x_i is the midpoint of the i -th class and f_i is its frequency.

4.2.2 The Median and Mode

Definition 4.2.2. The **median** is the middle value when data are arranged in order. For n data points:

- If n is odd: $\text{median} = x_{\frac{n+1}{2}}$.
- If n is even: $\text{median} = \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}$.

Definition 4.2.3. The **mode** is the most frequently occurring value in the dataset. A dataset can be unimodal, bimodal, multimodal, or have no mode.

Choosing a Measure of Central Tendency

- The **mean** uses all data values but is affected by outliers.
- The **median** is robust to outliers and preferred for skewed distributions.
- The **mode** is the only measure applicable to categorical data.
- For a symmetric distribution: $\text{mean} \approx \text{median} \approx \text{mode}$.
- For right-skewed data: $\text{mode} < \text{median} < \text{mean}$.
- For left-skewed data: $\text{mean} < \text{median} < \text{mode}$.

4.2.3 Range and Interquartile Range

Range and IQR

$$\text{Range} = x_{\max} - x_{\min}$$

$$\text{IQR} = Q_3 - Q_1$$

The IQR is preferred over the range because it is not affected by extreme values.

4.2.4 Variance

Definition 4.2.4. **Variance** measures the average squared deviation from the mean. There are two versions depending on whether the data represent a population or a sample.

Population Variance

When the data represent the **entire population**:

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

Sample Variance (Unbiased Estimator)

When the data are a **sample** drawn from a larger population:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

Derivation: Computational Formula for Population Variance

Starting from the definition:

$$\begin{aligned}\sigma^2 &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= \frac{1}{n} \sum_{i=1}^n (x_i^2 - 2x_i\bar{x} + \bar{x}^2) \\ &= \frac{1}{n} \left[\sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + \sum_{i=1}^n \bar{x}^2 \right]\end{aligned}$$

Since $\sum_{i=1}^n x_i = n\bar{x}$ and $\sum_{i=1}^n \bar{x}^2 = n\bar{x}^2$:

$$\begin{aligned}\sigma^2 &= \frac{1}{n} \left[\sum_{i=1}^n x_i^2 - 2\bar{x}(n\bar{x}) + n\bar{x}^2 \right] \\ &= \frac{1}{n} \left[\sum_{i=1}^n x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 \right] \\ &= \frac{1}{n} \left[\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right] \\ &= \frac{\sum_{i=1}^n x_i^2}{n} - \bar{x}^2\end{aligned}$$

Computational Formula for Variance

$$\sigma^2 = \frac{\sum x_i^2}{n} - \bar{x}^2$$

Similarly, for the sample variance:

$$s^2 = \frac{\sum x_i^2 - n\bar{x}^2}{n - 1}$$

4.2.5 Standard Deviation

Standard Deviation

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}, \quad s = \sqrt{s^2} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

The standard deviation has the same units as the original data, making it more interpretable than variance.

4.2.6 Effect of Linear Transformations

Theorem 4.2.1 (Linear Transformation of Data). *If $Y = aX + b$ where $a, b \in \mathbb{R}$ are constants, then:*

$$\begin{aligned}\bar{y} &= a\bar{x} + b \\ \sigma_Y &= |a| \sigma_X \\ s_Y &= |a| s_X\end{aligned}$$

The standard deviation is scaled by $|a|$ but is unaffected by b (adding a constant shifts data but does not change spread).

Derivation: Transformation Properties

Let $y_i = ax_i + b$ for each observation. Then:

$$\begin{aligned}\bar{y} &= \frac{1}{n} \sum_{i=1}^n y_i = \frac{1}{n} \sum_{i=1}^n (ax_i + b) \\ &= \frac{1}{n} \left(a \sum_{i=1}^n x_i + nb \right) = a \cdot \frac{\sum x_i}{n} + b = a\bar{x} + b\end{aligned}$$

For the variance:

$$\begin{aligned}\sigma_Y^2 &= \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 \\ &= \frac{1}{n} \sum_{i=1}^n [(ax_i + b) - (a\bar{x} + b)]^2 \\ &= \frac{1}{n} \sum_{i=1}^n (ax_i - a\bar{x})^2 \\ &= \frac{1}{n} \sum_{i=1}^n a^2 (x_i - \bar{x})^2 \\ &= a^2 \cdot \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = a^2 \sigma_X^2\end{aligned}$$

Therefore $\sigma_Y = |a| \sigma_X$.

IB Exam Tip: n vs $n - 1$

- Use σ (dividing by n) when the data represent the **entire population**.
- Use s (dividing by $n - 1$) when the data are a **sample** from a larger population.

- **Critical:** On your GDC, the “sample standard deviation” function (usually denoted s_x or σ_{n-1}) divides by $n - 1$. The “population standard deviation” (σ_x or σ_n) divides by n .
- In IB exam questions, *always check the wording*: if the question says “calculate the standard deviation of these 20 marks” without mentioning a larger population, use n . If it says “20 students were selected from a school of 500 to estimate the population standard deviation”, use $n - 1$.
- When in doubt, read the question carefully. IB questions are usually explicit about whether the data is a sample or the population.

Worked Example: Mean, Variance, and Transformations

The marks of 6 students in a test (out of 40) are: 28, 35, 18, 32, 25, 22.

(a) Find the mean and standard deviation.

(b) The teacher decides to convert the marks to a percentage by multiplying by 2.5 and adding 0. Find the new mean and standard deviation.

Solution:

$$\bar{x} = \frac{28 + 35 + 18 + 32 + 25 + 22}{6} = \frac{160}{6} = 26.67$$

Using the computational formula:

$$\begin{aligned}\sum x_i^2 &= 28^2 + 35^2 + 18^2 + 32^2 + 25^2 + 22^2 \\ &= 784 + 1225 + 324 + 1024 + 625 + 484 = 4466 \\ \sigma^2 &= \frac{4466}{6} - 26.67^2 = 744.33 - 711.11 = 33.22 \\ \sigma &= \sqrt{33.22} \approx 5.76\end{aligned}$$

(b) Here $Y = 2.5X + 0$:

$$\begin{aligned}\bar{y} &= 2.5 \times 26.67 = 66.67\% \\ \sigma_Y &= 2.5 \times 5.76 = 14.41\end{aligned}$$

4.3 Correlation and Regression

4.3.1 Scatter Diagrams

Definition 4.3.1. A **scatter diagram** (scatter plot) displays pairs of bivariate data points (x_i, y_i) on a Cartesian plane. It is the first step in investigating a relationship between two variables.

- **Positive correlation:** as x increases, y tends to increase.
- **Negative correlation:** as x increases, y tends to decrease.
- **No correlation:** no apparent linear relationship.
- **Strong vs weak:** describes how tightly clustered points are around a line.

4.3.2 Pearson’s Correlation Coefficient

Definition 4.3.2. *Pearson’s product-moment correlation coefficient* r measures the strength and direction of the linear association between two variables.

Pearson's Correlation Coefficient

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$$

where:

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - n\bar{x}\bar{y}$$

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2$$

$$S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum y_i^2 - n\bar{y}^2$$

Properties of r

- $-1 \leq r \leq 1$.
- $r = 1$: perfect positive linear correlation.
- $r = -1$: perfect negative linear correlation.
- $r = 0$: no linear correlation (but there may be a non-linear relationship).
- r is dimensionless (unit-free).

Interpreting r in Context

In IB exams, you must interpret r *in context*. For example, do not write “ $r = 0.85$ means strong correlation.” Instead write: “There is a strong positive linear correlation between the temperature and the number of ice creams sold, suggesting that as temperature increases, ice cream sales tend to increase.” Always address both **strength** and **direction**, and note that correlation does *not* imply causation.

4.3.3 Regression Line of y on x

Definition 4.3.3. The **least squares regression line of y on x** is the line $y = ax + b$ that minimises the sum of squared vertical deviations $\sum (y_i - \hat{y}_i)^2$.

Regression Line of y on x

$$y = ax + b$$

where:

$$a = \frac{S_{xy}}{S_{xx}}, \quad b = \bar{y} - a\bar{x}$$

The line always passes through the mean point $(\bar{x}, \bar{y})$.

Derivation: Least Squares Regression Formulas

We wish to find a and b that minimise:

$$S(a, b) = \sum_{i=1}^n (y_i - ax_i - b)^2$$

Substituting $b = \bar{y} - a\bar{x}$ (which minimises S with respect to b when we set $\frac{\partial S}{\partial b} = 0$):

$$\begin{aligned} S(a) &= \sum_{i=1}^n [y_i - ax_i - (\bar{y} - a\bar{x})]^2 \\ &= \sum_{i=1}^n [(y_i - \bar{y}) - a(x_i - \bar{x})]^2 \\ &= \sum_{i=1}^n [(y_i - \bar{y})^2 - 2a(x_i - \bar{x})(y_i - \bar{y}) + a^2(x_i - \bar{x})^2] \\ &= S_{yy} - 2a S_{xy} + a^2 S_{xx} \end{aligned}$$

This is a quadratic in a . The minimum occurs at:

$$\begin{aligned} \frac{dS}{da} &= -2S_{xy} + 2a S_{xx} = 0 \\ a &= \frac{S_{xy}}{S_{xx}} \end{aligned}$$

To verify this gives a minimum: $\frac{d^2 S}{da^2} = 2S_{xx} > 0$ (since $S_{xx} > 0$ unless all x_i are equal).

For b , setting $\frac{\partial S}{\partial b} = 0$:

$$\begin{aligned} \frac{\partial S}{\partial b} &= -2 \sum_{i=1}^n (y_i - ax_i - b) = 0 \\ \sum y_i &= a \sum x_i + nb \\ \bar{y} &= a\bar{x} + b \\ b &= \bar{y} - a\bar{x} \end{aligned}$$

Interpretation of a and b

- b is the y -intercept: the predicted value of y when $x = 0$. However, $x = 0$ may not be within the data range, making b meaningless in some contexts.
- a is the gradient (slope): the predicted *change in y* for a one-unit increase in x .
- In IB exams, always interpret a and b in context with appropriate units.

4.3.4 Regression Line of x on y (HL)

Definition 4.3.4. The **regression line of x on y** minimises the sum of squared horizontal deviations. It is used when we wish to predict x from y .

Regression Line of x on y

$$x = a'y + b'$$

where:

$$a' = \frac{S_{xy}}{S_{yy}}, \quad b' = \bar{x} - a'\bar{y}$$

This can be rearranged to: $y = \frac{1}{a'}x - \frac{b'}{a'}$.

Why Two Regression Lines?

The regression line of y on x minimises vertical residuals (used when predicting y from x).
 The regression line of x on y minimises horizontal residuals (used when predicting x from y).
 The two lines are different unless $|r| = 1$. Both pass through the point $(\bar{x}, \bar{y})$.

IB Exam Tip: Interpolation vs Extrapolation

- **Interpolation:** predicting within the range of observed data. This is reliable.
- **Extrapolation:** predicting outside the range of observed data. This is unreliable and should be treated with caution.
- In IB mark schemes, you may lose marks for making extrapolations without commenting on their unreliability.
- Always check whether a requested prediction falls within the data range before using the regression line.

Worked Example: Correlation and Regression

The data below show the temperature (x , in $^{\circ}\text{C}$) and ice cream sales (y , in hundreds) for 5 days:

x	15	20	25	30	35
y	2.1	3.5	5.0	6.2	8.1

- (a) Find the correlation coefficient r .
 (b) Find the equation of the regression line of y on x .
 (c) Estimate the sales when the temperature is 28°C .

Solution:

$$\bar{x} = \frac{15 + 20 + 25 + 30 + 35}{5} = 25$$

$$\bar{y} = \frac{2.1 + 3.5 + 5.0 + 6.2 + 8.1}{5} = 4.98$$

$$\sum x_i^2 = 225 + 400 + 625 + 900 + 1225 = 3375$$

$$\sum y_i^2 = 4.41 + 12.25 + 25 + 38.44 + 65.61 = 145.71$$

$$\sum x_i y_i = 31.5 + 70 + 125 + 186 + 283.5 = 696$$

$$S_{xx} = 3375 - 5(25)^2 = 3375 - 3125 = 250$$

$$S_{yy} = 145.71 - 5(4.98)^2 = 145.71 - 124.002 = 21.708$$

$$S_{xy} = 696 - 5(25)(4.98) = 696 - 622.5 = 73.5$$

(a)

$$r = \frac{73.5}{\sqrt{250 \times 21.708}} = \frac{73.5}{\sqrt{5427}} = \frac{73.5}{73.67} \approx 0.998$$

(b)

$$a = \frac{S_{xy}}{S_{xx}} = \frac{73.5}{250} = 0.294$$

$$b = \bar{y} - a\bar{x} = 4.98 - 0.294(25) = 4.98 - 7.35 = -2.37$$

So $y = 0.294x - 2.37$ (sales in hundreds).

(c) At $x = 28$:

$$\hat{y} = 0.294(28) - 2.37 = 8.232 - 2.37 = 5.862$$

Estimated sales: 586 ice creams (since y is in hundreds). This is an interpolation, so the estimate is reliable.

4.4 Probability Fundamentals

4.4.1 Sample Space, Events, and Outcomes

Definition 4.4.1. • An **experiment** (or trial) is any process that produces an outcome.

- The **sample space** U (or Ω or S) is the set of all possible outcomes.
- An **event** A is a subset of the sample space: $A \subseteq U$.
- An **outcome** is a single element of the sample space.

4.4.2 Basic Probability

Probability of an Event

For an event A in a sample space U where all outcomes are equally likely:

$$P(A) = \frac{n(A)}{n(U)} = \frac{\text{number of outcomes in } A}{\text{total number of outcomes}}$$

- $0 \leq P(A) \leq 1$ for any event A .
- $P(U) = 1$, $P(\emptyset) = 0$.

4.4.3 Complementary Events

Complement Rule

The complement of event A , denoted A' (or $\bar{A}$), is the event that A does *not* occur:

$$P(A') = 1 - P(A)$$

4.4.4 Combined Events

Addition Rule (General)

For any two events A and B :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Mutually Exclusive Events

Events A and B are **mutually exclusive** (disjoint) if they cannot both occur, i.e., $A \cap B = \emptyset$. In this case:

$$P(A \cap B) = 0 \quad \text{and} \quad P(A \cup B) = P(A) + P(B)$$

4.4.5 Venn Diagrams, Tree Diagrams, and Sample Space Diagrams

Visualising Probability

- **Venn diagrams:** best for showing relationships between 2–3 events, especially combined events and complements.
- **Tree diagrams:** best for sequential experiments with two or more stages. Multiply along branches, add between branches.
- **Sample space diagrams (grids):** best for two-stage experiments where each stage has a small number of equally likely outcomes (e.g., rolling two dice).

Tree Diagram Rules

- The probability on each branch = $P(\text{that outcome} \mid \text{previous outcome})$.
- Multiply probabilities **along** branches to get the probability of a path.
- Add probabilities of mutually exclusive paths to get the total probability of an event.

IB Exam Tips on Probability Questions

- Always check whether events are mutually exclusive before using $P(A \cup B) = P(A) + P(B)$.
- For “at least one” problems, use the complement: $P(\text{at least one}) = 1 - P(\text{none})$.
- For “or” problems, use the addition rule. For “and” problems, use multiplication (or conditional probability).
- Read carefully: “given that” or “knowing that” signals conditional probability (Section 4.5).
- IB exams often combine a Venn diagram question with a follow-up conditional probability question.

Worked Example: Probability with Venn Diagram

In a class of 30 students, 18 play football, 15 play basketball, and 5 play neither. A student is chosen at random.

(a) Draw a Venn diagram and find the probability the student plays both sports.

(b) Find the probability the student plays exactly one sport.

Solution:

(a) Let F = plays football, B = plays basketball.

$$n(F \cup B) = 30 - 5 = 25$$

$$n(F \cup B) = n(F) + n(B) - n(F \cap B)$$

$$25 = 18 + 15 - n(F \cap B)$$

$$n(F \cap B) = 8$$

So: $n(F \cap B) = 8$, $n(F \cap B') = 18 - 8 = 10$, $n(F' \cap B) = 15 - 8 = 7$, $n(F' \cap B') = 5$.

$$P(F \cap B) = \frac{8}{30} = \frac{4}{15}$$

(b)

$$P(\text{exactly one sport}) = \frac{10 + 7}{30} = \frac{17}{30}$$

4.5 Conditional Probability and Independence

4.5.1 Conditional Probability

Definition 4.5.1. The **conditional probability** of A given B , denoted $P(A|B)$, is the probability that A occurs given that B has already occurred.

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad \text{provided } P(B) > 0$$

Understanding Conditional Probability Intuitively

$P(A|B)$ restricts the sample space to only those outcomes where B occurs. It asks: “Among all outcomes where B happens, what fraction also have A happening?”

If using a Venn diagram: $P(A|B) = \frac{n(A \cap B)}{n(B)}$ (provided equally likely outcomes).

4.5.2 Independent Events

Definition 4.5.2. Two events A and B are **independent** if the occurrence of one does not affect the probability of the other:

$$P(A \cap B) = P(A) \times P(B)$$

Equivalently:

$$P(A|B) = P(A)$$

$$P(B|A) = P(B)$$

Theorem 4.5.1 (Independence of A and B'). If A and B are independent events, then A and B' are also independent.

Proof. We need to show that $P(A \cap B') = P(A) \times P(B')$.

From the Venn diagram (or addition rule), since $A = (A \cap B) \cup (A \cap B')$ and these are mutually exclusive:

$$\begin{aligned} P(A) &= P(A \cap B) + P(A \cap B') \\ P(A \cap B') &= P(A) - P(A \cap B) \end{aligned}$$

Since A and B are independent, $P(A \cap B) = P(A) \times P(B)$:

$$\begin{aligned} P(A \cap B') &= P(A) - P(A) \times P(B) \\ &= P(A)[1 - P(B)] \\ &= P(A) \times P(B') \end{aligned}$$

Therefore A and B' are independent. By the same argument, A' and B are independent, and A' and B' are independent. $\square$

Mutually Exclusive vs Independent

- **Mutually exclusive:** $A \cap B = \emptyset$, so $P(A \cap B) = 0$.
- **Independent:** $P(A \cap B) = P(A) \times P(B)$.
- If A and B are both mutually exclusive and $P(A) > 0$ and $P(B) > 0$, they **cannot** be independent (since $P(A) \times P(B) \neq 0$).
- Conversely, independent events with positive probability are **never** mutually exclusive.

- The only way events can be both independent and mutually exclusive is if at least one has probability zero.

Worked Example: Conditional Probability and Independence

A bag contains 5 red and 3 blue marbles. Two marbles are drawn **without replacement**.

(a) Find the probability that both are red.

(b) Find the probability that the second is red given that the first is red.

(c) Are the events “first marble is red” and “second marble is red” independent?

Solution:

Let R_1 = first marble is red, R_2 = second marble is red.

(a)

$$P(R_1 \cap R_2) = P(R_1) \times P(R_2|R_1) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$$

(b)

$$P(R_2|R_1) = \frac{4}{7}$$

(c) For independence, we need $P(R_2|R_1) = P(R_2)$. Since $P(R_2|R_1) = \frac{4}{7}$ but $P(R_2) = \frac{5}{8}$ (by symmetry), $\frac{4}{7} \neq \frac{5}{8}$, so the events are **not independent**. This makes sense: drawing without replacement changes the composition of the bag.

Note: if drawing **with replacement**, the events *would* be independent since $P(R_2|R_1) = \frac{5}{8} = P(R_2)$.

4.6 Bayes' Theorem

4.6.1 Bayes' Theorem Formula

Theorem 4.6.1 (Bayes' Theorem). For events A and B with $P(B) > 0$:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Derivation of Bayes' Theorem

Starting from the definition of conditional probability:

For $P(A|B)$:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

For $P(B|A)$:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \implies P(A \cap B) = P(B|A) \cdot P(A)$$

Substituting:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

4.6.2 Extended (Law of Total Probability) Form

Law of Total Probability

If $A_1, A_2, \dots, A_n$ are mutually exclusive and exhaustive events (i.e., they partition the sample space), then for any event B :

$$P(B) = \sum_{i=1}^n P(B|A_i) \cdot P(A_i)$$

Bayes' Theorem: Extended Form

For a partition A_1, A_2, A_3 of the sample space:

$$P(A_k|B) = \frac{P(B|A_k) \cdot P(A_k)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + P(B|A_3)P(A_3)}$$

For two events A and A' :

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B|A)P(A) + P(B|A')P(A')}$$

4.6.3 Tree Diagram Approach

Tree Diagrams vs Bayes' Formula

For IB exams, the **tree diagram approach** is often simpler and less error-prone than using Bayes' formula directly. The procedure is:

1. Draw a tree diagram with the **prior** probabilities on the first set of branches (e.g., $P(A)$, $P(B)$, $P(C)$).
2. On the second set of branches, write the **conditional** probabilities (e.g., $P(\text{positive}|A)$, $P(\text{positive}|B)$).
3. For “reverse” questions (finding $P(A|\text{positive})$):
 - (a) Find the probability of each path leading to the observed outcome.
 - (b) Sum these to get $P(\text{observed outcome})$.
 - (c) Divide the relevant path probability by the sum.

This is equivalent to Bayes' theorem but less prone to algebraic errors.

Worked Example: Bayes' Theorem (Medical Testing)

A disease affects 1% of a population. A test for the disease has a 95% true positive rate ($P(+|D) = 0.95$) and a 10% false positive rate ($P(+|D') = 0.10$).

A person tests positive. Find the probability they actually have the disease.

Solution using a tree diagram:

First branches (prior): $P(D) = 0.01$, $P(D') = 0.99$.

Second branches (conditional):

$$\begin{aligned} P(+|D) &= 0.95, & P(-|D) &= 0.05 \\ P(+|D') &= 0.10, & P(-|D') &= 0.90 \end{aligned}$$

Path probabilities for testing positive:

$$\begin{aligned} P(D \cap +) &= 0.01 \times 0.95 = 0.0095 \\ P(D' \cap +) &= 0.99 \times 0.10 = 0.0990 \end{aligned}$$

$$P(+) = 0.0095 + 0.0990 = 0.1085$$

$$P(D|+) = \frac{P(D \cap +)}{P(+)} = \frac{0.0095}{0.1085} \approx 0.0876$$

Interpretation: Even with a positive test result, there is only about an 8.8% chance the person has the disease! This is because the disease is rare (low prior probability) and the false positive rate is relatively high compared to the prevalence.

Solution using Bayes' formula:

$$P(D|+) = \frac{P(+|D) P(D)}{P(+|D) P(D) + P(+|D') P(D')} = \frac{0.95 \times 0.01}{0.95 \times 0.01 + 0.10 \times 0.99} = \frac{0.0095}{0.1085} \approx 0.0876$$

4.7 Discrete Random Variables

4.7.1 Probability Distributions

Definition 4.7.1. A **discrete random variable** X is a variable that can take a countable number of values $x_1, x_2, \dots$, each with an associated probability $P(X = x_i) = p_i$.

Requirements for a Probability Distribution

1. $0 \leq p_i \leq 1$ for all i .
2. $\sum_{\text{all } i} p_i = 1$.

A probability distribution is often displayed as a table:

x	x_1	x_2	$\dots$	x_n
$P(X = x)$	p_1	p_2	$\dots$	p_n

4.7.2 Expected Value

Definition 4.7.2. The **expected value** (or *expectation* or *mean*) of a discrete random variable X is:

Expected Value

$$E(X) = \sum_i x_i p_i = \sum_{\text{all } x} x \cdot P(X = x)$$

Also denoted μ_X .

Understanding $E(X)$

$E(X)$ is the long-run average value of X if the experiment were repeated many times. It is also called the **mean** of the probability distribution. Note that $E(X)$ need not be one of the possible values of X .

4.7.3 Variance of a Discrete Random Variable

Variance of a Discrete Random Variable

$$\text{Var}(X) = \sum_i (x_i - \mu)^2 p_i$$

where $\mu = E(X)$.

Computational Formula for Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

where:

$$E(X^2) = \sum_i x_i^2 p_i$$

(Derivation: $\text{Var}(X) = E(X^2) - [E(X)]^2$) Starting from the definition of variance:

$$\begin{aligned} \text{Var}(X) &= E[(X - \mu)^2] \\ &= E[X^2 - 2\mu X + \mu^2] \\ &= E(X^2) - 2\mu E(X) + \mu^2 \end{aligned}$$

Since $\mu = E(X)$:

$$\begin{aligned} \text{Var}(X) &= E(X^2) - 2\mu \cdot \mu + \mu^2 \\ &= E(X^2) - 2\mu^2 + \mu^2 \\ &= E(X^2) - \mu^2 \\ &= E(X^2) - [E(X)]^2 \end{aligned}$$

4.7.4 Linear Transformations of Random Variables

Theorem 4.7.1 (Effect of Linear Transformations on $E(X)$ and $\text{Var}(X)$). If $Y = aX + b$ where a, b are constants:

$$\begin{aligned} E(Y) &= a E(X) + b \\ \text{Var}(Y) &= a^2 \text{Var}(X) \end{aligned}$$

The standard deviation transforms as: $\sigma_Y = |a| \sigma_X$.

Derivation: Transformation Properties for Random Variables

Expected value:

$$\begin{aligned} E(Y) &= E(aX + b) = \sum (ax_i + b) p_i \\ &= a \sum x_i p_i + b \sum p_i \\ &= a E(X) + b \cdot 1 = a E(X) + b \end{aligned}$$

Variance:

$$\begin{aligned}
\text{Var}(Y) &= E(Y^2) - [E(Y)]^2 \\
&= E[(aX + b)^2] - [aE(X) + b]^2 \\
&= E[a^2X^2 + 2abX + b^2] - [a^2[E(X)]^2 + 2abE(X) + b^2] \\
&= a^2E(X^2) + 2abE(X) + b^2 - a^2[E(X)]^2 - 2abE(X) - b^2 \\
&= a^2[E(X^2) - [E(X)]^2] \\
&= a^2 \text{Var}(X)
\end{aligned}$$

Adding a Constant Does Not Affect Variance

Adding a constant b shifts the entire distribution but does not change its spread, so $\text{Var}(X + b) = \text{Var}(X)$. Only multiplying by a (stretching or compressing) changes the variance, and it does so by a factor of a^2 (not a).

Worked Example: Discrete Random Variable

A biased die has the probability distribution:

x	1	2	3	4	5	6
$P(X = x)$	0.1	0.2	0.3	0.15	0.15	0.1

(a) Find $E(X)$ and $\text{Var}(X)$.

(b) If $Y = 2X + 3$, find $E(Y)$ and $\text{Var}(Y)$.

Solution:

(a)

$$\begin{aligned}
E(X) &= 1(0.1) + 2(0.2) + 3(0.3) + 4(0.15) + 5(0.15) + 6(0.1) \\
&= 0.1 + 0.4 + 0.9 + 0.6 + 0.75 + 0.6 = 3.35
\end{aligned}$$

$$\begin{aligned}
E(X^2) &= 1(0.1) + 4(0.2) + 9(0.3) + 16(0.15) + 25(0.15) + 36(0.1) \\
&= 0.1 + 0.8 + 2.7 + 2.4 + 3.75 + 3.6 = 13.35
\end{aligned}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 13.35 - 3.35^2 = 13.35 - 11.2225 = 2.1275$$

(b)

$$\begin{aligned}
E(Y) &= 2(3.35) + 3 = 9.7 \\
\text{Var}(Y) &= 2^2 \times 2.1275 = 4 \times 2.1275 = 8.51
\end{aligned}$$

4.8 Binomial Distribution

4.8.1 Definition and Formula

Definition 4.8.1. A random variable X has a **binomial distribution** with parameters n and p , written $X \sim B(n, p)$, if X counts the number of successes in n independent trials, each with probability of success p .

Conditions for a Binomial Distribution

The following four conditions must all be satisfied (often remembered as BINS):

1. **B**inary outcomes: each trial has exactly two outcomes (success/failure).
2. **I**ndependent trials: the outcome of one trial does not affect another.
3. **N** fixed: the number of trials n is fixed in advance.
4. **S**ame probability: the probability of success p is constant for each trial.

Binomial Probability Mass Function

If $X \sim B(n, p)$, then:

$$P(X = r) = \binom{n}{r} p^r (1 - p)^{n-r}, \quad r = 0, 1, 2, \dots, n$$

where $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ is the binomial coefficient.

4.8.2 Mean and Variance

Mean and Variance of $B(n, p)$

If $X \sim B(n, p)$:

$$\begin{aligned} E(X) &= np \\ \text{Var}(X) &= np(1 - p) \end{aligned}$$

The standard deviation is $\sigma = \sqrt{np(1 - p)}$.

Derivation: Mean of the Binomial Distribution

Let $X \sim B(n, p)$. We can write $X = X_1 + X_2 + \dots + X_n$ where each X_i is an indicator variable for the i -th trial:

$$X_i = \begin{cases} 1 & \text{if trial } i \text{ is a success} \\ 0 & \text{if trial } i \text{ is a failure} \end{cases}$$

Each $X_i \sim B(1, p)$, so $E(X_i) = 1 \cdot p + 0 \cdot (1 - p) = p$.

By linearity of expectation:

$$E(X) = \sum_{i=1}^n E(X_i) = n \cdot p = np$$

Derivation: Variance of the Binomial Distribution

Since the trials are independent, the X_i are independent, so:

$$\text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i)$$

For each $X_i \sim B(1, p)$:

$$\begin{aligned} E(X_i^2) &= 1^2 \cdot p + 0^2 \cdot (1 - p) = p \\ \text{Var}(X_i) &= E(X_i^2) - [E(X_i)]^2 = p - p^2 = p(1 - p) \end{aligned}$$

Therefore:

$$\text{Var}(X) = n \cdot p(1 - p) = np(1 - p)$$

Using Your GDC for Binomial Calculations

- **Individual probability** $P(X = r)$: use the binomial PDF function.
- **Cumulative probability** $P(X \leq r)$: use the binomial CDF function.
- $P(X \geq r)$: calculate as $1 - P(X \leq r - 1)$.
- $P(a \leq X \leq b)$: calculate as $P(X \leq b) - P(X \leq a - 1)$.
- Know the menu path on your specific GDC model. On TI-Nspire: Menu $\rightarrow$ Probability $\rightarrow$ Distributions. On Casio fx-CG50: STAT $\rightarrow$ DIST $\rightarrow$ BINOMIAL.

IB Exam Tip: Identifying Binomial Context

In IB exams, binomial questions often use wording like:

- “A fair coin is tossed 10 times...” ($n = 10, p = 0.5$)
- “65% of people prefer...A random sample of 20 people is selected...” ($n = 20, p = 0.65$)
- “On average, 8% of items are defective. Find the probability that in a batch of 50...” ($n = 50, p = 0.08$)

Always verify the four conditions (BINS) before using the binomial distribution. If the question says “without replacement”, the trials are *not* independent and you should use the hypergeometric distribution (not required for IB, but worth recognising).

Worked Example: Binomial Distribution

A multiple choice test has 10 questions, each with 4 options (only one correct). A student guesses every answer.

- Find the probability the student gets exactly 3 correct.
- Find the probability the student gets at least 6 correct.
- Find the expected number and standard deviation of correct answers.

Solution: Here $X \sim B(10, 0.25)$.

(a)

$$\begin{aligned} P(X = 3) &= \binom{10}{3} (0.25)^3 (0.75)^7 \\ &= 120 \times 0.015625 \times 0.13348 \\ &= 120 \times 0.002087 \\ &\approx 0.2503 \end{aligned}$$

(b)

$$\begin{aligned} P(X \geq 6) &= 1 - P(X \leq 5) \\ &= 1 - \sum_{r=0}^5 \binom{10}{r} (0.25)^r (0.75)^{10-r} \\ &\approx 1 - 0.9803 = 0.0197 \end{aligned}$$

(c)

$$\begin{aligned} E(X) &= 10 \times 0.25 = 2.5 \\ \text{Var}(X) &= 10 \times 0.25 \times 0.75 = 1.875 \\ \sigma &= \sqrt{1.875} \approx 1.369 \end{aligned}$$

4.9 Normal Distribution

4.9.1 The Normal Distribution

Definition 4.9.1. A continuous random variable X has a **normal distribution** with parameters μ (mean) and σ^2 (variance), written $X \sim N(\mu, \sigma^2)$, if its probability density function is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}$$

4.9.2 Properties of the Normal Curve

Bell Curve Properties

- **Symmetric** about the mean μ .
- **Bell-shaped**: maximum at $x = \mu$, tails approach but never reach the x -axis.
- The **inflection points** of the curve are at $x = \mu \pm \sigma$.
- **Total area** under the curve = 1.
- Mean = median = mode = μ .

The 68–95–99.7 Rule (Empirical Rule)

For $X \sim N(\mu, \sigma^2)$:

$$\begin{aligned} P(\mu - \sigma < X < \mu + \sigma) &\approx 0.6827 \\ P(\mu - 2\sigma < X < \mu + 2\sigma) &\approx 0.9545 \\ P(\mu - 3\sigma < X < \mu + 3\sigma) &\approx 0.9973 \end{aligned}$$

The 68–95–99.7 Rule in Exams

This rule is useful for quick estimates and for checking whether GDC answers are reasonable. However, for exact answers in IB exams, always use your GDC's normal distribution functions.

4.9.3 Standardization

Definition 4.9.2. Standardization converts any normal variable to the standard normal $Z \sim N(0, 1)$:

Standardization Formula

$$Z = \frac{X - \mu}{\sigma}, \quad Z \sim N(0, 1)$$

Derivation of Standardization

Let $X \sim N(\mu, \sigma^2)$ and define $Z = \frac{X - \mu}{\sigma}$.

Showing $E(Z) = 0$:

$$E(Z) = E\left(\frac{X - \mu}{\sigma}\right) = \frac{E(X) - \mu}{\sigma} = \frac{\mu - \mu}{\sigma} = 0$$

Showing $\text{Var}(Z) = 1$:

$$\begin{aligned} \text{Var}(Z) &= \text{Var}\left(\frac{X - \mu}{\sigma}\right) = \frac{1}{\sigma^2} \text{Var}(X - \mu) \\ &= \frac{1}{\sigma^2} \text{Var}(X) = \frac{\sigma^2}{\sigma^2} = 1 \end{aligned}$$

Since any linear transformation of a normal variable is also normal, and we have shown Z has mean 0 and variance 1, we conclude $Z \sim N(0, 1)$.

4.9.4 Finding Probabilities

Normal Probabilities

For $X \sim N(\mu, \sigma^2)$, use your GDC to find:

- $P(X < a)$: use normal CDF with lower bound $-\infty$ (or a very small number) and upper bound a .
- $P(a < X < b)$: use normal CDF with lower bound a and upper bound b .
- $P(X > a) = 1 - P(X < a)$.
- By symmetry: $P(X < \mu - a) = P(X > \mu + a)$.

4.9.5 Inverse Normal

Inverse Normal

Given a probability p , the **inverse normal** function finds the value a such that:

$$P(X < a) = p$$

Use the inverse normal function on your GDC with the given μ , σ , and area p .

4.9.6 Normal Approximation to the Binomial

Theorem 4.9.1 (Normal Approximation to the Binomial Distribution). *If $X \sim B(n, p)$ and both $np > 5$ and $n(1 - p) > 5$, then X can be approximated by:*

$$X \approx Y \sim N(np, np(1 - p))$$

Conditions for Normal Approximation

$$np > 5 \quad \text{and} \quad n(1 - p) > 5$$

These ensure the binomial distribution is sufficiently symmetric. If p is close to 0 or 1, a larger n is needed.

Continuity Correction

Since the binomial distribution is discrete and the normal is continuous, we apply a **continuity correction**:

$$\begin{aligned} P(X \leq k) &\approx P(Y < k + 0.5) \\ P(X \geq k) &\approx P(Y > k - 0.5) \\ P(X = k) &\approx P(k - 0.5 < Y < k + 0.5) \\ P(a \leq X \leq b) &\approx P(a - 0.5 < Y < b + 0.5) \end{aligned}$$

Continuity Correction Rule of Thumb

The continuity correction extends by 0.5 in the direction that includes the relevant discrete value(s). For “ $\leq k$ ”, include k by going up to $k + 0.5$. For “ $\geq k$ ”, include k by going down to $k - 0.5$.

IB Exam Tip: When to Use Normal Approximation

- If n is small enough that your GDC can compute exact binomial probabilities (typically $n < 1000$), use the **exact binomial** — it is always more accurate.
- Normal approximation is mainly useful when n is very large or when the question explicitly asks for it.
- Always check the conditions $np > 5$ and $n(1-p) > 5$ *before* applying the approximation.
- Never forget the continuity correction — marks are deducted for omitting it.
- Common IB exam wording: “Using the normal approximation to the binomial distribution, find...” This signals that you must use the approximation.

Worked Example: Normal Distribution

The heights of IB students are normally distributed with mean 165 cm and standard deviation 8 cm. Let $H \sim N(165, 64)$.

(a) Find $P(H > 175)$.

(b) Find $P(160 < H < 172)$.

(c) The top 10% of students are selected for a basketball team. Find the minimum height required.

Solution:

(a) Using the GDC: $P(H > 175) = 1 - P(H < 175) \approx 1 - 0.8944 = 0.1056$.

Alternatively, using standardization:

$$z = \frac{175 - 165}{8} = 1.25$$

$$P(Z > 1.25) = 1 - \Phi(1.25) \approx 1 - 0.8944 = 0.1056$$

(b)

$$\begin{aligned} P(160 < H < 172) &= P\left(\frac{160 - 165}{8} < Z < \frac{172 - 165}{8}\right) \\ &= P(-0.625 < Z < 0.875) \\ &= \Phi(0.875) - \Phi(-0.625) \\ &\approx 0.8092 - 0.2660 = 0.5432 \end{aligned}$$

(c) We need h such that $P(H < h) = 0.90$. Using the inverse normal on the GDC:

$$h \approx 175.25 \text{ cm}$$

So the minimum height is approximately 175.3 cm.

Worked Example: Normal Approximation to Binomial

A fair coin is tossed 200 times. Using the normal approximation, find the probability of getting between 90 and 110 heads (inclusive).

Solution: Let $X \sim B(200, 0.5)$.

Check conditions: $np = 100 > 5$ and $n(1-p) = 100 > 5$. Both satisfied.

Normal approximation: $Y \sim N(100, 50)$ where $\mu = 100$ and $\sigma^2 = 50$, so $\sigma = \sqrt{50} \approx 7.071$.

With continuity correction:

$$\begin{aligned}
 P(90 \leq X \leq 110) &\approx P(89.5 < Y < 110.5) \\
 &= P\left(\frac{89.5 - 100}{7.071} < Z < \frac{110.5 - 100}{7.071}\right) \\
 &= P(-1.485 < Z < 1.485) \\
 &= \Phi(1.485) - \Phi(-1.485) \\
 &\approx 0.9312 - 0.0688 = 0.8624
 \end{aligned}$$

4.10 Continuous Random Variables and PDFs

4.10.1 Probability Density Functions

Definition 4.10.1. A **probability density function** (PDF) $f(x)$ for a continuous random variable X satisfies:

1. $f(x) \geq 0$ for all $x \in \mathbb{R}$.
2. $\int_{-\infty}^{\infty} f(x) \, dx = 1$.

Probability from a PDF

For a continuous random variable X with PDF $f(x)$:

$$P(a \leq X \leq b) = \int_a^b f(x) \, dx$$

Note: for continuous variables, $P(X = a) = 0$ for any single value a , so $P(a \leq X \leq b) = P(a < X < b) = P(a \leq X < b) = P(a < X \leq b)$.

Cumulative Distribution Function (CDF)

The CDF $F(x)$ of a continuous random variable X is:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) \, dt$$

- $F'(x) = f(x)$ (the PDF is the derivative of the CDF).
- $F(-\infty) = 0$, $F(+\infty) = 1$.
- $P(a < X \leq b) = F(b) - F(a)$.

4.10.2 Mode and Median of a Continuous Random Variable

Mode and Median

- **Mode:** the value of x at which $f(x)$ is maximised. Find by setting $f'(x) = 0$ and checking $f''(x) < 0$ (or by inspection for simple PDFs).
- **Median:** the value m such that $P(X \leq m) = 0.5$, i.e., solve $\int_{-\infty}^m f(x) \, dx = 0.5$.

4.10.3 Expected Value and Variance

Expected Value of a Continuous Random Variable

$$E(X) = \int_{-\infty}^{\infty} x f(x) \, dx$$

For a PDF defined on $[a, b]$:

$$E(X) = \int_a^b x f(x) \, dx$$

Variance of a Continuous Random Variable

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) \, dx = E(X^2) - [E(X)]^2$$

where:

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) \, dx$$

Strategy for IB Questions on Continuous Random Variables

Typical IB exam questions on this topic follow a pattern:

1. **Find k :** use the condition $\int f(x) \, dx = 1$ to determine an unknown constant.
2. **Find a probability:** compute a definite integral of $f(x)$.
3. **Find the median:** solve $\int f(x) \, dx = 0.5$ for the upper limit.
4. **Find $E(X)$ and $\text{Var}(X)$:** compute the appropriate integrals.
5. **Find the mode:** maximise $f(x)$ using calculus or inspection.

Worked Example: Continuous Random Variable

A continuous random variable X has PDF:

$$f(x) = \begin{cases} kx(4-x) & 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the value of k .
- (b) Find $P(1 < X < 3)$.
- (c) Find the median of X .
- (d) Find $E(X)$ and $\text{Var}(X)$.

Solution:

- (a) Using $\int_{-\infty}^{\infty} f(x) \, dx = 1$:

$$\begin{aligned} \int_0^4 kx(4-x) \, dx &= 1 \\ k \int_0^4 (4x - x^2) \, dx &= 1 \\ k \left[2x^2 - \frac{x^3}{3} \right]_0^4 &= 1 \\ k \left(32 - \frac{64}{3} \right) &= 1 \\ k \left(\frac{96 - 64}{3} \right) &= 1 \\ k \left(\frac{32}{3} \right) &= 1 \implies k = \frac{3}{32} \end{aligned}$$

(b)

$$\begin{aligned}
 P(1 < X < 3) &= \int_1^3 \frac{3}{32} x(4-x) \, dx \\
 &= \frac{3}{32} \int_1^3 (4x - x^2) \, dx \\
 &= \frac{3}{32} \left[2x^2 - \frac{x^3}{3} \right]_1^3 \\
 &= \frac{3}{32} \left[(18 - 9) - \left(2 - \frac{1}{3} \right) \right] \\
 &= \frac{3}{32} \left[9 - \frac{5}{3} \right] = \frac{3}{32} \times \frac{22}{3} = \frac{22}{32} = \frac{11}{16}
 \end{aligned}$$

(c) Find m such that $\int_0^m \frac{3}{32} x(4-x) \, dx = 0.5$:

$$\begin{aligned}
 \frac{3}{32} \left[2x^2 - \frac{x^3}{3} \right]_0^m &= 0.5 \\
 \frac{3}{32} \left(2m^2 - \frac{m^3}{3} \right) &= \frac{1}{2} \\
 2m^2 - \frac{m^3}{3} &= \frac{16}{3} \\
 6m^2 - m^3 &= 16
 \end{aligned}$$

By trial or GDC, $m \approx 2.30$ (to 3 s.f.). (This cubic has one root in $[0, 4]$, which is the median.)(d) $E(X)$:

$$\begin{aligned}
 E(X) &= \int_0^4 x \cdot \frac{3}{32} x(4-x) \, dx = \frac{3}{32} \int_0^4 (4x^2 - x^3) \, dx \\
 &= \frac{3}{32} \left[\frac{4x^3}{3} - \frac{x^4}{4} \right]_0^4 = \frac{3}{32} \left(\frac{256}{3} - 64 \right) \\
 &= \frac{3}{32} \left(\frac{256 - 192}{3} \right) = \frac{3}{32} \times \frac{64}{3} = 2
 \end{aligned}$$

 $E(X^2)$:

$$\begin{aligned}
 E(X^2) &= \int_0^4 x^2 \cdot \frac{3}{32} x(4-x) \, dx = \frac{3}{32} \int_0^4 (4x^3 - x^4) \, dx \\
 &= \frac{3}{32} \left[x^4 - \frac{x^5}{5} \right]_0^4 = \frac{3}{32} \left(256 - \frac{1024}{5} \right) \\
 &= \frac{3}{32} \left(\frac{1280 - 1024}{5} \right) = \frac{3}{32} \times \frac{256}{5} = \frac{768}{160} = \frac{24}{5} = 4.8
 \end{aligned}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 4.8 - 2^2 = 4.8 - 4 = 0.8$$

Chapter 5

Calculus

Calculus is the single most important topic in IB Mathematics AA HL. It accounts for a substantial proportion of marks across both Paper 1 and Paper 2, and it appears in the Internal Assessment and Extended Essay with great frequency. This chapter provides a comprehensive, rigorous treatment of every calculus topic in the syllabus, complete with proofs, derivations, worked examples, and exam strategies.

5.1 Limits

5.1.1 Intuitive Understanding

The concept of a limit concerns the behavior of a function $f(x)$ as the input x approaches a particular value a . We are not interested in the value of the function *at* $x = a$ (it may not even be defined there), but rather what value $f(x)$ gets arbitrarily close to as x gets arbitrarily close to a .

Definition 5.1.1. We write $\lim_{x \rightarrow a} f(x) = L$ and say “the limit of $f(x)$ as x approaches a equals L ” if we can make the values of $f(x)$ as close to L as we like by taking x sufficiently close to a (but $x \neq a$).

Evaluating a Simple Limit

Evaluate $\lim_{x \rightarrow 2} (3x^2 - 4x + 1)$.

Since the function $f(x) = 3x^2 - 4x + 1$ is a polynomial (and therefore continuous everywhere), we can simply substitute:

$$f(2) = 3(4) - 4(2) + 1 = 12 - 8 + 1 = 5$$

Therefore $\lim_{x \rightarrow 2} (3x^2 - 4x + 1) = 5$.

Limit at a Hole

Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$.

The function is undefined at $x = 3$ (division by zero), so we cannot simply substitute. Instead, we factor and simplify:

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \lim_{x \rightarrow 3} \frac{(x - 3)(x + 3)}{x - 3} = \lim_{x \rightarrow 3} (x + 3) = 6$$

The limit exists and equals 6, even though $f(3)$ is undefined.

5.1.2 One-Sided Limits

Sometimes the behavior of a function differs depending on whether x approaches a from the left or from the right.

Definition 5.1.2. The *right-hand limit* is

$$\lim_{x \rightarrow a^+} f(x) = L$$

if $f(x)$ approaches L as x approaches a from values greater than a .

The *left-hand limit* is

$$\lim_{x \rightarrow a^-} f(x) = L$$

if $f(x)$ approaches L as x approaches a from values less than a .

Existence of a Two-Sided Limit

$$\lim_{x \rightarrow a} f(x) = L \quad \text{if and only if} \quad \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = L$$

If the one-sided limits are not equal, then the (two-sided) limit does not exist.

One-Sided Limits with Piecewise Functions

Let $f(x) = \begin{cases} x^2 & \text{if } x \leq 1 \\ 3x - 2 & \text{if } x > 1 \end{cases}$. Find $\lim_{x \rightarrow 1} f(x)$ if it exists.

Left-hand limit: $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x^2 = 1$.

Right-hand limit: $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (3x - 2) = 1$.

Since both one-sided limits equal 1, we have $\lim_{x \rightarrow 1} f(x) = 1$.

Note that $f(1) = 1^2 = 1$, so f is continuous at $x = 1$ in this case.

5.1.3 Infinite Limits and Vertical Asymptotes

When $f(x)$ grows without bound as x approaches a , we say the limit is infinite.

Infinite Limits

$$\lim_{x \rightarrow a} f(x) = +\infty \quad \text{or} \quad \lim_{x \rightarrow a} f(x) = -\infty$$

This means $f(x)$ increases or decreases without bound. The line $x = a$ is a **vertical asymptote** of f .

Vertical Asymptotes

Find the vertical asymptotes of $f(x) = \frac{1}{(x-2)(x+5)}$.

The function is undefined at $x = 2$ and $x = -5$.

$$\lim_{x \rightarrow 2^+} \frac{1}{(x-2)(x+5)} = +\infty, \quad \lim_{x \rightarrow 2^-} \frac{1}{(x-2)(x+5)} = -\infty$$

$$\lim_{x \rightarrow (-5)^+} \frac{1}{(x-2)(x+5)} = -\infty, \quad \lim_{x \rightarrow (-5)^-} \frac{1}{(x-2)(x+5)} = +\infty$$

Both $x = 2$ and $x = -5$ are vertical asymptotes.

5.1.4 Limits at Infinity and Horizontal Asymptotes

Definition 5.1.3.

$$\lim_{x \rightarrow \infty} f(x) = L$$

means that $f(x)$ approaches L as x grows without bound. If such an L exists, the line $y = L$ is a **horizontal asymptote**.

Key Limit at Infinity

$$\lim_{x \rightarrow \infty} \frac{1}{x^n} = 0 \quad \text{for any } n > 0$$

Similarly, $\lim_{x \rightarrow -\infty} \frac{1}{x^n} = 0$ for even n ; for odd n the limit is 0 as well (the sign is accounted for by the sign of x in the numerator of the original expression).

Finding Horizontal Asymptotes

Find the horizontal asymptotes of $f(x) = \frac{3x^2 - 2x + 1}{5x^2 + 7}$.

Divide numerator and denominator by x^2 :

$$\lim_{x \rightarrow \infty} \frac{3 - \frac{2}{x} + \frac{1}{x^2}}{5 + \frac{7}{x^2}} = \frac{3 - 0 + 0}{5 + 0} = \frac{3}{5}$$

The horizontal asymptote is $y = \frac{3}{5}$.

Strategy for Rational Function Limits at Infinity

For $\lim_{x \rightarrow \infty} \frac{P(x)}{Q(x)}$ where P and Q are polynomials:

- If $\deg P < \deg Q$: the limit is 0.
- If $\deg P = \deg Q$: the limit is $\frac{\text{leading coefficient of } P}{\text{leading coefficient of } Q}$.
- If $\deg P > \deg Q$: the limit is $\pm\infty$.

5.1.5 Special Trigonometric Limit

Fundamental Trigonometric Limit

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \quad (\text{where } \theta \text{ is in radians})$$

Equivalently, $\lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} = 1$.

A direct consequence: $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta} = 0$ and $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = 1$.

Using the Trigonometric Limit

Evaluate $\lim_{x \rightarrow 0} \frac{\sin 5x}{3x}$.

Rewrite to use the standard form:

$$\lim_{x \rightarrow 0} \frac{\sin 5x}{3x} = \lim_{x \rightarrow 0} \frac{5}{3} \cdot \frac{\sin 5x}{5x} = \frac{5}{3} \cdot 1 = \frac{5}{3}$$

Trigonometric Limit with Multiplication by Conjugate

Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$.

Multiply numerator and denominator by $1 + \cos x$:

$$\lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{x^2(1 + \cos x)} = \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2(1 + \cos x)} = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^2 \cdot \frac{1}{1 + \cos x} = 1^2 \cdot \frac{1}{2} = \frac{1}{2}$$

5.1.6 L'Hôpital's Rule

When a limit produces an indeterminate form such as $\frac{0}{0}$ or $\frac{\infty}{\infty}$, L'Hôpital's Rule provides a powerful technique.

Theorem 5.1.1 (L'Hôpital's Rule). *If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$ (or both $= \pm\infty$), and $g'(x) \neq 0$ near a , then*

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

provided the limit on the right exists (or is $\pm\infty$).

Applicable Indeterminate Forms

L'Hôpital's Rule applies directly to:

$$\frac{0}{0} \quad \text{and} \quad \frac{\infty}{\infty}$$

For other forms ($0 \cdot \infty$, $\infty - \infty$, 0^0 , 1^∞ , ∞^0), rewrite the expression as a quotient first.

Repeated Application of L'Hôpital's Rule

Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x - \frac{x^2}{2}}{x^3}$.

First attempt (0/0 form):

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x - \frac{x^2}{2}}{x^3} \xrightarrow{\text{L'H}} \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{3x^2}$$

Still 0/0, apply again:

$$\xrightarrow{\text{L'H}} \lim_{x \rightarrow 0} \frac{e^x - 1}{6x}$$

Still 0/0, apply once more:

$$\xrightarrow{\text{L'H}} \lim_{x \rightarrow 0} \frac{e^x}{6} = \frac{1}{6}$$

This took **three** applications of L'Hôpital's Rule.

L'Hôpital with ∞/∞

Evaluate $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$.

This is ∞/∞ . Applying L'Hôpital's Rule:

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

IB Exam Tip: L'Hôpital's Rule vs. Algebraic Manipulation

On IB exams, L'Hôpital's Rule is a valid technique. However, consider:

- **Algebraic manipulation** (factoring, rationalizing, simplifying) is often faster for simple $0/0$ forms and is always acceptable.
- **L'Hôpital's Rule** is essential when algebraic methods are extremely cumbersome or when dealing with exponential/logarithmic limits.
- **Warning:** Always verify the indeterminate form before applying L'Hôpital. Applying it to a non-indeterminate form will give a wrong answer.
- **For Paper 1** (no calculator), L'Hôpital is often the best approach for limits involving e^x , $\ln x$, or trigonometric functions.

5.2 Differentiation: First Principles

5.2.1 The Definition of the Derivative

Definition 5.2.1. The *derivative* of f at x is defined as

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

provided this limit exists. If it exists, we say f is **differentiable** at x .

The quantity $\frac{f(x+h) - f(x)}{h}$ is the slope of the secant line through $(x, f(x))$ and $(x+h, f(x+h))$. As $h \rightarrow 0$, this secant approaches the tangent line, so the derivative gives the slope of the tangent line at the point $(x, f(x))$.

An equivalent form (sometimes more convenient) is:

$$f'(x) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

5.2.2 Deriving the Power Rule from First Principles

Derivative of $f(x) = x^n$ for positive integer n

We prove $\frac{d}{dx}[x^n] = nx^{n-1}$ by induction on n .

Base case ($n = 1$):

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1 = 1 \cdot x^0 \checkmark$$

Inductive step: Assume $\frac{d}{dx}[x^k] = kx^{k-1}$ for some $k \geq 1$. We show $\frac{d}{dx}[x^{k+1}] = (k+1)x^k$.

Let $f(x) = x^{k+1} = x \cdot x^k$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^{k+1} - x^{k+1}}{h}$$

Using the identity $(x+h)^{k+1} - x^{k+1} = h[(x+h)^k + (x+h)^{k-1}x + \cdots + (x+h)x^{k-1} + x^k]$ (factorisation of $a^{k+1} - b^{k+1}$):

$$f'(x) = \lim_{h \rightarrow 0} [(x+h)^k + (x+h)^{k-1}x + \cdots + (x+h)x^{k-1} + x^k]$$

There are $k + 1$ terms, each tending to x^k as $h \rightarrow 0$:

$$f'(x) = (k + 1)x^k \checkmark$$

Alternatively, using the product rule (once we have proved it): $\frac{d}{dx}[x \cdot x^k] = 1 \cdot x^k + x \cdot kx^{k-1} = (k + 1)x^k$.

Conclusion: By mathematical induction, $\frac{d}{dx}[x^n] = nx^{n-1}$ for all positive integers n .

5.2.3 Derivative of $\sin x$ from First Principles

Derivative of $f(x) = \sin x$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sin(x + h) - \sin x}{h}$$

Using the angle addition formula $\sin(x + h) = \sin x \cos h + \cos x \sin h$:

$$= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin x(\cos h - 1) + \cos x \sin h}{h}$$

$$= \lim_{h \rightarrow 0} \left[\sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h} \right]$$

Using the standard limits $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$ and $\lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0$:

$$= \sin x \cdot 0 + \cos x \cdot 1 = \cos x$$

Therefore $\frac{d}{dx}[\sin x] = \cos x$.

IB Exam Tip: First Principles on Paper 1

First principles differentiation questions are **common on Paper 1** (no calculator). Typical formats:

- “Find $f'(x)$ from first principles” for a given function.
- “Use the definition of the derivative to show that ...”
- The key is to set up the limit carefully, simplify the numerator (factor, expand, or use trigonometric identities), and then evaluate the limit.
- Common functions tested: x^n for small n , $\sqrt{x}$ (write as $x^{1/2}$), $\frac{1}{x}$, $\sin x$, $\cos x$.

First Principles: $f(x) = \frac{1}{x}$

Find $f'(x)$ from first principles for $f(x) = \frac{1}{x}$, $x \neq 0$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \lim_{h \rightarrow 0} \frac{\frac{x - (x+h)}{x(x+h)}}{h} = \lim_{h \rightarrow 0} \frac{-h}{hx(x+h)} = \lim_{h \rightarrow 0} \frac{-1}{x(x+h)} = -\frac{1}{x^2}$$

This confirms $\frac{d}{dx}[x^{-1}] = -x^{-2}$, consistent with the power rule.

5.3 Differentiation Rules and Proofs

5.3.1 Basic Rules

Power Rule

$$\frac{d}{dx}[x^n] = nx^{n-1} \quad \text{for all } n \in \mathbb{R}$$

This holds for any real exponent n : positive integers, negative integers, fractions, and irrationals.

Constant Multiple Rule

$$\frac{d}{dx}[cf(x)] = c \cdot f'(x)$$

where c is a constant. The derivative of a constant multiple is the constant times the derivative.

Sum and Difference Rule

$$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$$

The derivative of a sum (or difference) is the sum (or difference) of the derivatives. This extends to any finite number of terms.

5.3.2 Proof of the Product Rule

Proof of the Product Rule

Theorem: If f and g are differentiable at x , then

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$$

Proof (using first principles):

Let $h(x) = f(x)g(x)$. Then:

$$h'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x)g(x + \Delta x) - f(x)g(x)}{\Delta x}$$

We use the trick of adding and subtracting $f(x + \Delta x)g(x)$ in the numerator:

$$= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x)g(x + \Delta x) - f(x + \Delta x)g(x) + f(x + \Delta x)g(x) - f(x)g(x)}{\Delta x}$$

$$\begin{aligned}
&= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x)[g(x + \Delta x) - g(x)] + g(x)[f(x + \Delta x) - f(x)]}{\Delta x} \\
&= \lim_{\Delta x \rightarrow 0} f(x + \Delta x) \cdot \frac{g(x + \Delta x) - g(x)}{\Delta x} + g(x) \cdot \frac{f(x + \Delta x) - f(x)}{\Delta x}
\end{aligned}$$

Since f is differentiable, it is continuous, so $\lim_{\Delta x \rightarrow 0} f(x + \Delta x) = f(x)$:

$$= f(x) \cdot g'(x) + g(x) \cdot f'(x)$$

5.3.3 Proof of the Quotient Rule

Proof of the Quotient Rule

Theorem: If f and g are differentiable at x and $g(x) \neq 0$, then

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Proof: Write $h(x) = \frac{f(x)}{g(x)}$, so $f(x) = h(x) \cdot g(x)$. Applying the product rule:

$$f'(x) = h'(x)g(x) + h(x)g'(x)$$

Solving for $h'(x)$:

$$\begin{aligned}
h'(x)g(x) &= f'(x) - h(x)g'(x) \\
h'(x) &= \frac{f'(x)}{g(x)} - \frac{h(x)g'(x)}{g(x)} = \frac{f'(x)}{g(x)} - \frac{f(x)g'(x)}{[g(x)]^2}
\end{aligned}$$

Combining over a common denominator $[g(x)]^2$:

$$h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

5.3.4 Proof of the Chain Rule

Proof of the Chain Rule

Theorem: If g is differentiable at x and f is differentiable at $g(x)$, then

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$$

Proof (using first principles):

Let $h(x) = f(g(x))$. Define $\Delta u = g(x + \Delta x) - g(x)$, so $g(x + \Delta x) = g(x) + \Delta u$.

$$h'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(g(x + \Delta x)) - f(g(x))}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(g(x) + \Delta u) - f(g(x))}{\Delta x}$$

Multiply and divide by Δu (when $\Delta u \neq 0$):

$$= \lim_{\Delta x \rightarrow 0} \frac{f(g(x) + \Delta u) - f(g(x))}{\Delta u} \cdot \frac{\Delta u}{\Delta x}$$

As $\Delta x \rightarrow 0$, we have $\Delta u \rightarrow 0$ (since g is continuous). Therefore:

$$= \lim_{\Delta u \rightarrow 0} \frac{f(g(x) + \Delta u) - f(g(x))}{\Delta u} \cdot \lim_{\Delta x \rightarrow 0} \frac{g(x + \Delta x) - g(x)}{\Delta x}$$

$$= f'(g(x)) \cdot g'(x)$$

In Leibniz notation: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$, where $y = f(u)$ and $u = g(x)$.

Mnemonic for Differentiation Rules

- **Product Rule:** “ d -first times second, plus first times d -second.”
- **Quotient Rule:** “ d -top times bottom, minus top times d -bottom, over bottom squared.” (“Low d -High minus High d -Low, over the square of what’s below.”)
- **Chain Rule:** Differentiate the outside, leave the inside alone, then multiply by the derivative of the inside.

5.4 Derivatives of Elementary Functions

5.4.1 Complete Derivative Table

Derivatives of Elementary Functions

$f(x)$	$f'(x)$	Condition
x^n	nx^{n-1}	$n \in \mathbb{R}$
$\sin x$	$\cos x$	
$\cos x$	$-\sin x$	
$\tan x$	$\sec^2 x$	$x \neq \frac{\pi}{2} + k\pi$
$\sec x$	$\sec x \tan x$	$x \neq \frac{\pi}{2} + k\pi$
$\csc x$	$-\csc x \cot x$	$x \neq k\pi$
$\cot x$	$-\csc^2 x$	$x \neq k\pi$
e^x	e^x	
a^x	$a^x \ln a$	$a > 0, a \neq 1$
$\ln x$	$\frac{1}{x}$	$x > 0$
$\log_a x$	$\frac{1}{x \ln a}$	$x > 0$
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$	$ x < 1$
$\arccos x$	$\frac{-1}{\sqrt{1-x^2}}$	$ x < 1$
$\arctan x$	$\frac{1}{1+x^2}$	

5.4.2 Derivations

Derivative of $\cos x$ from First Principles

$$\frac{d}{dx}[\cos x] = \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$$

Using $\cos(x+h) = \cos x \cos h - \sin x \sin h$:

$$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} = \lim_{h \rightarrow 0} \frac{\cos x(\cos h - 1) - \sin x \sin h}{h} \\ &= \cos x \cdot \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} - \sin x \cdot \lim_{h \rightarrow 0} \frac{\sin h}{h} = \cos x \cdot 0 - \sin x \cdot 1 = -\sin x \end{aligned}$$

Derivative of $\tan x$ using the Quotient Rule

$\tan x = \frac{\sin x}{\cos x}$, so by the quotient rule:

$$\frac{d}{dx}[\tan x] = \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$$

Derivative of $\sec x$

$\sec x = \frac{1}{\cos x} = (\cos x)^{-1}$. By the chain rule:

$$\frac{d}{dx}[\sec x] = -(\cos x)^{-2} \cdot (-\sin x) = \frac{\sin x}{\cos^2 x} = \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} = \sec x \tan x$$

Derivative of $\csc x$

$\csc x = (\sin x)^{-1}$. By the chain rule:

$$\frac{d}{dx}[\csc x] = -(\sin x)^{-2} \cdot \cos x = -\frac{\cos x}{\sin^2 x} = -\frac{1}{\sin x} \cdot \frac{\cos x}{\sin x} = -\csc x \cot x$$

Derivative of $\cot x$

$\cot x = \frac{\cos x}{\sin x}$. By the quotient rule:

$$\frac{d}{dx}[\cot x] = \frac{(-\sin x)(\sin x) - (\cos x)(\cos x)}{\sin^2 x} = \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} = \frac{-1}{\sin^2 x} = -\csc^2 x$$

Derivative of $\arcsin x$ using Implicit Differentiation

Let $y = \arcsin x$, so $x = \sin y$ where $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.

Differentiate both sides with respect to x :

$$1 = \cos y \cdot \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{1}{\cos y}$$

Since $x = \sin y$, we have $\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$ (positive because $\cos y \geq 0$ for $y \in [-\pi/2, \pi/2]$):

$$\frac{d}{dx}[\arcsin x] = \frac{1}{\sqrt{1 - x^2}}$$

Derivative of $\arccos x$

Let $y = \arccos x$, so $x = \cos y$ where $0 \leq y \leq \pi$.

$$1 = -\sin y \cdot \frac{dy}{dx} \implies \frac{dy}{dx} = -\frac{1}{\sin y} = -\frac{1}{\sqrt{1-x^2}}$$

Here $\sin y \geq 0$ on $[0, \pi]$.

Alternatively, note that $\arccos x = \frac{\pi}{2} - \arcsin x$, so $\frac{d}{dx}[\arccos x] = -\frac{1}{\sqrt{1-x^2}}$.

Derivative of $\arctan x$

Let $y = \arctan x$, so $x = \tan y$ where $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

$$1 = \sec^2 y \cdot \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{\sec^2 y} = \cos^2 y$$

Using the identity $\cos^2 y = \frac{1}{1 + \tan^2 y}$:

$$\frac{d}{dx}[\arctan x] = \frac{1}{1+x^2}$$

Derivative of a^x

Let $y = a^x$. Take $\ln$ of both sides: $\ln y = x \ln a$.

Differentiate implicitly with respect to x :

$$\frac{1}{y} \cdot \frac{dy}{dx} = \ln a \implies \frac{dy}{dx} = y \ln a = a^x \ln a$$

Derivative of $\log_a x$

Let $y = \log_a x$, so $a^y = x$. Take $\ln$ of both sides: $y \ln a = \ln x$, giving $y = \frac{\ln x}{\ln a}$.

$$\frac{dy}{dx} = \frac{1}{\ln a} \cdot \frac{1}{x} = \frac{1}{x \ln a}$$

5.5 Implicit Differentiation

5.5.1 Concept

When y is defined implicitly as a function of x (e.g., $x^2 + y^2 = r^2$), it may be difficult or impossible to solve for y explicitly. **Implicit differentiation** allows us to find $\frac{dy}{dx}$ by differentiating both sides of the equation with respect to x , treating y as a function of x .

Key Idea of Implicit Differentiation

When differentiating a term involving y with respect to x , apply the chain rule:

$$\frac{d}{dx}[y^n] = ny^{n-1} \frac{dy}{dx}$$

Similarly:

$$\begin{aligned}\frac{d}{dx}[\sin y] &= \cos y \cdot \frac{dy}{dx} \\ \frac{d}{dx}[e^y] &= e^y \cdot \frac{dy}{dx} \\ \frac{d}{dx}[\ln y] &= \frac{1}{y} \cdot \frac{dy}{dx}\end{aligned}$$

Circle: $x^2 + y^2 = 25$

Find $\frac{dy}{dx}$ for the circle $x^2 + y^2 = 25$.

Differentiate both sides with respect to x :

$$2x + 2y \frac{dy}{dx} = 0$$

$$2y \frac{dy}{dx} = -2x \implies \frac{dy}{dx} = -\frac{x}{y}$$

Find $\frac{dy}{dx}$ at the point $(3, 4)$: $\frac{dy}{dx} = -\frac{3}{4}$.

This makes sense geometrically: the tangent to the circle at $(3, 4)$ has slope $-\frac{3}{4}$ (it is perpendicular to the radius, which has slope $\frac{4}{3}$).

Ellipse: $\frac{x^2}{9} + \frac{y^2}{4} = 1$

Find $\frac{dy}{dx}$ for $\frac{x^2}{9} + \frac{y^2}{4} = 1$.

Differentiate:

$$\frac{2x}{9} + \frac{2y}{4} \cdot \frac{dy}{dx} = 0 \implies \frac{2x}{9} + \frac{y}{2} \cdot \frac{dy}{dx} = 0$$

$$\frac{y}{2} \cdot \frac{dy}{dx} = -\frac{2x}{9} \implies \frac{dy}{dx} = -\frac{4x}{9y}$$

5.5.2 Second Derivatives Implicitly

To find $\frac{d^2y}{dx^2}$, differentiate $\frac{dy}{dx}$ with respect to x , remembering that $\frac{dy}{dx}$ may contain y (which is a function of x).

Second Derivative: $x^2 + y^2 = 25$

We found $\frac{dy}{dx} = -\frac{x}{y}$. Now find $\frac{d^2y}{dx^2}$.

Using the quotient rule (remembering y depends on x):

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{(-1)(y) - (-x)\frac{dy}{dx}}{y^2} = \frac{-y + x \cdot \left(-\frac{x}{y}\right)}{y^2} = \frac{-y - \frac{x^2}{y}}{y^2} \\ &= \frac{\frac{-y^2 - x^2}{y}}{y^2} = \frac{-(x^2 + y^2)}{y^3} = \frac{-25}{y^3}\end{aligned}$$

$$\text{At } (3, 4): \frac{d^2y}{dx^2} = \frac{-25}{64}.$$

5.6 Logarithmic Differentiation

5.6.1 When to Use Logarithmic Differentiation

Logarithmic differentiation is useful in two situations:

1. When the function is of the form $y = f(x)^{g(x)}$ (variable in both base and exponent).
2. When the function is a product or quotient of many factors (taking $\ln$ converts products to sums and quotients to differences).

Method of Logarithmic Differentiation

1. Take natural logarithms of both sides: $\ln y = \ln[f(x)^{g(x)}] = g(x) \ln f(x)$.
2. Differentiate both sides with respect to x , using implicit differentiation on the left.
3. Solve for $\frac{dy}{dx}$.

Variable Base and Exponent: $y = x^x$

Find $\frac{dy}{dx}$ for $y = x^x$, $x > 0$.

Take $\ln$ of both sides: $\ln y = x \ln x$.

Differentiate: $\frac{1}{y} \cdot \frac{dy}{dx} = 1 \cdot \ln x + x \cdot \frac{1}{x} = \ln x + 1$.

Therefore:

$$\frac{dy}{dx} = y(\ln x + 1) = x^x(\ln x + 1)$$

Product of Many Factors

Find $\frac{dy}{dx}$ for $y = \frac{x^3 e^{2x} (x+1)^5}{\sqrt{x^2+1}}$.

Take $\ln$:

$$\ln y = 3 \ln x + 2x + 5 \ln(x+1) - \frac{1}{2} \ln(x^2+1)$$

Differentiate:

$$\begin{aligned} \frac{1}{y} \cdot \frac{dy}{dx} &= \frac{3}{x} + 2 + \frac{5}{x+1} - \frac{x}{x^2+1} \\ \frac{dy}{dx} &= \frac{x^3 e^{2x} (x+1)^5}{\sqrt{x^2+1}} \left(\frac{3}{x} + 2 + \frac{5}{x+1} - \frac{x}{x^2+1} \right) \end{aligned}$$

5.7 Applications of Differentiation

5.7.1 Tangents and Normals

Equation of a Tangent Line

The tangent to $y = f(x)$ at $x = x_0$ has equation

$$y - y_0 = f'(x_0)(x - x_0)$$

where $y_0 = f(x_0)$ and $f'(x_0)$ is the gradient (slope) of the tangent.

Equation of a Normal Line

The normal to $y = f(x)$ at $x = x_0$ is perpendicular to the tangent. Its gradient is

$$m_{\text{normal}} = -\frac{1}{f'(x_0)} \quad (\text{provided } f'(x_0) \neq 0)$$

and its equation is

$$y - y_0 = -\frac{1}{f'(x_0)}(x - x_0)$$

If $f'(x_0) = 0$, the tangent is horizontal ($y = y_0$) and the normal is vertical ($x = x_0$).

Tangent and Normal

Find the equations of the tangent and normal to $y = x^3 - 3x + 2$ at $x = 1$.

Point: $(1, 1^3 - 3 + 2) = (1, 0)$.

Gradient: $y' = 3x^2 - 3$, so $y'(1) = 3 - 3 = 0$.

Tangent: Since the gradient is 0, the tangent is horizontal: $y = 0$.

Normal: Since the tangent is horizontal, the normal is vertical: $x = 1$.

5.7.2 Stationary Points

Definition 5.7.1. A **stationary point** (or **critical point**) of f is a point where $f'(x) = 0$.

There are three types of stationary points: **local maxima**, **local minima**, and **horizontal points of inflection**.

First Derivative Test

At a stationary point $x = a$ where $f'(a) = 0$:

- **Local maximum:** f' changes from positive to negative as x increases through a .
- **Local minimum:** f' changes from negative to positive as x increases through a .
- **Horizontal point of inflection:** f' does not change sign (e.g., $f(x) = x^3$ at $x = 0$).

Second Derivative Test

At a stationary point $x = a$ where $f'(a) = 0$:

- $f''(a) > 0 \implies$ **local minimum** (function is concave up).
- $f''(a) < 0 \implies$ **local maximum** (function is concave down).
- $f''(a) = 0 \implies$ the test is **inconclusive**. Use the first derivative test instead.

Classifying Stationary Points

Find and classify all stationary points of $f(x) = x^3 - 6x^2 + 9x + 1$.

$$f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3)$$

Stationary points at $x = 1$ and $x = 3$.

Second derivative: $f''(x) = 6x - 12$.

At $x = 1$: $f''(1) = 6 - 12 = -6 < 0$, so **local maximum**. Value: $f(1) = 1 - 6 + 9 + 1 = 5$.

Point: $(1, 5)$.

At $x = 3$: $f''(3) = 18 - 12 = 6 > 0$, so **local minimum**. Value: $f(3) = 27 - 54 + 27 + 1 = 1$.

Point: $(3, 1)$.

5.7.3 Increasing and Decreasing Functions

Monotonicity

- f is **strictly increasing** on an interval if $f'(x) > 0$ for all x in the interval.
- f is **strictly decreasing** on an interval if $f'(x) < 0$ for all x in the interval.
- f is **constant** on an interval if $f'(x) = 0$ for all x in the interval.

5.7.4 Concavity and Points of Inflection

Definition 5.7.2. • f is **concave up** (convex) on an interval if $f''(x) > 0$ on that interval. The graph curves upward (like a cup $\cup$).

- f is **concave down** (concave) on an interval if $f''(x) < 0$ on that interval. The graph curves downward (like a cap $\cap$).

Definition 5.7.3. A **point of inflection** is a point on the curve where the concavity changes. At a point of inflection $x = a$:

1. $f''(a) = 0$ (or $f''(a)$ does not exist), AND
2. $f''(x)$ changes sign as x passes through a .

Critical IB Distinction: $f''(a) = 0$ is NOT Sufficient

A very common IB exam trap: $f''(a) = 0$ does **not** guarantee a point of inflection.

Counter-example: $f(x) = x^4$. We have $f'(x) = 4x^3$ and $f''(x) = 12x^2$.

- $f''(0) = 0$, but $f''(x) \geq 0$ for all x , so there is **no sign change** in f'' .
- The function is concave up on both sides of $x = 0$.
- $x = 0$ is a local minimum, **not** a point of inflection.

Always check for a sign change in $f''(x)$ to confirm a point of inflection.

Finding Points of Inflection

Find all points of inflection of $f(x) = \frac{x^4}{4} - 2x^3 + 6x^2 - 8x + 3$.

$$f'(x) = x^3 - 6x^2 + 12x - 8 = (x - 2)^3$$

$$f''(x) = 3x^2 - 12x + 12 = 3(x^2 - 4x + 4) = 3(x - 2)^2$$

$f''(x) = 3(x - 2)^2 \geq 0$ for all x , and $f''(x) = 0$ only at $x = 2$.

Since $f''(x)$ does not change sign at $x = 2$ (it is positive on both sides), $x = 2$ is **not** a point of inflection. The function has no points of inflection.

Note: $f'(2) = 0$ and $f''(2) = 0$, so this is a horizontal point of inflection for f' , meaning $x = 2$ is a **stationary point of inflection** of f ? No — it is a stationary point where the second derivative test is inconclusive, but the first derivative test shows $f'(x) = (x - 2)^3$ changes from negative to positive, so $x = 2$ is actually a **local minimum**.

True Point of Inflection

Find the point of inflection of $f(x) = x^3 - 3x^2 + 4$.

$$f'(x) = 3x^2 - 6x, \quad f''(x) = 6x - 6 = 6(x - 1).$$

$f''(x) = 0$ when $x = 1$. Check the sign change:

- For $x < 1$: $f''(x) < 0$ (concave down).
- For $x > 1$: $f''(x) > 0$ (concave up).

Since f'' changes sign, $x = 1$ is a point of inflection. The point is $(1, 1 - 3 + 4) = (1, 2)$.

5.7.5 Optimization Problems

Optimization involves finding the maximum or minimum value of a quantity subject to given constraints.

General Strategy for Optimization

1. **Identify** the quantity to be maximized or minimized. Call it Q .
2. **Write** Q as a function of one variable.
3. **Find** the domain (the feasible region).
4. **Differentiate** Q and set $Q' = 0$ to find stationary points.
5. **Classify** each stationary point (first or second derivative test).
6. **Check** endpoints of the domain if it is closed and bounded.
7. **Answer** the question in context, with appropriate units.

Optimization: Minimizing Surface Area

A closed cylindrical can must hold 500 cm^3 of liquid. Find the dimensions that minimize the surface area.

Let r = radius and h = height.

$$\text{Volume: } V = \pi r^2 h = 500, \text{ so } h = \frac{500}{\pi r^2}.$$

$$\text{Surface area: } S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + \frac{1000}{r}.$$

$$\text{Differentiate: } \frac{dS}{dr} = 4\pi r - \frac{1000}{r^2}.$$

$$\text{Set to zero: } 4\pi r = \frac{1000}{r^2} \implies 4\pi r^3 = 1000 \implies r^3 = \frac{250}{\pi}.$$

$$r = \sqrt[3]{\frac{250}{\pi}} \approx 4.30 \text{ cm}$$

$$h = \frac{500}{\pi r^2} = \frac{500}{\pi \cdot \left(\frac{250}{\pi}\right)^{2/3}} = 2\sqrt[3]{\frac{250}{\pi}} \approx 8.61 \text{ cm}$$

Note: $h = 2r$, so the optimal can has height equal to diameter.

Second derivative: $\frac{d^2S}{dr^2} = 4\pi + \frac{2000}{r^3} > 0$ for all $r > 0$, confirming a minimum.

5.7.6 Related Rates of Change

Definition 5.7.4. Related rates problems involve finding the rate of change of one quantity given the rate of change of a related quantity, using the chain rule:

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

Strategy for Related Rates

1. **Draw** a diagram and identify all variables.
2. **Write** an equation relating the relevant quantities.
3. **Differentiate** both sides with respect to time t (using the chain rule on all spatial variables).
4. **Substitute** the known values (including the rates) and solve for the unknown rate.
5. **Include** units in your answer.

Related Rates: Expanding Circle

A circular oil spill is expanding. When the radius is 50 m, the radius is increasing at a rate of 3 m/s. How fast is the area increasing?

The area is $A = \pi r^2$. Differentiate with respect to t :

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

Substitute $r = 50$ and $\frac{dr}{dt} = 3$:

$$\frac{dA}{dt} = 2\pi(50)(3) = 300\pi \text{ m}^2/\text{s}$$

Related Rates: Ladder Sliding

A 5 m ladder leans against a wall. The base slides away at 1 m/s. How fast is the top descending when the base is 3 m from the wall?

By Pythagoras: $x^2 + y^2 = 25$, where x = distance of base from wall, y = height of top.

Differentiate: $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$.

When $x = 3$: $y = \sqrt{25 - 9} = 4$.

$$2(3)(1) + 2(4) \frac{dy}{dt} = 0 \implies 6 + 8 \frac{dy}{dt} = 0 \implies \frac{dy}{dt} = -\frac{6}{8} = -0.75 \text{ m/s}$$

The top is descending at 0.75 m/s.

IB Exam Tip: “Show That” Questions

When an IB question says “show that,” you must provide **full working** with every step justified. For differentiation problems:

- State which rule you are using.
- Show the intermediate steps clearly.
- Do not skip algebraic simplification steps.
- For “show that $f'(x) = \dots$ ” questions, the answer is given — focus on demonstrating **how** to arrive at it.
- For “find” questions, you need the correct final answer, but showing working is essential for method marks.

5.8 Kinematics

5.8.1 Displacement, Velocity, and Acceleration

Kinematic Relationships

For a particle moving in a straight line:

Displacement: $s(t)$

Velocity: $v(t) = \frac{ds}{dt} = s'(t)$

Acceleration: $a(t) = \frac{dv}{dt} = v'(t) = s''(t)$

Recovery of Displacement from Velocity

$$s(t) = \int v(t) dt + C$$

The constant C is determined by an initial condition (e.g., $s(0) = s_0$).

5.8.2 Total Distance vs. Displacement

Definition 5.8.1. • **Displacement** is the net change in position: $s(t_2) - s(t_1) = \int_{t_1}^{t_2} v(t) dt$.

- **Total distance** travelled is the integral of speed: $\int_{t_1}^{t_2} |v(t)| dt$.
- **Speed** is the magnitude of velocity: $\text{speed}(t) = |v(t)|$.

Total Distance Travelled

$$\text{Total distance} = \int_{t_1}^{t_2} |v(t)| dt$$

Method: Find when $v(t) = 0$ (the particle changes direction). Split the integral at these times and integrate the absolute value on each subinterval.

Distance vs. Displacement

A particle moves with velocity $v(t) = t^2 - 4t + 3$ for $0 \leq t \leq 5$. Find the total distance travelled and the displacement.

Factor: $v(t) = (t - 1)(t - 3)$. The particle changes direction at $t = 1$ and $t = 3$.

Sign of v : positive for $t < 1$, negative for $1 < t < 3$, positive for $t > 3$.

Displacement:

$$s(5) - s(0) = \int_0^5 (t^2 - 4t + 3) dt = \left[\frac{t^3}{3} - 2t^2 + 3t \right]_0^5 = \frac{125}{3} - 50 + 15 = \frac{40}{3}$$

Total distance:

$$\begin{aligned} \int_0^1 v(t) dt &= \left[\frac{t^3}{3} - 2t^2 + 3t \right]_0^1 = \frac{1}{3} - 2 + 3 = \frac{4}{3} \\ \int_1^3 (-v(t)) dt &= - \left[\frac{t^3}{3} - 2t^2 + 3t \right]_1^3 = - \left(9 - 18 + 9 - \frac{4}{3} \right) = \frac{4}{3} \\ \int_3^5 v(t) dt &= \left[\frac{t^3}{3} - 2t^2 + 3t \right]_3^5 = \frac{40}{3} - 0 = \frac{40}{3} - 0 = \frac{125}{3} - 50 + 15 - (9 - 18 + 9) = \frac{40}{3} \end{aligned}$$

$$\text{Total distance} = \frac{4}{3} + \frac{4}{3} + \frac{40}{3} = \frac{48}{3} = 16 \text{ units.}$$

5.9 Curve Sketching

5.9.1 Systematic Approach to Curve Sketching

Complete Curve Sketching Checklist

Follow this systematic approach to sketch any curve:

1. **Domain:** Find all x -values where f is defined.
2. **Intercepts:**
 - y -intercept: set $x = 0$.
 - x -intercepts: set $y = 0$ (solve $f(x) = 0$).
3. **Symmetry:** Check for even/odd symmetry, or periodicity.
4. **Asymptotes:**
 - Vertical: values where f is undefined (or limits are infinite).
 - Horizontal: $\lim_{x \rightarrow \pm\infty} f(x)$.
 - Oblique/slant: when degree of numerator = degree of denominator + 1.
5. **First derivative:** Find $f'(x) = 0$ for stationary points, determine intervals of increase/decrease.
6. **Second derivative:** Find $f''(x) = 0$ for potential inflection points, determine concavity.

7. **Behavior at extremes:** What happens as x approaches domain boundaries?

8. **Sketch:** Plot key points and asymptotes, then connect smoothly.

Comprehensive Curve Sketch

Sketch $f(x) = \frac{x^2}{x^2 - 4}$.

Domain: $x^2 - 4 \neq 0$, so $x \neq \pm 2$. Domain = $\mathbb{R} \setminus \{-2, 2\}$.

Intercepts: y -intercept: $f(0) = 0$. x -intercepts: $x^2 = 0 \implies x = 0$ (origin).

Symmetry: $f(-x) = f(x)$, so even (symmetric about y -axis).

Asymptotes:

- Vertical: $x = -2$ and $x = 2$.
- Horizontal: $\lim_{x \rightarrow \infty} \frac{x^2}{x^2 - 4} = 1$, so $y = 1$.

First derivative:

$$f'(x) = \frac{2x(x^2 - 4) - x^2(2x)}{(x^2 - 4)^2} = \frac{2x^3 - 8x - 2x^3}{(x^2 - 4)^2} = \frac{-8x}{(x^2 - 4)^2}$$

$f'(x) = 0 \implies x = 0$. $f'(x) > 0$ for $x < -2$ and $-2 < x < 0$; $f'(x) < 0$ for $0 < x < 2$ and $x > 2$.

Stationary point: $(0, 0)$. Since f' changes from $+$ to $-$, it is a local maximum.

Second derivative: $f''(0) < 0$ confirms the maximum. Checking for inflection points is also possible but involved.

Sketch: The curve passes through the origin (local max), approaches $y = 1$ as $x \rightarrow \pm\infty$, and has vertical asymptotes at $x = \pm 2$.

5.10 Integration: Anti-Derivatives

5.10.1 The Indefinite Integral

Definition 5.10.1. The *indefinite integral* (or *anti-derivative*) of $f(x)$ is a function $F(x)$ such that $F'(x) = f(x)$. We write

$$\int f(x) \, dx = F(x) + C$$

where C is the *constant of integration*.

5.10.2 Standard Integral Table

Standard Integrals

$f(x)$	$\int f(x) \, dx$
x^n	$\frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
$\frac{1}{x}$	$\ln x + C$
e^x	$e^x + C$
a^x	$\frac{a^x}{\ln a} + C \quad (a > 0, a \neq 1)$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\sec^2 x$	$\tan x + C$
$\csc^2 x$	$-\cot x + C$
$\sec x \tan x$	$\sec x + C$
$\csc x \cot x$	$-\csc x + C$
$\frac{1}{\sqrt{1-x^2}}$	$\arcsin x + C$
$\frac{1}{1+x^2}$	$\arctan x + C$
$\tan x$	$-\ln \cos x + C = \ln \sec x + C$
$\cot x$	$\ln \sin x + C$
$\csc x$	$-\ln \csc x + \cot x + C$
$\sec x$	$\ln \sec x + \tan x + C$

Basic Integration

$$\int (3x^4 - 2x^2 + 5x - 7) \, dx = \frac{3x^5}{5} - \frac{2x^3}{3} + \frac{5x^2}{2} - 7x + C$$

Integrating with Fractional Powers

$$\int \frac{2}{\sqrt{x}} \, dx = \int 2x^{-1/2} \, dx = 2 \cdot \frac{x^{1/2}}{1/2} + C = 4\sqrt{x} + C$$

5.11 Definite Integrals and the Fundamental Theorem

5.11.1 The Definite Integral

Definition 5.11.1. The *definite integral* of f from a to b is

$$\int_a^b f(x) \, dx = F(b) - F(a)$$

where F is any anti-derivative of f .

Theorem 5.11.1 (Fundamental Theorem of Calculus, Part 1). *If f is continuous on $[a, b]$ and $F(x) = \int_a^x f(t) dt$, then $F'(x) = f(x)$ for all $x \in [a, b]$.*

In other words, integration and differentiation are inverse processes: the derivative of an integral gives back the original function.

Theorem 5.11.2 (Fundamental Theorem of Calculus, Part 2). *If f is continuous on $[a, b]$ and F is any anti-derivative of f , then*

$$\int_a^b f(x) dx = F(b) - F(a)$$

This connects the definite integral (area-like concept) with the anti-derivative (algebraic concept).

5.11.2 Properties of Definite Integrals

Properties of Definite Integrals

$$\begin{aligned}\int_a^a f(x) dx &= 0 \\ \int_a^b f(x) dx &= -\int_b^a f(x) dx \quad (\text{Reversal}) \\ \int_a^b [f(x) \pm g(x)] dx &= \int_a^b f(x) dx \pm \int_a^b g(x) dx \quad (\text{Linearity}) \\ \int_a^b c \cdot f(x) dx &= c \int_a^b f(x) dx \quad (\text{Constant multiple}) \\ \int_a^b f(x) dx &= \int_a^c f(x) dx + \int_c^b f(x) dx \quad (\text{Additivity})\end{aligned}$$

Evaluating a Definite Integral

$$\int_1^3 (x^2 + 2x) dx = \left[\frac{x^3}{3} + x^2 \right]_1^3 = \left(\frac{27}{3} + 9 \right) - \left(\frac{1}{3} + 1 \right) = 18 - \frac{4}{3} = \frac{50}{3}$$

Definite Integral with Trigonometric Functions

$$\int_0^{\pi/2} \sin^2 x dx = \int_0^{\pi/2} \frac{1 - \cos 2x}{2} dx = \left[\frac{x}{2} - \frac{\sin 2x}{4} \right]_0^{\pi/2} = \left(\frac{\pi}{4} - 0 \right) - (0 - 0) = \frac{\pi}{4}$$

5.12 Integration Techniques

5.12.1 Substitution (Reverse Chain Rule)

Integration by Substitution

If $u = g(x)$, then $du = g'(x) dx$, and

$$\int f(g(x)) g'(x) dx = \int f(u) du = F(u) + C = F(g(x)) + C$$

For a **definite integral**, change the limits:

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

When to Use Substitution

- Look for a function and its derivative (or a multiple thereof) in the integrand.
- If the integrand has a “composite function” structure, try substituting the inner function.
- For $\int f'(x) \cdot [f(x)]^n dx$, let $u = f(x)$.
- Sometimes a substitution transforms a seemingly complex integral into a standard form.

Substitution: $\int 2xe^{x^2} dx$

Let $u = x^2$, so $du = 2x dx$.

$$\int 2xe^{x^2} dx = \int e^u du = e^u + C = e^{x^2} + C$$

Substitution with Definite Integral

Evaluate $\int_0^1 x\sqrt{1+x^2} dx$.

Let $u = 1 + x^2$, so $du = 2x dx$.

When $x = 0$: $u = 1$. When $x = 1$: $u = 2$.

$$\int_0^1 x\sqrt{1+x^2} dx = \frac{1}{2} \int_1^2 u^{1/2} du = \frac{1}{2} \left[\frac{2u^{3/2}}{3} \right]_1^2 = \frac{1}{3} (2^{3/2} - 1) = \frac{2\sqrt{2} - 1}{3}$$

Trigonometric Substitution

Evaluate $\int \frac{x}{\sqrt{1-x^2}} dx$.

Let $u = 1 - x^2$, so $du = -2x dx$, i.e., $-\frac{1}{2}du = x dx$.

$$\int \frac{x}{\sqrt{1-x^2}} dx = -\frac{1}{2} \int u^{-1/2} du = -u^{1/2} + C = -\sqrt{1-x^2} + C$$

5.12.2 Integration by Parts

Integration by Parts

$$\int u dv = uv - \int v du$$

Equivalently, for definite integrals:

$$\int_a^b u dv = [uv]_a^b - \int_a^b v du$$

The LIATE Rule for Choosing u

When choosing which part of the integrand is u and which is dv , use the **LIATE** priority (highest to lowest):

1. **L** — Logarithmic functions ($\ln x$, $\log_a x$)
2. **I** — Inverse trigonometric functions ($\arcsin x$, $\arctan x$, etc.)
3. **A** — Algebraic functions (polynomials, x^n , rational functions)
4. **T** — Trigonometric functions ($\sin x$, $\cos x$, $\tan x$, etc.)
5. **E** — Exponential functions (e^x , a^x)

Choose u from the highest priority category available. The remaining factor is dv .

Integration by Parts: Polynomial times Exponential

Evaluate $\int xe^x dx$.

By LIATE, let $u = x$ (Algebraic) and $dv = e^x dx$.
Then $du = dx$ and $v = e^x$.

$$\int xe^x dx = xe^x - \int e^x dx = xe^x - e^x + C = e^x(x - 1) + C$$

Integration by Parts: Polynomial times Trigonometric

Evaluate $\int x^2 \sin x dx$.

This requires **repeated** integration by parts.

First application: $u = x^2$, $dv = \sin x dx$, $du = 2x dx$, $v = -\cos x$.

$$\int x^2 \sin x dx = -x^2 \cos x + \int 2x \cos x dx$$

Second application (on $\int 2x \cos x dx$): $u = 2x$, $dv = \cos x dx$, $du = 2 dx$, $v = \sin x$.

$$\int 2x \cos x dx = 2x \sin x - \int 2 \sin x dx = 2x \sin x + 2 \cos x + C$$

Combining:

$$\int x^2 \sin x dx = -x^2 \cos x + 2x \sin x + 2 \cos x + C$$

Integration by Parts: $\int \ln x dx$

Let $u = \ln x$ and $dv = dx$. Then $du = \frac{1}{x} dx$ and $v = x$.

$$\int \ln x dx = x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - \int 1 dx = x \ln x - x + C$$

Cyclic Integration by Parts: $\int e^x \sin x dx$

Let $I = \int e^x \sin x dx$.

First application: $u = \sin x$, $dv = e^x dx$, $du = \cos x dx$, $v = e^x$.

$$I = e^x \sin x - \int e^x \cos x dx$$

Second application: $u = \cos x$, $dv = e^x dx$, $du = -\sin x dx$, $v = e^x$.

$$I = e^x \sin x - \left(e^x \cos x + \int e^x \sin x dx \right) = e^x \sin x - e^x \cos x - I$$

Solving for I :

$$2I = e^x(\sin x - \cos x) \implies I = \frac{e^x(\sin x - \cos x)}{2} + C$$

Tabular Method for Repeated Integration by Parts

When integrating $x^n \cdot f(x)$ where $f(x)$ is e^x , $\sin x$, or $\cos x$ (functions that cycle upon repeated differentiation), use the **tabular method**:

Sign	Differentiate (u)	Integrate (dv)
+	x^n	$f(x)$
−	nx^{n-1}	$\int f(x) dx$
+	$n(n-1)x^{n-2}$	$\iint f(x) dx^2$
$\vdots$	$\vdots$	$\vdots$

Multiply diagonally with alternating signs. Stop when the “Differentiate” column reaches 0 or a repeat of $f(x)$.

5.12.3 Partial Fractions for Integration

When the integrand is a rational function with distinct linear factors in the denominator, partial fractions can decompose it into simpler terms.

Partial Fraction Decomposition

For $\frac{1}{(x-a)(x-b)}$ with $a \neq b$:

$$\frac{1}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$$

where $A = \frac{1}{a-b}$ and $B = \frac{1}{b-a}$.

Integration Using Partial Fractions

Evaluate $\int \frac{1}{(x-1)(x-3)} dx$.

Write: $\frac{1}{(x-1)(x-3)} = \frac{A}{x-1} + \frac{B}{x-3}$.

$$1 = A(x-3) + B(x-1).$$

$$x = 1: 1 = -2A \implies A = -\frac{1}{2}.$$

$$x = 3: 1 = 2B \implies B = \frac{1}{2}.$$

$$\begin{aligned}\int \frac{1}{(x-1)(x-3)} dx &= -\frac{1}{2} \int \frac{1}{x-1} dx + \frac{1}{2} \int \frac{1}{x-3} dx = -\frac{1}{2} \ln|x-1| + \frac{1}{2} \ln|x-3| + C \\ &= \frac{1}{2} \ln \left| \frac{x-3}{x-1} \right| + C\end{aligned}$$

Integration with Repeated Linear Factor

Evaluate $\int \frac{2x+1}{x(x^2+1)} dx$.

$$\frac{2x+1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}.$$

$$2x+1 = A(x^2+1) + (Bx+C)x = (A+B)x^2 + Cx + A.$$

Comparing: $A+B=0$, $C=2$, $A=1$. So $B=-1$.

$$\int \left(\frac{1}{x} + \frac{-x+2}{x^2+1} \right) dx = \ln|x| - \frac{1}{2} \ln(x^2+1) + 2 \arctan x + C$$

5.12.4 Trigonometric Integrals

Reduction Formulas for Powers of Sine and Cosine

Odd power of sin: $\int \sin^{2k+1} x dx = \int \sin^{2k} x \cdot \sin x dx = \int (1 - \cos^2 x)^k \cdot \sin x dx$

Use $u = \cos x$, $du = -\sin x dx$.

Odd power of cos: Similar strategy with $u = \sin x$.

Even powers: Use the half-angle (power-reducing) identities:

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

Integral of $\sin^3 x$

$$\int \sin^3 x dx = \int \sin^2 x \cdot \sin x dx = \int (1 - \cos^2 x) \sin x dx$$

Let $u = \cos x$, $du = -\sin x dx$:

$$= - \int (1 - u^2) du = - \left(u - \frac{u^3}{3} \right) + C = -\cos x + \frac{\cos^3 x}{3} + C$$

Integral of $\cos^2 x$

$$\int \cos^2 x dx = \int \frac{1 + \cos 2x}{2} dx = \frac{x}{2} + \frac{\sin 2x}{4} + C$$

5.13 Applications of Integration

5.13.1 Area Under a Curve

Area Under a Curve (above x -axis)

If $f(x) \geq 0$ on $[a, b]$, the area between the curve $y = f(x)$ and the x -axis from $x = a$ to $x = b$ is

$$A = \int_a^b f(x) \, dx$$

Area Between Curve and x -axis (general case)

When $f(x)$ changes sign on $[a, b]$, the total area is

$$A = \int_a^b |f(x)| \, dx$$

Method: Find where $f(x) = 0$, split the integral at these points, and ensure each sub-integral is positive.

Area Below the x -axis

Find the area enclosed by $y = x^2 - 4$ and the x -axis.

x -intercepts: $x^2 = 4 \implies x = \pm 2$.

On $[-2, 2]$, $x^2 - 4 \leq 0$, so:

$$A = \int_{-2}^2 |x^2 - 4| \, dx = \int_{-2}^2 (4 - x^2) \, dx = \left[4x - \frac{x^3}{3} \right]_{-2}^2 = \left(8 - \frac{8}{3} \right) - \left(-8 + \frac{8}{3} \right) = \frac{32}{3}$$

5.13.2 Area Between Two Curves

Area Between Two Curves

The area between $y = f(x)$ and $y = g(x)$ from $x = a$ to $x = b$ (where $f(x) \geq g(x)$) is

$$A = \int_a^b [f(x) - g(x)] \, dx$$

In general, if the curves cross:

$$A = \int_a^b |f(x) - g(x)| \, dx$$

Area Between Two Curves

Find the area between $y = x^2$ and $y = x + 2$.

Find intersection points: $x^2 = x + 2 \implies x^2 - x - 2 = 0 \implies (x - 2)(x + 1) = 0$, so $x = -1$ and $x = 2$.

On $[-1, 2]$, $x + 2 \geq x^2$ (since the line is above the parabola in this interval):

$$A = \int_{-1}^2 [(x + 2) - x^2] \, dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2 = \left(2 + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) = \frac{9}{2}$$

5.13.3 Area Between a Curve and the y -axis

Area Between Curve and y -axis

The area between the curve $x = g(y)$ and the y -axis from $y = c$ to $y = d$ is

$$A = \int_c^d |g(y)| \, dy$$

This is useful when it is easier to express x as a function of y .

5.13.4 Volume of Revolution

Volume of Revolution about the x -axis (Disk Method)

When the region bounded by $y = f(x)$, the x -axis, $x = a$, and $x = b$ is rotated about the x -axis:

$$V = \pi \int_a^b [f(x)]^2 \, dx$$

Volume of Revolution about the y -axis

Method 1 (using x as a function of y):

$$V = \pi \int_c^d [g(y)]^2 \, dy \quad \text{where } x = g(y)$$

Method 2 (shell method):

$$V = 2\pi \int_a^b x|f(x)| \, dx$$

Derivation of the Disk Formula

Consider rotating the region under $y = f(x)$ from $x = a$ to $x = b$ about the x -axis. Partition $[a, b]$ into subintervals of width Δx . The i -th subinterval $[x_{i-1}, x_i]$ sweeps out a **disk** (or washer) with:

- Outer radius: $R_i = f(x_i^*)$ (where $x_i^* \in [x_{i-1}, x_i]$)
- Thickness: Δx
- Volume of disk: $\pi R_i^2 \Delta x$

The total volume is approximated by the Riemann sum:

$$V \approx \sum_i \pi [f(x_i^*)]^2 \Delta x$$

Taking the limit as $\Delta x \rightarrow 0$ (as the number of subintervals $\rightarrow \infty$):

$$V = \lim_{n \rightarrow \infty} \sum_{i=1}^n \pi [f(x_i^*)]^2 \Delta x = \pi \int_a^b [f(x)]^2 \, dx$$

Volume of Revolution

Find the volume generated when $y = \sqrt{x}$, $x = 0$, $x = 4$ is rotated about the x -axis.

$$V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \int_0^4 x dx = \pi \left[\frac{x^2}{2} \right]_0^4 = \pi \cdot 8 = 8\pi$$

Volume of Revolution about the y -axis

Find the volume when $y = x^2$ ($0 \leq x \leq 2$) is rotated about the y -axis.

Using the shell method:

$$V = 2\pi \int_0^2 x \cdot x^2 dx = 2\pi \int_0^2 x^3 dx = 2\pi \left[\frac{x^4}{4} \right]_0^2 = 2\pi \cdot 4 = 8\pi$$

Using Method 1: $x = \sqrt{y}$, y ranges from 0 to 4.

$$V = \pi \int_0^4 (\sqrt{y})^2 dy = \pi \int_0^4 y dy = \pi \left[\frac{y^2}{2} \right]_0^4 = 8\pi \checkmark$$

5.13.5 Average Value of a Function**Average Value**

The **average (mean) value** of f on $[a, b]$ is

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx$$

Derivation of the Average Value Formula

Partition $[a, b]$ into n equal subintervals of width $\Delta x = \frac{b-a}{n}$. Sample points $x_1^*, x_2^*, \dots, x_n^*$.

The average of the sample values is:

$$\frac{f(x_1^*) + f(x_2^*) + \dots + f(x_n^*)}{n} = \frac{1}{b-a} \sum_{i=1}^n f(x_i^*) \Delta x$$

As $n \rightarrow \infty$, this becomes:

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx$$

Average Value

Find the average value of $f(x) = \sin x$ on $[0, \pi]$.

$$\bar{f} = \frac{1}{\pi - 0} \int_0^\pi \sin x dx = \frac{1}{\pi} [-\cos x]_0^\pi = \frac{1}{\pi} (-\cos \pi + \cos 0) = \frac{1}{\pi} (1 + 1) = \frac{2}{\pi}$$

5.14 Differential Equations

5.14.1 Separation of Variables

Definition 5.14.1. A **separable** differential equation is one that can be written in the form

$$\frac{dy}{dx} = f(x) \cdot g(y)$$

i.e., the right-hand side is a product of a function of x and a function of y .

Method of Separation of Variables

1. Separate: $\frac{1}{g(y)} dy = f(x) dx$
2. Integrate both sides: $\int \frac{1}{g(y)} dy = \int f(x) dx$
3. Solve for y (if possible) and include the constant of integration.
4. If an initial condition is given, use it to find the constant.

Separable Differential Equation

Solve $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 2$.

Separate: $y dy = x dx$.

Integrate: $\int y dy = \int x dx \implies \frac{y^2}{2} = \frac{x^2}{2} + C$.

Using $y(0) = 2$: $\frac{4}{2} = 0 + C \implies C = 2$.

So $\frac{y^2}{2} = \frac{x^2}{2} + 2 \implies y^2 = x^2 + 4 \implies y = \sqrt{x^2 + 4}$ (taking positive root since $y(0) = 2 > 0$).

Separable DE with Exponential

Solve $\frac{dy}{dx} = xy$, given $y(0) = 5$.

Separate: $\frac{1}{y} dy = x dx$.

Integrate: $\ln|y| = \frac{x^2}{2} + C$.

$y(0) = 5$: $\ln 5 = 0 + C \implies C = \ln 5$.

$\ln|y| = \frac{x^2}{2} + \ln 5 \implies \ln\left|\frac{y}{5}\right| = \frac{x^2}{2} \implies y = 5e^{x^2/2}$.

5.14.2 Exponential Growth and Decay

Exponential Growth and Decay

If $\frac{dN}{dt} = kN$ (rate of change proportional to the current amount), then

$$N(t) = N_0 e^{kt}$$

where $N_0 = N(0)$ is the initial amount and k is the constant of proportionality:

- $k > 0$: exponential **growth**
- $k < 0$: exponential **decay**

The **half-life** is $t_{1/2} = \frac{\ln 2}{|k|}$.

Derivation of the Exponential Growth/Decay Formula

Given $\frac{dN}{dt} = kN$, separate variables:

$$\frac{1}{N} dN = k dt$$

Integrate both sides:

$$\ln |N| = kt + C$$

Exponentiate:

$$|N| = e^{kt+C} = e^C \cdot e^{kt}$$

Let $N_0 = \pm e^C$ (sign determined by the sign of N):

$$N(t) = N_0 e^{kt}$$

At $t = 0$: $N(0) = N_0 e^0 = N_0$, confirming N_0 is the initial value.

Half-Life Problem

The half-life of carbon-14 is approximately 5730 years. A fossil contains 30% of the original carbon-14. How old is the fossil?

$$k = -\frac{\ln 2}{5730}$$

$N(t) = N_0 e^{kt}$. When $N(t) = 0.3 N_0$:

$$0.3 = e^{-t \ln 2 / 5730} \implies \ln 0.3 = -\frac{t \ln 2}{5730}$$

$$t = \frac{5730 \cdot \ln 0.3}{-\ln 2} = \frac{5730 \cdot \ln(10/3)}{\ln 2} \approx 9953 \text{ years}$$

5.14.3 Euler's Method

Euler's Method for Numerical Solutions

Given $\frac{dy}{dx} = f(x, y)$ with initial condition $y(x_0) = y_0$, approximate y at $x_0 + h, x_0 + 2h, \dots$ using:

$$y_{n+1} = y_n + h \cdot f(x_n, y_n)$$

$$x_{n+1} = x_n + h$$

where h is the step size.

Euler's Method

Use Euler's method with $h = 0.5$ to approximate $y(2)$ for $\frac{dy}{dx} = x + y$, $y(1) = 1$.

Step 1: $x_0 = 1$, $y_0 = 1$.

$$y_1 = y_0 + h \cdot f(x_0, y_0) = 1 + 0.5(1 + 1) = 2, \quad x_1 = 1.5$$

Step 2: $x_1 = 1.5$, $y_1 = 2$.

$$y_2 = y_1 + h \cdot f(x_1, y_1) = 2 + 0.5(1.5 + 2) = 2 + 1.75 = 3.75, \quad x_2 = 2$$

So $y(2) \approx 3.75$ by Euler's method with $h = 0.5$.

IB Exam Tip: Identifying the Type of Differential Equation

On IB exams, you may be asked to identify the type of DE or choose the correct method:

- **Separable:** Look for $\frac{dy}{dx} = f(x)g(y)$ or equivalent forms that can be rearranged so all y -terms are with dy and all x -terms are with dx .
- **Not separable:** If $\frac{dy}{dx}$ cannot be factored into a product of a function of x and a function of y , other methods (like Euler's method or integrating factors, though the latter is not in the syllabus) may be needed.
- **Always check** whether an initial condition is given — it allows you to determine C .
- **Domain consideration:** After solving, state the domain on which the solution is valid.

5.15 Maclaurin Series

5.15.1 Definition

Definition 5.15.1. The **Maclaurin series** of a function f is the Taylor series centered at $x = 0$:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$$

Validity and Convergence

- The Maclaurin series represents $f(x)$ only within its **interval of convergence**.
- For the standard IB functions (e^x , $\sin x$, $\cos x$, $\ln(1+x)$, $(1+x)^p$), the series converges for $|x| < 1$ (or for all x in the case of e^x , $\sin x$, $\cos x$).
- Always check convergence at the endpoints if needed.

5.15.2 Standard Maclaurin Expansions

Standard Maclaurin Series

$f(x)$	Maclaurin Series	Convergence
e^x	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$	all x
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$	all x
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$	all x
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$	$-1 < x \leq 1$
$(1+x)^p$	$1 + px + \frac{p(p-1)}{2!}x^2 + \frac{p(p-1)(p-2)}{3!}x^3 + \dots$	$ x < 1$

For $(1+x)^n$ where n is a positive integer, the series terminates (it is the binomial expansion) and converges for all x .

5.15.3 Derivations of Standard Series

Maclaurin Series for e^x

Let $f(x) = e^x$. Then $f^{(n)}(x) = e^x$ for all n , so $f^{(n)}(0) = 1$ for all n .

$$e^x = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

This series converges for all $x \in \mathbb{R}$ (by the ratio test, $\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/(n+1)!}{x^n/n!} \right| = \lim_{n \rightarrow \infty} \frac{|x|}{n+1} = 0 < 1$).

Maclaurin Series for $\sin x$

Let $f(x) = \sin x$. Compute derivatives at $x = 0$:

n	$f^{(n)}(x)$	$f^{(n)}(0)$
0	$\sin x$	0
1	$\cos x$	1
2	$-\sin x$	0
3	$-\cos x$	-1
4	$\sin x$	0

The pattern repeats with period 4. Only odd-order terms survive:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

Converges for all $x \in \mathbb{R}$.

Maclaurin Series for $\cos x$

Let $f(x) = \cos x$. Compute derivatives at $x = 0$:

n	$f^{(n)}(x)$	$f^{(n)}(0)$
0	$\cos x$	1
1	$-\sin x$	0
2	$-\cos x$	-1
3	$\sin x$	0
4	$\cos x$	1

Only even-order terms survive:

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$$

Alternative derivation: Since $\frac{d}{dx}[\sin x] = \cos x$, differentiate the series for $\sin x$ term by term:

$$\cos x = \frac{d}{dx} \left[x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \right] = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

Maclaurin Series for $\ln(1+x)$

Let $f(x) = \ln(1+x)$. Then $f(0) = 0$.

$$f'(x) = \frac{1}{1+x} = (1+x)^{-1}$$

$$f''(x) = -(1+x)^{-2}, \quad f'''(x) = 2(1+x)^{-3}, \quad f^{(4)}(x) = -6(1+x)^{-4}$$

In general: $f^{(n)}(x) = (-1)^{n-1}(n-1)!(1+x)^{-n}$ for $n \geq 1$.

So $f^{(n)}(0) = (-1)^{n-1}(n-1)!$ for $n \geq 1$.

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}(n-1)!}{n!} x^n = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

Converges for $-1 < x \leq 1$.

Alternative derivation: Since $\frac{d}{dx}[\ln(1+x)] = \frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots$ for $|x| < 1$, integrate term by term:

$$\ln(1+x) = \int (1 - x + x^2 - x^3 + \cdots) dx = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

5.15.4 Using Substitution in Known Series**Substitution in Maclaurin Series**

Replace x with an expression to obtain new series. For example, in the series for e^x :

- Replace x with $-x$: $e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots$

- Replace x with x^2 : $e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \dots$
- Replace x with $2x$: $e^{2x} = 1 + 2x + 2x^2 + \frac{4x^3}{3} + \dots$

Series for e^{x^2}

Find the Maclaurin series for e^{x^2} up to the term in x^6 .

Replace x with x^2 in the series for e^x :

$$e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \dots = 1 + x^2 + \frac{x^4}{2} + \frac{x^6}{6} + \dots$$

Series for $\ln(1 - x^2)$

Replace x with $(-x^2)$ in the series for $\ln(1 + x)$:

$$\ln(1 - x^2) = (-x^2) - \frac{(-x^2)^2}{2} + \frac{(-x^2)^3}{3} - \dots = -x^2 - \frac{x^4}{2} - \frac{x^6}{3} - \dots$$

valid for $|x| < 1$.

5.15.5 Products of Series

Product of Two Maclaurin Series

Find the Maclaurin series of $e^x \cos x$ up to x^4 .

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots$$

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + \dots$$

Multiply and collect terms up to x^4 :

$$\begin{aligned} e^x \cos x &= \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}\right) \left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right) + \dots \\ &= 1 \left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right) + x \left(1 - \frac{x^2}{2}\right) + \frac{x^2}{2}(1) + \frac{x^3}{6}(1) + \frac{x^4}{24}(1) + \dots \\ &= 1 - \frac{x^2}{2} + \frac{x^4}{24} + x - \frac{x^3}{2} + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots \\ &= 1 + x + \frac{x^4}{24} + \frac{x^4}{24} - \frac{x^3}{2} + \frac{x^3}{6} + \dots \\ &= 1 + x - \frac{x^3}{3} + \frac{x^4}{12} + \dots \end{aligned}$$

Note: the x^2 terms cancel ($-\frac{1}{2} + \frac{1}{2} = 0$).

5.15.6 Differentiating and Integrating Series

Term-by-Term Differentiation and Integration

If $f(x) = \sum_{n=0}^{\infty} a_n x^n$ for $|x| < R$, then:

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \quad (\text{same radius of convergence})$$

$$\int_0^x f(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1} \quad (\text{same radius of convergence})$$

Series from Integration

Find the Maclaurin series for $\arctan x$.

We know $\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \dots$ for $|x| < 1$.

Since $\frac{d}{dx}[\arctan x] = \frac{1}{1+x^2}$, integrate term by term:

$$\arctan x = \int_0^x (1 - t^2 + t^4 - t^6 + \dots) dt = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

valid for $|x| \leq 1$ (converges at the endpoints by the alternating series test).

5.15.7 Series from Differential Equations

Finding a Series from a Differential Equation

Find the first three non-zero terms of the Maclaurin series for the solution to $\frac{dy}{dx} = y + x$, with $y(0) = 1$.

Method 1: Solve the DE directly.

This is a linear DE: $y' - y = x$. The integrating factor is e^{-x} .

$e^{-x}y' - e^{-x}y = xe^{-x}$, so $\frac{d}{dx}[e^{-x}y] = xe^{-x}$.

Integrating (by parts): $e^{-x}y = -xe^{-x} - e^{-x} + C$, so $y = -x - 1 + Ce^x$.

$y(0) = 1$: $1 = -0 - 1 + C \implies C = 2$.

$$y = 2e^x - x - 1 = 2 \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots \right) - x - 1 = 1 + x + x^2 + \frac{x^3}{3} + \dots$$

Method 2: Use successive differentiation.

$y(0) = 1$.

From the DE at $x = 0$: $y'(0) = y(0) + 0 = 1$.

Differentiate the DE: $y'' = y' + 1$. At $x = 0$: $y''(0) = 1 + 1 = 2$.

Differentiate again: $y''' = y''$. At $x = 0$: $y'''(0) = 2$.

$$y = 1 + x + \frac{2}{2!}x^2 + \frac{2}{3!}x^3 + \dots = 1 + x + x^2 + \frac{x^3}{3} + \dots$$

5.15.8 General Binomial Expansion

General Binomial Theorem

For any real p and $|x| < 1$:

$$(1+x)^p = \sum_{n=0}^{\infty} \binom{p}{n} x^n = 1 + px + \frac{p(p-1)}{2!} x^2 + \frac{p(p-1)(p-2)}{3!} x^3 + \dots$$

where $\binom{p}{n} = \frac{p(p-1)(p-2)\cdots(p-n+1)}{n!}$.

When p is a positive integer, the series terminates at $n = p$ (finite sum).

Binomial Expansion for Non-Integer Exponent

Find the Maclaurin series of $\frac{1}{\sqrt{1-x}} = (1-x)^{-1/2}$ up to x^3 .

$$\begin{aligned} (1-x)^{-1/2} &= 1 + \left(-\frac{1}{2}\right)(-x) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2!}(-x)^2 + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{3!}(-x)^3 + \dots \\ &= 1 + \frac{x}{2} + \frac{3x^2}{8} + \frac{5x^3}{16} + \dots \end{aligned}$$

Note: $\frac{1}{\sqrt{1-x^2}}$ is obtained by replacing x with x^2 :

$$\frac{1}{\sqrt{1-x^2}} = 1 + \frac{x^2}{2} + \frac{3x^4}{8} + \frac{5x^6}{16} + \dots$$

This is useful because $\frac{d}{dx}[\arcsin x] = \frac{1}{\sqrt{1-x^2}}$, so integrating gives:

$$\arcsin x = x + \frac{x^3}{6} + \frac{3x^5}{40} + \dots$$

IB Exam Tip: Maclaurin Series Strategy

- **Know the standard series by heart.** The IB data booklet provides some, but fluency saves time.
- **IB questions often ask for** the series “up to the term in x^n ” — write out enough terms and collect like powers.
- **Common IB tasks:**
 - Find the Maclaurin series of a composite function (use substitution in a known series).
 - Find the Maclaurin series of a product (multiply two known series).
 - Find the Maclaurin series of a quotient or a function defined by a differential equation.
 - Use a Maclaurin series to evaluate a limit (e.g., $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$).
 - Use a Maclaurin series to approximate a definite integral.
- **Using series for limits:** Substitute the series and simplify — often the result is immediate.

- **Using series for integrals:** Integrate the series term by term, then evaluate.
- **Error bounds** are sometimes asked: the remainder after n terms for an alternating series is bounded by the first omitted term.

Using Maclaurin Series to Evaluate a Limit

Evaluate $\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{x^5}$.

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$$

$$\sin x - x + \frac{x^3}{6} = \left(x - \frac{x^3}{6} + \frac{x^5}{120} - \dots \right) - x + \frac{x^3}{6} = \frac{x^5}{120} - \dots$$

$$\lim_{x \rightarrow 0} \frac{\frac{x^5}{120} - \dots}{x^5} = \frac{1}{120}$$

Using Maclaurin Series to Approximate an Integral

Use the Maclaurin series for e^{-x^2} to approximate $\int_0^{0.5} e^{-x^2} dx$.

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2} - \frac{x^6}{6} + \dots$$

Integrate term by term:

$$\begin{aligned} \int_0^{0.5} e^{-x^2} dx &= \left[x - \frac{x^3}{3} + \frac{x^5}{10} - \frac{x^7}{42} + \dots \right]_0^{0.5} \\ &= 0.5 - \frac{0.125}{3} + \frac{0.03125}{10} - \frac{0.0078125}{42} + \dots \\ &\approx 0.5 - 0.04167 + 0.003125 - 0.000186 \approx 0.46127 \end{aligned}$$

Chapter 6

IB Exam Mastery

This chapter is your practical guide to maximising marks on the IB Mathematics AA HL examination. It covers the precise meaning of every command term you will encounter, how markschemes are structured, paper-specific strategies, proof-writing technique, common mark-losing errors, calculator skills, and time management. Master this chapter and you will approach every exam paper with confidence and precision.

6.1 Understanding IB Command Terms

Every IB Mathematics question begins with a **command term** that tells you exactly what the examiner expects. Misunderstanding a command term is one of the most reliable ways to lose marks that you otherwise deserve. Below is a complete reference for every command term that appears on AA HL papers.

6.1.1 Hence

Command Term: Hence

When a question says **Hence**, you **must** use the result you obtained in the immediately preceding part of the question. You are *not* permitted to use an alternative method, even if it is valid.

Example: Hence

Question. (a) Find $\frac{d}{dx}(x^2e^x)$. (b) **Hence** evaluate $\int_0^1 2xe^x + x^2e^x dx$.

Part (a): $\frac{d}{dx}(x^2e^x) = 2xe^x + x^2e^x$.

Part (b) — correct approach:

$$\int_0^1 (2xe^x + x^2e^x) dx = [x^2e^x]_0^1 = e - 0 = e.$$

This uses the result from (a) by the Fundamental Theorem of Calculus.

What loses marks: Integrating by parts from scratch, or using any method that does not reference the derivative found in part (a). The examiner provided part (a) specifically so you would use it. If you ignore it, you lose all method marks for part (b), even if your final answer is correct.

6.1.2 Hence or Otherwise

Command Term: Hence or Otherwise

Hence or otherwise means you *may* use the previous result, but you are also free to use any other valid mathematical method. Both approaches can earn full marks. However, using the “hence” route is almost always faster and less error-prone.

Example: Hence or Otherwise

Question. (a) Show that $\sin 2\theta = 2 \sin \theta \cos \theta$. (b) **Hence or otherwise**, solve $\sin 2\theta = \sin \theta$ for $0 \leq \theta \leq 2\pi$.

Using “hence”: Substitute from (a): $2 \sin \theta \cos \theta = \sin \theta$, so $\sin \theta(2 \cos \theta - 1) = 0$, giving $\theta = 0, \pi, 2\pi, \frac{\pi}{3}, \frac{5\pi}{3}$.

Using “otherwise”: Rewrite $\sin 2\theta - \sin \theta = 0$ and use the compound-angle formula independently. This is valid but repeats working already done in (a).

6.1.3 Show That

Command Term: Show That

Show that means the answer is already given to you. You must present a complete, logical chain of working that *leads to* the stated result. The final answer itself carries **no marks** — all marks are for the working. Omitting steps, even obvious ones, will cost marks.

Example: Show That

Question. Show that $\frac{x^2 - 9}{x^2 + 6x + 9} = \frac{x - 3}{x + 3}$, for $x \neq -3$.

Full-credit working:

$$\begin{aligned} \frac{x^2 - 9}{x^2 + 6x + 9} &= \frac{(x - 3)(x + 3)}{(x + 3)(x + 3)} && \text{(factorizing numerator and denominator)} \\ &= \frac{(x - 3)(x + 3)}{(x + 3)^2} && \text{(rewriting denominator)} \\ &= \frac{x - 3}{x + 3} && \text{(cancelling } x + 3 \neq 0). \end{aligned}$$

What loses marks: Writing only the answer $\frac{x - 3}{x + 3}$ with no working. Writing “by factorisation” without showing the factors. Skipping the condition $x \neq -3$ when it is relevant to the domain.

6.1.4 Find

Command Term: Find

Find means you must calculate or determine the answer. Unlike “show that”, the **final answer carries marks**. You still need to show sufficient working, but the examiner is primarily interested in whether you arrive at the correct result.

Example: Find

Question. Find the y -intercept of the graph of $f(x) = \frac{2x-1}{x+3}$.

Solution: The y -intercept occurs at $x = 0$:

$$f(0) = \frac{2(0)-1}{0+3} = -\frac{1}{3}.$$

The y -intercept is $\left(0, -\frac{1}{3}\right)$.

What loses marks: Writing only $-1/3$ without stating the coordinate pair. Arithmetic errors in substitution.

6.1.5 Determine**Command Term: Determine**

Determine is a stronger form of “find”. It requires not just a calculation but also **reasoning, justification, or a decision**. You may need to consider multiple cases, test conditions, or explain why a particular result holds. The reasoning itself attracts marks.

Example: Determine

Question. Determine the number of solutions to $x^3 - 3x + 1 = 0$.

Solution: Let $f(x) = x^3 - 3x + 1$. Then $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$. The critical points are at $x = -1$ and $x = 1$. Evaluating:

$$f(-1) = 3 > 0, \quad f(1) = -1 < 0, \quad \lim_{x \rightarrow -\infty} f(x) = -\infty, \quad \lim_{x \rightarrow +\infty} f(x) = +\infty.$$

Since f changes sign once between $(-\infty, -1)$, once between $(-1, 1)$, and once between $(1, \infty)$, by the Intermediate Value Theorem there are **three** real solutions.

What loses marks: Simply stating “three solutions” without justification. Using the GDC to find approximate roots without explaining the reasoning process.

6.1.6 Prove**Command Term: Prove**

Prove demands a rigorous, complete, and logically sound argument. Every step must be justified. You cannot use the result you are trying to prove at any point in your working (circular reasoning). The level of rigour expected is higher than for “show that”.

Example: Prove

Question. Prove that the product of two odd integers is odd.

Solution: Let $a = 2m + 1$ and $b = 2n + 1$, where $m, n \in \mathbb{Z}$ (since a and b are odd). Then:

$$\begin{aligned} ab &= (2m+1)(2n+1) \\ &= 4mn + 2m + 2n + 1 \\ &= 2(2mn + m + n) + 1. \end{aligned}$$

Since $2mn + m + n \in \mathbb{Z}$, the product ab is of the form $2k + 1$ for some integer k , and is therefore odd.

What loses marks: Giving a single numerical example (e.g., $3 \times 5 = 15$, which is odd). Not defining m and n as integers. Skipping algebraic expansion steps.

6.1.7 Solve

Command Term: Solve

Solve means find *all* values that satisfy the given equation or inequality under the stated conditions. You must consider the domain, check for extraneous solutions, and present the complete solution set.

Example: Solve

Question. Solve $\log_2(x) + \log_2(x - 2) = 3$.

Solution:

$$\begin{aligned}\log_2(x(x - 2)) &= 3 \\ x(x - 2) &= 2^3 = 8 \\ x^2 - 2x - 8 &= 0 \\ (x - 4)(x + 2) &= 0 \\ x &= 4 \quad \text{or} \quad x = -2.\end{aligned}$$

Checking: $x = -2$ is invalid because $\log_2(-2)$ is undefined. Therefore $x = 4$ is the only solution.

What loses marks: Failing to reject $x = -2$. Not stating the domain requirement $x > 2$. Missing one of the potential solutions before checking validity.

6.1.8 Verify

Command Term: Verify

Verify means substitute the given value(s) into the given expression or equation and **confirm** that the stated result holds. This is *not* the same as proving — you are simply checking a claim, not establishing it from first principles. Verification typically carries fewer marks than proof.

Example: Verify

Question. Verify that $x = 3$ is a solution of $x^3 - 6x^2 + 11x - 6 = 0$.

Solution:

$$\begin{aligned}(3)^3 - 6(3)^2 + 11(3) - 6 &= 27 - 54 + 33 - 6 \\ &= 0. \quad \checkmark\end{aligned}$$

Since the left-hand side equals zero when $x = 3$, the value $x = 3$ is verified as a solution.

What loses marks: Attempting to solve the equation from scratch (while not wrong, it wastes time and is not what was asked). Making an arithmetic error in the substitution.

6.1.9 Deduce

Command Term: Deduce

Deduce means derive a conclusion from the information or results already given or established in previous parts. You must show a clear logical chain from the given information to your conclusion. You should *not* introduce new, unexplained assumptions.

Example: Deduce

Question. (a) Find $f'(x)$ where $f(x) = x \ln x - x$. (b) **Deduce** that $f(x)$ has a minimum at $x = 1$.

Solution: From (a), $f'(x) = \ln x + 1 - 1 = \ln x$.

Setting $f'(x) = 0$: $\ln x = 0$, so $x = 1$.

For $0 < x < 1$: $\ln x < 0$, so $f'(x) < 0$ (function decreasing).

For $x > 1$: $\ln x > 0$, so $f'(x) > 0$ (function increasing).

Since f' changes from negative to positive at $x = 1$, by the first derivative test, f has a **minimum** at $x = 1$.

What loses marks: Not connecting the result of part (a) to part (b). Simply stating “ $x = 1$ is the minimum” without using the derivative found in (a).

6.1.10 Write Down

Command Term: Write Down

Write down means state the answer directly, with minimal or no working. This usually applies when the answer can be read off, recalled from memory, or obtained by direct substitution. However, you should still show brief working if there is any doubt — examiners can award *no* marks for an incorrect answer with no working, even on “write down” questions.

Example: Write Down

Question. Write down the value of $\log_2 8$.

Solution: $\log_2 8 = 3$, since $2^3 = 8$.

What loses marks: Writing an incorrect value. If the question is less trivial, writing nothing because you are unsure — always attempt an answer.

6.1.11 Sketch

Command Term: Sketch

Sketch means draw a graph showing the **general shape** and **key features** (intercepts, asymptotes, maxima, minima, end behaviour). The graph does **not** need to be to scale, but all key features must be clearly labelled and correctly positioned relative to each other.

Example: Sketch

Question. Sketch the graph of $f(x) = \frac{1}{x-2} + 1$.

Key features to show:

- Vertical asymptote at $x = 2$

- Horizontal asymptote at $y = 1$
- y -intercept at $\left(0, \frac{1}{2}\right)$
- x -intercept at $(1, 0)$
- Correct general shape (rectangular hyperbola, shifted)
- Behaviour on each side of the asymptote

What loses marks: Drawing the graph to scale but missing key features. Forgetting to label asymptotes. Drawing the wrong branch orientation. Not indicating intercepts.

6.1.12 Draw

Command Term: Draw

Draw means produce an **accurate, to-scale** diagram using a ruler and compass (on Paper 1) or using your GDC (on Paper 2). Unlike “sketch”, the relative positions and proportions must be correct. This command term is rare in AA HL but may appear in geometry questions.

6.1.13 State

Command Term: State

State means give a specific name, value, theorem, or expression. No working is required, but the answer must be precise. This is similar to “write down” but often used when the answer is a definition, a theorem name, or a specific mathematical object.

Example: State

Question. State the Intermediate Value Theorem.

Solution: If f is continuous on the closed interval $[a, b]$ and k is any number between $f(a)$ and $f(b)$, then there exists at least one $c \in (a, b)$ such that $f(c) = k$.

What loses marks: Giving an informal or imprecise statement. Missing the condition of continuity. Confusing it with a different theorem.

6.1.14 Explain

Command Term: Explain

Explain means give a mathematical reason or justification for a statement or result. You must use mathematical language and reasoning, not informal English. The explanation itself carries marks.

Example: Explain

Question. Explain why $f(x) = x^2 + 1$ has no real zeros.

Solution: For all $x \in \mathbb{R}$, $x^2 \geq 0$, so $x^2 + 1 \geq 1 > 0$. Since $f(x) > 0$ for all real x , the equation $x^2 + 1 = 0$ has no real solutions.

What loses marks: Writing “because the parabola doesn’t cross the x -axis” without mathematical justification. Saying “you can’t square root a negative number” without connecting it to the specific function.

6.1.15 Justify

Command Term: Justify

Justify is stronger than “explain”. It requires you to provide **mathematical evidence** — calculations, theorems, or logical deductions — that definitively support your answer. Simply giving a reason is not enough; the evidence must be sufficient to convince a sceptical reader.

Example: Justify

Question. The roots of $x^2 - 5x + 6 = 0$ are α and β . Justify that $\alpha + \beta = 5$ without solving the equation.

Solution: By Vieta’s formulas, for a quadratic $ax^2 + bx + c = 0$ with roots α and β :

$$\alpha + \beta = -\frac{b}{a} = -\frac{(-5)}{1} = 5.$$

What loses marks: Solving the equation to find $\alpha = 2$ and $\beta = 3$, then adding them. While the numerical answer is correct, the question explicitly asked for justification *without solving*, so this approach does not earn full credit.

6.2 Markscheme Structure and Method Marks

Understanding how the markscheme works is essential for maximising your score. The IB uses a precise system of letter-coded marks.

6.2.1 Types of Marks

Mark Types in IB Markschemes

M — Method marks. Awarded for selecting and applying the correct method or approach. If you use the wrong method, you typically lose all M marks for that part, even if your algebra is flawless.

A — Answer marks. Awarded for obtaining the correct answer *following correct working*. If you make an arithmetic error but used the right method, you may still earn M marks but lose the A mark.

R — Reasoning marks. Awarded for correct mathematical reasoning, justification, or explanation. These appear in “explain”, “justify”, “prove”, and “determine” questions.

C — Communication marks. Awarded for clear, logical, and well-organised presentation of your working. This includes using correct notation, labelling steps, and writing in coherent sentences where required.

6.2.2 How Marks Are Distributed

Multi-part questions in IB Mathematics typically follow a predictable pattern:

Typical Multi-Part Question Structure

Part (a) — Foundation (easy marks): Tests a basic skill or recall. Usually 1–3 marks. This part often establishes a result used later.

Part (b) — Application (moderate): Uses the result from part (a) or applies a concept in a straightforward way. Usually 3–5 marks.

Part (c) — Extension (challenging): Combines concepts, requires deeper reasoning, or extends the problem. Usually 4–8 marks. This is where grade boundaries are most affected.

Example: Mark Distribution

Question (7 marks total).

- (a) Find $f'(x)$ where $f(x) = xe^{-x}$. [2 marks: M1, A1]
 (b) Hence find the coordinates of the stationary point of f . [3 marks: M1, A1, A1]
 (c) Determine whether this stationary point is a maximum or a minimum. [2 marks: R1, R1]

In part (c), both marks are for reasoning: you must use the second derivative *or* a sign test and clearly state your conclusion with justification.

6.2.3 Follow-Through Marks

Follow-Through (FT) Marks — When They Apply

Follow-through marks allow you to earn credit in later parts of a question even if you made an error in an earlier part. The examiner follows your **incorrect** answer forward and awards marks for correct *method* applied to your wrong value.

Example: If you incorrectly find $f'(x) = 2xe^{-x} - xe^{-x}$ in part (a) (sign error), but then correctly set this equal to zero, solve for x , and find the y -coordinate in part (b), you can still earn **M marks** in part (b) for the correct method. You would lose the **A marks** because the final numerical answer is wrong.

Follow-Through Marks — When They Do NOT Apply

Follow-through marks are **NOT** awarded in these situations:

- The error makes the subsequent working *trivially simple* (e.g., you get $x = 0$ by mistake and the rest of the question collapses).
- The question uses the command term “hence” and your part (a) answer is wrong — you cannot “follow through” a method you were supposed to use from (a).
- The error is in a “show that” or “prove” question, where the result is already given and you are demonstrating the method.
- The question is essentially a new start, not dependent on the previous part.
- Your working contains a **conceptual** error (not just an arithmetic slip). For example, integrating instead of differentiating is a conceptual error, not a follow-through situation.

6.3 Paper 1: Non-Calculator Strategies

6.3.1 Paper 1 Format

Paper 1 Overview

- **Duration:** 120 minutes
- **Calculator:** NOT permitted
- **Short questions:** Section A (approximately 60 marks)
- **Extended questions:** Section B (approximately 60 marks)
- **Total:** 120 marks
- **Weighting:** 30% of total IB grade

6.3.2 Essential Non-Calculator Skills

The following algebraic skills must be **fluent** — you should be able to perform them quickly and accurately without hesitation.

Algebraic Fluency Checklist

- Expanding brackets: $(ax + b)(cx + d)$, $(a + b)^n$ for small n
- Factorising quadratics: $ax^2 + bx + c$ by inspection
- Completing the square: $ax^2 + bx + c = a(x - h)^2 + k$
- Simplifying algebraic fractions
- Working with surds: $\sqrt{a} + \sqrt{b}$, rationalising denominators
- Laws of indices including negative and fractional powers
- Logarithm laws: $\log(ab)$, $\log(a/b)$, $\log(a^b)$
- Exact trigonometric values: $\sin(\pi/6)$, $\cos(\pi/4)$, $\tan(\pi/3)$, etc.
- Solving simultaneous equations (substitution and elimination)
- Partial fraction decomposition (for integration)

6.3.3 Speed Techniques

Factorising Quadratics Quickly

For $x^2 + bx + c$: find two numbers that *multiply* to c and *add* to b .

Speed Factorisation

$x^2 - 7x + 12$: factors of 12 that sum to -7 are -3 and -4 . Therefore $x^2 - 7x + 12 = (x - 3)(x - 4)$.

$6x^2 + x - 2$: multiply $6 \times (-2) = -12$. Factors of -12 that sum to 1 are 4 and -3 . Split the middle term:

$$6x^2 + 4x - 3x - 2 = 2x(3x + 2) - 1(3x + 2) = (2x - 1)(3x + 2).$$

Completing the Square Mentally

Mental Completing the Square

For $x^2 + bx$: add and subtract $\left(\frac{b}{2}\right)^2$.

$$x^2 + 6x = (x + 3)^2 - 9$$

$$x^2 - 10x = (x - 5)^2 - 25$$

$$x^2 + 5x = \left(x + \frac{5}{2}\right)^2 - \frac{25}{4}$$

For $ax^2 + bx + c$: factor out a first, then complete the square inside.

Recognising Special Forms

Forms to Recognise Instantly

$a^2 - b^2 = (a - b)(a + b)$	(difference of squares)
$a^2 + 2ab + b^2 = (a + b)^2$	(perfect square)
$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$	(sum of cubes)
$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$	(difference of cubes)
$a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + ab^{n-2} + b^{n-1})$	

Efficient Simplification

Efficient Simplification

Simplify $\frac{\sqrt{3} + 1}{\sqrt{3} - 1}$.

Multiply numerator and denominator by $\sqrt{3} + 1$:

$$\frac{(\sqrt{3} + 1)^2}{(\sqrt{3})^2 - 1^2} = \frac{3 + 2\sqrt{3} + 1}{3 - 1} = \frac{4 + 2\sqrt{3}}{2} = 2 + \sqrt{3}.$$

6.3.4 Common Paper 1 Mistakes

Top Paper 1 Errors

- Sign errors in expansion.** Expanding $-(2x - 3)$ as $-2x - 3$ instead of $-2x + 3$. Always use brackets and distribute carefully.
- Forgetting domain restrictions.** When simplifying $\frac{x^2 - 4}{x - 2} = x + 2$, forgetting to state $x \neq 2$. When taking square roots, forgetting that $\sqrt{x^2} = |x|$, not x .
- Not showing enough working for proof questions.** Proof questions on Paper 1 require *every* step. Skipping algebraic steps like “by inspection” or “obviously” will lose marks.
- Arithmetic errors.** Without a calculator, errors in basic arithmetic ($7 \times 8 = 54$, $\sqrt{144} = 11$) are common. Double-check all calculations. Write intermediate steps — do not attempt multi-step arithmetic mentally.
- Incorrect exact values.** Confusing $\sin(\pi/3) = \frac{\sqrt{3}}{2}$ with $\sin(\pi/6) = \frac{1}{2}$. Memorise the exact value table thoroughly.

6. Losing the constant of integration. Writing $\int 2x \, dx = x^2$ instead of $x^2 + C$.

6.3.5 Exact Values Reference

Exact Trigonometric Values for Paper 1

	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

6.4 Paper 2: GDC (Calculator) Strategies

6.4.1 Paper 2 Format

Paper 2 Overview

- **Duration:** 120 minutes
- **Calculator:** Permitted (GDC required)
- **Short questions:** Section A (approximately 60 marks)
- **Extended questions:** Section B (approximately 60 marks)
- **Total:** 120 marks
- **Weighting:** 30% of total IB grade

6.4.2 When to Use the GDC vs. Algebraic Methods

GDC Decision Guide

Use the GDC when:

- The question says “find” or “determine” and the algebra is tedious
- You need numerical solutions to equations that cannot be solved algebraically
- You need to evaluate definite integrals (especially with no elementary antiderivative)
- You need statistical calculations: mean, standard deviation, binomial/normal probabilities
- You need to find intersections, maxima, or minima graphically
- You need matrix operations (multiplication, inverse, determinant)

Use algebraic methods when:

- The question says “show that” or “prove” (the GDC alone is insufficient)
- The question says “hence” and expects you to use a previous algebraic result
- Exact answers are required (the GDC gives decimal approximations)
- The working is straightforward and algebraic is faster
- You are doing proof or derivation (GDC cannot do this)

6.4.3 GDC Techniques

Solving Equations Numerically

Solving Equations on the GDC

Question. Find the solution to $e^x = x + 3$.

Method: Use the solver or find the intersection of $y = e^x$ and $y = x + 3$.

On TI-84: Press **MATH** → **0:Solver**, enter $e^X - X - 3 = 0$, provide an initial guess, and solve. Alternatively, graph both functions and use **2nd** → **CALC** → **5:intersect**.

Result: $x \approx 1.50524$ (to 5 s.f.)

What to write: “Using the GDC solver, $x \approx 1.51$ (3 s.f.)”

Finding Intersections

Finding Intersections

Graph both functions, then use the intersection tool. On most GDCs:

1. Enter both functions in the $Y =$ editor
2. Press **GRAPH**
3. Use the **CALC** or **TRACE** menu
4. Select “Intersect” and specify both curves
5. Move the cursor near the intersection point and confirm

Always verify the intersection visually before accepting the GDC result.

Calculating Definite Integrals

Numerical Integration

Question. Evaluate $\int_0^2 e^{-x^2} dx$.

This integral has no elementary antiderivative. Use the GDC:

On TI-84: **MATH** → **9:fnInt**(, enter e^{-X^2} , X , 0, 2.

Result: $\int_0^2 e^{-x^2} dx \approx 0.88208$ (5 s.f.)

What to write: State the integral, write “using GDC”, then give the answer with appropriate precision.

Statistical Calculations

GDC Statistical Functions

Binomial distribution:

- $\text{binompdf}(n, p, x)$ — probability of exactly x successes
- $\text{binomcdf}(n, p, x)$ — probability of at most x successes ($X \leq x$)
- For $P(X \geq x)$: use $1 - \text{binomcdf}(n, p, x - 1)$
- For $P(a \leq X \leq b)$: use $\text{binomcdf}(n, p, b) - \text{binomcdf}(n, p, a - 1)$

Normal distribution:

- $\text{normalcdf}(a, b, \mu, \sigma) = P(a < X < b)$
- $\text{invNorm}(\text{area}, \mu, \sigma)$ — inverse normal
- For $P(X < a)$: use $\text{normalcdf}(-10^{99}, a, \mu, \sigma)$

Matrix Operations**Matrix Operations on the GDC**

Most GDCs support matrix entry, multiplication, inversion, and determinant calculation. On the TI-84:

- $2^{\text{nd}} \rightarrow \text{MATRIX}$ to access the matrix editor
- Define matrices $[A]$, $[B]$, etc. with appropriate dimensions
- Use $[A]^{-1}$ for the inverse
- Use $\det([A])$ for the determinant
- Use $[A] \times [B]$ for multiplication

Tip: Always store your matrices in named variables rather than retyping them.

6.4.4 Common GDC Mistakes**GDC Errors That Lose Marks**

- 1. Radian vs. degree mode.** This is the single most common GDC error. Before every exam, check your angle mode. If the question involves π or trigonometric functions in calculus, you are almost certainly in radian mode. Set your GDC to radian mode at the start of the exam.
- 2. Rounding too early.** Store intermediate GDC values in variables (e.g., $\text{STO} \rightarrow X$) rather than writing down a rounded value and re-entering it. Rounding to 3 significant figures mid-calculation can change your final answer.
- 3. Wrong window settings.** If you cannot see a graph, check your window. Use ZoomFit or manually set appropriate ranges. For trigonometric functions, ensure the x -range covers at least $[0, 2\pi]$ in radian mode.
- 4. Misinterpreting GDC output.** The GDC gives *decimal approximations*. If the question asks for an exact answer, the GDC result alone is insufficient. Also, the GDC may give multiple solutions — ensure you identify the correct one for the given domain.
- 5. Not writing down GDC input/output.** The markscheme often awards marks for stating what you entered into the GDC and what it returned. Write: “Using the GDC solver on $\cos x = 0.3x$, $x \approx 1.23$ (3 s.f.)”

Golden Rule for Paper 2

Always write down what you entered into the GDC and what it gave you. The examiner must be able to follow your method. A correct answer with no working earns zero marks if the method is not evident.

6.4.5 When the GDC Cannot Help

Situations Where the GDC Is Insufficient

- **“Show that” questions:** The GDC can *verify* a result numerically, but verification is not a demonstration. You must show algebraic working.
- **“Prove” questions:** The GDC cannot construct a proof. You must provide a rigorous logical argument.
- **“Hence” questions:** If the question directs you to use a specific previous result, the GDC approach may not earn method marks.
- **Exact form answers:** When the answer must be in exact form (e.g., $\ln 3$, $\pi/4$, $\sqrt{5}$), the GDC’s decimal output is not acceptable as the final answer, though it can help you check your algebraic result.
- **Derivation questions:** If you need to derive a formula or expression, the GDC cannot do this for you.
- **Paper 2 Section B proof parts:** Extended proof questions within Paper 2 still require full algebraic proof.

6.5 Paper 3: Extended Response and Investigation

6.5.1 Paper 3 Format

Paper 3 Overview

- **Duration:** 75 minutes
- **Calculator:** Permitted (GDC required)
- **Format:** 2–3 extended-response questions (investigation style)
- **Total:** typically 55 marks
- **Weighting:** 20% of total IB grade
- **Nature:** Unseen contexts, combining multiple topics, requiring investigation and generalisation

6.5.2 Nature of Paper 3

Paper 3 is fundamentally different from Papers 1 and 2. Instead of testing individual topics in isolation, it presents **unfamiliar mathematical contexts** that require you to:

- **Investigate** patterns and relationships
- **Generalise** from specific examples
- **Prove** results you have conjectured
- **Combine** techniques from different areas of the syllabus

The mathematics itself is within the syllabus, but the *context* and *presentation* will be unfamiliar. This tests your ability to **think mathematically**, not just recall formulas.

6.5.3 Strategies for Paper 3

Paper 3 Strategy: Read First, Then Act

Step 1: Read the entire question. Before writing anything, read the whole question (all parts, a through f or however many there are). This gives you the overall picture: where the question is going, what results you will need, and what the final goal is. Many students start part (a) without realising that part (c) gives a crucial hint.

Step 2: Identify the mathematical structure. Look past the unfamiliar context. Is this really about arithmetic sequences? Binomial coefficients? Complex numbers on the unit circle? Once you recognise the underlying mathematics, the problem becomes manageable.

Step 3: Work through the early parts carefully. Early parts are designed to guide you. They often provide results or patterns you will need later. Do not rush them.

Step 4: Generalise from specific cases. If the question asks you to compute specific values and then “generalise” or “conjecture”, look for the pattern. Express it in terms of n (or whatever variable is used).

Step 5: Test your conjecture. Before attempting a proof, verify your conjecture with one more specific case that you have not yet used. If it holds, you have more confidence in your proof attempt.

Step 6: Prove your conjecture rigorously. Use the techniques from Section 6.6. State what is given, state what is to be proved, provide clear logical steps, and match your conclusion to the conjecture.

Step 7: If stuck, use the GDC. For investigation questions, generating numerical examples with the GDC can reveal patterns you might otherwise miss.

6.5.4 Common Paper 3 Themes

Frequent Paper 3 Themes

- 1. Extending familiar concepts:** Applying a technique you know (e.g., induction) to a new type of statement (e.g., a divisibility result you have not seen before).
- 2. Combining topics:** A question might require calculus *and* complex numbers, or probability *and* series, or vectors *and* trigonometry. Look for connections.
- 3. Algebraic manipulation with parameters:** Instead of numerical coefficients, you may work with parameters (a, b, k, α). Keep track of conditions on these parameters (e.g., $k > 0$, $a \neq 1$).
- 4. Proof in unfamiliar contexts:** You may be asked to prove a result about a number-theoretic pattern, a geometric property, or a combinatorial identity that you have not encountered before. The proof techniques (induction, contradiction, direct) are the same; only the context is new.
- 5. Investigating families of functions:** You might be asked to explore the graphs of $f_n(x) = x^n e^{-x}$ for different values of n , find patterns in their stationary points, and prove a general result.

6.5.5 Time Management for Paper 3

Paper 3 Time Allocation

With 55 marks in 75 minutes, you have approximately **1.36 minutes per mark**, slightly more generous than Papers 1 and 2. However, the questions are longer and more complex.

Recommended approach:

- **First 5 minutes:** Read both (or all three) questions completely. Decide which to attempt first.
- **Next 30 minutes:** Work on your stronger question. Aim to complete it fully.
- **Next 30 minutes:** Work on the second question. Complete as much as possible.
- **Last 10 minutes:** Review your work, check calculations, ensure all parts are attempted.

Important: On Paper 3, partial marks are extremely valuable. Even if you cannot complete a proof, writing out the correct approach, the base case, or the first few steps of algebraic manipulation will earn marks. **Never leave a part blank.**

6.6 How to Write Proofs for Full Credit

Proof questions are among the most discriminating on the IB exam. They test not only your mathematical knowledge but also your ability to construct a clear, rigorous, and logically sound argument. This section provides a comprehensive guide.

6.6.1 Structure of a Good Proof

Anatomy of a Complete Proof

A well-structured proof follows these steps:

Step 1: State what is given. Begin by clearly identifying the hypotheses or given conditions. Use “Given...” or “Let...” or “Suppose...”

Step 2: State what is to be proved. Restate the proposition or theorem. This shows the examiner you understand the goal.

Step 3: Present logical steps with connectives. Each step should follow from the previous one using a logical connective: **therefore, since, because, hence, but, so, thus, we have, it follows that**. Never leave the examiner to guess how you got from one line to the next.

Step 4: Conclude. End with a statement that matches what was to be proved. Use “Therefore...” or “Hence, as required.” or “This completes the proof.”

6.6.2 Common Proof Mistakes That Lose Marks

Proof Errors Checklist

1. **Circular reasoning.** Starting with what you need to prove and working backwards, then presenting it as if it were forward reasoning.

Example of circular reasoning:

$$\begin{aligned}x^2 + 2x + 1 &= 0 \\(x + 1)^2 &= 0 \\x + 1 &= 0 \\x &= -1\end{aligned}$$

“Therefore $x = -1$ is a root of $x^2 + 2x + 1 = 0$.”

The problem: You assumed the equation has a root and derived $x = -1$. To *show* $x = -1$ is a root, you must *substitute* $x = -1$ and verify it satisfies the equation.

2. Missing cases. For example, proving a result for $n \geq 0$ without checking $n = 0$. Assuming $x > 0$ without justification. Not considering both even and odd cases for integer proofs.

3. Vague language. Writing “it’s obvious” or “clearly” or “we can see”. Every claim must be justified mathematically. If it is truly obvious, write a one-line justification (“since $x^2 \geq 0$ for all $x \in \mathbb{R}$ ”).

4. Skipping algebraic steps. Even if a step seems simple to you, write it out. The markscheme allocates marks to individual steps. Missing a line of algebra may cost you a method mark.

5. Not defining variables. Writing “Let $k = n + 1$ ” without stating what n is. Always define every variable before using it.

6.6.3 Proof by Induction: Complete Checklist

Mathematical Induction Checklist

To earn full marks on a proof by induction, ensure every one of the following is present:

- 1. Basis step.** Clearly stated and verified. Write: “For $n = 1$: LHS = ... = ..., RHS = ... = Since LHS = RHS, the statement is true for $n = 1$.”
- 2. Inductive hypothesis.** Clearly stated using the correct notation. Write: “**Assume** the statement is true for $n = k$, where $k \in \mathbb{N}$, $k \geq 1$. That is, assume ...” (write the full statement with k).
- 3. Inductive step.** Must *start* from the statement for $n = k + 1$ (the LHS for $n = k + 1$) and use the hypothesis to transform it. Write: “Consider the statement for $n = k + 1$.”
- 4. Logical connection.** Use the word “therefore” or “hence” at the point where the hypothesis is applied. Show the examiner exactly where the assumption is used.
- 5. Conclusion.** State that since both the basis step and the inductive step hold, “by the Principle of Mathematical Induction, the statement is true for all $n \in \mathbb{N}$, $n \geq 1$.”

Worked Example: Proof by Induction — Full Marks

Question (6 marks). Prove by mathematical induction that

$$\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}, \quad \text{for all } n \in \mathbb{N}.$$

Solution:

Basis step: For $n = 1$:

$$\text{LHS} = 1^2 = 1, \quad \text{RHS} = \frac{1(2)(3)}{6} = 1. \quad \text{LHS} = \text{RHS}. \quad \checkmark$$

Inductive hypothesis: Assume the statement is true for $n = k$, where $k \in \mathbb{N}$. That is, assume

$$\sum_{r=1}^k r^2 = \frac{k(k+1)(2k+1)}{6}. \quad (\text{IH})$$

Inductive step: Consider the statement for $n = k + 1$:

$$\begin{aligned} \text{LHS} &= \sum_{r=1}^{k+1} r^2 = \sum_{r=1}^k r^2 + (k+1)^2 \\ &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 && (\text{by the inductive hypothesis (IH)}) \\ &= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \\ &= \frac{(k+1)[k(2k+1) + 6(k+1)]}{6} \\ &= \frac{(k+1)(2k^2 + k + 6k + 6)}{6} \\ &= \frac{(k+1)(2k^2 + 7k + 6)}{6} \\ &= \frac{(k+1)(k+2)(2k+3)}{6} \\ &= \frac{(k+1)((k+1)+1)(2(k+1)+1)}{6}. \end{aligned}$$

This is the statement for $n = k + 1$. **Therefore**, if the statement is true for $n = k$, it is also true for $n = k + 1$.

Conclusion: Since the statement is true for $n = 1$ (basis step) and the inductive step holds, by the Principle of Mathematical Induction, the statement is true for all $n \in \mathbb{N}$.

Markscheme annotation: M1 (basis step correct), M1 (hypothesis stated), M1 (starts $n = k + 1$ from sum), M1 (uses hypothesis), A1 (algebra to final form), R1 (conclusion with PMI reference).

6.6.4 Proof by Contradiction: Complete Checklist

Proof by Contradiction Checklist

To earn full marks on a proof by contradiction:

- 1. State the assumption.** Begin: “Assume, for contradiction, that ...” or “Suppose, to the contrary, that ...” State the *negation* of what you want to prove.
- 2. Derive a contradiction.** Using logical deduction from your assumption (and any given conditions), arrive at a statement that is clearly false or contradicts a known fact.
- 3. State what contradicts what.** Explicitly identify the contradiction: “This contradicts the fact that...” or “But this is impossible because...”
- 4. Conclude.** “Therefore, our assumption must be false, and hence [original statement] is true.”

Worked Example: Proof by Contradiction — Full Marks

Question (5 marks). Prove by contradiction that $\sqrt{2}$ is irrational.

Solution:

Assumption: Assume, for contradiction, that $\sqrt{2}$ is rational. Then $\sqrt{2}$ can be written as $\sqrt{2} = \frac{a}{b}$, where $a, b \in \mathbb{Z}$, $b \neq 0$, and $\gcd(a, b) = 1$ (i.e., the fraction is in its lowest terms).

Derivation:

$$\begin{aligned}\sqrt{2} &= \frac{a}{b} \\ 2 &= \frac{a^2}{b^2} && \text{(squaring both sides)} \\ a^2 &= 2b^2.\end{aligned}$$

Since $a^2 = 2b^2$ and b^2 is an integer, a^2 is even, and therefore a is even. (If a were odd, then $a = 2m + 1$ and $a^2 = 4m^2 + 4m + 1$ would be odd, a contradiction.)

So let $a = 2k$ for some $k \in \mathbb{Z}$. Substituting:

$$\begin{aligned}(2k)^2 &= 2b^2 \\ 4k^2 &= 2b^2 \\ b^2 &= 2k^2.\end{aligned}$$

By the same reasoning, since $b^2 = 2k^2$, b^2 is even, and therefore b is even.

Contradiction: We have shown that both a and b are even. But this contradicts our initial assumption that $\gcd(a, b) = 1$.

Conclusion: Therefore, our assumption that $\sqrt{2}$ is rational must be false. Hence, $\sqrt{2}$ is irrational.

Markscheme annotation: M1 (assumes rational, defines a/b in lowest terms), M1 (deduces a is even), M1 (deduces b is even), R1 (identifies contradiction with $\gcd(a, b) = 1$), R1 (concludes $\sqrt{2}$ is irrational).

6.7 Common Mistakes That Lose Marks

This section catalogues the most frequent errors organised by topic. Use it as a checklist when reviewing your practice papers.

6.7.1 Algebra

Common Algebra Mistakes

1. Sign errors in expansion.

$$-(2x - 3) \neq -2x - 3 \quad (\text{wrong}); \quad -(2x - 3) = -2x + 3 \quad (\text{correct})$$

Prevention: Always write brackets and distribute the negative sign term by term.

2. Dropping terms in long expressions. When expanding $(x+1)^5$, omitting the x^3 term or writing the wrong coefficient.

Prevention: Use Pascal's triangle or the binomial theorem systematically. Write all terms before simplifying.

3. Not distributing correctly.

$$3(x+2)^2 \neq 3x^2 + 12 \quad (\text{wrong}); \quad 3(x+2)^2 = 3(x^2 + 4x + 4) = 3x^2 + 12x + 12 \quad (\text{correct})$$

Prevention: Expand the brackets *first*, then multiply by the coefficient.

4. Cancelling incorrectly.

$$\frac{x+3}{x+5} \neq \frac{3}{5} \quad (\text{wrong!})$$

Prevention: Only cancel *factors*, never *terms*. You can cancel $(x+3)$ only if it appears as a factor in both numerator and denominator.

5. Squaring negatives.

$$(-x)^2 \neq -x^2 \quad (\text{wrong}); \quad (-x)^2 = x^2 \quad (\text{correct})$$

6.7.2 Functions

Common Function Mistakes

1. Forgetting domain restrictions. The domain of $f(x) = \ln(x-1)$ is $x > 1$, not $\mathbb{R}$. The domain of $f(x) = \sqrt{x^2 - 4}$ is $|x| \geq 2$. The domain of $f(x) = \frac{1}{x^2 - 9}$ is $x \neq \pm 3$.

2. Wrong inverse function. To find f^{-1} , swap x and y and solve for y . A common error is to simply replace $f(x)$ with $f^{-1}(x)$ without swapping. Also, always state the domain and range of the inverse.

3. Confusing the range of f with the range of f^{-1} . The range of f^{-1} equals the domain of f , not the range of f .

4. Composition order. $f \circ g$ means “ f of g ”, i.e., first apply g , then apply f . Writing $f(g(x))$ as $g(f(x))$ is a common error.

5. Even/odd function confusion. f is even if $f(-x) = f(x)$; f is odd if $f(-x) = -f(x)$. A function can be neither. Checking only one value of x is not sufficient.

6.7.3 Calculus

Common Calculus Mistakes

1. Forgetting $+C$ in indefinite integration.

$$\int 2x \, dx = x^2 + C \quad (\text{correct, always include } C)$$

This is one of the most common errors in all of IB Mathematics.

2. Chain rule errors.

$$\frac{d}{dx}[\sin(3x)] \neq \cos(3x) \quad (\text{wrong}); \quad \frac{d}{dx}[\sin(3x)] = 3 \cos(3x) \quad (\text{correct})$$

Prevention: Always identify the “inner function” and multiply by its derivative.

3. Product rule vs. chain rule confusion.

$$\frac{d}{dx}[x^2 \sin x] \neq 2x \cos x \quad (\text{this applies the chain rule, not the product rule})$$

$$\text{Correct: } \frac{d}{dx}[x^2 \sin x] = 2x \sin x + x^2 \cos x.$$

4. Wrong limits of integration. When finding the area between two curves, using the wrong x -values for the limits. When integrating by substitution, forgetting to change the limits.

5. Integration by parts — wrong choice of u and dv . Choosing u as the more complicated part leads to a harder integral. Remember LIATE (Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential) as a guide for choosing u .

6. Confusing definite and indefinite integrals. Writing $+C$ in a definite integral (incorrect) or forgetting $+C$ in an indefinite integral (incorrect).

7. Second derivative test errors. Computing $f''(x)$ incorrectly or evaluating it at the wrong point. Also, $f''(x_0) = 0$ is *inconclusive*, not “inflection point”.

6.7.4 Trigonometry

Common Trigonometry Mistakes

1. Degree/radian mix-up. Using degrees in a calculus context (where radians are required) or vice versa. The derivative of $\sin x$ is $\cos x$ *only in radians*.

2. Wrong quadrant for inverse trig.

$$\arcsin\left(-\frac{1}{2}\right) \neq \frac{7\pi}{6} \quad (\text{wrong}); \quad \arcsin\left(-\frac{1}{2}\right) = -\frac{\pi}{6} \quad (\text{correct})$$

Remember the ranges: $\arcsin \in [-\pi/2, \pi/2]$, $\arccos \in [0, \pi]$, $\arctan \in (-\pi/2, \pi/2)$.

3. Trigonometric identity errors.

$$\sin(A + B) \neq \sin A + \sin B \quad (\text{wrong})$$

$$\cos 2\theta \neq 2 \cos \theta \quad (\text{wrong}); \quad \cos 2\theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta \quad (\text{correct})$$

4. Missing solutions in trigonometric equations. When solving $\sin \theta = \frac{1}{2}$ for $\theta \in [0, 2\pi]$, giving only $\theta = \pi/6$ and forgetting $\theta = 5\pi/6$.

5. Ambiguous case in non-right triangles. When using the sine rule, forgetting that there may be two possible angles (the ambiguous case). Always check whether the supplement also satisfies the conditions.

6.7.5 Probability

Common Probability Mistakes

- 1. Not reading the question carefully.** Confusing “at least” ($\geq$) with “at most” ($\leq$). Confusing “exactly 3” with “at least 3”. Misreading the distribution parameters.
- 2. Wrong distribution.** Using the binomial distribution when the trials are *not* independent (e.g., sampling without replacement — use hypergeometric or normal approximation). Using Poisson when the binomial is appropriate.
- 3. Conditional probability errors.** Confusing $P(A | B)$ with $P(B | A)$. Forgetting that $P(A \cap B) = P(A) \cdot P(B | A)$, not $P(A) \cdot P(B)$ (unless A and B are independent).
- 4. Normal distribution — wrong standardisation.**

$$Z = \frac{X - \mu}{\sigma} \quad (\text{correct}); \quad Z = \frac{X - \sigma}{\mu} \quad (\text{common error — swapped})$$

- 5. Forgetting that probabilities must sum to 1.** When constructing a probability distribution, the sum of all probabilities must equal 1. Always verify this.

6.7.6 Vectors

Common Vector Mistakes

- 1. Confusing position and displacement vectors.** A position vector is measured from the origin. A displacement (or direction) vector is the difference between two position vectors. They are not interchangeable in all formulas.

- 2. Wrong angle formula.**

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} \quad (\text{correct})$$

A common error is to forget to divide by the magnitudes, or to use the cross product magnitude instead.

- 3. Scalar triple product sign errors.** The volume of the parallelepiped is $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$. Forgetting the absolute value means you might report a negative volume.

- 4. Vector equation of a line — confusing point and direction.** The vector equation $\mathbf{r} = \mathbf{a} + t\mathbf{d}$ requires $\mathbf{a}$ to be a point on the line and $\mathbf{d}$ to be the direction vector. Swapping these gives a different line.

- 5. Cross product — wrong order.** $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$. The order matters for the direction of the resulting vector.

6.7.7 Complex Numbers

Common Complex Number Mistakes

1. Wrong argument quadrant. When finding $\arg(z)$, always sketch the complex number to identify the correct quadrant. The GDC function $\angle(z)$ or manual calculation $\arctan(y/x)$ may give the wrong angle if z is in the second or third quadrant.

Prevention: Use $\arg(z) = \arctan(y/x)$ only for the first quadrant. For other quadrants, add or subtract π as appropriate.

2. De Moivre's theorem errors.

$$[r \operatorname{cis} \theta]^n = r^n \operatorname{cis}(n\theta) \quad (\text{correct})$$

Common error: writing $\operatorname{cis}(\theta)^n = \operatorname{cis}(n\theta)$ without the r^n factor, or writing $r^n \operatorname{cis}(\theta^n)$.

3. Roots of unity — missing roots. An n th root equation has exactly n distinct roots (counting multiplicity). Forgetting to list all n roots loses marks.

4. Complex conjugate errors. $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ (correct), but $\overline{z_1 z_2} \neq \bar{z}_1 \cdot \bar{z}_2$ (wrong); the correct formula is $\overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2$.

5. Modulus-argument form to Cartesian — arithmetic errors. When converting $z = r \operatorname{cis} \theta$ to $a + bi$, errors in evaluating $r \cos \theta$ or $r \sin \theta$.

6.7.8 Statistics

Common Statistics Mistakes

1. Using n instead of $n - 1$ for sample variance.

$$s_n^2 = \frac{\sum (x_i - \bar{x})^2}{n} \quad (\text{population variance})$$

$$s_{n-1}^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} \quad (\text{unbiased estimate of population variance})$$

The IB almost always uses s_{n-1}^2 (the unbiased estimate). Check which formula the question expects.

2. Wrong formula for standard deviation. The variance is the *square* of the standard deviation, not the standard deviation itself. The standard deviation is the square root of the variance. Do not confuse the two.

3. Interpreting the p -value incorrectly. The p -value is the probability of observing a test statistic at least as extreme as the one computed, *assuming H_0 is true*. It is *not* the probability that H_0 is true.

4. Type I and Type II error confusion. Type I error: rejecting H_0 when H_0 is true (false positive). Type II error: failing to reject H_0 when H_0 is false (false negative).

5. Confidence interval — wrong critical value. Using $z = 1.96$ when the population standard deviation is unknown and the sample size is small. In this case, the t -distribution should be used (though the IB syllabus focuses primarily on the normal distribution for confidence intervals).

6.7.9 General Errors

Common General Mistakes

- 1. Not reading the question.** Answering a different question from the one asked. For example, finding the x -intercept when asked for the y -intercept.
- 2. Not answering what was asked.** Finding the derivative when asked to find the *equation of the tangent*. Computing the integral when asked to find the *area* (which may require taking absolute values or subtracting curves).
- 3. Units.** Forgetting to include units in the final answer when the question specifies them (e.g., metres, seconds, kilograms). Mixing units (e.g., km/h vs. m/s).
- 4. Rounding.** Rounding to the wrong number of significant figures or decimal places. The question specifies the precision; follow it exactly. If not specified, give 3 significant figures unless the answer is exact.
- 5. Not attempting a question.** A blank answer earns zero marks. Even a few lines of correct working can earn 1–2 marks. Always attempt every part of every question.
- 6. Illegible handwriting.** If the examiner cannot read your answer, they cannot award marks. Write clearly and organise your working neatly.
- 7. Crossing out correct work.** If you write a correct solution and then cross it out to try a different (incorrect) method, the examiner will mark the visible (incorrect) solution. Clearly indicate which solution you want marked.

6.8 Calculator Techniques Reference

This section provides a practical reference for the most important GDC operations on the two most common calculator models: the TI-84 series and the TI-Nspire series.

6.8.1 TI-84 Series Tips

TI-84 Essential Operations

Solving equations:

- MATH $\rightarrow$ 0:Solver — enter equation, set bounds, solve
- Alternatively: graph the function and use 2nd $\rightarrow$ CALC $\rightarrow$ 2:zero

Finding intersections:

- Graph both functions, then 2nd $\rightarrow$ CALC $\rightarrow$ 5:intersect
- Select first curve, second curve, and guess

Numerical derivative:

- MATH $\rightarrow$ 8:nDeriv(— nDeriv(expression, variable, value)
- Example: nDeriv($X^2 \cdot e^X$, X, 1) gives $f'(1)$ for $f(x) = x^2 e^x$
- **Warning:** nDeriv uses a symmetric difference approximation and can be inaccurate near sharp turns or discontinuities

Definite integrals:

- MATH $\rightarrow$ 9:fnInt(— fnInt(expression, variable, lower, upper)
- Example: fnInt(X^2 , X, 0, 3) gives $\int_0^3 x^2 dx$
- Alternatively: graph and use 2nd $\rightarrow$ CALC $\rightarrow$ 7: $\int f(x)dx$

Statistical distributions:

- 2nd $\rightarrow$ VARS $\rightarrow$ DISTR
- A:binompdf(n, p, x) — $P(X = x)$
- B:binomcdf(n, p, x) — $P(X \leq x)$
- 2:normalcdf(a, b, μ, σ) — $P(a < X < b)$
- 3:invNorm(area, μ, σ) — inverse normal

Storing values:

- After a calculation, press STO $\rightarrow$ ALPHA and choose a letter
- Example: 1.2345 STO $\rightarrow$ ALPHA A stores 1.2345 as A

6.8.2 TI-Nspire Tips

TI-Nspire Essential Operations**Solving equations:**

- In a Calculator page: menu $\rightarrow$ **Algebra** $\rightarrow$ **Solve** (or type solve())
- Example: solve($e^x = x + 3, x$)
- For numerical solving when algebraic solve fails: nSolve($e^x = x + 3, x$)

Finding intersections:

- Graph both functions on a Graphs page
- menu $\rightarrow$ **Analyze Graph** $\rightarrow$ **Intersection**
- Click on both curves and set the lower/upper bounds

Numerical derivative:

- Type nDeriv(expression, variable, point)
- Example: nDeriv($x^2 \cdot e^x, x, 1$)

Definite integrals:

- Type the integral template: CTRL + I (or use the template from the palette)
- Fill in the integrand, variable, and limits

Statistical distributions:

- menu $\rightarrow$ **Probability** $\rightarrow$ **Distributions**
- **Binomial Pdf** and **Binomial Cdf**
- **Normal Cdf** and **Inverse Normal**

Defining functions and variables:

- Define a function: Define $f(x) = x^2 \cdot e^x$
- Store a value: 1.2345 $\rightarrow$ a
- Use these in subsequent calculations for precision

6.8.3 Storing Values for Precision

The Golden Rule of GDC Precision

Never round intermediate values. Store exact GDC outputs in variables and use those variables in subsequent calculations.

Example: If your GDC gives $x = 1.386294361$ as the solution to an equation, do NOT write $x = 1.386$ and then use 1.386 in the next part. Instead:

1. Compute x on the GDC
2. Store it: $\text{STO} \rightarrow X$
3. In the next part, use X directly: $X^2 + 3 \cdot X$

This preserves the full precision of the GDC and prevents accumulation of rounding errors.

On Paper 2: Write “Let $a = 1.39$ (3 s.f.)” but use the stored full-precision value in your GDC for further calculations. Round *only* your final answer to the required precision.

6.8.4 Using Tables on the GDC

Table Feature for Pattern Recognition

The table feature (TI-84: $2\text{nd} \rightarrow \text{GRAPH}$; TI-Nspire: $\text{CTRL} + T$ on a Graphs page) is invaluable for:

- Generating sequences and spotting patterns (useful for Paper 3 investigations)
- Evaluating a function at many points quickly
- Checking your algebraic answers numerically
- Finding approximate roots by looking for sign changes
- Verifying that a function satisfies a given condition at specific points

Tip: Set the table start value and step size appropriately. For trigonometric functions in radians, a step of $\pi/12$ or $\pi/6$ is often useful.

6.8.5 Common GDC Models and Their Quirks

GDC Model Quirks

TI-84 Plus CE:

- Battery can drain during long exams — start with a full charge
- The nDeriv function can give inaccurate results near cusps or discontinuities
- Press MODE to check angle units and numerical format
- The $\text{STAT} \rightarrow \text{EDIT}$ screen is easily overwritten — be careful when entering data

TI-84 Plus (non-CE):

- Uses AAA batteries — bring spares
- Slower processor; complex graphs take longer to render
- Same functionality as CE for mathematical operations

TI-Nspire CX (non-CAS):

- Larger screen and better resolution — easier to read graphs
- The nSolve function is more robust than the TI-84 solver
- Can have multiple pages open simultaneously (Calculator, Graphs, Lists)
- Press $\text{CTRL} \rightarrow \text{DOC}$ to switch between pages
- **Important:** Ensure you have the non-CAS version; CAS calculators are *not* permitted

TI-Nspire CX II (non-CAS):

- Same features as CX but faster
- Improved keyboard layout
- De-interlacing feature improves graph readability

All models:

- Press CLEAR (TI-84) or ESC (Nspire) to return to the home screen
- **Reset your calculator before the exam** to clear stored variables and programs (unless programs are permitted)
- Familiarise yourself with *your specific model* well before the exam

6.9 Time Management and Exam Strategy

Effective time management is the difference between a grade 6 and a grade 7. This section provides a practical framework.

6.9.1 Papers 1 and 2: The One-Minute-Per-Mark Rule

Time Budget: Papers 1 and 2

Paper 1: 120 marks in 120 minutes = **1 minute per mark.**

Paper 2: 120 marks in 120 minutes = **1 minute per mark.**

Paper 3: 55 marks in 75 minutes $\approx$ **1.36 minutes per mark.**

Practical Time Allocation for Papers 1 and 2

Section A (short questions, approximately 60 marks):

- Target: 50–55 minutes
- This gives you 5–10 minutes of buffer for harder short questions
- If a 2-mark question takes more than 3 minutes, move on and return later

Section B (extended questions, approximately 60 marks):

- Target: 55–60 minutes
- Read the entire section before starting
- Begin with the question you find most approachable
- Within each question, attempt every part — partial marks add up

Review time:

- Reserve 5–10 minutes at the end
- Check that all parts have been attempted
- Verify answers using alternative methods or the GDC (Paper 2)
- Ensure all answers are rounded to the required precision

6.9.2 When to Skip and Return

The Skip Strategy

Skip a question temporarily if:

- You have spent more than 1.5 minutes per mark without progress
- You cannot see how to start the question at all
- You feel confused or anxious — move on to regain composure

Before skipping, write down:

- Any relevant formula you can identify
- A diagram if applicable
- The first step of the method, even if you cannot continue

This ensures you earn partial marks and gives you a starting point when you return.

When you return:

- Re-read the question carefully — you may notice something you missed
- Look at later parts of the question for hints (later parts often require earlier results, which tells you what approach to use)
- Try a different method from the one you attempted first

6.9.3 How to Check Your Work Efficiently

Efficient Checking Strategies

- 1. Substitute your answer back.** If you solved an equation, substitute your solution back into the original equation. If you found a derivative, check it against the GDC's numerical derivative.
- 2. Use the GDC to verify (Paper 2).** For algebraic results, use the GDC to check numerically. For example, if you found that the stationary point is at $x = \ln 3$, verify that $f'(\ln 3) \approx 0$ on the GDC.
- 3. Check dimensions and units.** If the answer is a length, is it positive? If it is a probability, is it between 0 and 1? If it is an area, does it have square units?
- 4. Check reasonableness.** Does the magnitude of the answer make sense? A probability of 2.3 or a length of -5 metres is clearly wrong.
- 5. Check against your estimate.** Before solving, make a rough estimate. If your exact answer is orders of magnitude different from your estimate, investigate.
- 6. Re-read the question.** The most common source of errors is not the mathematics but the reading. Re-read the question to confirm you answered what was asked.

6.9.4 Partial Marks Strategies

Maximising Partial Marks When Stuck

When you cannot solve a problem completely, use these strategies to earn every possible mark:

- 1. Write the relevant formula.** If you know the formula but cannot complete the calculation, write the formula with the correct values substituted. This typically earns 1 mark.
- 2. Draw a diagram.** For geometry, vectors, or probability questions, a clearly labelled diagram can earn marks even without calculations.
- 3. State what you know.** If a question asks you to prove something and you cannot complete the proof, write the given conditions and the first few logical steps. The basis step of an induction proof, for example, typically earns 1–2 marks on its own.
- 4. Attempt every part.** Even if part (a) is wrong, attempt part (b) using your (wrong) answer from (a). You may earn follow-through method marks.
- 5. Show your working, always.** A correct answer with no working earns full marks if the answer is correct — but if there is any error, no working means zero marks. Working provides a safety net.
- 6. Use the answer space fully.** If you are unsure, write multiple approaches. The examiner will mark the best one. Clearly label different attempts so the examiner can follow your reasoning.

6.9.5 Final Exam Day Checklist

Pre-Exam Checklist

The night before:

- Ensure your GDC is fully charged (or has fresh batteries)
- Reset your GDC to clear stored variables (or check your school's policy)
- Set your GDC to radian mode
- Pack: GDC, spare batteries, pens, pencils, ruler, protractor, compass, eraser
- Review the command terms (Section 6.1) and common mistakes (Section 6.7)
- Do NOT stay up late cramming — sleep is essential for mathematical reasoning

In the exam room:

- Write your candidate number on every page
- Read the entire paper before starting (especially Paper 3)
- Start with questions you are confident about to build momentum
- Keep track of time — check the clock every 15–20 minutes
- Show all working, even for “write down” questions
- Round final answers to the specified precision
- If you finish early, use the time to check every answer

Chapter 7

Ultra-Condensed Formula Sheet

*Every formula from the entire IB Mathematics AA HL course,
compressed into a single rapid-reference appendix.
This sheet alone is sufficient for full-course revision.*

7.1 Number and Algebra – Quick Reference

Scientific Notation

$a = m \times 10^n$, $1 \leq |m| < 10$, $n \in \mathbb{Z}$.
Significant figures: count from the first non-zero digit.

Arithmetic Sequences & Series

Term: $u_n = u_1 + (n-1)d$
Sum: $S_n = \frac{n}{2}(u_1 + u_n) = \frac{n}{2}[2u_1 + (n-1)d]$

Geometric Sequences & Series

Term: $u_n = u_1 r^{n-1}$
Sum (finite): $S_n = \frac{u_1(r^n - 1)}{r - 1} = \frac{u_1(1 - r^n)}{1 - r}$
Sum (infinite): $S_\infty = \frac{u_1}{1 - r}$, $|r| < 1$

Compound Interest & Finance

Compound interest: $A = P(1 + r)^n$
Compounding k times/yr: $A = P\left(1 + \frac{r}{k}\right)^{kn}$
Continuous: $A = Pe^{rt}$
Depreciation: $A = P(1 - r)^n$

Exponent Laws

$a^m \cdot a^n = a^{m+n}$ $(a^m)^n = a^{mn}$
 $\frac{a^m}{a^n} = a^{m-n}$ $(ab)^n = a^n b^n$
 $a^0 = 1$ $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
 $a^{-n} = \frac{1}{a^n}$ $a^{1/n} = \sqrt[n]{a}$

Logarithm Laws

$\log_a(xy) = \log_a x + \log_a y$ $\log_a(x/y) = \log_a x - \log_a y$
 $\log_a(x^n) = n \log_a x$ $\log_a a = 1$, $\log_a 1 = 0$
Change of base: $\log_a b = \frac{\log_c b}{\log_c a} = \frac{\ln b}{\ln a}$
Inverse: $a^{\log_a x} = x$, $\log_a(a^x) = x$

Binomial Theorem

Expansion: $(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$

General term: $T_{r+1} = \binom{n}{r} a^{n-r} b^r$

Binomial coefficient: $\binom{n}{r} = \frac{n!}{r!(n-r)!} = {}^nC_r$

Pascal identity: $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$

Counting Principles

Fundamental: $n(A \times B) = n(A) \cdot n(B)$

Permutations: ${}_nP_r = P(n, r) = \frac{n!}{(n-r)!}$

Combinations: $\binom{n}{r} = C(n, r) = \frac{n!}{r!(n-r)!}$

Relation: $\binom{n}{r} = \frac{{}_nP_r}{r!}$

Circular permutations: $(n-1)!$

Permutations with repetition: $\frac{n!}{n_1! n_2! \cdots n_k!}$

Complex Numbers – Forms

Rectangular: $z = a + bi$, $a = \operatorname{Re}(z)$,
 $b = \operatorname{Im}(z)$

Modulus: $|z| = \sqrt{a^2 + b^2}$

Argument: $\arg(z) = \arctan\left(\frac{b}{a}\right)$, adjusted for quadrant

Polar: $z = r(\cos \theta + i \sin \theta) = r \operatorname{cis} \theta$

Exponential: $z = r e^{i\theta}$

Complex Numbers – Operations

Conjugate: $\bar{z} = a - bi$; $z\bar{z} = |z|^2 = a^2 + b^2$

Division: $\frac{z_1}{z_2} = \frac{z_1 \bar{z}_2}{z_2 \bar{z}_2} = \frac{z_1 \bar{z}_2}{|z_2|^2}$

Multiplication (polar): $z_1 z_2 = r_1 r_2 \operatorname{cis}(\theta_1 + \theta_2)$

Division (polar): $\frac{z_1}{z_2} = \frac{r_1}{r_2} \operatorname{cis}(\theta_1 - \theta_2)$

De Moivre: $z^n = r^n \operatorname{cis}(n\theta) = r^n e^{in\theta}$

Complex Roots

n th roots of unity ($z^n = 1$):

$$z_k = \operatorname{cis}\left(\frac{2k\pi}{n}\right), \quad k = 0, 1, 2, \dots, n-1$$

n th roots of any complex number ($z^n = w = R e^{i\alpha}$):

$$z_k = R^{1/n} \operatorname{cis}\left(\frac{\alpha + 2k\pi}{n}\right), \quad k = 0, 1, \dots, n-1$$

Roots are equally spaced on a circle of radius $R^{1/n}$.

Proof by Mathematical Induction

Step 1: Prove the statement is true for $n = 1$ (base case).

Step 2: Assume true for $n = k$ (inductive hypothesis).

Step 3: Prove true for $n = k + 1$ using the hypothesis.

Step 4: Conclude the statement holds for all $n \in \mathbb{Z}^+$.

Partial Fractions

Case 1 – Distinct linear factors: $\frac{px + q}{(ax + b)(cx + d)} = \frac{A}{ax + b} + \frac{B}{cx + d}$

Case 2 – Repeated linear factor: $\frac{px + q}{(ax + b)^2(cx + d)} = \frac{A}{ax + b} + \frac{B}{(ax + b)^2} + \frac{C}{cx + d}$

Case 3 – Irreducible quadratic: $\frac{px^2 + qx + r}{(ax + b)(x^2 + cx + d)} = \frac{A}{ax + b} + \frac{Bx + C}{x^2 + cx + d}, \quad (c^2 - 4d < 0)$

Absolute Value & Inequalities

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$|x| \leq a \iff -a \leq x \leq a, \quad |x| \geq a \iff x \leq -a \text{ or } x \geq a$$

$$|ab| = |a||b|, \quad |a + b| \leq |a| + |b| \quad (\text{triangle inequality})$$

Quadratic inequality $ax^2 + bx + c > 0$: roots of $= 0$ divide real line into intervals; test one point per interval. If $a > 0$: positive outside the roots (if real). If $\Delta < 0$ and $a > 0$: always positive.

Systems of Linear Equations

2 equations, 2 unknowns: substitution or elimination.

Cramer's rule: For $Ax + By = E, Cx + Dy = F$:

$$x = \frac{\begin{vmatrix} E & B \\ F & D \end{vmatrix}}{\begin{vmatrix} A & B \\ C & D \end{vmatrix}}, \quad y = \frac{\begin{vmatrix} A & E \\ C & F \end{vmatrix}}{\begin{vmatrix} A & B \\ C & D \end{vmatrix}}$$

System has unique solution iff determinant $\neq 0$ (i.e. lines are not parallel).

7.2 Functions – Quick Reference

Linear Function

$$y = mx + c, \quad \text{gradient } m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Point-gradient: } y - y_1 = m(x - x_1)$$

$$x\text{-intercept: set } y = 0; \quad y\text{-intercept: } c$$

$$\text{Parallel: } m_1 = m_2; \quad \text{Perpendicular: } m_1 m_2 = -1$$

Quadratic Function – Three Forms

$$\text{General: } f(x) = ax^2 + bx + c$$

$$\text{Vertex: } f(x) = a(x - h)^2 + k, \quad \text{vertex } (h, k)$$

$$\text{Factored: } f(x) = a(x - p)(x - q), \quad \text{roots } p, q$$

$$\text{Axis of symmetry: } x = -\frac{b}{2a}$$

$$\text{Discriminant: } \Delta = b^2 - 4ac$$

$\Delta > 0$: two real roots; $\Delta = 0$: one repeated root; $\Delta < 0$: no real roots

$$\text{Sum/product of roots: } \alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

Polynomial Functions

General degree n : $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

Remainder theorem: $f(x) = (x - c)Q(x) + R \implies R = f(c)$

Factor theorem: $(x - c)$ is a factor of $f(x) \iff f(c) = 0$

Fundamental theorem of algebra: A polynomial of degree n has exactly n roots in $\mathbb{C}$ (counting multiplicity).

Vieta's Formulas – Cubic

$$ax^3 + bx^2 + cx + d = a(x - \alpha)(x - \beta)(x - \gamma):$$

$$\alpha + \beta + \gamma = -\frac{b}{a}, \quad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}, \quad \alpha\beta\gamma = -\frac{d}{a}$$

Rational Functions – Asymptotes

Vertical asymptote: set denominator = 0 (after cancelling common factors).

Horizontal asymptote ($y = \frac{p(x)}{q(x)}$):

- $\deg(p) < \deg(q)$: $y = 0$
- $\deg(p) = \deg(q)$: $y = \frac{\text{leading coefficient of } p}{\text{leading coefficient of } q}$
- $\deg(p) > \deg(q)$: oblique/slant asymptote (polynomial long division)

Oblique asymptote: divide numerator by denominator; the quotient (ignoring remainder) gives the asymptote.

Function Transformations

Transformation	Effect on $y = f(x)$	Order
Vertical translation	$y = f(x) + k$	Shift up k
Horizontal translation	$y = f(x - h)$	Shift right h
Vertical stretch	$y = a f(x)$	Stretch by $ a $ from x -axis
Reflection in x -axis	$y = -f(x)$	$a = -1$ case
Horizontal stretch	$y = f(bx)$	Stretch by $\frac{1}{ b }$ from y -axis
Reflection in y -axis	$y = f(-x)$	$b = -1$ case
Combined	$y = a f(b(x - h)) + k$	Apply in reverse: h, b, a, k

Inverse Functions

Condition: f must be one-to-one (passes horizontal line test).

Procedure: Write $y = f(x)$, swap x and y , solve for y .

Domain/range swap:

$$\text{Domain of } f^{-1} = \text{Range of } f, \quad \text{Range of } f^{-1} = \text{Domain of } f$$

Key identities: $f^{-1}(f(x)) = x, \quad f(f^{-1}(x)) = x$

Reflection: $y = f^{-1}(x)$ is the reflection of $y = f(x)$ in $y = x$.

Derivative: $(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$

Composite Functions

Definition: $(f \circ g)(x) = f(g(x))$

Domain of $f \circ g$: all x in domain of g such that $g(x)$ is in domain of f .

Key caution: $f \circ g \neq g \circ f$ in general.

Inverse of composite: $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$

Exponential & Logarithmic Functions

Exponential: $f(x) = a^x$, $a > 0$, $a \neq 1$. Always positive. y -intercept: $(0, 1)$. Asymptote: $y = 0$.

Natural exponential: $f(x) = e^x$. Gradient at any point $= e^x$.

Logarithmic: $f(x) = \log_a x$, inverse of a^x . x -intercept: $(1, 0)$. Asymptote: $x = 0$.

$$\log_a x = y \iff a^y = x$$

$$\log_a x > 0 \iff x > 1; \quad \log_a x < 0 \iff 0 < x < 1 \quad (\text{for } a > 1).$$

Modulus Function

$$f(x) = |x|: \quad f(x) = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Graph properties: V-shape, vertex at origin, never negative.

Equation $|f(x)| = a$: solve $f(x) = a$ or $f(x) = -a$.

Equation $|f(x)| = |g(x)|$: solve $f(x) = g(x)$ or $f(x) = -g(x)$.

Inequality $|f(x)| < a$ ($a > 0$): $-a < f(x) < a$.

7.3 Geometry and Trigonometry – Quick Reference**3D Geometry – Volume and Surface Area**

Solid	Volume	Surface Area
Cuboid	lwh	$2(lw + lh + wh)$
Cube (side s)	s^3	$6s^2$
Cylinder	$\pi r^2 h$	$2\pi r(r + h)$
Cone	$\frac{1}{3}\pi r^2 h$	$\pi r(r + l)$, $l = \sqrt{r^2 + h^2}$
Sphere	$\frac{4}{3}\pi r^3$	$4\pi r^2$
Hemisphere	$\frac{2}{3}\pi r^3$	$3\pi r^2$
Pyramid	$\frac{1}{3}Bh$	$B + \text{sum of lateral faces}$

Distance and Midpoint

2D Distance: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

3D Distance: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

Midpoint (2D): $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

Midpoint (3D): $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$

Vectors – Operations

Equality: $\mathbf{a} = \mathbf{b} \iff$ same magnitude and direction.

Magnitude: $|\mathbf{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$

Unit vector: $\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$

Addition/Subtraction: $\mathbf{a} \pm \mathbf{b} = \langle a_1 \pm b_1, a_2 \pm b_2, a_3 \pm b_3 \rangle$

Scalar multiplication: $k\mathbf{a} = \langle ka_1, ka_2, ka_3 \rangle$, $|k\mathbf{a}| = |k| |\mathbf{a}|$

Dot Product and Cross Product**Dot product:**

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}| |\mathbf{b}| \cos \theta$$

$$\mathbf{a} \cdot \mathbf{b} = 0 \iff \mathbf{a} \perp \mathbf{b}. \quad \cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}. \quad \text{Projection: } \text{proj}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2} \mathbf{b}$$

Cross product (3D):

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle$$

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}| |\mathbf{b}| \sin \theta. \quad \mathbf{a} \times \mathbf{b} = \mathbf{0} \iff \mathbf{a} \parallel \mathbf{b}.$$

$$\text{Area of parallelogram: } |\mathbf{a} \times \mathbf{b}|. \quad \text{Area of triangle: } \frac{1}{2} |\mathbf{a} \times \mathbf{b}|.$$

Lines in 3D**Vector form:** $\mathbf{r} = \mathbf{a} + t\mathbf{d}, \quad t \in \mathbb{R}$ **Parametric form:**

$$x = x_0 + ta,$$

$$y = y_0 + tb,$$

$$z = z_0 + tc$$

$$\text{Cartesian (symmetric) form: } \frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

Two lines intersect if the system of parametric equations has a consistent solution.

$$\text{Skew lines: not parallel, no intersection.} \quad \text{Angle between lines: } \cos \theta = \frac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{|\mathbf{d}_1| |\mathbf{d}_2|}$$

$$\text{Shortest distance (point to line): } d = \frac{|\overrightarrow{AP} \times \mathbf{d}|}{|\mathbf{d}|}, \quad A \text{ on line, } P \text{ external point.}$$

$$\text{Shortest distance (skew lines): } d = \frac{|(\mathbf{a}_2 - \mathbf{a}_1) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|}{|\mathbf{d}_1 \times \mathbf{d}_2|}$$

Planes in 3D**Vector equation:** $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}, \quad \mathbf{n} = \langle A, B, C \rangle$ (normal vector)**Cartesian equation:** $Ax + By + Cz = D$ **Parametric:** $\mathbf{r} = \mathbf{a} + s\mathbf{u} + t\mathbf{v}, \quad s, t \in \mathbb{R}$

$$\text{Angle between planes: } \cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1| |\mathbf{n}_2|}$$

$$\text{Angle between line and plane: } \sin \theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}| |\mathbf{n}|}$$

$$\text{Distance (point to plane): } d = \frac{|Ax_0 + By_0 + Cz_0 - D|}{\sqrt{A^2 + B^2 + C^2}}$$

Line-plane intersection: substitute parametric line into plane equation, solve for t .**Exact Trigonometric Values**

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undef.	0

2D Coordinate Geometry**Distance:** $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ **Midpoint:** $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$ **Gradient:** $m = \frac{y_2 - y_1}{x_2 - x_1}$ **Perpendicular bisector:** line through midpoint, gradient $= -\frac{1}{m}$ (where m is gradient of segment).**Area of triangle (coordinates):** $\Delta = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$ **Trigonometric Identities – Pythagorean**

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

Double Angle Identities

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Compound Angle Identities

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

Reciprocal Trig Functions & Identities

$$\csc \theta = \frac{1}{\sin \theta}, \quad \sec \theta = \frac{1}{\cos \theta}, \quad \cot \theta = \frac{\cos \theta}{\sin \theta} = \frac{1}{\tan \theta}$$

Reduction formulas ($n \in \mathbb{Z}$):

$$\begin{array}{lll} \sin(\pi - \theta) = \sin \theta & \cos(\pi - \theta) = -\cos \theta & \tan(\pi - \theta) = -\tan \theta \\ \sin(\pi + \theta) = -\sin \theta & \cos(\pi + \theta) = -\cos \theta & \tan(\pi + \theta) = \tan \theta \\ \sin(\frac{\pi}{2} - \theta) = \cos \theta & \cos(\frac{\pi}{2} - \theta) = \sin \theta & \tan(\frac{\pi}{2} - \theta) = \cot \theta \end{array}$$
Sine Rule, Cosine Rule, Area Formula**Sine rule:** $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$ **Cosine rule:** $c^2 = a^2 + b^2 - 2ab \cos C$, $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$ **Area of triangle:** $\Delta = \frac{1}{2} ab \sin C = \frac{1}{2} bc \sin A = \frac{1}{2} ca \sin B$ **Heron's formula:** $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$, $s = \frac{a+b+c}{2}$

General Solutions of Trigonometric Equations

$$\sin \theta = k \quad (k \neq \pm 1) \implies \theta = \arcsin(k) + 2n\pi \quad \text{or} \quad \pi - \arcsin(k) + 2n\pi$$

$$\sin \theta = 1 \implies \theta = \frac{\pi}{2} + 2n\pi$$

$$\sin \theta = -1 \implies \theta = \frac{3\pi}{2} + 2n\pi$$

$$\cos \theta = k \quad (k \neq \pm 1) \implies \theta = \pm \arccos(k) + 2n\pi$$

$$\cos \theta = 1 \implies \theta = 2n\pi$$

$$\cos \theta = -1 \implies \theta = \pi + 2n\pi$$

$$\tan \theta = k \implies \theta = \arctan(k) + n\pi$$

where $n \in \mathbb{Z}$ in all cases.

7.4 Statistics and Probability – Quick Reference

Measures of Central Tendency & Spread

Mean (ungrouped): $\bar{x} = \frac{\sum x_i}{n}$

Mean (grouped): $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$

Variance (population): $\sigma^2 = \frac{\sum (x_i - \mu)^2}{n} = \frac{\sum x_i^2}{n} - \mu^2$

Variance (sample): $s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{\sum x_i^2 - n\bar{x}^2}{n-1}$

Standard deviation: $\sigma = \sqrt{\sigma^2}, \quad s = \sqrt{s^2}$

Interquartile range: $\text{IQR} = Q_3 - Q_1$

Outliers: below $Q_1 - 1.5 \text{IQR}$ or above $Q_3 + 1.5 \text{IQR}$

Linear Transformations of Data

For $Y = aX + b$:

$$\bar{y} = a\bar{x} + b$$

$$s_Y = |a| s_X$$

$$\sigma_Y = |a| \sigma_X$$

Regression & Correlation

PMCC: $r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}} = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$

Regression line of y on x : $y - \bar{y} = \frac{S_{xy}}{S_{xx}}(x - \bar{x})$

Regression line of x on y : $x - \bar{x} = \frac{S_{xy}}{S_{yy}}(y - \bar{y})$

Coefficient of determination: $r^2 = \frac{\text{explained variation}}{\text{total variation}}$

Probability Axioms and Rules

$$0 \leq P(A) \leq 1$$

$$P(A') = 1 - P(A)$$

$$P(\emptyset) = 0, \quad P(\Omega) = 1$$

$$P(A \cup B) = P(A) +$$

$$P(B) - P(A \cap B)$$

Mutually excl.: $P(A \cup B) = P(A) + P(B)$

Indep.: $P(A \cap B) = P(A)P(B)$

Conditional Probability & Bayes' Theorem

Conditional: $P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$

Multiplication rule: $P(A \cap B) = P(A | B)P(B) = P(B | A)P(A)$

Independent check: $A \perp B \iff P(A | B) = P(A)$

Total probability: $P(A) = \sum_{i=1}^n P(A | B_i) P(B_i), \quad \{B_i\} \text{ partition of } \Omega$

Bayes' theorem: $P(B_k | A) = \frac{P(A | B_k) P(B_k)}{\sum_{i=1}^n P(A | B_i) P(B_i)}$

Discrete Random Variables

Expected value: $E(X) = \mu = \sum x_i p_i$

Variance: $\text{Var}(X) = \sigma^2 = E(X^2) - [E(X)]^2 = \sum x_i^2 p_i - \mu^2$

Standard deviation: $\sigma = \sqrt{\text{Var}(X)}$

Linear transformation: $E(aX + b) = aE(X) + b, \quad \text{Var}(aX + b) = a^2 \text{Var}(X)$

Sum of independent r.v.s: $E(X + Y) = E(X) + E(Y), \quad \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

Properties: $E(a) = a, \quad E(X + a) = E(X) + a, \quad \text{Var}(a) = 0, \quad \text{Var}(X + a) = \text{Var}(X)$

Binomial Distribution $X \sim B(n, p)$

PMF: $P(X = r) = \binom{n}{r} p^r (1 - p)^{n-r}, \quad r = 0, 1, \dots, n$

Mean: $E(X) = np$

Variance: $\text{Var}(X) = np(1 - p)$

Standard deviation: $\sigma = \sqrt{np(1 - p)}$

Conditions: fixed n , independent trials, two outcomes, constant p .

Poisson Distribution $X \sim \text{Po}(\lambda)$

PMF: $P(X = r) = \frac{\lambda^r e^{-\lambda}}{r!}, \quad r = 0, 1, 2, \dots$

Mean: $E(X) = \lambda$

Variance: $\text{Var}(X) = \lambda$

Conditions: events occur independently at a constant average rate in a fixed interval.

Grouped Data and Cumulative Frequency

Estimated mean (grouped): $\bar{x} \approx \frac{\sum f_i m_i}{\sum f_i}, \quad m_i = \text{class midpoint}$

Cumulative frequency polygon (ogive): plot cumulative frequency vs. upper class boundary. Median/quartiles from ogive.

Histogram: frequency density = $\frac{f}{\text{class width}}$; area of bar = frequency.

Normal Distribution $X \sim N(\mu, \sigma^2)$

PDF: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

Standardization: $Z = \frac{X - \mu}{\sigma}, \quad Z \sim N(0, 1)$

68–95–99.7 rule:

$$P(\mu - \sigma < X < \mu + \sigma) \approx 0.6827$$

$$P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.9545$$

$$P(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.9973$$

Inverse: given $P(X < x_0) = p$, find z_0 from $N(0, 1)$ table, then $x_0 = \mu + z_0\sigma$.

7.5 Calculus – Quick Reference

Limits – Key Results

L'Hôpital's rule: If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ yields $\frac{0}{0}$ or $\frac{\infty}{\infty}$:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Special limits:

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1, \quad \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta} = 0, \quad \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

Limit laws: $\lim(f + g) = \lim f + \lim g$, $\lim(cf) = c \lim f$, $\lim(fg) = \lim f \cdot \lim g$,
 $\lim(f/g) = \frac{\lim f}{\lim g}$ (if $\lim g \neq 0$).

Derivative from First Principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Complete Derivative Table

Function $f(x)$	Derivative $f'(x)$
c (constant)	0
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
a^x ($a > 0$)	$a^x \ln a$
$\ln x$	$\frac{1}{x}$
$\log_a x$	$\frac{1}{x \ln a}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\csc x$	$-\csc x \cot x$
$\sec x$	$\sec x \tan x$
$\cot x$	$-\csc^2 x$
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$
$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$
$\arctan x$	$\frac{1}{1+x^2}$
$ x $	$\frac{x}{ x } = \operatorname{sgn}(x)$

Differentiation Rules

Linearity: $(af + bg)' = af' + bg'$

Product rule: $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$

Quotient rule: $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$

Chain rule: $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$

Implicit & Logarithmic Differentiation

Implicit: Differentiate both sides w.r.t. x ; treat y as a function of x so $\frac{d}{dx}[y^n] = ny^{n-1}\frac{dy}{dx}$.

Logarithmic differentiation (for $y = [f(x)]^{g(x)}$):

1. Take $\ln$ of both sides: $\ln y = g(x) \ln f(x)$
2. Differentiate implicitly: $\frac{y'}{y} = g'(x) \ln f(x) + \frac{g(x) f'(x)}{f(x)}$
3. Solve: $y' = y \left[g'(x) \ln f(x) + \frac{g(x) f'(x)}{f(x)} \right]$

Second Derivative & Stationary Points

Second derivative test:

- $f''(x_0) > 0$: local minimum
- $f''(x_0) < 0$: local maximum
- $f''(x_0) = 0$: test inconclusive (use first derivative sign chart)

Point of inflexion: $f''(x_0) = 0$ and f'' changes sign at x_0 .

Complete Integral Table

Integrand	Antiderivative
$\int x^n dx$	$\frac{x^{n+1}}{n+1} + C, \quad n \neq -1$
$\int \frac{1}{x} dx$	$\ln x + C$
$\int e^{kx} dx$	$\frac{1}{k}e^{kx} + C$
$\int a^x dx$	$\frac{a^x}{\ln a} + C$
$\int \sin(kx) dx$	$-\frac{1}{k}\cos(kx) + C$
$\int \cos(kx) dx$	$\frac{1}{k}\sin(kx) + C$
$\int \sec^2(kx) dx$	$\frac{1}{k}\tan(kx) + C$
$\int \csc^2(kx) dx$	$-\frac{1}{k}\cot(kx) + C$
$\int \frac{1}{\sqrt{a^2 - x^2}} dx$	$\arcsin\left(\frac{x}{a}\right) + C$
$\int \frac{1}{a^2 + x^2} dx$	$\frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$
$\int \frac{1}{x^2 - a^2} dx$	$\frac{1}{2a}\ln\left \frac{x-a}{x+a}\right + C$
$\int \tan x dx$	$-\ln \cos x + C$
$\int \cot x dx$	$\ln \sin x + C$

Integration Techniques

Substitution: Let $u = g(x)$, $du = g'(x) dx$. Rewrite integral entirely in terms of u .

Integration by parts:

$$\int u dv = uv - \int v du$$

Use LIATE priority for choosing u : **L**ogarithmic, **I**nverse trig, **A**lgebraic, **T**rigonometric, **E**xponential.

Definite integration by parts:

$$\int_a^b u dv = [uv]_a^b - \int_a^b v du$$

Partial fractions: decompose rational function, then integrate term by term using

$$\int \frac{1}{ax + b} dx = \frac{1}{a} \ln|ax + b|.$$

Fundamental Theorem of Calculus**FTC Part 1:** $\frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x)$ **FTC Part 2:** $\int_a^b f(x) dx = F(b) - F(a), \quad F'(x) = f(x)$ **Properties:**

$$\int_a^a f(x) dx = 0 \qquad \int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$\int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx \qquad \int_a^b k f(x) dx = k \int_a^b f(x) dx$$

Area Formulas**Area under curve** ($f(x) \geq 0$):

$$A = \int_a^b f(x) dx$$

Area between two curves ($f(x) \geq g(x)$ on $[a, b]$):

$$A = \int_a^b [f(x) - g(x)] dx$$

Area between curve and y-axis ($f(y) \geq 0$):

$$A = \int_c^d f(y) dy$$

Signed area: If f crosses the x -axis, integrate $|f(x)|$ or split the integral.**Volume of Revolution****About the x -axis:**

$$V = \pi \int_a^b [f(x)]^2 dx$$

About the y -axis:

$$V = \pi \int_c^d [f(y)]^2 dy$$

About a horizontal line $y = k$:

$$V = \pi \int_a^b [f(x) - k]^2 dx$$

About a vertical line $x = h$:

$$V = \pi \int_c^d [f(y) - h]^2 dy$$

Average Value of a Function

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx$$

Kinematics – Calculus-Based

Position: $s(t)$; **Velocity:** $v(t) = s'(t) = \frac{ds}{dt}$; **Acceleration:** $a(t) = v'(t) = s''(t)$

Displacement: $\Delta s = \int_{t_1}^{t_2} v(t) dt$

Total distance: $D = \int_{t_1}^{t_2} |v(t)| dt$

Speed: $|v(t)|$

Differential Equations – Separation of Variables

If $\frac{dy}{dx} = f(x)g(y)$, separate and integrate:

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$

General solution contains an arbitrary constant C .

Particular solution found using an initial/boundary condition.

Growth/decay: $\frac{dN}{dt} = kN \implies N(t) = N_0 e^{kt}$ ($k > 0$: growth, $k < 0$: decay)

Newton's law of cooling: $\frac{dT}{dt} = -k(T - T_{\text{env}})$

Logistic: $\frac{dP}{dt} = kP \left(1 - \frac{P}{L}\right)$

Maclaurin Series

General formula: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$

Function	Maclaurin Series	Interval of Convergence
e^x	$\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$	all $x \in \mathbb{R}$
$\sin x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$	all $x \in \mathbb{R}$
$\cos x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$	all $x \in \mathbb{R}$
$\ln(1+x)$	$\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$	$-1 < x \leq 1$
$\frac{1}{1-x}$	$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots$	$ x < 1$
$(1+x)^k$	$\sum_{n=0}^{\infty} \binom{k}{n} x^n = 1 + kx + \frac{k(k-1)}{2!}x^2 + \dots$	$ x < 1$

Maclaurin Series – Operations

Differentiation: Differentiate the series term by term (within radius of convergence).

Integration: Integrate the series term by term (within radius of convergence).

Substitution: Replace x with an expression, e.g. $e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \dots$

Multiplication: Multiply two series and collect like powers of x .

Standard limits from series:

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

Absolute and Conditional Convergence

Ratio test: $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$.

$L < 1$: converges; $L > 1$: diverges; $L = 1$: inconclusive.

Alternating series test: If $a_n \rightarrow 0$ and a_n is decreasing, then $\sum (-1)^n a_n$ converges.

Absolute convergence: $\sum |a_n|$ converges $\implies \sum a_n$ converges.

Conditional convergence: $\sum a_n$ converges but $\sum |a_n|$ diverges.

Error bound (alternating): $|R_N| \leq |a_{N+1}|$

Comparison test: If $0 \leq a_n \leq b_n$ and $\sum b_n$ converges, then $\sum a_n$ converges. If $\sum a_n$ diverges, then $\sum b_n$ diverges.

Limit comparison test: If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$ with $0 < L < \infty$, then $\sum a_n$ and $\sum b_n$ both converge or both diverge.

Integral test: If f is positive, continuous, and decreasing for $x \geq 1$, then $\sum a_n$ and $\int_1^\infty f(x) dx$ both converge or both diverge.

Continuity and Differentiability

Continuity at $x = a$: $\lim_{x \rightarrow a} f(x) = f(a)$

Differentiability at $x = a$: $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ exists

Key fact: Differentiable $\implies$ Continuous. Continuous $\nRightarrow$ Differentiable.

Intermediate value theorem (IVT): If f continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$, then $f(c) = 0$ for some $c \in (a, b)$.

Eigenvalues and Eigenvectors

For an $n \times n$ matrix A :

Eigenvalue equation: $A\mathbf{v} = \lambda\mathbf{v}$, $\mathbf{v} \neq \mathbf{0}$

Characteristic equation: $\det(A - \lambda I) = 0$

Finding eigenvectors: For each λ , solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$

Diagonalization: $A = PDP^{-1}$, where D is the diagonal matrix of eigenvalues and P has corresponding eigenvectors as columns.

Trace: $\text{tr}(A) = \sum \lambda_i = \sum a_{ii}$

Determinant: $\det(A) = \prod \lambda_i$

Matrix Operations

Addition: $(A + B)_{ij} = a_{ij} + b_{ij}$ (same dimensions)

Scalar multiplication: $(kA)_{ij} = k a_{ij}$

Matrix multiplication: $(AB)_{ij} = \sum_k a_{ik} b_{kj}$; A is $m \times n$, B is $n \times p$, AB is $m \times p$.

Identity: $AI = IA = A$

Inverse: $AA^{-1} = A^{-1}A = I$. $(AB)^{-1} = B^{-1}A^{-1}$.

Determinant of 2×2 : $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Determinant of 3×3 : expand by cofactors along any row/column.

Inverse of 2×2 : $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$